Texts for Curriculum Renewal

Quantitative Argumentation

Making Sense with Numbers

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Preface

Notes to Pre-publication Reviewers

In 2023, I was challenged to develop two new core quantitative courses. This was part of a broader project to rethink the liberal arts curriculum from the ground up, and "core" meant something more stringent than usual. The courses would serve a requirement for *all* entering students to take exactly *those* courses. The courses filled a slot that, at almost all universities and colleges, is occupied by introductory statistics and pre-calculus/intro-calculus.

Students come from high school with widely varying expectations about where they will be placed in the mathematics curriculum, anywhere from Calc III to no place at all. This pair of courses was to be the common placement for *all* students, regardless of background.

The courses were to be closely aligned with a mission to foster critical thinking generally, as well as leadership in both the public and entrepreneurial realms. They were not to be a mere sample of college-level topics but a coherent introduction to methods genuinely useful in policy debates, commerce, and science.

This book is the text for one of the two courses. The paired course covers data science and statistics, including methods common to contemporary science as well as causal inference and the Bayesian framework.

Outline of follow-up data science course



A free, online version of the book will be published alongside this print edition. Exercises, discussion materials, and supplementary resources will also be available online.

Preface (for the to-be released version)

You, the reader, likely fall into one of three groups: students in college, students bound for college, or college instructors. A goal often claimed

by colleges is that, among other things, they foster the development of *critical thinking* skills. While often used merely as a marketing slogan or mission statement, many college faculty press for the explicit teaching of critical thinking to be taught explicitly. This book is allied is allied with that goal.

This book has a particular emphasis on critical thinking with quantitative information. But there are broad themes in the explicit critical thinking movement that are worth keeping in mind when considering how to teach critical thinking. Among these are 1) finding, evaluating, and using information effectively, 2) translating data and knowledge into formats that highlight key relationships, 3) building sound arguments based on evidence and knowledge, and 4) being aware of your own thinking patterns, including the development of new patterns and finding the appropriate level of skepticism and to your own and other's conclusions.

There is an argument to be made that traditional mathematics education serves these themes, but I think it is a weak argument. When it comes to (1), statistics is a much better fit than algebra. And while algebra does provide a *format for knowledge* as in (2), it is very limited, leaving no room for uncertainty or even data. The mathematical approach to *building sound arguments* is centered on proof and symbolic manipulation. But there are few settings in the real world where mathematical proof is applicable. Building sound arguments in practice often requires reasoning with incomplete information, uncertainty, and incomplete understanding.

The traditional approach is strongest in providing mechanisms for new patterns of thinking, for instance the idea of marginal or instantaneous change, which has many applications in describing and analyzing the world around us. But the structure of the traditional curriculum puts the mathematical thinking patterns most central to making use of information out of reach for most students.

This book seeks to provide access to the quantitative concepts and skills that can support critical thinking. A central theme is *modeling*, the process of building quantitative representations of real-world systems. We make use of a concrete framework for modeling that can be taught and learned. In this framework, the system components are **quantities**. As a concept, *quantities* are akin to *number* but involve additional information that provides basic facts about how different kinds of quantities are related to one another. There are almost no equations used. Instead, the relationships between quantities in models are represented with **functions** and **spaces**. Equally important are the ideas of **rates** and **rates of change**.

These modeling tools are also the elements of an important language to describe **uncertainty and risk** quantitatively. Increasingly, particularly over the last 40 years with the mainstream adoption of **Bayesian inference**, this language provides the means to use new **observations and evidence** to update our existing models.

Naturally, the point is not to build models for their own sake. They are built to allow us to extract information and conclusions from them that can inform our understanding of the real world and inform decision-making. Another important use for models is to foster communication between individuals and groups, allowing us to try to resolve through principled argument the uncertainties and disagreements that often paralyze or divide us.

Comments to instructors

The instructor interested in using this book or drawing from it to design a course is entitled to know why I have broken with many established conventions for teaching mathematics. The motivation has been a skeptical attitude: Are these conventions useful? What alternatives may be more effective for demonstrating that the mathematical topics are useful and provide new ways to think about important issues outside of mathematics?

Consider, for instance, the central role of algebra in the established curriculum. Mathematics instructors have generally been educated in the established curriculum and have come to think of algebra as an essential component of advanced mathematical thinking. We treat algebra as a way of explaining, and a framework for student evaluation. We often define student success in terms of skills in algebraic manipulation and whether the student is ready to take on a course in calculus. Some students, particularly the students we encounter in our university-level mathematics courses, have gained enough proficiency in algebraic manipulation to encourage us to use these skills as the foundation for further mathematical learning.

From the student point of view, however, the situation is different. A large fraction of students do not establish sufficient algebra skills to participate successfully in the university-level curriculum. Studies from professional organizations such as the Mathematical Association of America suggest that "a large fraction" means something like 80-90%. (To satisfy yourself that this is accurate, prompt an AI with this: "What fraction of college-bound students succeed in mastering algebra to an extent that they can use it effectively in university-level mathematics such as calculus?")

We temper our disappointment with this lack of success with assertions like, "Calculus is hard!" and "Not everyone is cut out for mathematics." And we make the most of our limited successes by accepting the contention that passing Calc I constitutes success.

Before accepting this state of affairs, it is worth questioning whether another approach to presenting mathematical concepts could lead to greater understanding of the quantitative concepts used throughout our economy, society, and science. My experience over 30 years of college teaching is that de-emphasizing algebraic manipulation can enable students to access not just anecdotes of mathematical analysis but the very topics that appear in the third and later semesters of the traditional calculus curriculum:

functions of multiple variables, partial derivatives (and their uses), linear algebra, and differential equations.

Naturally, I am not suggesting that the students following an alternative pathway are able to do the problem sets associated with linear algebra or differential equations. But they can understand how dynamics are described and trajectories calculated. They can create an accurate mental picture of what it means to "train" a model or to evaluate trade-offs in realistic scenarios.

What are the alternatives to algebra-focused instruction? We can start by noting that most of the topics taught in a high-school level algebra course are not relevant to the skills needed for quantitative analysis of systems and data. Factoring polynomials? Solving quadratic equations? Division of polynomials? You won't find them in this book. Better to build on foundational skills like understanding functions as a transformation of inputs to an output, and graphical interpretation of relationships described by functions.

Some, but not much, algebra is needed to understand rates of change, and an entirely adequate understanding can be developed from function graphics without formulas. Some, but not much, algebra is needed to work with dimensional analysis. The algebra of logarithms and e^t need not interfere with the development of understanding of magnitudes and the ever-increasing rate of change of exponential growth. Emphasizing functional notation rather than the idiosyncrasies of x^{-2}, $\sqrt{3}t$, or $\sin^{-1}(x)$ removes many pitfalls and connects better to the computational environments (even in spreadsheets) that students will encounter in their careers.

Famously, word problems are not a strong motivator of interest in algebra. Better to set the goal as building representations of real-world systems and using these representations to extract meaningful insights and make informed decisions.

A good approximation to the convention curriculum is "teaching algebra to reach calculus." The standard, and often encountered alternative is, "teaching algebra for its own sake." The alternative used in this book is to teach the fundamental ideas of calculus first, with a particular emphasis on why these ideas are important and useful. One such fundamental idea is "rates of change." This book introduces it early (chapter 4) and with only elementary algebra. The topic is used over and over again in various contexts before using the nomenclature of derivatives and accumulation:

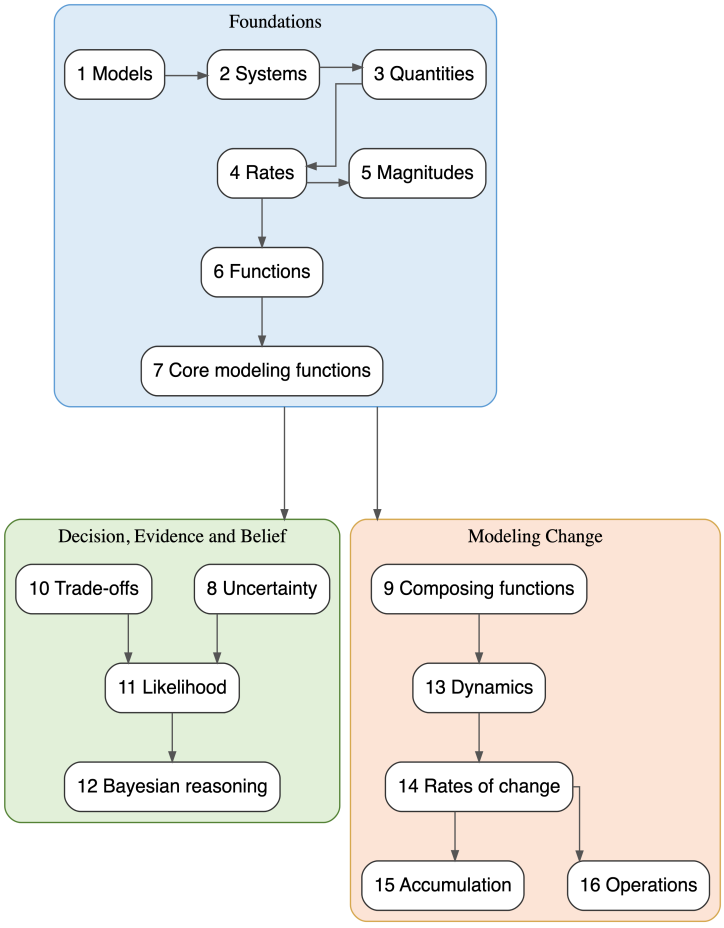
describing functions, dimensional analysis, handling trade-offs, dealing with data and hypotheses using likelihood, and modeling dynamic systems.

Overall, the style of this book is to *use* mathematical concepts first, and minimize or even exclude the bathwater of mathematical formalisms such as limits. The flow of topics emphasizes mathematical description rather than the precise definition of ideas in mathematical terms. The students working through the book will develop an intuitive and practical understanding of the concepts and an understanding that, with further work, they could use them in their own work and applications. And, those students who have succeeded in the conventional mathematics curriculum general find themselves with a greatly enriched understanding of the mechanisms and applied purpose of the mathematical tools they have previously learned.

- Daniel Kaplan, October 2026, Saint Paul, Minnesota

Sequencing of chapters

In general, each chapter builds on the previous chapters. But there is some flexibility; not every chapter uses concepts from *all* the previous chapters.



1. Argument and reasoning

The second word in *Quantitative Argumentation* comes from the simpler word "argument." In everyday speech, an argument is a quarrel. The root of "argument" goes back to pre-history, as documented, for instance, in the Mirriam-Webster dictionary. In the hypothesized proto-Indo-European language, *arg* meant shiny or bright. Latin inherited this sense of shining a light; the Latin *arguere* means "to make clear," "enlighten," "prove," or "demonstrate." Entering English in the 1300s, "argument" means a reason or proof used to settle a doubt. ☞**Argumentation**☜ refers to the process of constructing good arguments, a central theme of this book.

New vocabulary

- argumentation
- formal reasoning
- givens
- heuristic reasoning
- models
- quantitative models
- SIR model
- word or phrase

"Quantitative" for most people means using numbers. This is a good start, but "quantity" is a richer concept than "number." *Quantitative argumentation* is using the power of quantitative reasoning to enhance our broader, native reasoning ability. Those enhancements allow us to frame compelling arguments about diverse real-world situations, arguments that can foster consensus and guide useful debate with the goal of making informed decisions and effective actions.

This chapter examines reasoning and how we can combine different forms of reasoning to improve our understanding and decision making.

Reasoning transforms ☞**givens**☜ into new beliefs and understandings. "Givens" include facts, beliefs, assumptions, observations, knowledge, premises, perceptions, and impressions as we have them at the start of a reasoning process. Reasoning is the extraction of meaning from the givens.

The notation ☞**word or phrase**☜ highlights that the word or phrase has a technical meaning and is not just used in a casual, everyday sense. At the end of each chapter, make sure that you feel comfortable with the technical meaning so that it will come to mind when you encounter the word again. As a reminder, the technical words will be highlighted in the margin in each chapter that they appear.

Two forms of reasoning will be important to us. ☞**Heuristic reasoning**☜ is the sort widely used because of its efficiency and applicability throughout our interaction with the world. We could define it simply as common-sense reasoning, but "common" hides as much as it reveals. The second form of reasoning, ☞**formal reasoning**☜, has a much smaller range of applicability in the world but gives (when correctly applied) more certainty and the persuasiveness that goes along with certainty.

Combining the two types of reasoning gives us the benefits of both: the broad applicability of heuristic reasoning and the reliability of formal reasoning. ☞**Quantitative models**☜ bridge the gap between the two forms of reasoning. Understanding the process of modeling, and using models well and deliberately, is what quantitative argumentation is all about.

1.1. Reasoning from givens

Consider a familiar example: a rooster's crowing awakens you. You reason that it is dawn. One *given* is an observation: you heard a rooster sound. Another given is that you were awakened. But "dawn" is not a given. You did not directly observe dawn, it comes through reasoning from the givens.

Hearing a rooster sound is not the same thing as "a rooster is crowing." For instance, you might have been dreaming about the rooster noise. The observation itself is uncertain. This uncertainty propagates to the conclusion. So we don't know for sure that it is dawn.

What links "rooster sound" with the reasonable belief that it is dawn? The link is a small bit of knowledge: "Roosters crow at sunrise."

Common sense says that the other given—you were awakened—supports the belief that it is dawn. This supporting role does not come from either "a rooster is crowing" or "roosters crow at sunrise."

There are many words that describe the creation of a new belief or fact from givens: a deduction, an inference, a conclusion, a finding, etc. A logician—that is, someone who strictly enforces the rules of formal logic—would call a foul for claiming that "it is dawn" is a conclusion from hearing a rooster sound. She can argue, "That could be right, but it's not necessarily so." For instance, the crowing sound might have come from a social media feed, not a rooster. Without the rooster involved, the link "roosters crow at sunrise" would be irrelevant. It's also fair to say that, although roosters stereotypically crow at dawn, they don't *always* do so; there can be crowless dawns. And roosters sometimes crow at other times of day or night.

A simple metaphor illuminates the process of reasoning: a chain. Physical chains are made by joining elements together. Each link ties together the surrounding links. The link between "a rooster is crowing" and "it is dawn" is the knowledge or belief that "roosters crow at dawn."

Consider a richer example. You are driving on a highway. The brake lights on the car ahead of you flash on. Naturally, you touch your own brakes to slow down. This is a matter of experience and common sense. But let's treat it as a chain of links between the observation ("brake lights flash") and the action ("touch your brakes").

Here are three beliefs that might participate in the chain:

i. Brake lights imply rapid slowing down.
ii. A difference in speeds between two cars can lead to a collision, while matching speeds avoids collisions.
iii. To match a slowing speed, use your brakes.

There are overlapping uses of words or synonyms in these three statements such as "speed" and "slowing" or "difference" and "match". Now imagine a mental process that even unconsciously notes such overlaps and uses them to link the beliefs together. Seeing brake lights flash might, for instance, activate belief (i), which then activates belief (ii), which in turn activates belief (iii). In this way, a mental connection is made between observing brake lights and the action of touching your brakes.

We will leave the biology of this to neuroscientists. But computer scientists have long studied reasoning, especially machine reasoning. The last few years have seen breakthroughs in building such machines, resulting in the so-called "Large Language Models" (LLMs) or "Artificial Intelligence" (AI) that you may be using on an everyday basis.

LLMs rely on finding alignments between text segments and using such alignments to build reasoning chains. Later chapters outline how such machines work. For now, we will stick to the chain metaphor.

1.2. Heuristic reasoning

A heuristic is a useful reasoning link but is not guaranteed to be correct. The phrase "rule of thumb" captures the role of a heuristic. Before standardized rulers were readily available to craftsmen, the length of a thumb provided a handy way to mark of an inch in measurement. (Jokingly, I'll point out that people working in the metric system might have invented a "rule of ring finger" to measure a centimeter. But the metric system was introduced in the 1790s. By then, genuine rulers were commonplace.)

Some of the examples used in the previous section contain heuristics. For instance, "a rooster sound" is a pretty good indication that a rooster is crowing. "Roosters crow at dawn" may be a stereotype, but it explains a lot about roosters. "Brake-lights flashing on" is a sensible way to detect whether the car ahead is slowing rapidly, but the driver ahead might just be tapping her brakes to signal to you to stop tailgating.

Heuristic reasoning is serious business. Beyond providing a framework for "common sense," three Nobel prizes in Economics have been awarded for descriptions of real-world human and corporate reasoning in terms of heuristics: Herbert Simon in 1978, Daniel Kahneman in 2002, and Richard Thaler in 2017.

The topic of their work is often labelled as concerning "bounded rationality" or "behavioural economics."

Traditionally, what came to be called heuristic reasoning was often lumped under the idea of "logical fallacy" and sometime pejoratively called "irrationality." This hostile attitude stems from the misplaced view that *formal reasoning*, the sort taught in mathematics classes, is the only valid form of reasoning.

Heuristic reasoning aligns well with the actual process of much human reasoning. It is valid in the sense that it enables us to respond quickly and helpfully to inputs in the world around us. Some speculate that heuristic reasoning is a kind of reflex that evolved to support survival.

1.3. Formal reasoning

The word "argument" is a little misplaced when it comes to formal reasoning. More commonly, formal reasoning is called "deduction," "derivation," "logical reasoning," or "proof." But simple arithmetic is also a form of formal reasoning.

The distinguishing features of formal reasoning are, first, that a proposed link in the chain of formal reasoning is either correct or in error; true or false. Second, each link is based on a "law" or "rule" that is certain. Naturally, people make mistakes in attempting to follow formal reasoning rules; this is why students' math homework often has many red marks to indicate an error. If a reasoning chain has any fallacious links, the entire chain is fallacious. The conclusion at the end of the chain, even if a useful guide for action, is not yet demonstrably true. Many readers are likely familiar with attempted mathematics derivations which contain two fallacious links that happen to "cancel out" so that the conclusion at the end of the chain

is true and could have been demonstrated by an entirely correct chain of reasoning.

A correct application of formal reasoning is not subject to meaningful dispute, although people unskilled in formal reasoning may not see the correctness of the chain. Nobody should argue about whether $7 + 11 = 18$ nor that 6.83 is bigger than 3.79. Formal reasoning is completely reliable, except when humans make an error applying it.

In discussing heuristic reasoning, we have used "givens" as the name for the statements at the start of a chain in reasoning. Formal reasoning also has starting points conventionally called "premises," but "givens" works just as well. "

Other synonyms include "assumptions," "axioms," and "postulates.

A premise is a statement assumed to be true for the purpose of reasoning. It is a starting point for a chain of reasoning. The "formal" in "formal reasoning" means that it is whether a premise is a true or useful description of the world is technically irrelevant to the correctness of a chain of formal reasoning. For instance, Euclidian geometry is based on a set of premises that are indisputable within the framework of Euclidian geometry, for instance that "a straight line is the shortest path between two points." Denying this in a Euclidean geometry class is inadmissible. But there are other frameworks for geometry—to so-called non-Euclidean geometries—where curves can be the shortest path between points.

Despite the certain strengths of careful formal reasoning, it can give misleading results when applied to many real-world situations. The starting points for formal reasoning—premises—do not necessarily reflect the realities of the world. For instance, airplane pilots and mariners learn to deal with the spherical shape of Earth, travelling along "great circle" routes. But even if using spherical geometry correctly, the notion of "shortest distance" may not reflect the actual goal of navigation. It might be more appropriate to think about paths with the shortest journey *time* or *energy cost* which can depend on factors such as winds and currents.

In use formal reasoning to move from real-world *givens* to useful conclusions, we need a bridge the gap between formal premises and the real world. Formal reasoning is reliable but narrow in application. Heuristic reasoning is widely applicable, but not completely reliable.

1.4. Models as bridges to formal reasoning

We will use ☞**models**☜ as bridges between real-world givens and the power and reliability of formal reasoning. For now, consider models are used in everyday life, commerce, and science.

A model is a *representation* of an item, process, or setting of interest. We build models because they are easier to use or modify than the reality. For instance, a blueprint is a model of a building; model houses or apartments let people see a building at full scale without intruding on neighbors' privacy. The blueprint helps us to think about the situation: Is there enough space for the needed furniture; will windows bring in enough light? Medical research often uses animal models to study the physiology of disease progression or to test potential treatments ultimately intended for use in humans. And there are model ships, trains, trucks, and planes that represent large real-world objects and bring them within a child's reach.

For quantitative argumentation, appropriate models are built from mathematical components. Some of the tools and materials from which quantitative models are built will be familiar from your previous studies and experiences. Other modeling materials and common model-building patterns will be new to you. As with so many things, you learn at first from copying and modifying models from textbooks and other sources, that is, models already built by other people. With growing practice and confidence, you may find that you can create your own useful models. But knowing about the structure of quantitative models will help you make better sense of quantitative arguments, so that you can apply your own judgment and knowledge when deciding whether to accept or rebut an argument.

Once built, the model becomes a *premise*, the starting point in a chain of formal reasoning. The model, however, may not be a true description of the real-world situation. The model is a representation, not the real world itself. If the model is faulty, the conclusions from the formal reasoning chain may be faulty as well.

Building a model is *not* an exercise in formal reasoning. Instead, model construction uses our knowledge, understanding, and belief about real-world matters, together with heuristic reasoning. Many different models may suit a given situation. The choice among them depends on what we hope to accomplish, whether that be decision-making, prediction, or testing and augmenting our understanding of the situation. Appropriate models are therefore built to suit our real-world purposes.

The following chapters will cover concepts, techniques, and methods that have been found useful in constructing quantitative models. By analogy to carpentry, learning to model starts with exposure to important kinds of tools. For instance, you can drive a nail by using a rock, but you will likely bend the nail and not be able to remove it. A hammer is a purpose-designed tool for driving nails. The length of a hammer's handle, the shape of the business end of the hammer head, the curved claw at the other end of the head, and technique—how to hold the hammer, how to swing your arm—make the hammer effective. Probably you learned how to use a hammer by watching others and applying intuition. You did not need to study extensively the properties of wood or the metallurgy of nails.

A final point. Recall that we build quantitative models using heuristic reasoning. There is no such thing as proving the correctness of a model. Indeed, a famous truism is

> "All models are wrong. Some models are useful."

A model's usefulness depends partly on the modeler's purpose. Importantly, modeling is usually an *iterative* process. We build a model, then check the extent to which it captures the essential aspects of reality. When it does not, careful analysis can reveal where the model fails. Even if the first model proposed is not adequate, cycles of testing and improvement can eventually lead to a model that is useful for the purpose at hand.

1.5. Example: Modeling an epidemic

We begin with an accessible example. The features of an epidemic, such as COVID-19, are familiar. A pathogen, such as SARS-CoV-2, emerges from a typically unknown origin and infects a handful of people. The pathogen is contagious: infective people transmit it somewhat at random to susceptible people. Some of the susceptibles become infective and can transmit the virus to other people. At the same time, infective people recover, losing their ability to convert susceptibles into new infectives.

The verbal description in the previous paragraph is a model of an epidemic. The given mechanism is called "Susceptible, Infective, Recovered," or 👉**SIR**👈 for short. Later techniques, particularly in Chapter 9, translate the verbal description into a quantitative model that allows us to simulate the outbreak. The form of the model is bookkeeping that, based on today's

numbers of susceptible and infective people, calculates tomorrow's tally of new cases and recoveries.

The number of new cases is a function of the current size of the outbreak—that is, the current number of infectives—and the number of susceptible people who can come into contact with infectives. As infective people recover, the number of infectives goes down. However, this may be more than balanced by the new cases coming from the susceptible group in the population. As the susceptible group decreases in size, few new cases will develop and the epidemic will wane.

Note that bookkeeping is an instance of formal reasoning. In this example, the formal reasoning will be handled by a computer program which can calculate the day-to-day numbers of susceptibles, infectives, and recovered people as the epidemic evolves. The mechanism implemented by the computer program is strongly analogous to the way a river current, with its eddies and backflows, transports piece of flotsom, say, a fallen leaf. (Chapter 13 introduces the largely intuitive mechanics of such flows.)

Figure 1.1 shows an epidemic in a population of 100,000 people. It starts on day 0 with the arrival of the initial cases and progresses day by day for 50 days. The number of infectives (red) grows at first, then declines. The susceptible population (black) decreases in proportion to the number of infectives.

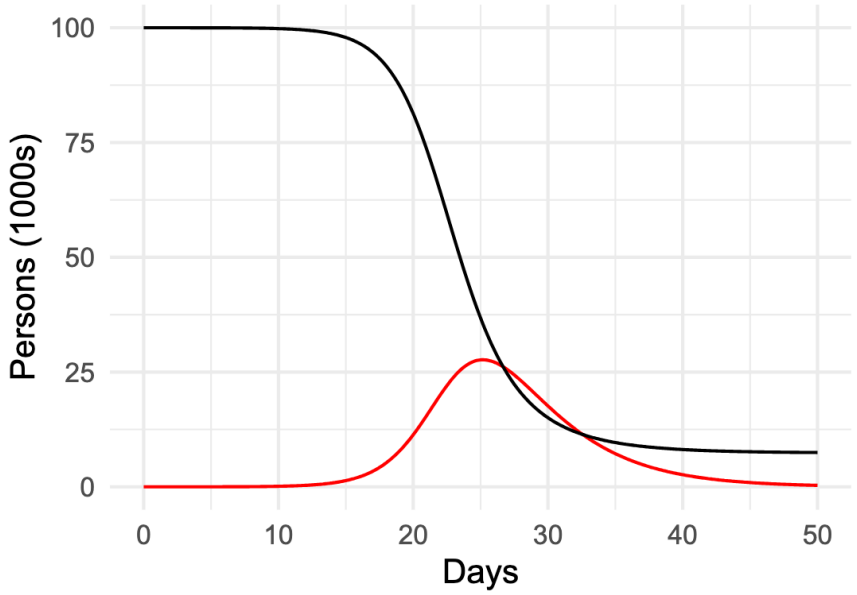


Figure 1.1.: The progression of a simulated epidemic among 100,000 susceptibles in the 50 days after the initial cases appear. The black curve shows the number of susceptibles; the red curve shows the number of infectives.

Use the interactive version of the model to explore scenarios for epidemics. For instance, you might be interested to know the relationship between the size of the outbreak and the number of initial cases. Similarly, you can explore whether social distancing is effective or the possible importance of isolating infectives. "Play" is a light-hearted word, but such activity is genuinely useful for drawing conclusions about the epidemic and the effectiveness of various interventions.

Interactive SIR model



2. Systems, dimensions, and data

Even in informal speech, we use the word "system" a lot. In this book, we will build and use models of … well, what? We could provide a several nouns to complete "models of …", for instance, "things," "objects," "processes," or "phenomena." In this book, we will use the abstract word 👉**system**👈 as a general way to refer to what is being modeled. The reason has to do with a specific meaning for "system" that will help us understand a very useful structure for models used in argumentation.

> **System**: *a framework or an organized structure with interconnected elements.* Or, from the Merriam-Webster dictionary: "a regularly interacting or interdependent group of items forming a unified whole."

This chapter sets up a framework for thinking about "elements" or "items" productively, with an eye toward what goes into a model used in quantitative argumentation. To give you an idea of what's coming, consider the following thought, which at this point may seem like a mere jumble of words

> "Systems are composed of dimensions and the relationships which connect those dimensions. In models, dimensions are represented by quantities."

In some cases, being consistent in terminology means avoiding some words with which you are undoubtedly familiar from mathematics classes. The issue here is that words like "variable" are used for so many unrelated purposes that they lose specific meaning. For instance, in statistics and data science, the noun variable refers to an aspect or quality of an object, for instance temperature or humidity or comfort. But in mathematics, "variable" has several distinct meanings, such as an "unknown" or a symbol, as in "x is a variable. Find x."

Vocabulary Review

- argumentation (1)
- models (1)

New

- data
- data frame
- dimension
- object type
- operationalization
- sample
- specimen
- value
- variable

Merriam-Webster



In the traditional curriculum, mathematics and statistics are treated largely independently and in different courses. Quantitative argumentation, however, encompasses both. Conflicts and ambiguities of meaning therefore need to be dealt with carefully and explicitly. Let's get started!

2.1. Dimension

There are several synonyms for scientist's or statistician's ☞**variable**☜" Familiar among these are "characteristic", "factor," and "aspect" as well as "trait," "feature," "attribute," and "property." In this book, all such synonyms will be grouped under the word ☞**dimension**☜. An example sentence: "Her personality has several dimensions: intelligence, honesty, anxiety, and diligence."

Chapter 3 uses the geometrical idea of "space." As you know, objects or spaces are often described using their "dimensions." For instance, a book has three spatial dimensions: width, length, and height. Of course, a book has other attributes as well, each of which we will call a dimension: weight, price, page-count, genre, and so on.

Many of the dimensions of a book are easily measured. Others, such as "genre," require not a measurement but a description. This is true as well for the dimensions of, say, personality. How do you measure honesty? Or anxiety? Or even intelligence?

Often, to construct a mathematical model, we will need to quantify a dimension such as anxiety, honesty, or genre. That is, we need to translate a dimension into some measurable quantity.

The way that the dimension (as in aspect or property) is translated is called ☞**operationalization**☜. We can operationalize length using a ruler. We can operationalize weight using a scale. Such operationalizations are obvious and familiar. But what's familiar to us in the 21st century was not always familiar. The measurement of temperature is a case in point. Before 1600, "temperature" was sensed by skin or the glowing color of hot metal, but not measured quantitatively. Around 1610, the "thermoscope" enabled visualization of temperature using the level of water in a tube. Not until more than a decade later was a quantitative scale—a ruler—added to the thermoscope, turning it into a thermometer. From a modeling perspective, a thermometer is a device that operationalizes the quantitative measurement

of temperature. Today, of course, thermometers are the obvious and familiar ways to measure temperature.

Age and height are also familiar dimensions. We operationalize height, for instance, by marking the top of a person's head on a wall or door frame, then using a ruler to measure (that is, "quantify") the distance from the floor to the mark. We operationalize age—this will be obvious—by calculating the time difference between birth and the moment of measurement. (More commonly, we ask the person!)

Sometimes an operationalization is technical, obscure, and to some extent arbitrary. For instance, the economic variables "cost of living" and "poverty level" have official operationalizations in the US. An officially defined, standardized "market basket" serves as the basis for calculating the cost of living. The government defines poverty level by a formula that involves the cost of living, family size, and composition, and is based on pre-tax income and does not depend on location, even though housing costs, food, energy, and such vary markedly from one place to another.

In a quantitative model of a system, each dimension of the system is operationalized into a quantity that measures that characteristic. Being attentive to the details of the operationalization can be an important component of critical thinking.

Quantities have 👉**values**👈, that is, a specific numerical level along with the appropriate unit. (See Chapter 3.) For many system dimensions, the value of the operationalized quantity will differ across contexts, say, from hour to hour or place to place. For example, as I write this, the outdoor temperature is 52°F, as operationalized by the US Weather Service from a thermometer in a ventilated, shielded box at a specific location in my city. Values can change; the temperature tomorrow will likely be different. In reading "the temperature tomorrow will be different," the underlying idea is not that the variable or operationalization will change—the Weather Service will keep doing what it has done for many years. Instead, the idea of "different" is that the value of the quantity will be different.

2.2. Example: Operationalizing comfort

Imagine a hot, humid, sultry summer afternoon in the US. The relative humidity is 90%, the temperature 95°F (equivalent to 35°C). Sultry weather makes us uncomfortable.

It is helpful to provide people with a quantitative scale of comfort so they can make decisions about outdoor activities. The context is a system composed of three quantities: humidity, temperature, and comfort. (There may be other quantities we will eventually need to tell a complete story—wind speed, for instance—but for simplicity, we work with just the three.)

It may be evident that humidity and temperature are inherently quantities, but what about comfort? In modeling, we often deal with concepts that are not inherently quantitative and may be vague. There are many ways to operationalize comfort to a quantitative scale, each imperfect in its own way. A favored operationalization of weather-related comfort is the "feels like" scale, known as the "heat index." The heat index derives from a technical model of convective heat transfer from the skin. The heat index is not a comprehensive operationalization of the general concept of comfort. Still, the heat index captures an important aspect of comfort quantitatively.

Another example of an imperfect, but practical operationalization is the pain-rating scale shown in Figure 2.1.

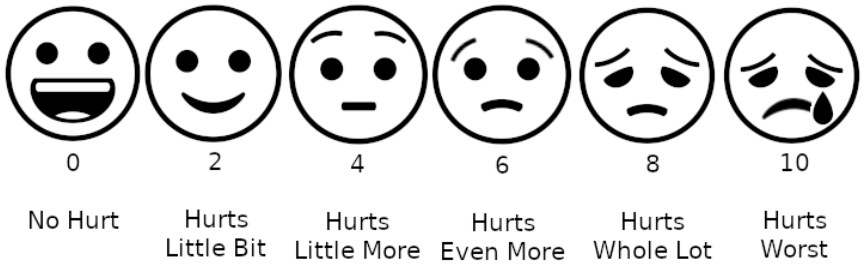


Figure 2.1.: The Wong-Baker FACES pain scale, which turns a subjective feeling into a quantitative indicator.

A common criticism of quantitative analysis is that any particular quantitative operationalization is imperfect and incomplete. This criticism can be valid and serve as a prompt to refine the operationalization and the analysis method, but it by no means invalidates quantitative analysis. The test of operationalization is not whether it is perfect, but whether it is good enough to contribute meaningfully to the purposes for which we are conducting the quantitative analysis.

2.3. Data

We are using "givens" to refer to the entire scope of facts, knowledge, and belief about the world. Some people informally use "data" here. Indeed

"Data" is rooted in a Latin root: "to give" or "given." But for us 👉**data**👈 will stand for recorded facts, findings, and measurements.

A broad, modern definition of "data" might be "anything that can be stored on a computer," the sort of material mined when training AI systems. Such data can take many forms: video, audio, maps and other images, text. Even the contents of a personal diary recording fleeting thoughts and perceptions is data in the can-be-stored-on-a-computer sense.

A 👉**data frame**👈 is a particular, simple but highly structured organization of data that is similar to a spreadsheet format. Data frames are at the core of much data storage and use in science, government, industry, finance, and so on. Software for handling data, especially databases that encompass multiple data frames, is the basis of a hundreds-of-billions-dollar segment of the worldwide economy and will likely continue to be so despite the explosion in AI usage.

It pays to be familiar with the specific structure of data frames. Even if you don't work with data, the vocabulary used to describe data frames corresponds to a set of concepts that are ubiquitous in quantitative argumentation. For reference, Figure 2.2 shows a schematic diagram of a data frame. The data frame itself is just the text and numbers in the rectangular array. The contents of the data frame are measurements made in the 1880s of the heights (in inches) of adult children, along with the child's sex and the heights of the child's mother and father.

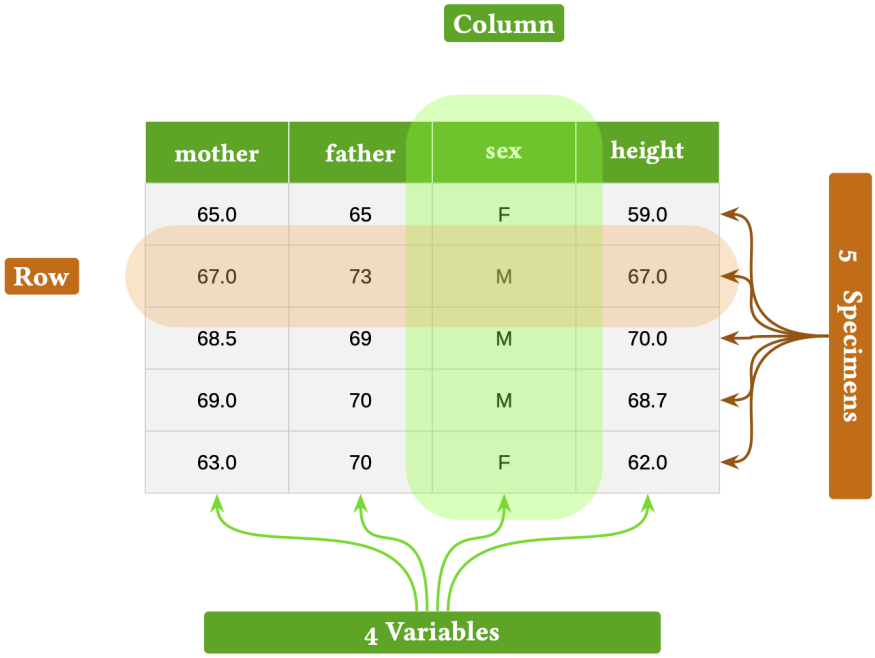


Figure 2.2.: A schematic picture of a small data frame.

A data frame is a rectangular array of cells, organized into rows and columns. Each row represents the records for one specific object or event, which we will call a ☞**specimen**☜. In Figure 2.2, the specimen is an adult child. The five rows of Figure 2.2 correspond to five different children.

All the objects/rows/specimens in a data frame should be of the same kind, which we will call the ☞**object type**☜. So, the object type in Figure 2.2 is an adult child. The object type in any data frame depends on the setting and purpose for which the data frame is being collected. For instance, a data frame used by a real-estate agent might have an object type of individual houses for sale. A data frame used by an astronomer might have an object type of planets. The object type in a data frame used by a grocery store would be a product or object sold by the store.

In statistics, it is conventional to use the term ☞**unit of observation**☜ instead of "object type." We don't follow this convention in this book because the word "unit" will be used in an utterly different sense as in "meters" or "miles per hour."

A data frame can record several or many different dimensions for each object. In Figure 2.2, the dimensions are the child's sex and height, as well as the heights of the child's parents. Almost universally, the word "variable" is used for the columns of a data frame. We will honor that convention. For instance, the data frame in Figure 2.2 has four variables: mother, father, sex, and height.

At the intersection of each variable and each row, such as the single "M" highlighted in both green and orange in Figure 2.2, is a ☞**value**☜. In a

data frame, the values in each variable must be recorded as the same type. For instance, all the values under mother are inches of height. Similarly, the values under sex are all either "M" or "F." (This narrow set reflects the social conventions of the 1880s.)

It will suffice for our purposes to emphasize two different types of variables: quantitative variables versus categorical variables. You can think of a categorical variable as one whose values come from a finite set of words or letters. An ID (such as the SKU/bar-code of a grocery item or a registrar's unique identifier for individual students) is a categorical variable even if it consists only of digits. Other types of variables in widespread use are images (such as a picture of the grocery item), videos, dates, and timestamps, etc.

The entries in each row of a data frame record the values of the variables contained in the data frame for an individual specimen. Typically, a data frame contains many specimens, such as the several children displayed in Figure 2.2. The complete 1880s data frame covers 898 specimens, that is, 898 children. The small data frame is a 👉**sample**👈 from the original larger data.

⚠ Sample vs example

In everyday English, a "sample" is sometimes used synonymously with "example." For instance, a grocery store staffer might give you a sample of some product, like a small cup of apple cider. Such a sample is intended to give the customer an idea of the quality or taste of the product. The paper cup of apple cider might also be called an "example" of the product, although this is not idiomatic in normal English. In contrast, nobody would use the word "example" to refer to a blood sample taken at a doctor's office.

In this book, however, we use "sample" in its statistical sense. If you were studying fruit juice, say, you might select some bottles of juice from the larger collection on the grocery shelf. This would be a sample from those bottles.

In statistics, a sample of size one, for instance, a single bottle pulled off the shelf at random, is not a useful sample. To be useful, samples must contain more than one specimen, usually many more. This is because a central question in all statistical inquiry is the extent to which the sample can be considered representative of the larger population from which it was taken. The logic of "Is the sample representative?" is

based on comparing *between specimens* in the sample. With just one specimen, such comparison is impossible.
In medical research, samples typically consist of dozens or hundreds of people, although many situations require much larger samples of, say 100,000 people.

It's easy to imagine how to generate a sample from a large data frame. Scroll through the large data frame, pointing now and again to an individual specimen and copying the values from the different dimensions over to the smaller data frame that will eventually contain the sample.

The special word "census" is used when the data frame gives a complete enumeration of all possible specimens from the entity.

The specimens in every data frame are often a sample from a potentially larger entity. For instance, you might record the dimensions of color, shape, size, location, and date of acquisition of seashells in the data frame that contains the catalog of your hobby seashell collection.

You might have assembled your collection in the usual way, by wandering along a beach and picking up the shells that caught your eye. The abstract word for such a process is 👉**sampling**👈. It is helpful to have a term for the larger entity from which each specimen was sampled. The word "population" is often used here. Even though the root of "population" is the same as for the word "people," when it comes to data frames we use "population" to refer to any source of specimens. For instance, we might have a population of stars or a population of events such as daily weather or forest fires.

In this book, when we say that a population has dimensions, we mean the set of aspects or qualities of the elements in the population. Those dimensions may be related to one another, for example the relationship between age and height of young children. But there is something subtle going on. When it comes to data, any detected relationships are collective properties across all the specimens in your sample. You can only discern a relationship from data when you have multiple specimens to compare to one another.

3. Quantities

In Chapter 2 we used the word "dimension" to stand for any of the properties or attributes of an object or system. For example, among the many dimensions of a person are age, eye color, sex, strength, health, and so on. We also discussed "operationalization," which mean the process or procedure for converting a dimension into a quantity. That is, operationalization tells how to *measure* a dimension, turning it into one or more quantities. For instance, we might operationalize "health" using measures such as blood pressure, cholesterol level, blood-sugar level, and so on.

I would be understandable to think that "quantity" is simple another word for "number." Quantitative argumentation, however, needs to take careful account of the *kind* of quantity being considered. For instance, a quantity of 5 could be 5 seconds, 5 dollars, 5 miles, or 5 gallons. The number 5 is the same in each case, but the *kind* of quantity is different. This is likely obvious to you, but there is more to be said.

In this book, we will use "*quantity*" within a specific framework of surrounding concepts. Mastering this framework will make clear many things that are confusing to untrained people, which is to say, most everybody. The framework is not complex, but it requires some discipline and precision of thought.

Vocabulary Review

- dimension (2)
- operationalization (2)
- value (2)

New

- analysis
- base dimensions
- base units
- capita
- controlling for
- conversion factor
- coordinate plane
- derived dimensions
- derived units
- dimensionless
- flavors of one
- number line
- partial change
- per
- point
- pure number
- quantity
- si
- space
- units
- with respect to
- worth

3.1. The quantity framework

☞**Quantity**☜ and its framework concern anything measurable in a standardized way. Here are a few examples of *standardized measures*: foot, meter, degrees F, pints, hours, acres, miles, kilograms, pounds, dollars, miles per hour. Scientists and engineers, craftsmen, business people, health-care workers, and others learn more specialized standard units, such as joules, amps, coulombs, moles, watts, kilowatt-hours, board feet, and many, many more. They also encounter words for the *kind of thing* being

The word ☞**per**☜ is used technically. See Chapter 4.

measured: length, time, mass, weight, energy, power, volume, flow, speed, costs, prices, and such.

"Standardized measures" are better known, in both technical and everyday speech, as ☞**units**☜. Our technical definition of quantity is a combination of two or (optionally) three elements:

Definition 3.1. A *quantity* is composed of these elements:

i. A **pure number** such as 7 or 2.15 or 6.1×10^{-8}
ii. A **unit** such as "meter," "bushel," or "acre."
iii. (Optionally) additional words describing the stuff more specifically, for instance, "a bushel *of apples*" or "a pint *of blueberries*."

The primary use of quantities is to state or record *how much* there is of something. Most people are well acquainted with this use, having learned it about the same time as they learned arithmetic. As a reminder, we turn to a nice book from 1931, *Arithmetic for the Practical Man* by J.E. Thompson. He wrote:

> Common measurements are of several kinds .... The chief of the common measurements are those of
>
> i. Length
> ii. Surface, or area
> iii. Volume, or capacity
> iv. Weight, or force of gravity
> v. Time
> vi. Angles, latitude and longitude
> vii. Temperature, or intensity of heat
> viii. Money, or value

The "several kinds" of things in Thompson's list are mutually ☞**incommensurate**☜: They cannot be arithmetically compared to one another or added to (or subtracted from) one another. Is a mile larger than a cubic centimeter? Invalid comparison. Is 212°F bigger than 45 degrees of latitude? Don't go there, even though both units are named "degrees." How much is 12 seconds plus 4 dollars? No.

3.2. Base units

A *unit* is a way of quantifying "how much." To say, "this video covers 30" involves a number but not a "how much." For that, we have to specify what the 30 refers to. Thirty seconds or 30 minutes are both reasonable durations for videos, but one can also find videos that cover high-speed events, like the flare-up of a match, with much of the action occurring over 30 milliseconds. At the other extreme, this time-lapse video of a bean sprouting covers about 30 days. Milliseconds, seconds, minutes, hours, days, months, years, centuries, millennia, etc. are different standard units of time.

Video of flare-up



Units are part of everyday life: liters, cups, acres, degrees C, miles, cubits, meters, microns, feet, carats, drams, and so on. The quantitative thinker must have in mind, for each quantity under consideration, what the units of measure are. The particular units employed vary from country to country. Different professions use different units for the same kind of quantity. For instance, a jeweler uses a carat as a measure of mass, while a shipping company might use a ton (or, slightly different, a "tonne").

Time-lapse video of a bean sprouting



Le Système international d'unités, usually referred to as 👉**SI**👈, is the primary official definition for units. SI distinguishes between two classes of units: seven 👉**base units**👈 and many, many 👉**derived units**👈. Examples of the kinds of things measured in derived units: force, energy, pressure, power, and dose of radioactivity. Each base unit (see Table 2.1) provides a "letter" in a seven-letter alphabet for describing all SI units. The different base units cannot be defined in terms of one another, but derived units can.

Information about the International System of Units (SI)



Kind of quantity	Unit name	Unit symbol	Dimension symbol
time	second	s	T
length	meter	m	L
mass	kilogram	kg	M
temperature	kelvin	K	Θ

Table 2.1: The four (of seven) base units in SI most often encountered in everyday life. For completeness, note that the other three base units are the ampere (electric current), the mole (amount of a substance), and the candela (luminous intensity).

The SI base units have a strongly scientific or technological flavor. But quantitative reasoning is important in other domains as well. Consequently, this book frequently use two more base units that are not covered by SI: people and 👉**worth**👈.

We count people not by their mass or volume, but *by the head*. The Latin word for head is ☞**capita**☜, and the phrase *per capita* is common in quantitative descriptions of economies, educational systems, and the like. There are alternatives to *capita* when counting people. It depends on the context: deaths, births, passengers, students, children, patients, victims, fatalities, and so on.

"Subjective": relating to a personal viewpoint.

As for "worth," this is a rich and subjective concept. And there are many kinds of worth. The notion of worth that is most widely considered in quantitative reasoning is, effectively, monetary worth. The meaning of a given quantity of money is what it can buy, which changes from place to place and time to time. A second is a fixed amount of time, wherever and whenever, but a dollar corresponds to different amounts of stuff depending on where and when it is spent.

Economists go so far as to identify inflation-adjusted prices as "real prices." The price actually paid is called a "nominal price."

Economists, to deal with changing value over time, often measure money in terms of inflation-adjusted currency, for instance, "☞**constant dollars**☜." As regards place-to-place variation, especially between countries, a concept called "☞**purchasing power parity**☜" (PPP) is applied. This operationalization of money is not perfect, nor can it be, considering the broad set of contexts in which money is a factor. However, the quality of a model reflects its *utility* in providing information. Not including money because of the imperfection of its operationalization would undercut such utility.

Theoreticians have created a base unit for money called the "International Dollar." In practice, however, the US Dollar (USD) is widely used, augmented by terms such as "constant 2020" or "adjusted for PPP."

Table 2.2: Three dimensions not included in the SI but which are useful for general quantitative reasoning.

Kind of quantity	Unit name	Unit symbol	Dimension symbol
people	capita	person	C (for "capita")
money	USD	USD	W (for "worth)
pure number	"dimensionless"	none	[1]

We have made up the dimension symbols C and W. There is no standard in general use, though perhaps there should be.

3.3. Derived units

Arithmetic multiplication or division of base quantities produces many of the quantities we deal with regularly. The most familiar examples involve division: a unit for velocity is "miles *per* hour"; a unit for flow of a liquid (as in the output of a shower head) is "liters *per* minute." The word ☞**per**☜ is shorthand for "divided by."

Multiplication is also extremely common. An *area* is a length multiplied by a length. If the lengths are in inches, the area is in "square inches," that is, an inch multiplied by an inch. The units "square mile" or "square foot" are the same, with the base unit inch replaced by mile or foot, respectively. Similarly, volume is area multiplied by length. For instance, a liter is 10 cm $\times$ 10 cm $\times$ 10 cm, or, according to convention, 1000 cm^3 (pronounced "cubic centimeters").

We use the term "**derived units**" for the kinds of units created by multiplication or division of the base units. A *meter* is a base SI unit; a *cubic meter* is a derived unit. The SI derived unit for velocity is meters per second. Both meters and seconds are SI base units. Meters-per-second, like miles-per-hour, is a *derived unit*.

"Per" is used in units to signify division. In contrast, there is no word to indicate multiplication. Instead, the two base units involved are merely named one after the other, as in "man-hours" (a unit for effort) or "acre-feet" (a unit of volume used in hydrology).

We often use simple names for units consisting of complicated assemblies of multiplied or divided base units. For example, a watt (unit symbol W) is an SI derived unit of power. In terms of SI base units, it is kg $\cdot$ m^2 per s^3.

3.4. Standard dimensions

Since the unit is an essential component of any quantity, quantitative reasoning requires care in working with units. A matter of considerable importance when working with units is ☞**commensurability**☜, the ability to measure by the same standard. For example, all of these units refer to *volume*: liters, cups, cc, pints, gallons, bushels, barrels, cubic meters, and acre-feet. These volume units are *commensurate*, meaning that one can measure the same quantity using any of them. It might make the most sense

The word *commensurate* is from the Latin *com* (together) and *mensurare* (to measure).

to measure the volume of water in a swimming pool using cubic meters—the volume is 2500 m^3—since the meter is the SI base unit for length. (A standard pool is 50 m long by 25 m wide by 2 m deep.) Nevertheless, it is possible—if frivolous—to measure the pool's volume in cups, bushels, or acre-feet. (Just FYI, there are 10,567,000 cups or 70,943.98 bushels or 2.026786 acre-feet in a pool.)

Only when the quantities are commensurate can we add or subtract them. For instance, the sum of 3 cups and 0.5 gallons is a valid quantity equivalent to 2.60247 liters, 11 cups, 0.6875 gallons, etc. However, "3 cups plus 20 seconds" makes no sense, since volume and time are not commensurate.

☞**Dimensional analysis**☜ is an important method to keep track of how quantities are related. Tables 2.1 & 2.2 show the symbol for the dimension for each basic unit. These are called ☞**base dimensions**☜. Any unit that is commensurate with a base unit—for instance, mile and yard are commensurate with the base unit meter—shares that base unit's dimension.

Each quantity expressible in a derived unit has a corresponding ☞**derived dimensions**☜. Just as derived units are composed by multiplication and division, derived dimensions are composed by multiplying and dividing the basic dimensions. For instance, the dimension of *surface area* is $\mathrm{L} \times \mathrm{L}$. Some derived dimensions are complicated, for example, the dimension of *power*: $\mathrm{M\,L^2\,T^{-3}}$. When writing such dimensions, we tend to use exponents when a basic dimension appears more than once, such as the -2 on T^2.

Just looking at $\mathrm{M\,L^2\,T^{-3}}$ does not easily call to mind the word "power." It helps to have a notation for translating names into the corresponding composition of base dimensions. For this purpose, we use square brackets. For instance, we can write the dimension of *power* using square braces: [power]. In this shorthand, [power] means the same thing as $\mathrm{M\,L^2\,T^{-3}}$. Similarly, if we are working with a derived unit like miles per hour (mph), we will write that unit in square braces when referring to the corresponding dimension. For instance, $[\mathrm{mph}] = \mathrm{L\,T^{-1}}$.

In addition to the six dimensions listed above—T, L, M, Θ, C, and W—it will be helpful to add one more: the dimension of a ☞**pure number**☜. A pure number is a special kind of quantity in that it carries *no unit designation*. 7 is a pure number, as is π and most of the numbers you considered in high-school mathematics. All pure numbers share the same dimension, which we will write by putting the number 1 in square brackets, like this, [1].

All the base dimensions have names—e.g. time, worth, capita, length—and [1] is certainly an important and useful dimension deserving of a name. The word equivalent for [1] is "☞**dimensionless**☜." Admittedly, using "dimensionless" as a type of dimension takes some getting used to.

Pure numbers are not the only kinds of dimensionless quantities. Importantly, angles are dimensionless but are measured in units such as degrees or radians. An angle of 90 degrees is the same as an angle $\pi/2$ radians. Different units, same quantity. Students may recall from trigonometric studies the definition of the unit "*radian*" as the L along the arc of a circle divided by the L of the radius, as in Figure 3.1. A length divided by another length is dimensionless.

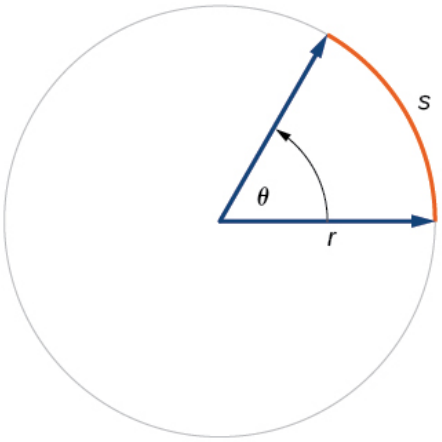


Figure 3.1.: An angle (denoted θ here) is the ratio of the length s of the arc subtended by the angle to the radius r of the circle centered on the vertex of the angle: $\theta = s/r$. [Image source: UTSA Libraries and Museums]

Steep roads often have "grade markers" indicating how steep they are. In the US, interstate highways may not be steeper than 6%. The *steepness* is operationalized as the change in elevation divided by the horizontal distance travelled. Driving the steepest mile along Lincoln Gap Road in Vermont (USA) involves a net change in elevation of about 800 feet. The road grade is therefore 800 ft/mile. We can write the ratio more explicitly in terms of the two quantities involved:

Image source

$$\frac{800\,\text{ft}}{1\,\text{mile}} = \frac{800\,\text{ft}}{5280\,\text{ft}} = \boxed{\frac{800}{5280}}\,\boxed{\frac{\cancel{\text{ft}}}{\cancel{\text{ft}}}}^{1} = 0.1515\,. \tag{3.1}$$

In the boxed quantity in Math expression 3.1, we have converted 1 mile to

its equivalent: 5280 feet. The result is a unit ft-per-ft, which is simply the number 1. So the grade is $800/5280 \approx 0.15 = 15\%$. Percent is the standard dimensionless unit to report grades. ß

3.5. Unit conversion

To facilitate addition, subtraction, and comparison of quantities, we often need to convert them to a common unit, as in Math expression 3.1. For conversion, the units must be commensurate. Except for temperatures, we convert between commensurate units by multiplication with a ☞**conversion factor**☜.

The units for representing [temperature] are commensurate: °C, °F, °K. However, conversion among them involves not just multiplication but also addition or subtraction. The reason has to do with the meaning of zero. While 0 s means no time at all and 0 m means no length at all, the commonly used temperature units differ on what zero means. In degrees C, zero is the freezing point of water. In degrees F, zero marks a temperature well below freezing. In °K, zero means no thermal energy, corresponding to -273.15°C.

Every conversion problem involves a "from" quantity and a "to" quantity. In converting 12.5 miles to meters, the "from" quantity is in units of miles and the "to" quantity is in meters. The conversion factor *always* multiplies the "from" quantity. The result of the multiplication is the "to" quantity. The appropriate conversion factor for 12.5 miles to meters is 1609.34 m/mile. The conversion is

$$\begin{aligned} 1609.34\,\frac{\text{m}}{\text{mile}} \times 12.5\,\text{miles} &= 1609.34 \times 12.5 \boxed{\text{m} \times \frac{\cancel{\text{miles}}^{\,1}}{\cancel{\text{miles}}}} \\ &= 20116.75\,\text{m}. \end{aligned} \tag{3.2}$$

The key to the conversion is in the box of Math expression 3.2. The "mile" appears on both the top and bottom of the quantity. This cancels out the two occurrences of "mile" to leave us with m.

Conversion factors are always dimensionless; that is, [any conversion factor] = [1]. Even so, keeping the *units* of a conversion factor in mind substantially reduces the possibility of confusion or error. We call conversion factors ☞**flavors of one**☜ because every conversion factor is equivalent to the number 1. Every conversion factor is the ratio of *two equal quantities*, although the quantities are expressed in different units.

To illustrate, we will rewrite statements like "an inch is 0.0254 m" as flavors of one. Naturally, the word "is" in "1 inch is 0.0254 m" means that "1 inch" is the same quantity as "0.0254 m." Dividing one by the other gives:

$$\frac{1\,\text{inch}}{0.0254\,\text{m}} = 39.37\,\frac{\text{inches}}{\text{m}} = \text{flavor of } 1\ .$$

Similarly, "1 day is 86,400 s" can be translated to a ratio with "1 day" on top and "86,400 s" on the bottom:

$$\frac{1\,\text{day}}{86400\,\text{s}} = 0.0000157407\,\frac{\text{days}}{\text{s}} = \text{flavor of } 1\ .$$

Writing conversion factors as ratios *with units* signals clearly which is the "from" quantity and which is the "to" quantity. Not writing the units ratio explicitly invites confusion. For example, when converting 1 second to days, does one multiply by 86,400 or divide by 86,400? Remember that the conversion factor always *multiplies* the "from" quantity. Therefore, the correct conversion factor will be the flavor of one with the "from" unit on the bottom. Accidentally flipping the top and bottom of the conversion factor results in large errors.

Both $39.37\,\frac{\text{inches}}{\text{m}}$ and $0.0000157407\,\frac{\text{days}}{\text{s}}$ are quantities that are flavors of one. The numerical parts of the quantity—39.37 and 0.0000157407—are not 1, but the units—for instance, $\frac{\text{days}}{\text{s}}$—balance this out to give an overall "unit-full," dimensionless quantity that equals 1.

3.6. Change as a quantity

You walk to work on a crisp, autumn day. On your return home, it's uncomfortably warm. What happened? The temperature changed. It was 41°F in the morning and 75°F in the late afternoon. That's a **change** of 34°F *from* morning *to* afternoon.

Many people find this obvious, but it is worth pointing out. The change—34°F in the example—is calculated by taking the *to* value and subtracting the *from* value: 75°F - 41°F = 34°F.

That night, the temperature drops to 52°F. The temperature change *from* late afternoon to nighttime is 52°F - 75°F = -23°F. Change can be either positive or negative, depending on whether the *to* value is larger or smaller than the *from* value.

For a quantitative change to be meaningful, both the *to* and *from* quantities must have the same dimension; they must be the same *kind of thing*. In the previous examples, the variable is temperature for both *to* and *from*.

It is natural to think about change in terms of *change over time*, e.g., from morning to afternoon. However, quantitative reasoning often uses "*change*" in a more general sense: any difference in the value of a quantity from one setting to another. For instance, air pressure changes with elevation: at sea level, it is relatively high compared to the pressure at the top of a mountain. We call this *change over elevation*. Another example: look at a map of average temperature. The average temperature at 45° latitude is typically different from that at 35° latitude: a *change over latitude*. Still another example: the relative humidity is, all other things being equal, lower at higher temperatures: a *change over temperature*.

Although we quantify change by looking at the shifting value of one variable, there is always a *second variable*. The second variable sets the context for the change. In the previous paragraph, we used the word "over" as the marker for the second variable. To illustrate, in the change in air pressure over elevation, we are primarily interested in the variable *air pressure*. The second context-setting variable is *elevation*.

Recognizing that quantitative change involves *two* variables, we will highlight the second, context-setting variable with a more memorable phrase: "☞**with respect to**☜," as in "change in humidity with respect to temperature" or "change in GDP with respect to tariff levels" or "change in blood pressure with respect to dosage of some drug."

Often, several context-setting variables contribute at the same time to a change in the primary variable. Imagine, for instance, that the primary variable is the dollar value of sales in an ice-cream shop. Typically, sales varies from one time to another, that is, *with respect to time*. But underlying the change with respect to time are other context-setting variables: the change with respect to outdoor temperature or sunniness level. Sales may shift with respect to price; perhaps the shop raised prices. Moreover, we might gain an advantage in understanding sales patterns by treating *time* as two distinct variables: sales with respect to the *hour of the day* or with respect to the *day of the week*.

A distinctive concept in quantitative reasoning is ☞**partial change**☜. We signalled partial change very subtly in an earlier example with the phrase "☞**all other things being constant**☜." In a partial change, we examine the change in the primary variable with respect to one context-setting

variable, while holding the other context-setting variables constant. The 👉**total change**👈 in the primary variable is regarded as being the sum of contributions over all the context-setting variables from the individual partial changes with respect to one context-setting variable at a time. (Economists sometimes use the Latin phrase *ceteris paribus* to express the idea of holding all other things constant.)

Reasoning with partial change is one facet of quantitative 👉**analysis**👈. (The word "analysis" stems from two Greek roots which together mean "loosen up.") breaking up the overall system into contributing components. But how to hold the other context-setting variables constant? Experimentalists do this by imposing constant, laboratory control conditions. There are also statistical techniques that can be applied even when a good experiment is not feasible. The reports of results from such statistical studies often contain the phrase "👉**controlling for**👈 or"👉**holding constant**👈" even though even though data were collected outside a controlled laboratory environment. How to accomplish "controlling for" is a central and important issue in statistical and econometric method; important enough that the 2021 Nobel prize in economics was awarded for work on this issue.

ceteris-paribus



2021 Nobel prize in economics



3.7. Spaces

For understandable reasons, humans have a strong intuitive sense of spatial relationships. Some (but not all) quantitative reasoners instinctively apply spatial metaphors to mathematical concepts and perform aspects of mathematical reasoning in terms of spatial relationships. A key example is the 👉**number line**👈, which simplifies reasoning about numbers.

A number line is a valid representation of a quantity. In this metaphor, a 👉**point**👈 on the line stands for a value for the quantity. The number line itself is the set of all admissible values for the quantity. (Many kinds of quantities do not admit negative values in which case we start the line at zero and extend it in only the positive direction.)

Often, quantitative reasoners must work with multiple quantities simultaneously. For instance, in the epidemic model introduced in Chapter 1, the numbers of susceptibles (S) and infectives (I) change simultaneously in a coordinated manner.

One could think of this pair of quantities as represented by a pair of number lines, but there is a more powerful spatial representation: a mathematical ☞**space**☜. For the (S, I) pair, the space is analogous to the physical space of points on a table top. Each *point* on the table top corresponds to a specific, unique (S, I) value pair. To read off the quantitative values, we place two perpendicular, intersecting number lines on the table. The two number lines form the ☞**Cartesian coordinate system**☜, named after the French philosopher René Descartes (1596-1650), who contributed substantially to methods for spatial reasoning about mathematical objects. The *space* denominated by the coordinate system is the ☞**coordinate plane**☜.

The coordinate plane is said to be *two-dimensional*, since each point corresponds uniquely to two coordinate values. Many people are comfortable thinking about *three-dimensional* coordinate space, marked out by three mutually perpendicular number lines. For instance, three quantities specify an airplane's instantaneous position: latitude, longitude, and altitude. The everyday space that we (and airplanes) move around in is familiar: no math training necessary!

Most people find it hard to intuitively grasp the meaning of spaces of dimension higher than 3. Indeed, many people insist that there is no such thing. However, the physical existence of high-dimensional spaces is not the issue; it turns out to be incredibly powerful to *think* in terms of high-dimensional spaces. Almost always, such thinking involves training and mastering a fluent switching back and forth between sets of quantities and positions in space. In this introductory-level book, we will use only a handful of examples involving high-dimensional spaces. The examples we use will be one- or two-dimensional so that intuition can guide.

The take-home point is that we use the word *space* to refer to a collection of quantities. If the collection has just a single quantity, the space is the number line. A set of two quantities corresponds to the coordinate plane, and a set of three quantities to a point in regular 3-dimensional space. In this sense, a set of ten quantities corresponds to a point in 10-dimensional space, and on and on to higher dimensions.

4. Rates

☞**Rates**☜ appear so often in quantitative reasoning that it is worth taking pains to be explicit about what they are and when they should be used. Recall that a *ratio* is simply one quantity divided by another. A rate is a kind of ratio, but in a rate the division reflects a particular intent to describe the relationship between the *dimension* underlying the quantities.

Everyday conversation is peppered with rates, but we don't always use the word "rate" when mentioning them. The situation is like the comedic character[1] who is informed, to his pleasure, that he had been speaking *prose* for his entire life. Likewise, automobile fuel economy is a kind of *rate*. It relates how far a car goes to how much fuel is consumed. Electricity prices are *rates*; they relate how much electricity is used to the amount of the bill. Acceleration is a *rate*. Money inflation is a *rate*. And, naturally, so are bank or credit card interest *rates*, although with interest the word "rate" appears explicitly. Paying attention to when and whether the thing you are talking about is a rate pays dividends (also a kind of rate) to the quantitative reasoner. And always take note of the dimensions being related by a rate.

To illustrate a rate that doesn't look like one, consider this everyday example. I walk for exercise, so I prompted an AI,

Prompt: Walking consumes calories, but how much?

AI Response: Walking a mile burns approximately 60-to-100 calories, but this varies based on individual weight, pace, terrain, and fitness level. The word "calorie" is common in everyday speech, so it's understandable that the AI is using it. However, the quantitative unit is a "cal." One-thousand cal is, in metric convention, a kcal, but it is often written "Cal," with a capital C. Dietary information is usually presented in this large unit. A cal or Cal is commensurate with the metric unit Joule (J). We will use "Cal" in this dialog.

The AI agent might have responded, "How should I know? You didn't tell me how many miles you walk." But the actual response was clever, as if

Vocabulary Review

- conversion factor (3)
- data (2)
- dimension (2)
- dimensionless (3)
- flavors of one (3)
- per (3)
- pure number (3)
- quantity (3)
- units (3)
- value (2)
- with respect to (3)
- worth (3)

New

- compound rates
- rate
- rate of change

Just FYI ... The *dimension* of cal, Cal, and J is [energy]. [Energy] in terms of base dimensions is written $M\,L^2\,T^{-2}$

[1] Monsieur Jourdain in *The Would-be Gentleman* by Molière, 1622-1673

the AI were thinking, "He didn't tell me his walking distance. What kind of answer can I give that will be meaningful even without the distance information?" That is where *rates* come into the matter.

The uncertainty stems from not knowing my individual weight, pace, terrain, and fitness, all of which the AI mentions as playing a role.

The AI response contains two quantities: 1 mile and 60-to-100 Cals. As you may know, 60-to-100 is a way of describing *uncertainty*, which won't be covered until Chapter 8, so for the present we simplify to "80 Cals." Do these two quantities—1 mile and 80 Cals—specify a rate? Not until we go through the trouble of dividing one by the other, for instance 80 Cals per mile.

By giving a rate, the AI is providing a DIY answer. My role, as the you-it-yourselfer, is to calculate the calories burned in my daily walk by *multiplying* the rate by my walking distance:

The word **per** in the expressions "80 Cals per mile" and "0.0125 miles per Cal" plays the same role as the ÷ symbol: to signify division. We could equally well write 80 $\frac{\text{Cals}}{\text{mile}}$ or even 80 Cals mile^{-1}.

$$80\ \frac{\text{Cals}}{\text{mile}} \times 4\ \text{miles} = 320\ \text{Cals}$$

The rate is about *my* energy consumption, it is about the act of *walking*. A similar rate likely applies to your own walk, to your friend's walk, and even a complete stranger's walk.

When we talk about rates in this book we will tend to be explicit. We will (in this chapter) use the words "rate" and "per" to signal that the quantity we mean is a rate. For instance, at the risk of being long winded, we will say, "My car's fuel economy rate is 36 miles per gallon." [Returning to the point about a rate being a ratio of *two* quantities, we might even have said, "36 miles per 1 gallon." But this is not common practice.]

The two quantities in the AI response were 1 mile and 80 Cals. We formed these into a rate by dividing 80 Cals by 1 mile giving 80 Cals per mile. But we could equally well have done the dividing the other way round: divide 1 mile by 80 Cals giving a different rate: 0.0125 miles per Cal. These two rates are different, but they both describe the relationship between walking and energy consumption. But even though the rates *80 Cal per mile* and *0.0125 mile per Cal* describe the same situation, they are not just different formats; they are entirely different quantities. First, the *dimension* differs: Cal-per-mile (L M T^{-2}) versus mile-per-Cal (L^{-1} M^{-1} T^{2}).

Which of the two rates to prefer? The answer depends on the context. Here is one possible context:

Suppose you have just come home from a 3-mile walk and want to quench your thirst with a can of soda. In walking 3 miles, you (or, more precisely, the model person described by 80 Cal/mile) used up 240 Cal, that is

$$80 \frac{\text{Cal}}{\text{mile}} \times 3 \text{ miles} = 240 \text{ Cal}$$

.

Now consider this different context:

Last night your friend ate a lot of pasta—3000 Cal's worth—loading up for her participation in a walking marathon. How far will this get her towards the total distance of 26.219 miles? You want to know how many calories the runners will use up. This calculation involves the other rate:

$$0.0125 \frac{mile}{Cal} \times 3000 \; Cal = 37.5 \; miles,$$

More than enough!

You can also frame the relationship another way; how many Cals will propel you the whole distance, 26.219 miles. The end result is to be in Cals, so we need a rate that will transform 26.219 miles into Cals. Here we use the form of the rate with Cals on top and miles on the bottom: 80 Cals per mile. Multiplying the distance times the appropriate rate gives:

$$26.219 \text{ miles} \times 80 \frac{\text{Cals}}{\text{mile}} = 2097.5 \text{ Cals}.$$

4.1. Everyday rates

Some examples of rates encountered in everyday life:

- comparison shopping: dollars per oz.
- dollars per square foot
- earnings payment *per* money in an investment account
- flow versus rate of flow: water volume per minute used by a shower
- economic product per person
- miles per gallon (Compound: ton miles per gallon(truck) or passenger miles per gallon (car or plane).

4.2. Conversion: Flavors of 1

In Chapter 3 we referred to "dimensionless quantities." For instance, 60 is pure number; pure numbers lack units and are therefore always dimensionless. However, even dimensionless numbers can have *units*. For example, an angle is dimensionless: it is a ratio of two lengths.

See Figure 3.1. The dimension of the angle ratio is [length] / [length], that is, L / L or [1].

Dimensionless quantities (that is, quantities with dimension [1]) play an important role in converting from one unit to another. In general, to accomplish the conversion from one unit (say, miles) to another of the same dimension (say, km), multiply the first quantity by a ☞**conversion factor**☜ that is dimensionless. As introduced in Chapter 3, we call such conversion factors ☞**flavors of one**☜.

Temperature is an exception.

A flavor of one is a ratio formed by dividing a quantity by an *exactly equal* quantity. For example, 1 km is the same length as 0.621371 miles. The conversion factor that converts miles to km will be a ratio with km on top and miles on the bottom. Constructing this ratio from the two equal amounts gives:

$$\frac{1\text{ km}}{0.621371\text{ mi}} = \frac{1}{0.621371}\,\frac{\text{km}}{\text{mi}} = 1.609334\,\frac{\text{km}}{\text{mi}}$$

The conversion factor 1.609334 km/mi is a flavor of one; it is the ratio of two equal quantities.

A different, but closely related flavor of one converts km to miles: 0.621371 mi/km. In both cases, the **dimension** of the flavors of one are L / L = [1]. That is, "dimensionless."

Every conversion factor is a quantity, having both a pure number and units. Looking at the pure number in isolation won't tell you which way the conversion goes. For that, you need the units. If the units are mi/km, the conversion goes from km to miles. If the units are km/mi, the conversion goes from miles to km. In general, the units of the conversion factor that translates *from* into *to* has units *to*/*from*.

Figure 4.1 shows the result of a Google search on "kilometer to miles." The boxes labeled "kilometer" and "mile" indicate the units of the two main quantities provided. However, the value in the formula, 1.609, omits the units, which should be listed explicitly as mi/km. *Multiplying* the km value by 1.609 mi/km turns km into miles.

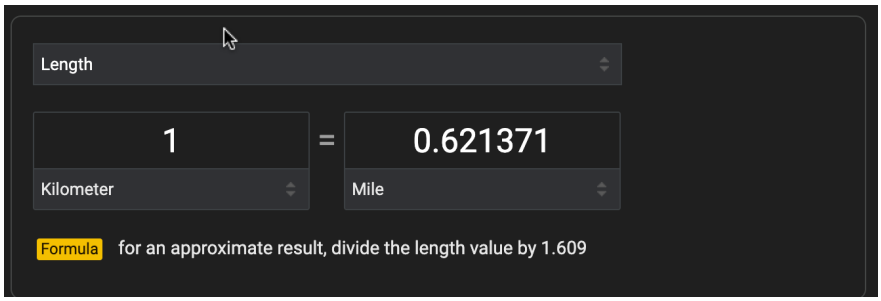


Figure 4.1.: The Google app for converting a kilometer into the equivalent distance in miles. The "formula" is correct, but it may be confusing because the units are missing from the quantity written as "1.609."

4.3. Percent

Section 3.4 introduced the notion that "percent" is a *unit* sometimes used to report dimensionless quantities. (See Section 4.3.) For instance, 13 percent—often written 13%—is the same dimensionless quantity as 0.13. The word "percent" is short for the Latin phrase "per centum." The word *centum* means 100. Since *per* means "divided by," *per centum*" means "divided by 100."

"C-note" is old American slang for a $100 bill, perhaps short for "*centum*-note. The Latin numeral C also means 100.

Most adults interpret "%" as meaning "a fraction *of*." The *whole* of something is 100% (which is, of course, just 1.00). A store discount of 20% means "we take 20% off *of* the price." Subtracting 20% from the whole means the new price is

$$100\% - 20\% = 1 - 0.2 = 0.8 = 80\%$$

of the original (non-discounted) price. A credit card interest rate of 28% per year means that after a year, if there are no new charges or payments, 28% *of* the original debt will be *added* to the original debt. In other words, the end-of-year debt will be $100\% + 28\% = 1 + 0.28 = 1.28 = 128\%$ *of* the start-of-year debt.

In the above paragraph, the word "*of*" has been *italicized*. "80% *of* ..." is perfectly idiomatic English. It means "0.80 multiplied by ...." The quantitative thinker has to pay careful attention to this use of idiom and translate it properly into the appropriate mathematical operation.

When it comes to *division*, standard English uses the keyword "*per*" to signal which quantity is being divided by which. Unfortunately, "*of*" is not a reliable signal for multiplication; "of" is used in so many other ways in English. By analogy to "*per*," we quantitative thinkers might push for general acceptance of a Latin word that provides a clearer signal. Appropriate choices are "*ex*," "*de*," and "*e*," all of which carry the meaning "out of."

With such a convention, we would say "20% *ex* price" instead of "20% of the price."

In mathematical theory, the use of percent is extraneous. Anything we write as a percent, we can write in decimal notation. In practice, people have a marked preference for quantities expressed with one or two digits, that is, in the range 0 to $\pm$ 100. This preference is why it would be absurd to report weather temperatures in the SI base unit kelvin. "Today's high temperature will be 303°K," violates the 2-digit preference, whereas the equivalent "30°C" or "86°F" remain within the preferred range.

Be on the lookout for purely *rhetorical* uses of units in percent. "Rhetoric" refers to techniques for speaking persuasively or compellingly. Everyone supports genuinely compelling quantitative arguments, but sometimes quantities are presented in ways that create a misleading impression. Here is an example from the *New York Times* in an article about the distinctive rhetorical style found in AI compositions:

> *"A.I.s like gesturing at complexity ("intricate" and "tapestry" have surged since 2022), as well as precision and speed: "swift," "meticulous," "adept." But "delve" — in particular the conjugation "delves" — is an extreme case. In 2022, the word appeared in roughly one in every 10,000 abstracts collected in PubMed. By 2024, usage had shot up by 2,700 percent."* — Sam Kriss, "Why does AI write like … that", New York Times, Dec. 3, 2025

Why does AI write like … that



An increase from *1 in 10,000* to 2700 is staggering and suggests that things are getting out of hand. But the quoted increase by 2700 *percent* corresponds to growing from 1 in 10,000 to 28 in 10,000. This is still a very small fraction. The double-zeros at the end of 2700 cancel out the "percent": 2700% is the same as 27. "Shooting up" from 1 by 27 gives 28.

Be aware that the practice of using percent as a unit can produce large numbers, like 2700%, that may create a misleading impression. The statement in the article would have been more meaningfully phrased as, "By 2024, 28 in every 10,000 abstracts contained 'delve'."

4.4. Rates of change

Earlier, in Section 3.6, we introduced *change* in a quantity as a simple difference in the value of the quantity. When modeling *systems*, we focus

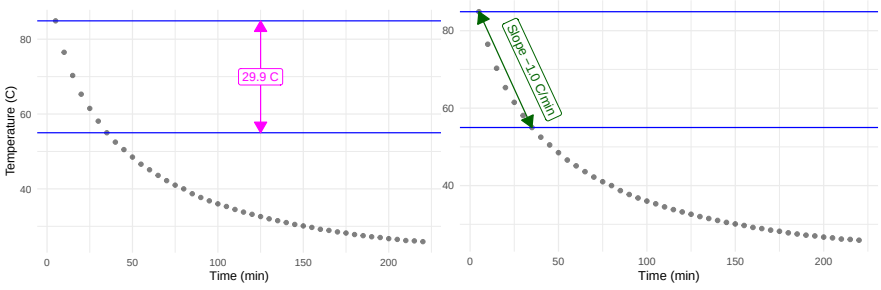
on the connections between the quantities involved. So why did the value of the quantity change? Typically, it is because something else changed in the system. We use the phrase "*with respect to ...*" to identify what may have changed. For instance, we can look at the change in temperature over an hour; the varying condition is then *time* and the change is described as being *with respect to time*. Sticking with the temperature example, we all know that temperature changes with respect to elevation (as we, say, climb a mountain), with respect to the amount of sunlight, or with respect to position.

The simple idea of *change* (Section 3.6) involves looking at the shifting value of one primary quantity. *Rate of change*, however, involves *two* quantities, a primary and a context-setting, "with respect to" quantity.

A 👉**rate of change**👈 includes both the primary and the context-setting quantity in the calculation. To form a rate of change, divide the quantitative difference in the primary quantity by the corresponding difference in the context-setting quantity.

"Primary" and "context-setting" can be thought of in terms of *functions*. (See Chapter 6.) The context-setting quantity is the input, while the "primary" quantity is the output.

To illustrate, consider the temperature of a cup of coffee over time as the cup sits at room temperature. "Over time" corresponds to the quantitative-thinking concept of *function*: temperature is a function of time. Figure 4.2 shows some data collected by Prof. Stan Wagon. He poured near-boiling coffee into a cup instrumented with a recording thermometer. Following the familiar pattern, the coffee cooled.



(a) Change in temperature with respect to time

(b) Rate of change of temperature with respect to time

Figure 4.2.: Coffee temperature recorded every five minutes for four hours (240 minutes).

Blue lines mark the temperature levels recorded at 5 and 35 minutes. The *change* in temperature with respect to time is the vertical distance shown

by the magenta arrow. The change over the 30-minute time interval is 29.9°C. The dimension of the change is simply Θ, that is, temperature, read directly from the vertical axis.

The *rate of change* over the 30-minute time interval is - 29.9°C / 30 min = -0.997°C/min. Notice that the dimension of the rate of change is ΘT^{-1}, that is, temperature *divided by time*.

Note that we could have calculated the rate of change for any interval, not just the 30-minute interval between times marked by the blue lines. Thus, the rate of change can itself change. The idea of changing rates of change takes some getting used to. But it forms a cornerstone of quantitative argumentation. We come back to it in Chapter 13.

4.5. Compound rates

As stated previously, a rate is a quantity. Calling the quantity a "rate" is merely a way of pointing out how the quantity was formed by the division of one quantity by another. Other than this history of the process of formation, a rate is no different than any other kind of quantity: a pure number plus a unit plus (optionally) a descriptive word.

Rates are often composed by arithmetic on other rates. We call these ☞**compound rates**☜, but this is merely to call attention to the formation of the rate. Like all rates, a compound rate is simply a quantity. Even so, in interpreting a quantity, it is helpful to keep in mind where it comes from. The words "rate" and "compound rate" are helpful in this.

Consider this example from an AI search on "running energy per mile":

Running energy per mile, or calories burned per mile, averages around 100 for a 150-pound person but varies significantly based on individual factors like body weight, fitness level, age, sex, and muscle mass. A commonly cited rule of thumb is to estimate about 100 calories burned per mile; a more accurate estimation involves multiplying body weight (in pounds) by 0.71 to get a general figure, for example, a 200-pound person burns around 142 calories per mile. — AI generated

You can also think of it as a rate of change, since "Calories burned" is a change in the runner's stock of energy and the mile is really a change in the distance run.

"Running energy per mile" is a rate. The AI response goes on to discuss the role of body weight in the number of calories burned. Part of this involves a *compound rate* 0.71 Calories *per* mile *per* pound of body weight. Note the

two uses of "per" in the phrase. The two occurrences of *per* signal that this is a compound rate.

The English grammar of *per-per*s creates an ambiguity. Two different mathematical interpretations are possible for the English phrase "per mile per pound." The parentheses of mathematical notation make this clearer. For instance, here are the two interpretations using mathematical parentheses to indicate the order of operations:

1. (Calories burned per mile) per pound, or
2. Calories burned per (mile per pound),

Alternatively, using the over-under notation of ratios:

1. $\frac{\text{Calories burned / mile}}{\text{pound}}$, which is $\frac{\text{Calories burned}}{\text{mile}\times\text{pounds}}$.
2. $\frac{\text{Calories burned}}{\text{miles/pound}}$ which resolves to $\frac{\text{Calories}\times\text{pound}}{\text{mile}}$.

Compound rates (1) and (2) are different quantities. To see this, imagine a 150-pound person running 3 miles, burning 300 Calories in the process. Note that there are three quantities involved:

i. 300 Calories burned
ii. 150 pounds of weight
iii. 3-mile length run

Compound rate (1) works out to be 300 Calories / (3 mile $\times$ 150 pounds), that is, 0.667 Calories per (pound-miles). In contrast, compound rate (2) is (300 Calories $\times$ 150 pounds) / 3 miles, or 15,000 Calorie-miles per pound. (1) and (2) are different numbers, different units, and different dimensions.

It is not possible to tell just from the English phrase "Calories per pound per mile" which compound rate is intended. Instead, we need to look for clues. The AI text gives a clue about the use of the quantity 0.71. It says, "for example, a 200-pound person burns around 142 calories per mile." Quantity (1) divides the calories-per-mile by pounds, whereas quantity (2) multiplies the calories-per-mile by pounds. We can try both:

1. $0.71\ \frac{\text{Cal}}{\text{mile-pounds}} \times 200\,\text{pounds}$ gives 141 Cal per mile.
2. $0.71\ \frac{\text{Cal}}{\text{mile-per-pounds}} \times 200\,\text{pounds}$ gives 141 Cal pounds^2 per mile.



Only format (1) gives the result cited in the AI example.

There is a more fundamental way to see why interpretation (1) is the right one. The person is using 0.71 Cals to carry out a task. Which task? To move one pound for one mile. The unit "pound-mile" measures the amount of the task; a 150-pound person needs to perform 150 pound-miles to get themselves a distance of one mile.

Consider this example involving an eco-science task: measuring the number of animals crossing different roads. The technique: position the observer along the road who will spend a couple of hours with binoculars, counting the crosses they see. Making a fair comparison between roads requires adjusting for the time duration of observation. Less obvious, the raw count also needs to be adjusted for the length of roadway visible in binocular range. (On a curved or hilly road, for instance, sightlines might be limited to, say, 100 yards, while on a straight road, a half-mile of road is visible.) The amount of monitoring done by the observer units of mile-hours, that is, miles of road visible *times* hours spent monitoring. The format for the compound rate that fairly compares roads is animal crossings per mile-hour.

Some other examples of compound rates:

- Shipping costs. Dollars per the amount of shipping. There are two components to "Shipping": the weight of the goods and the distance covered. The total amount of shipping is the weight times the distance. For railways, this is sometimes presented as "ton-miles."
- Medicine dosing. Nurses using an intravenous (IV) drip to administer medicine are given the dose as a compound rate: milligrams of medicine per kilogram-hour. The nurse mixing the medicine in the IV bottle needs to know how many hours the bottle will last. The bigger the bottle, or the bigger the patient, the more milligrams needed.
- Rent. Rents for commercial space are often specified in dollars per square-foot-months, that is, monthly rent per unit area. Multiply monthly rent per unit area by the number of square feet to get the rent payment needed each month.
- Work effort is sometimes tallied using units such as person-hours or person-months.
- Energy use. The amount of energy needed to heat a house depends on the size of the house, the length of the heating season, and the

temperature. An AI search on "amount of natural gas to heat a house" produced this response:

Another standard metric is the energy use per square foot per heating degree-day (HDD), where a typical range is 5.0 to 10.0 BTU/HDD/Sq.Ft.

BTU (British Thermal Unit) is a measure of energy. The amount of heating "service" required is the product of duration, coldness, and house size. The standard unit is degree-day-square-feet. So the rate of energy use is BTU per degree-day-square-feet. There are tables showing the average annual number of heating degree-days for various locales. A web search for "annual heating degree days for Minneapolis" yielded an estimate of 6000 degree-days. For a house of 1200 square feet, this comes to a demand of 1200 square-feet × 6000 degree-days, or just over 7 million degree-day-square-feet. The typical range for energy use over a year would be 35 to 70 million BTU.

Heating degree-days



Notice that the unit "BTU/HDD/Sq.Ft." quoted by the AI is ambiguous. It should be BTU per HDD-(ft^2). Suppose that we misinterpret the quoted range—5.0 to 10.0—as being in units of BTU/(HDD/ft^2). For the 1200-square-foot house in the example, we would have 6000 degree-days divided by 1200 square feet, resulting in 5 degree-days per square foot. The total BTU usage for the house would be (mis-)calculated as 5.0 to 10.0 BTU per (HDD/Sq.Ft.) × 5 HDD/Sq.Ft. = 25 to 50 BTU. For comparison, bringing 1 liter of water to a boil consumes about 250 BTU. It is hard to imagine that heating a house for a year would use less energy than boiling a liter of water. Mistaking HDD × ft^2. for HDD *per* ft^2 will cause a lot of trouble when designing or outfitting a house!

4.6. Multiplying quantities

This morning, the author went for his habitual walk, which he estimates consumed 300 Cal. For diet planning or indicating general health, 300 Cal can be a good summary of the exercise. But the estimate combines two different factors: the energetics of walking and the distance walked. We use a rate, Cals per mile, to describe the energetics of walking. For walking, the rate is approximately 80 Cal/mile. Just about anyone on just about any walk can use the rate to find the total calories consumed; multiply the walk's length by the energetic rate. A two-mile walk consumes 80 Cal $mile^{-1}$ × 2 mile = 160 Cal.

This is roughly the caloric content of one can of (non-diet) soda.

Using the rate, we can compare the energetics of walking and running, again using a Cal/mile rate.

> *Take a moment to guess whether running or walking consumes more calories.*

Many people guess that running consumes more energy. For instance, the author used to go for a 3-mile run each morning, and people assume that the author's 4-mile walk is less exercise than the 3-mile run. There is no reason to assume; an internet search shows that running consumes about 100 Cal/mile. Surprisingly, that is similar to the energy rate for walking.

The author's 4-mile walk uses about 300 Calories, roughly the same amount of energy as the 3-mile run: 100 Cal mile^{-1} $\times$ 3 mile = 300 Cal. Once again, knowing the energy *rate* enables us to calculate a *total energy*: multiply the rate by the distance.

The pattern that "rate $\times$ quantity gives total" is encountered every day. Some examples:

Context	Rate	Multiplier	Result
Buying groceries	12 dollars per pound	0.75 pounds	9 dollars
American driving	35 miles per gallon	10 gallons	350 miles
European driving	5 liters per 100 km	500 km	25 liters
Wages	17 dollars / hour	40 hours	680 dollars

The ratio/rate (price per pound, liters per 100 km, dollars per hour) describes the context. Multiplying the rate by an amount—in terms of the denominator of the ratio—gives a total in terms of the numerator of the ratio.

As previously stated, we use dimensionless flavors of one to convert units between two quantities with the *same* dimension. It is reasonable to view dimensionful rates similarly: as conversion factors between different dimensions. Have gallons and want miles: multiply by the rate miles-per-gallon. Have 3/4 of a pound of cheese and want the price: multiply by the rate dollars-per-pound.

For some people, the result of multiplying two quantities sometimes seems odd. For instance, units of *passenger-miles* denote the amount of transportation service. One passenger going one mile is one passenger-mile; an airplane carrying 150 passengers for 3000 miles is providing 150 passengers $\times$ 3000 miles of transit: 450,000 passenger-miles. A water-resources engineer might measure volume in acre-feet: multiplying area in acres by the depth in feet of water. Mechanics working on a car or a bike have to

apply the right amount of torque to tighten a bolt without overstressing it. Torque is measured in foot-pounds: a foot of length times a pound of force.

5. Order of magnitude

In his 2018 book *Factfulness*, physician and global health expert Hans Rosling skews a common misconception that the countries of the world are mainly split between rich, healthy countries and poor countries with ill living conditions. Figure 5.1 displays data comparing child mortality across countries over a wide range of GDP levels both in 1960 and 2020.

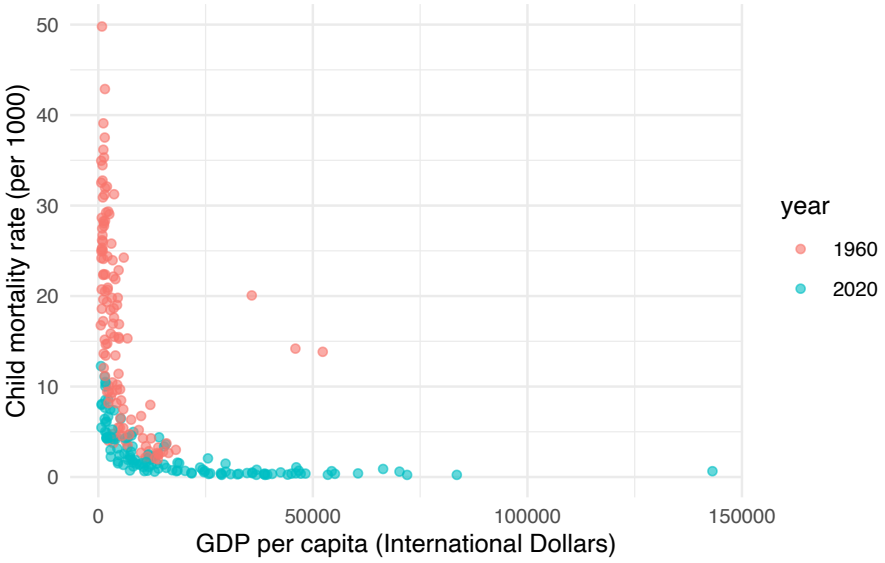


Figure 5.1: Child mortality (deaths per 1000) versus GDP for 1960 and 2020 for 125 different countries. Source: Bolt and van Zanden – Maddison Project Database 2023 – with minor processing by Our World in Data.

Figure 5.1 is interactive; hovering the cursor over a dot causes the country name to appear, along with the precise numerical value for mortality and GDP.

Interactive versions of the graphs identify each country

Are the data, as depicted in Figure 5.1, consistent with the rich vs poor split between countries? Or is there a smooth spectrum from low to high GDP and corresponding child mortality rates?

There is a tall vertical column of poor countries with high mortality rates and a long horizontal row of rich countries with low mortality. At the intersection between the row of points and the column of points is a knot of

Vocabulary Review

- argumentation (1)
- capita (3)
- data (2)
- dimension (2)
- per (3)
- point (3)
- quantity (3)
- rate (4)
- si (3)
- units (3)
- value (2)

New

- logarithm
- magnitude
- order of magnitude
- scientific notation
- sorting

"Child Mortality" is operationalized as the rate of death of children under five years in age. It ranges widely from country to country, from around 30 deaths per 1000 children down to almost zero.

countries with GDP much lower than the rich countries and child mortality lower than the poorest countries. Some economists justify the use of GDP as a measure of well-being by pointing to the horizontal row of countries to claim there is a strong relationship between health and wealth

Looking at the first few rows of the data behind Figure 5.1 shows the structure of the data:

Table 5.1: The first six rows (out of 250) of the GDP and child mortality data.

Entity	Year	Child mortality rate	GDP per capita
Afghanistan	1960	35.32	1326
Afghanistan	2020	6.13	1928
Algeria	1960	23.96	3328
Algeria	2020	2.29	13086
Australia	1960	2.49	14013
Australia	2020	0.38	48269

Each specimen—that is, each row—is a country in a given year (1960 or 2020).

Our World in Data



The three countries shown in the excerpt each seem to contradict the idea of a dominant correlation between health and wealth. The proportional improvement between mortality rates from 1960 to 2020 is much greater than the proportional increase in GDP. Figure 5.1 does not make such a pattern immediately evident. Perhaps the three countries in Table 5.1 are exceptional cases? Or, might the graphical format be hiding the pattern? Is there another way of looking at the data that might better reveal a pattern?

We will return to this matter in Figure 5.2 once we haveintroduced an important tool in quantitative argumentation: the comparison of *magnitudes*.

5.1. Counting digits

When quantities are about the same size, it is easy to compare them to see which is bigger or smaller. For most people, the cognitive load of such comparisons is very low for values in the range 1-100. In an instant, a person can pick the larger of two quantities. Try it, which is larger: 72 or 68? No work at all.

The cognitive load rises for quantities that span a much larger range, say 1 to 10,000 or larger.

When quantities are a mixture of the very large and the very small, comparisons require attention. For instance, consider Table 5.2, which shows the gross domestic product (GDP) of a handful of countries in 2024. The table standardizes values from different countries to a standard unit: United States dollars (USD).

Country	GDP in USD
Albania	27177740000.00
Australia	1752193000000
Bhutan	3019250000.00
China	18743803000000
Estonia	42764930000.00
Germany	4659929000000

Table 5.2: The 2024 gross domestic product of a handful of countries. The digits are correct, but the size of the numbers is hard to make out because the table ignores sensible formatting conventions. [Source: World Bank]

In Table 5.2, the numbers are "left justified", that is, their leading digits line up in a column. Unfortunately, the leading digits—27 for Albania and 18 for China—are little help in comparing the values.

Now consider the GDP numbers, written in a different format. Table 5.3 makes it easy to see which countries have very large values and which comparatively smaller values. First, we have sorted the values. Sorting helps the eye compare every value to every other value in an instant.

Country	GDP in USD
China	18,743,803,000,000
Germany	4,659,929,000,000
Australia	1,752,193,000,000
Estonia	42,764,930,000
Albania	27,177,740,000
Bhutan	3,019,250,000

World Bank data



Table 5.3: A better presentation of the GDP data from Table 5.2. Commas improve readability by separating thousands. The `monospace` font makes it easier to align the digits

The improved format of Table 5.3 also makes it clear why looking at the leading digits is not always helpful in comparing values. Instead, the first thing to look at is the *number of digits*. Because the values are all aligned to the right, we do not even need to count the number of digits; we can treat them as if we are looking at a bar graph. It is necessary to look to the leading digits *only when* the *number of digits* is the same.

5.2. Unit prefixes and scientific notation

As noted in the previous section, people are quick to compare or sort numbers with 1, 2, or even three digits. Bigger numbers, however, require shifting to another mode: counting digits. Similarly, comparing tiny numbers—such as 0.000032 and 0.00018—involves counting the zeros that immediately follow the decimal point.

There are two widely used techniques for formatting numbers that include information about the number of digits *directly* in the number. The logic of these two techniques will bring us very close to explaining how to construct graphics like Figure 5.2.

The most familiar of the two techniques involves unit prefixes. Every day, we deal with both very large and very small distances. For instance, using base SI units, the width of a typical room is about 3 m, while a bike ride to a hardware store is about 2500 m, depending on the neighborhood. The screw bought there to fix a bike could have a diameter of 0.004 m. If these numbers sound strange, it is because everyday practice is to change the units in order to avoid counting zeros. Instead of 0.004 meters, say 4 millimeters. Rather than 2500 meters, say 2.5 kilometers.

The standard prefixes "milli" and "kilo" mean, respectively, one-thousandth and one thousand. The *milli* prefix—often abbreviated simply as m—moves the decimal point 3 places to the right: 0.004 m is the same as 4 mm. Likewise, kilo (abbreviated as k) moves the decimal three places to the left: 2500 m becomes 2.5 km.

The system was introduced on a limited basis by the revolutionary government of France. (See Table 5.4.) It has been extended to handle larger and smaller quantities on several occasions.

Table 5.1.: Official unit prefixes that move the decimal point to avoid zero counting. The year of introduction of each prefix paints a picture of when people started to talk about quantities very large and very small. Additional prefixes, used only in very specialized situations, were added in 1964, 1991, and 2022.

Abbreviation	Prefix	Moves decimal point	Year of introduction
p	pico	12 places to right	1960
n	nano	9 places to right	1960
μ	micro	6 places to right	1883
m	milli	3 places to right	1794
c	centi	2 places to right	1795
d	deci	1 place to right	1795
....	...	...	...
da	deca	1 place to left	1795
h	hecto	2 places to left	1795
k	kilo	3 places to left	1795
M	mega	6 places to left	1873
G	giga	9 places to left	1960
T	terra	12 places to left	1960

Only a handful of prefixes are in everyday use: micro, milli, kilo, mega, giga. These cover a broad range, from a tiny dose of medicine to the wealth of ultra-rich individuals. Notice that the most used prefixes each shift the decimal point by three places, enabling the range of quantities to eliminate leading or trailing zeros and be represented as numbers with one, two, or three digits plus the prefix. The 1795 prefixes that move the decimal point by 1 or 2 places have mostly fallen out of use. The most common exception is *centi*, which is widely used for length: centimeter (cm). The prefix *hecto* is in regular use to refer to area; a "hectare" is an area of (100 m)-squared, that is, $(100\ \text{m})^2 = 100^2\ \text{m}^2 = 10{,}000\ \text{m}^2$.

For temperatures, on the other hand, the prefix system is not used. Even without prefixes, we can write everyday temperatures with two digits.

As regards time, we have an idiomatic system for avoiding zeros: seconds, minutes, hours, days, weeks, months, years, decades, centuries, millennia. The right-moving prefixes, however, are used for time: milliseconds, microseconds, and such.

When talking about money, it is common to replace the Latin prefixes with words like "thousand," "million," "billion," and "trillion." Unfortunately, such words differ from language to language. The French use "milliard" and "billion" to mean what in English is a billion and a trillion, respectively. The Japanese use 4-digit groups for monetary quantities. "Man" is 1,0000 (that is, 10,000), "oku" is 1,0000,0000 (that is, one-hundred million), "cho" is 1,0000,0000,0000, which aligns with English "billion."

People can easily confuse prefixes they do not use often. Such mistakes can lead to serious problems. Mistake milligrams for micrograms, and the dose of medicine will be 1,000 times too small or too large. News reports and politicians frequently confuse "million" with "billion."

The prefix system works best when applied to units with short names, such as meters, grams, watts, joules, and so on. Things can get out of hand with derived units. For instance, a fundamental quantity used in quantum physics is Planck's constant, which has the unusual dimension [energy] × [time].

Planck's constant: `0.00000000000000000000000000000000662607015` m^2 kg / s

Doing calculations with such numbers is error-prone, since it is easy to omit or add zeros and hard to see at a glance that this has occurred. Writing it more compactly with prefixes is not much help:

`0.663` femtometers^2 gm / s

☞**Scientific notation**☜ is helpful here. To illustrate, consider a "light year," often used to measure distances between stars. In the familiar units of kilometers, and using traditional mathematical notation, a light year is

$$9.461 \times 10^{12}.$$

Such notation involves a kind of punctuation used to organize two numbers. The punctuation framework is $\square \times 10^{\square}$. (The colors are not part of the notation. We use them to enhance clarity of what goes where.)

The two number-roles in scientific notation have standard, if obscure, names.

- The **mantissa**, 9.461 in the example, is conventionally written with a single 1-to-9 digit to the left of the decimal point, 9 here. There can be any number of digits, including 0, to the *right* of the decimal point.

- The **exponent** is placed as a superscript to 10. It is always an integer, that is, a round number. In the example, the exponent is 12.

Here are a few examples of large and small numbers in scientific notation:

- Planck's constant: $6.62607015 \times 10^{-34}$ m² kg / s
- China GDP for 2023: 1.87×10^{13} USD
- Value of German Mark in USD during an episode of hyperinflation: January 1923 1.4×10^{-4} USD; October 1923 2.0×10^{-10} USD;

A primary virtue of scientific notation is that the reader does not need to count digits in order to compare two numbers. The exponent takes care of the counting for you.

A flaw of this notation is that the most important number, the exponent, is written in the smallest font. In computer notation, however, the punctuation is dramatically streamlined: Planck `6.62607015e-34`; China `1.87e13`; German Mark `1.4e-4`. The `e` in the computer notation is shorthand for "exponent."

Public health workers and sociologists use a variant of scientific notation with the punctuation "per." A mortality rate of 0.0063 (deaths per person) will instead be presented as 63 deaths per 10,000 people, avoiding the need to count digits. For a rare disease, we present its prevalence across different units, e.g., cases per 100,000 people or 1,000,000 people. Naturally, 63 deaths per 10,000 people can also be sensibly written as 6.3 deaths per 1000, or, stretching things, 0.63 deaths per 100. Writing the quantity as, say, 0.063 deaths per 10 persons or 0.0063 deaths per person goes to far towards illegibility, even if mathematically justified.

The names "mantissa" and "exponent" are absolutely standard. However, the idea of "the exponent of a number" is flawed. Numbers don't have exponents, scientific notation describes them in an exponent-based way.

However, an important idea in quantitative argumentation is "the 👉**magnitude**👈 of a number." When comparing two numbers, the phrase 👉**order of magnitude**👈 difference can be very useful.

A good way to think about the magnitude of a number is by reference to scientific notation. The exponent in scientific notation gives a quick readout of the magnitude. For instance,

a. 394.32 (that is, 3.9432e2) is a magnitude-2 number
b. 0.14 (that is, 1.4e-1) is a magnitude-minus-1 number

Comparing such different sizes of numbers as (a) and (b) is often done using the magnitude. (a) and (b) are "3 orders of magnitude" different.

In Section 5.4 we'll consider a more precise way to specify the magnitude of a number. The exponent in scientific notation is a rounding off of the precise magnitude.

5.3. Example: The magnitude of child mortality

We return now to the patterns in child mortality across the decades and across the world. Figure 5.1 showed the numbers for mortality and GDP. Now we are going to look at the same data, but using magnitudes to represent the fact.

Figure 5.2 plots the mortality-vs-GDP data on axes that are arranged to highlight order-of-magnitude differences. In contrast to Figure 5.1 the data points are much more spread out across the graphics frame. Subtle patterns are visible that involve many more countries.



Figure 5.2.: The same data as in Figure 5.1, but placed on axes based on the *magnitude* of the number. For a handful of countries, black lines link the 1960 and 2020 data.

For instance, there are points from 1960 that are more or less directly above many points from 2020. "Directly above" means that there is little or no change in GDP but that there is a big change in mortality. This suggests that the mortality change stems to a great extent from factors other than GDP. One such factor is the introduction in the 1970s of oral rehydration therapy,

an inexpensive but effective treatment for diarrheal illnesses that were a major cause of child mortality. This style of quantitative analysis, examining the *partial change* in one quantity while holding another constant, is detailed in Section 4.4.

The method used to produce Figure 5.2, the more intelligible version of the mortality-versus-GDP graph, is to set the axes' scales to reflect the *magnitude* of the quantity. For ease of interpretation it is common to label the axes in a magnitude plot with corresponding numerical values rather than magnitudes. That is, the horizontal-axis labels are 1000, 10,000, and 100,000 rather than the corresponding magnitudes 3, 4, and 5. However, the magnitude value determines the spacing along the axis. Similarly, the vertical-axis labels are 0.3, 1, 3, 10, and 30. We already know that the magnitude of 1 is 0 and that the magnitude of 10 is 1. But what are the magnitudes of 0.3, 3, and 30?

Oral rehydration therapy



5.4. Logarithms: Just the magnitude, please

Computers happily take into account the two numerals in scientific notation when doing arithmetic. But this can be hard for people. There is, however, a way to condense the two components of scientific notation into a single number conveying the same information. The basic idea is to fold the mantissa into the magnitude by allowing non-integer values for the magnitude.

The name for this condensed form of scientific notation is probably familiar to you already: 👉**logarithm**👈. Try to put aside any anxiety you associate with that name from a previous algebra course. A logarithm is just a format for writing a number.

To illustrate, consider the scientific-notation form of the 2023 GDP of China: `1.87e13` USD. This value is somewhat greater than `1e13` USD but less than `10e13` USD. Of course, `10e13` ought to be written `1e14`. The magnitude of `1e13` is 13, that is, the value of the exponent. Similarly, the magnitude of `1e14` is 14.

So what is the logarithm of `1.87e13`? Since the original number is between `1e13` and `1e14`, its logarithm should be between 13 and 14: something like 13._ _. The digits to fill in the blanks are, to be honest, by no means obvious.

Indeed, the answer was invented as recently as 1600 by John Napier (1550-1617) as an arithmetical aid. The logarithm of `1.87e13` turns out to be 13.272, between 13 and 14, but closer to 13.

To help reinforce the connection between the magnitude in scientific notation and the logarithm, here are a few examples:

Scientific notation	Logarithm
`1.2325e4`	`4.0908`
`7.3e15`	`15.86`
`4.1e-8`	`-7.39`

🔥 Units and dimensions when using logarithms

A careful consideration of units and dimensions is crucial to quantitative argumentation. Logarithms, however, only apply to pure numbers. There is no use talking about the logarithm of a kg or the dimension of a magnitude.
Comparing a quantity to another naturally requires that they two quantities have the same dimension. To use logarithms sensibly, particularly when comparing quantities, make sure to convert all the quantities to the same *unit* before calculating the logarithm.
Occasionally you may encounter a phrase like "log dollars." This is not a genuine unit but a reminder that the values presented are magnitudes and that the original unit of measurement was dollars.

5.5. Order of magnitude phenomena

Many phenomena and their associated quantities are best thought of in terms of magnitude rather than number. An example: Animals, including humans, go about the world in varying states of illumination, from the bright sunlight of high noon to the dim shadows of a half-moon. To see in such diverse conditions, the eye needs to respond to light intensity over a broad range of magnitudes. When comparing two numbers of different magnitudes, the phrase used is 👉**order of magnitude.**👈

For instance, there is a difference of two orders of magnitude between 100 (`1e2` in scientific notation) and 10,000 (`1e4`).

The *lux* is the unit of illuminance in the Système international. Table 5.5 shows the illumination in a range of familiar outdoor settings.

Table 5.2.: Illuminance in lux in some outdoor settings. Source

Illuminance	Condition
110,000 lux	Bright sunlight
20,000 lux	Shade illuminated by clear blue sky, midday

Illuminance	Condition
1,000 lux	Typical overcast day, midday
400 lux	Sunrise or sunset on a clear day (ambient illumination)
0.25 lux	A full Moon, clear night sky
0.01 lux	A quarter Moon, clear night sky

For a creature active both night and day, the eye needs to be sensitive over a range of illumination spanning more than six orders of magnitude. To accomplish this, the eyes use several mechanisms: pupil contraction or dilation accounts for about one order of magnitude; photopic (color, cones) versus scotopic (black-and-white, rods, nighttime) accounts for about three orders of magnitude; adaptation over minutes (1 order); and squinting (1 order).

Illuminance in lux



More impressively, human perception of sound spans 11 orders of magnitude in the energy impinging on the eardrum. The energy density of perceptible sound ranges from the threshold of hearing at 0.000000000001 W/m² to a conversational level of 0.000001 W/m² to 0.1 W/m² in the front rows of a rock concert. However, in terms of our subjective perception of loudness, each order-of-magnitude change is perceived the same way, whether it is from street traffic to a vacuum cleaner or from a whisper to everyday conversation. (The unit of sound measurement is the decibel (dB), with 10 decibels corresponding to an order of magnitude in the energy density of sound.) See Table 5.6 for the dB level of a wide range of sound levels.

Table 5.3.: Energy density of sound in various situations. Sound at 85 dB, for extended periods, can cause permanent hearing loss. Exposure to 120 dB of sound for 30 seconds is dangerous.

Situation	Energy level (dB)
Rustling leaves	10 dB
Whisper	20 dB
Mosquito buzz	40 dB
Normal conversation	60 dB
Busy street traffic	70 dB
Vacuum cleaner	80 dB
Large orchestra	98 dB
Earphones (high level)	100 dB
Rock concert	110 dB

Situation	Energy level (dB)
Jackhammer	130 dB
Military jet takeoff	140 dB

6, 60, 600, and 6000 miles per hour are quantities that successively differ in size by orders of magnitude. Such differences often point to a substantial change in context. A jog is 6 mph, a car on a highway goes 60 mph, a cruising commercial jet goes 600 mph, and a rocket passes through 6000 mph on its way to orbital velocity. The range from an infant's crawl to highway cruising covers three orders of magnitude in speed.

Of course, many phenomena are **not** usefully represented by orders of magnitude. For example, the difference between normal body temperature and high fever is 0.01 orders of magnitude in temperature. An increase of 1 order of magnitude in blood pressure from the normal level would cause instant death! The difference between a very tall adult and a very short adult is about 1/4 of an order of magnitude.

We are using the Kelvin scale, which is a physically meaningful scale for a *ratio* of temperatures.

Comparing with orders of magnitude makes sense when the relevant comparison is a *ratio*. "A car is 10 times faster than a person" refers to the ratio of speeds. In contrast, we generally use *differences* to compare quantities such as body temperature, blood pressure, and adult height. Fever is only 2°C higher in temperature than normal. A 30 mmHg increase in blood pressure is worth noting as an indication of hypertension. A very tall and a very short adult differ by about 3 feet.

In medicine, blood pressure is measured using old-fashioned, non-SI units based on the height of a column of mercury that will balance blood pressure. The "mmHg" is short for "millimeters of mercury."

Thinking in terms of orders of magnitude is appropriate when working with a set of objects whose sizes span one or many factors of 2. Comparing baseball and basketball players? No need for orders of magnitude. However, in considering the weights of mammals, from bats to whales, orders of magnitude will reveal patterns that the whales would obscure on a number-line scale.

6. Introduction to Functions

In the nomenclature of this book, a 👉**system**👈 is a framework or an organized structure with **interconnected elements**. We have called these elements *dimensions* and *operationalize* them to translate them into quantities.

This chapter introduces an important modeling representation for the connections between system elements: 👉**functions**👈.

A function is a mathematical or computing concept, a way of organizing knowledge and encapsulating a process. A good way to think about a function is as a machine for transforming inputs into an output.

Vocabulary Review

- dimension (2)
- models (1)
- operationalization (2)
- quantity (3)
- rate (4)
- value (2)

New

- algorithm
- algorithms
- arg
- exogenous
- functions
- system

6.1. Example: A solar-panel system

To illustrate how functions come into play in modeling systems, consider this diagram of the relationships in a household solar-panel system:

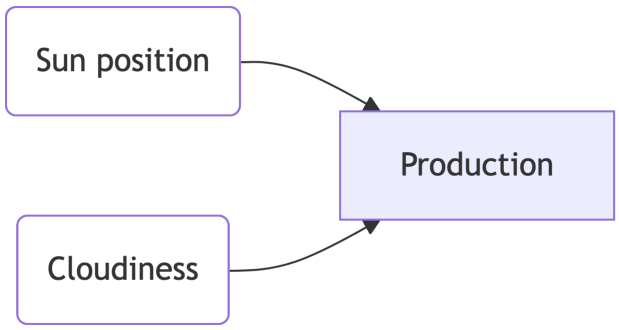


Figure 6.1.: A diagram of a simple system. The dimensions are named. A sharp-cornered box encloses a dimension that is the output from a function. The arrows impinging on the sharp-cornered box indicate inputs to each function.

Such diagrams name the dimension, e.g., "Cloudiness." The arrows pointing into the box surrounding the word indicate which other dimensions are having an influence. In Figure 6.1, "production" is in a box with sharp corners, with arrows pointing in, signifying that "production" is the *output* of a function. The inputs to the function are at the tail-end of the arrows:

"cloudiness" and "sun position." Taken as a whole, the diagram says that electricity *production* is a function of the *sun position* and the amount of *cloudiness*.

Models are always limited in scope; they don't try to account for every feature in a system. The notation used in Figure 6.1 makes it explicit what dimensions are being modeled and which dimensions are taken as givens. The sharp-cornered box denotes dimensions that are produced within the model. Economists call such dimensions ☞**endogenous**☜ meaning to "come from inside." The round-cornered boxes indicate which dimensions are taken as given, that is, "coming from outside" the model.

Economists use the word ☞**exogenous**☜, where the prefix *exo* means "outside of."

Figure 6.1 depicts a very simple model of a photovoltaic electricity generation system. But models can be much more complicated, with both sharp- and round-cornered boxes providing inputs to any given sharp-cornered box.

6.2. The organization of a function

In diagrams like Figure 6.1, each sharp-cornered box corresponds to a function. Specifying what are the inputs to a function is essential, but the complete story requires an essential additional feature: an ☞**algorithm**☜. The algorithm is a complete operational description of the process to be followed in turning the inputs into the output.

Mathematics classes often use functions without displaying the algorithm that implements the input-to-output process. For instance, you may remember the sin function, and even memorized the output produced for a handful of particular inputs. A typical classroom notation might look like $\sin \pi = 0$. This particular example denotes that when the input number is π, the output of the sin function will be zero.

For any input values that you haven't memorized, you need to refer to the algorithm behind sin to calculate the output. The algorithm, however, is hidden inside a calculator or the like. You provide the input using the keypad then press a labeled button to invoke the algorithm to work on that input. The output is displayed on the screen.

Functions like sin take only a single input. But the functions we use in modeling often take two or more inputs. The notation we use for functions therefore has to be able to handle multiple inputs and, of particular importance, say which input plays what role in the algorithm.

We will use a concrete example to demonstrate how this is done.

On a hot day, comfort is a function of temperature and humidity. That is, temperature and humidity are the inputs, comfort is the output. Hot and wet conditions produced discomfort; cool, dry conditions are more comfortable. The previous sentence corresponds to a function that takes temperature and humidity as inputs and produces "comfort" as an output.

Temperature and humidity are both quantities, so they are directly suited to be inputs to a function. It's less obvious how to operationalize *comfort*, which is a subjective and multi-dimensional concept.

One widely used operationalization of comfort is the so-called "heat index," which often appears in weather reports and forecasts.

Section 6.3 presents the heat-index algorithm and its origins.

It helps to give a function a name so that it can be referred to and distinguished from other functions. Using mnemonic names—that is, names that remind you what the function is about—is a good practice. We will call the function `heat_index()`.

The parentheses following the name are a convention to remind us that `heat_index` is a function. Otherwise, we might confuse it with a quantity. For the sake of clarity, this book will write function names followed by parentheses, e.g. `heat_index()`. In use, the parentheses provide punctuation for organizing the inputs. These will be placed inside the parentheses, separated by commas. For instance, the heat index on a day where the temperature is 30 °C and the humidity is 80% will be calculated as:

```
heat_index(30, 0.80)
```

```
[1] 37.66704
```

The process of calculating the output of a function for specific inputs, as above, has many names. Informally, we could say that we are "running the function." Some people prefer the verb to be "executing." In mathematics, the process is called "evaluating" the function or even "applying" the function to its inputs.

Here, "executing" is not about killing but instead about doing something, for instance the orders of an executive officer.

We have already mentioned that the `heat_index()` function takes two inputs: temperature and humidity. In the algorithm implemented by `heat_index()`, temperature and humidity play different roles. We identify where each comes in by using names. You don't need to know how to program computers to follow this book. Still, much of the notation can be

understood with just a little instruction. In the text of this book, we will use the conventions of a popular computer language: R.

Without going into detail, the definition of `heat_index()` looks like this:

```
heat_index <- function(temperature, humidity) {
  # Algorithm goes here.
}
```

This notation organizes four pieces of information:

1. this is a function;
2. the name of the function is `heat_index`;
3. it takes two inputs,
4. the first input is named temperature, the second is named humidity,

The names that go inside the `function()` parentheses are often called the *arguments* of the function (although we will also refer to them as *inputs* in this book). Arguments are like characters in a play. Inputs are the actors who play the characters. Note, however, that in this book we are already using "argument" with a specific meaning as in the title of this book. We will use the shorthand 👉**arg**👈 the names of inputs to a function, and reserve "argument" for the sense of Chapter 1, that is, a claim supported by evidence and a chain of reasoning. .

When using `heat_index()` to perform a calculation, we give the arg values inside parentheses as in `heat_index(30, 0.80)`.

If the user wrote the inputs in the wrong order, `heat_index(0.80, 30)` it would be as if the user wanted to specify a very cold temperature (0.70 °F) and a meaninglessly high humidity (8500%). To avoid such problems, the R language allows the user to refer to the args by their names rather than their positions:

```
heat_index(humidity = 0.80, temperature = 30)
```

```
[1] 37.66704
```

6.3. Operationalizing comfort: the heat index

The heat index is familiar to those living in hot climates; few realize that it originated as a careful quantitative operationalization of comfort. Key research into the manner was presented by R.G. Steadman in 1979 as the "sultriness index".

R.G. Steadman "The Assessment of Sultriness", *Journal of Applied Meteorology and Climatology.* July 1979 pp. 861-873

Figure 6.2 displays the sultriness function graphically, but doesn't convey the extensive quantitative modeling that went into its development.

TABLE 2. Temperature-humidity scale: Apparent temperature (°C) corresponding to each combination of dry-bulb temperature and relative humidity. Values in parentheses correspond to skin humidities above 90% and are approximate.

Dry bulb temperature (°C)	Relative humidity (%)										
	0	10	20	30	40	50	60	70	80	90	100
20	16	17	17	18	19	19	20	20	21	21	21
21	18	18	19	19	20	20	21	21	22	22	23
22	19	19	20	20	21	21	22	22	23	23	24
23	20	20	21	22	22	23	23	24	24	24	25
24	21	22	22	23	23	24	24	25	25	26	26
25	22	23	24	24	24	25	25	26	27	27	28
26	24	24	25	25	26	26	27	27	28	29	30
27	25	25	26	26	27	27	28	29	30	31	33
28	26	26	27	27	28	29	29	31	32	34	(36)
29	26	27	27	28	29	30	31	33	35	37	(40)
30	27	28	28	29	30	31	33	35	37	(40)	(45)
31	28	29	29	30	31	33	35	37	40	(45)	
32	29	29	30	31	33	35	37	40	44	(51)	
33	29	30	31	33	34	36	39	43	(49)		
34	30	31	32	34	36	38	42	(47)			
35	31	32	33	35	37	40	(45)	(51)			
36	32	33	35	37	39	43	(49)				
37	32	34	36	38	41	46					
38	33	35	37	40	44	(49)					
39	34	36	38	41	46						
40	35	37	40	43	49						
41	35	38	41	45							
42	36	39	42	47							
43	37	40	44	49							
44	38	41	45	52							
45	38	42	47								
46	39	43	49								
47	40	44	51								
48	41	45	53								
49	42	47									
50	42	48									

Figure 6.2.: The sultriness function. (Source: Assessment of sultriness, p. 862.) Note: Meteorologists measure the "dry bulb" temperature with an ordinary thermometer. In contrast, the "wet bulb" temperature measures evaporative cooling, operationalized by wrapping the thermometer with wet gauze.

6.4. Algorithms

We called Figure 6.2 the sultriness *function*. But the research article that introduced it did not write `sultriness(temperature, humidity)` or give an implementation as a formula. Figure 6.2, which is intended for a human reader, is the only presentation that was given in the research article.

It's likely that you can figure out how to find the output of `heat_index()` given the inputs. You can confirm that a temperature of 30 °C and humidity of 80% corresponds to a heat index of 37 °C.

But, imagining that you don't know how to use Figure 6.2, let's write out instructions. Such instructions have a more formal name: 👉**algorithms**👈. Admittedly, the name "algorithm" is offputting to many people. It comes from the name of a mathematician, Muhammad ibn Musa al-Khwarizmi (c. 780 – c. 850). He is famous for a book whose key word is "al Jabr," Arabic for "the reunion of broken parts." Our word "algebra" comes directly from "al Jabr." Al-Khwarizmi's book gives a description of the steps to follow when reuniting, say, the broken parts of what we now write as $a = x/b$ to get $x = a\,b$.

Here's a table-reading algorithm, in a form intended for an educated person:

1. Take the temperature and find the closest value in the left column of the table. That closest value marks the row of interest.
2. Take the humidity and find the closest value in the top row of the table. That closest value marks the column of interest.
3. Find the number at the junction of the row and the column of interest. That is output.

Like all algorithms, this one is written in terms of tasks that the processor already know how to perform, e.g. "find the closest value." If you didn't know how to perform that task, you would need a description of the steps involved. That is, algorithms are often written in terms of other algorithms.

The simplest algorithms are those where the answer is already known. For instance, to learn how to add numbers with multiple digits, you first had to memorize the basic facts of single-digit arithmetic: the *addition table*.

Chapter 7 introduces a small set of basic functions called the "core modeling functions." Each of these is pretty simple, and you will be expected to be able to sketch out a rough graph and know the values of certain special inputs and their output values. Of course, human memory is fallible. So you will have a computer implementation of each core modeling function to use and refer to.

6.5. What do you need to know about an algorithm?

In the modern world, many of the objects we use productively every day are so complicated that few people, if any, know exactly what is inside them. For example, telephoning a friend does not require knowing in detail how a phone or a telecommunications system works. Drivers typically do not know how a modern gasoline engine works. We can use these machines because they have a user interface that we know how to operate. Using the interface lets us confirm we are using it properly. For instance, confidence in a phone comes (in part) from the text messages sent and successfully received.

The same is true of functions. Using a function requires knowing what kinds of inputs it takes and observing what kind of output it produces. Although the "t" in "mortgage" is silent, it refers to an important aspect mortgage loans. The root *mort* means "death", as in "mortality." In a mortgage, the borrower pays the lender a fixed amount each month. This gradually reduces the amount of money owed. When that amount reaches zero, the mortgage is dead: no more payments are required. Some people even have funereal rites to celebrate the death of a mortgage. To illustrate, consider a situation faced by people buying a home. Almost always, they buyer needs to borrow money, a form of loan called a "mortgage."

The internet has apps that calculate monthly mortgage payments. Behind such apps is a function that takes three inputs: the amount borrowed (the principal), the annual interest rate on the loan, and the loan duration (that is, the number of monthly payments needed to pay off the loan.) The function's output is the amount of the monthly payment. In internet apps, inputs are often specified with sliders or check-boxes or drop-down menus. To stay close to the `function(arg1, arg2, arg3)` notation, we provide an implementation using that convention. In the interactive version of this book, you can even find the output for any inputs you care to specify.

```
monthly_payment(200000, 6, 30)
```

```
[1] 1199.101
```

Keep in mind that `monthly_payment()` is a *model* of what you have to pay. It takes into account the interest rate and the duration of the loan. But there are often other dimensions of the situation, for example, processing fees or mortgage insurance. You, or your real-estate agent, will need a more complete model of the situation. Even AI can easily get it wrong if it doesn't know the details of the mortgage contract you will eventually be signing.

Typically, computer languages accept purely numerical inputs rather than a complete description of the quantity. So, the first input is 200000 rather than $200,000, the second is 6 rather than 6%, and the third is 30 rather than 30 years.

What is needed to know to use this function successfully? Here is a list:

1. The name of the function: `monthly_payment()`.
2. The order in which to provide the inputs: `(principal, interest, duration)`
3. The meaning of each input and the eventual output:
    i. `principal`: a numerical amount, the currency does not matter.
    ii. `interest_rate`: a yearly interest rate
    iii. `duration`: the number of years
    iv. Output: A numerical amount. The currency will be the same as that of the `principal` input.
4. A way to check the function's action and whether the user correctly works the interface. Such checking can take several forms. The simplest is to give the same inputs to a trusted source, as in Figure 6.3, and compare the outputs.

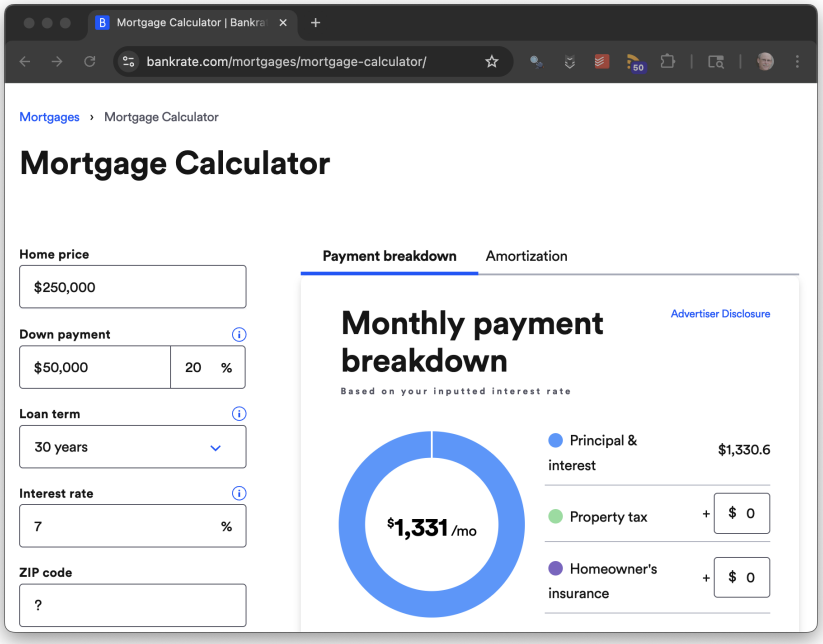


Figure 6.3.: A randomly selected mortgage payment calculator from the internet. The interface is different from the `monthly_payment()` function, with entry slots for house price and down payment rather than `principal`.

What about the algorithm behind the app? Does the ordinary user need to know that? Being able to see and interpret the algorithm—which is, in this case, a formula—would enhance mastery of the subject. However, in practice, even algebraically skilled people who might understand a derivation of the formula and be able to translate it into computer code will need to check the results. Will AI help? Maybe eventually. Sadly, the AI query I used in 2025, "How to calculate mortgage payment," returned a formula with a fatal error.

Derivation of a mortgage calculator formula

7. Core modeling functions

Vocabulary Review

- arg (6)
- data (2)
- functions (6)
- magnitude (5)
- pure number (3)
- quantity (3)
- rate (4)
- units (3)

New

- core modeling functions
- input and output scaling

The previous chapter introduced notation for functions and their args. Functions can have any number of args. In modeling, a function representing one of the sharp-cornered boxes—a quantity *endogenous* to the model, will have one arg for each of the incoming arrows originating from other boxes.

This chapter introduces and names a small set of *one-arg* functions that are particularly useful in modeling. We call these the 👉**core modeling functions**👈. In making selections for the core modeling functions, our primary goal is to provide a library of simple relationships of the sort that appear in many modeling contexts. Each core modeling function will have a name, so you can think of them as letting you refer to a relationship type by name: an aid to communication both with other people and with the computer.

7.1. Shapes of the core modeling functions

We display the shapes of the core modeling function in a conventional way: **graphing** the function on square-gridded graph paper. The core modeling functions have been specifically defined to take a pure number as input and to produce a pure number as output. To remind the reader of this, the graphs of the core modeling functions are placed on a small grid of unmarked graph paper with a major horizontal and a major vertical axis.

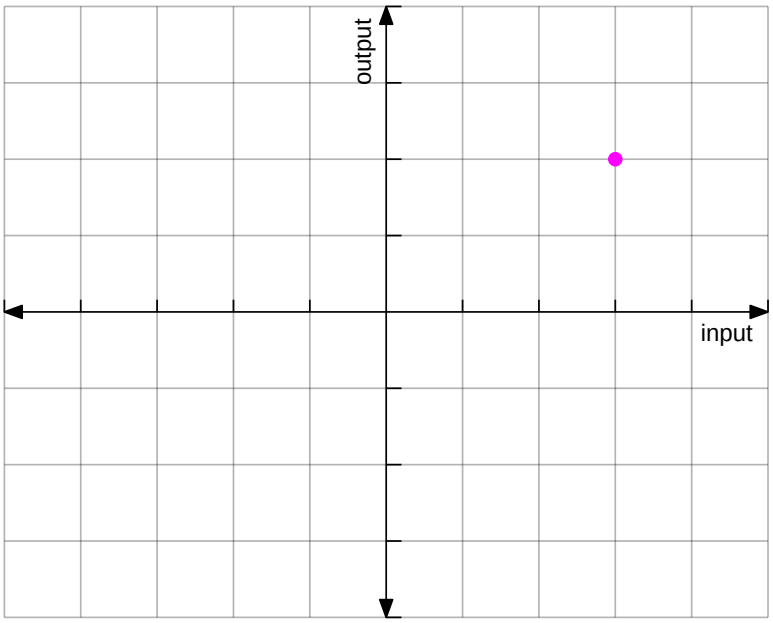


Figure 7.1.: Graph paper onto which the core modeling functions will be displayed. The two double-headed axes intersect at (input = 0, output = 0). You can read off any coordinate by counting the vertical and horizontal grid lines that separate it from (0,0). For instance, a small dot has been placed at coordinate (input = 3, output = 2).

The coordinates of the graph paper are used to determine the input-to-output values of the function. The grid lines in the display are not just decoration: they show how a numeric input gets translated to a numeric output.

In the displays of the core modeling functions, you can infer the numerical values of input and output by counting the vertical and horizontal grid lines respectively. The vertical axis marks zero on the horizontal scale. The successive vertical lines correspond to values separated by 1. The horizontal grid lines are interpreted in the same way.

Why not label the graph paper with the numerical values? It would be simple enough to do, but for the reasons described in Section 7.2 it would make the graphics too busy.

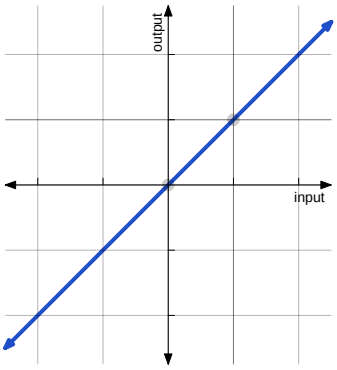
We will work with eight core modeling functions. Often, one or another of these eight will suffice for modeling the relationship between two quantities.

THE EIGHT CORE MODELING FUNCTIONS.

We have constructed the core modeling collection from the collective experience of modelers over many generations. We have intentionally violated convention by giving even famous functions new names that are more descriptive than the traditional names.

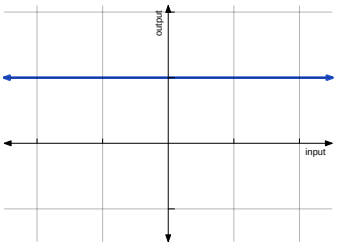
Often, the name of a core function is a reminder of the shape. For instance, `hill()` has a graph that looks like a hill (in cross-section), while `hillside()` is just one side of the hill. `Osc()` is oscillatory, that is, goes up and down repeatedly, like a long chain of hills. On the other hand, sometimes, the names—`recip()`, `exp2()`, and `log2()`—honor traditional nomenclature.

`same()`



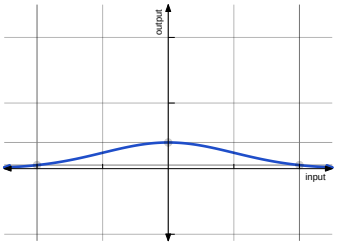
The name `same()` reflects that the *output* is the same as the *input*. On a graph this corresponds to a diagonal line.

`flat()`



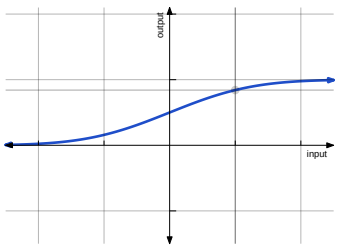
The output of `flat()` is always 1, regardless of the input.

`hill()`



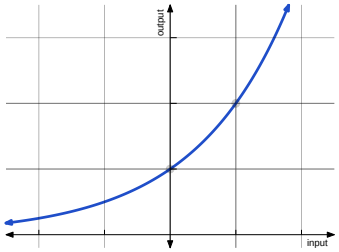
The function output rises gradually from zero output, then falls, returning gradually to zero output.

`hillside()`



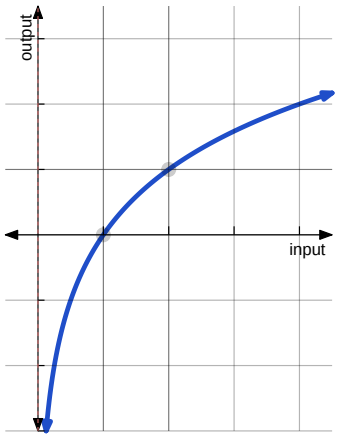
The function output rises gradually from zero, leveling off at an output of 1.

`exp2()`



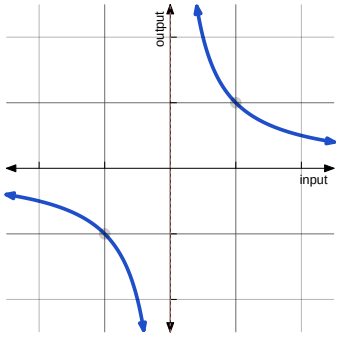
The function starts out almost flat then rises at an every-increasing rate. This pattern is called **exponential growth**. Another way to see this, exponential growth corresponds to steady, uniform growth in magnitude. (See Chapter 5) `exp10()` produces magnitude in the familiar steps by a factor of 10.

`log2()`



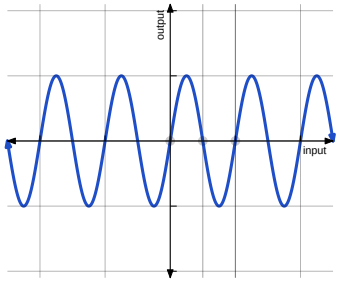
For each doubling of the input value, the function output goes up by 1 unit. Only positive inputs are allowed. Another way to see this: the logarithm reads off the magnitude of a number. `log10()` gives the scientific-notation format of magnitude.

`recip()`



The reciprocal function. The output is one over the input. For instance, if the input is 2, the output is 1/2. Negative inputs produce negative outputs.

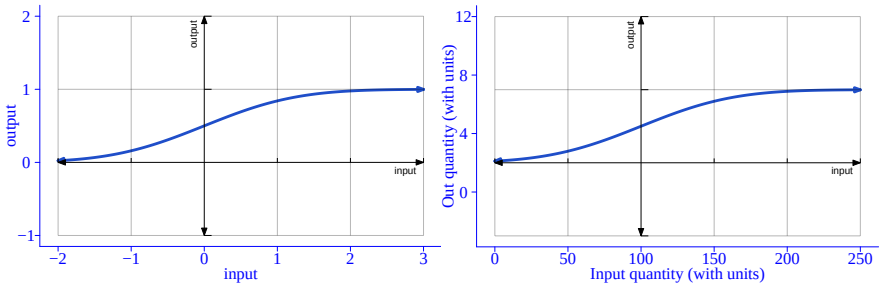
`osc()`



The name `osc()` is short for oscillatory; the function output oscillates up and down as the input increases. A complete cycle is accomplished in one unit of the input. Another way to say this: The period of `osc()` is 1.

7.2. Input and output scaling

The core modeling functions have been defined always to take a pure number as input and produce a pure-number output. In modeling, however, typically the inputs to a function are *quantities* and so is the output. There is a simple strategy for using the core modeling functions to represent quantity-to-quantity relationships. We call this ☞**input and output scaling**☜.



(a) No scaling

(b) With scaling

Figure 7.2.: Input and output scaling applied to the core function hillside().

Figure 7.2 shows the essence of the scaling technique. We apply a new set

of tick marks is applied to the modeling function. These are shown in blue in the figure.

In providing a quantity as input to the function, use the blue labels on the horizontal axis. In reading off the output, use the blue labels on the vertical axis. For the scaled function depicted in Figure 7.2, the input 150 (with units as appropriate) corresponds to the output 5 (again, with whatever are units of the output quantity).

In brief, the core modeling function takes pure-number input and output which you can read from the black grid lines. The input-and-output scaled function has quantity inputs which you read from the blue axes.

Once you have the blue axes arranged suitably for your modeling purpose, it's not hard to write the customized function in, for instance, a computer implementation. To illustrate for the scaled example in Figure 7.2(b), we'll call the customize quantity-to-quantity function `modelfun()`:

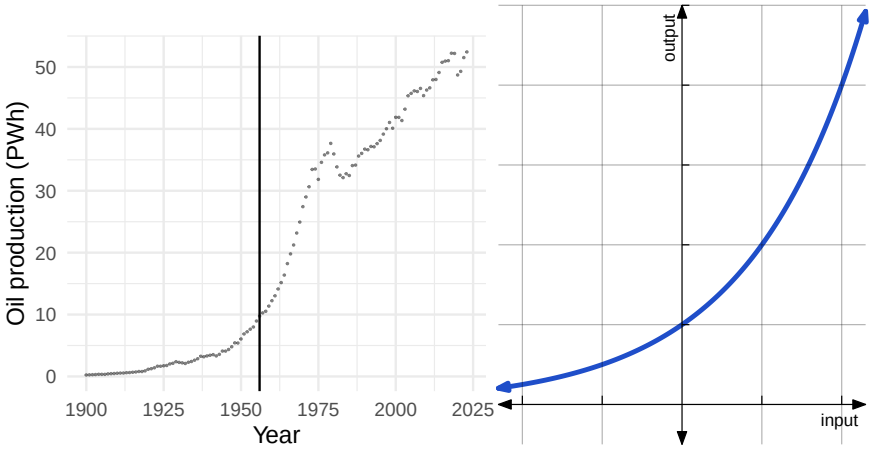
Input scaling facts: The math input 0 has the blue label 100. A step of one unit on the math axes corresponds to a step of 50 on the blue axes.

Output scaling facts: The math output 0 corresponds to about 2 on the blue axes; this means that the B output scaling parameter is 2. The math output 1 corresponds to 7 on the blue axes, so that 1 step in math input—from 0 to 1—corresponds to steping from 2 to 7 in the quantity output. That is, a step of one unit of math output is 5 units of quantity output. This corresponds to an A parameter of 5 for the output scaling.

7.3. Example: Global oil production

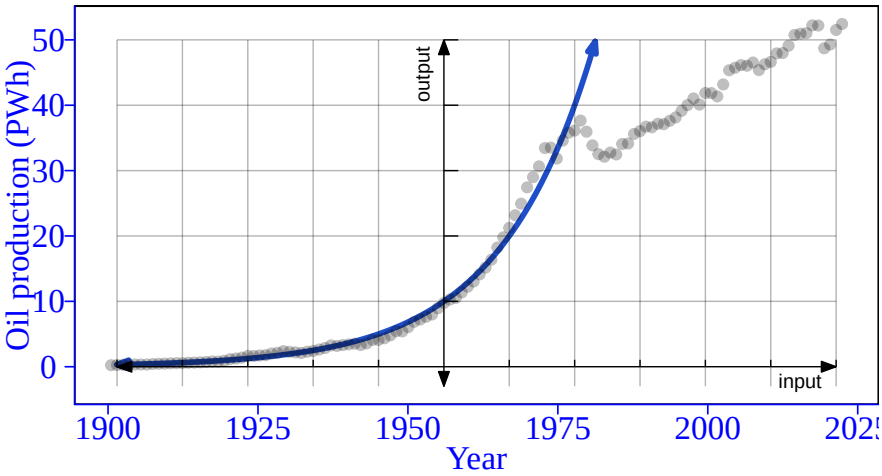
Figure **??** shows global oil production each year since 1900. Production is measured in terrawatt-hours. To put that in context, 16 PWh is enough fuel for all the world's 1.7 billion gasoline vehicles for a year's driving. [If the global fleet were electric, the energy consumption would be about 3.5 PWh.]

Oil production through 1975 follows an exponential growth pattern: increasing at an every-increasing rate.



(a) Oil production in petaawatt-hours of energy

(b) Core modeling function: `exp2()`



(c) Scaled model

The scales of the historical oil-production plot are straightforward. The data run from 1900 to 2023. Production was near zero in 1900 and reached about 50 PWh in 2023. The production growth pattern before about 1975 is reminiscent of the shape of the `exp2()` core modeling function, but the scale in the graph of `exp2()` is unsuited to the data scale.

By figuratively picking up the oil-data graph, along with its scales, and stretching and shifting it appropriately, we can adopt the scale to the shape of the exponential function, at least for the years before 1975. The result, in panel (c), is a graphic with two scales. The blue scales on the outside come from the oil data. The input scaling for this model of oil production comes

from the alignment of the mathematical gridding with the horizontal blue quantity scale.

For instance, zero on the mathematical horizontal grid—the location of the double-headed vertical "output" line—matches up to 1956. You can also see that year 2000 in the blue scale matches up closely to 4 on the horizontal grid scale. Consequently, the input scaling function will turn 1956 into 0 and year 2000 to 4.

7.4. Stories told by shape

For each core modeling function, we present the story in two ways: a narrative and a graph. The action of stories typically develops over time. In the language of functions, we will treat time as the input and examine the output over time. Of course, in any given modeling context, the input might be something other than time: for instance, the dose of a drug, or the speed of a vehicle.

To avoid unnecessary abstraction, our descriptions will assume that the function's output is the outdoor temperature. The brief stories we will tell to introduce each function will be about how temperature changes over time.

osc( ) A function that oscillates smoothly up and down, repeating itself over and over again. Such repeating functions are called ☞**periodic**☜. For example, the outdoor temperature rises and falls each day, repeating itself roughly the same way day after day.

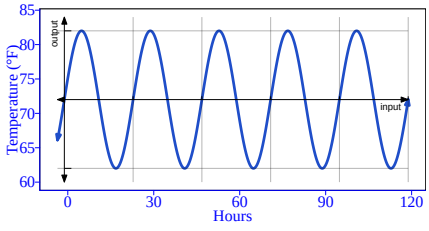


Figure 7.4.: osc( )

hillside( ) The hillside function starts steadily, then gradually increases before leveling off. For example, hillside() might represent the temperature as a warm front moves past at around 10 am: a steady high temperature in the afternoon having gradually replaced the the mornings low temperature.

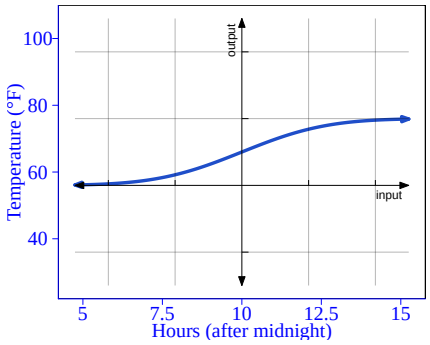


Figure 7.5.: hillside( )

hill() The hill() function goes up from a baseline and then goes back down to the baseline. For instance, the hill() can describe a heat wave lasting several days before relief and the return to normal temperature.

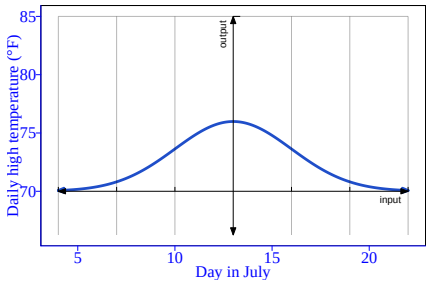


Figure 7.6.: hill( )

exp2( ) After each unit interval of time, the temperature *doubles* from what it was at the start of the interval. Think of an apocalyptic science-fiction story about climate change. The temperature is increasing over time, and as it does, various feedback mechanisms—e.g., the release of methane from melting permafrost or the increased absorption of sunlight by the ocean as an ice cap melts—strengthen. Eventually, of course, something has to give. The doubling pattern cannot continue indefinitely. But the `exp2()` core modeling function doesn't add that bit to the story.

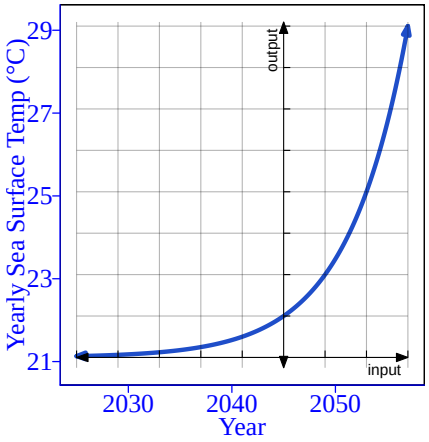


Figure 7.7.: exp2( )

log2() Picking up on the science fiction described under exp2(), imagine a crusty, old meteorologist who tells *how long* it will take for the doubling temperature to reach any given level.

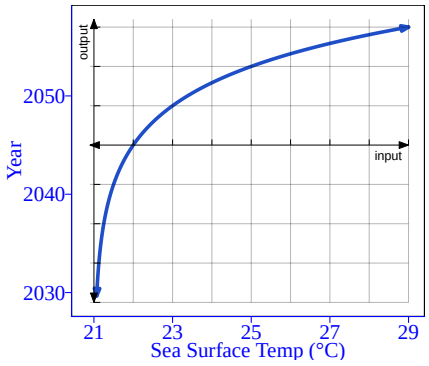


Figure 7.8.: log2( )

same( ) As time increases, the temperature increases likewise. Hour to hour, the temperature increases by the same amount.

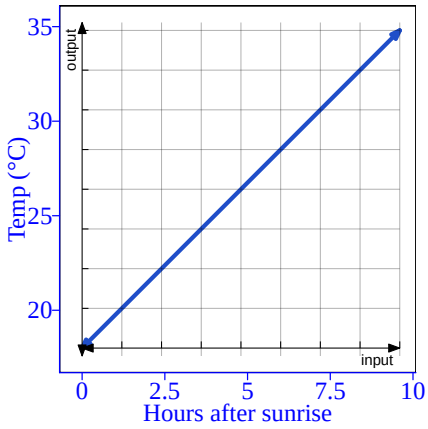


Figure 7.9.: same( )

flat() A story with no action. As time goes by, nothing changes! Daytime temperatures are the same as nighttime temperatures.

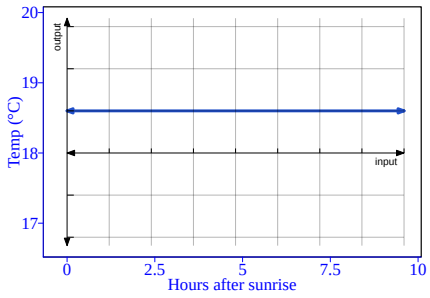


Figure 7.10.: flat( )

8. Uncertainty in quantities

We have looked at functions so far as our model representation of the relationship between one quantity (the function output) and one or more other quantities (the function inputs). This role for functions will remain central to how we use models for quantitative reasoning. Still, there are other important uses for functions.

This chapter considers a distinctly different use for functions: quantifying the 👉**uncertainty**👈 in the value of a single quantity.

Imagine a family in the process of constructing a budget for next year. Naturally, the family has a good idea about their expected income next year. Likewise, they can estimate next year's expenses for rent, utilities, food, transportation, and so on. This information guides them in planning for discretionary spending, such as vacations, clothing, and so on.

Suppose the family's current rent is 1500 dollars per month. The lease is up for renewal in 6 months; they are almost sure the rent will increase—their best guess: a 4% increase to $1560 per month.

Most people making budgets use a best guess for quantities whose exact value is uncertain. Doing this, the family figures next year's total rent expenditures as six monthly payments of $1,500 per month, followed by another six months at $ 1,560 per month. Total: $18,360 per year.

In situations where the stakes are higher, it can be worthwhile to keep track not just of the best guess but of the *uncertainty* in that guess. It is common to express uncertainty using 👉**ranges**👈 An equivalent format is "approximately 80 $\pm$ 20 Cals per mile." For instance, in Chapter 4, an AI quoted a range for the number of Cals burned per mile while walking: "approximately 60 to 100 Cals."

The shorthand notation for the range just mentioned is [60, 100]. In statistics, such a range is not intended as an absolute statement that the value can never be less than 60 or greater than 100. Rather, it is a statement that the value is *likely* to be in the range, and *unlikely* to be outside of it.

Vocabulary Review

- analysis (3)
- data (2)
- functions (6)
- models (1)
- number line (3)
- per (3)
- point (3)
- quantity (3)
- rate (4)
- sample (2)
- value (2)

New

- bayesian
- exponential distribution
- frequentism
- hypothesis
- mean
- normal distribution
- probability distributions
- random number generation
- range
- relative probability
- simulation
- standard deviation
- uncertainty
- uniform distribution
- variance

In the statistics interpretation of ranges like [60, 100], "likely" is interpreted as a 95% chance.

However, when doing calculations—such as adding up the uncertainties for each of the several budget categories—the range format does not suffice. To support quantitative reasoning, we need another format for expressing uncertainty.

8.1. Relative probability

The fundamental problem with the range format is this: It is not necessarily the case that uncertainty propagated through a function is correctly represented by applying the function to both the top and bottom of the range.

To illustrate, consider the number of Cals burned by a person in a walk of about 3 to 4 miles. In addition to this walk length information, we know that the energy consumption per unit distance is 60 to 100 Cal/mile. As described in Chapter 3, we combine quantities with different dimensions by either multiplication or division. The correct choice here is multiplication, since [Cal] = [miles] [Cal/mile]. This multiplication would directly provide the answer if there was no uncertainty attached to the distance or energy consumption per unit distance. But here, we are not just multiplying quantities but *uncertain quantities* described by ranges. Using the range notation the multiplication can be written like this:

$$[3 \text{ to } 4]\,\text{miles} \times [60 \text{ to } 100]\,\text{Cals/mile} = \textbf{?}\ . \tag{8.1}$$

The question mark in Math expression 8.1 is justified because there is no standard way to multiply uncertainty *ranges*. Anyone can invent a *reasonable-sounding* strategy for the calculation, for instance, multiply the lower ends of each range, and similarly for the upper end of the result:

$$[3 \times 60 \text{ to } 4 \times 100]\,\text{Cals} = [180 \text{ to } 400]\,\text{Cals}\ .$$

As reasonable as this looks, there is a problem with the strategy: how can we tell whether it gives the *right* answer to the underlying question?

Indeed, there are situations where the above strategy is clearly wrong. For instance, consider multiplying the two ranges [-3 to 5] by [-10 to 2]. Slavishly following the above procedure yields a range of [30 to 10]. Is this right? For both ranges, note that zero is well within the limits. Multiplying any

number by zero gives zero, so zero ought to be inside the range produced by the multiplication. However, zero is not in [30 to 10].

Figuring out how to reliably handle calculations involving uncertainty involved the work of many mathematicians and scientists over the 18th and 19th centuries. A good answer to the walking energetics question in Math expression 8.1 is [200 to 360] Cals. Similarly, a good answer to [-3,5]×[-10,2] is [-25 to 17], which correctly includes zero inside the range.

The advent of the computer has made such uncertainty calculations much more accessible. At the heart of the matter is the concept of 👉**probability distributions**👈. When we use a range (such as 3 to 4 miles or 60 to 100 Cals/mile) to represent uncertainty, we mention only two values for the quantity: the lower and upper bounds on the range. In contrast, probability distributions consider every possible value: the whole number line, not just the two points defining the range.

A probability distribution can be created by assigning a positive number to each point on the number line. This number is called a 👉**relative probability**👈. Relative probability numbers cannot be negative.

There are an infinite number of points on the number line, so to use probability distributions, we need to keep track of an infinite number of relative probabilities: one for each point on the number line. Fortunately, Chapter 6 has already provided a mechanism for handling the situation: *functions*.

To illustrate, consider the uncertainty we previously represented as the range 60 to 100 Cals/mile. Figure 8.1 shows the corresponding probability distribution as a *function*.

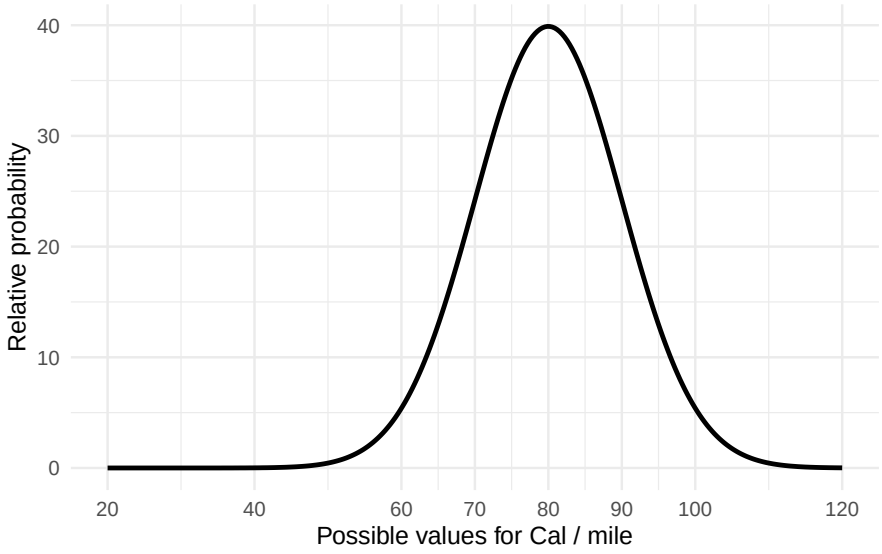


Figure 8.1.: A standard probability distribution corresponding to 60 to 100 Cal/mile.

The *input* to the function is drawn from the set of possible values for the uncertain quantity. The *output* from the function is the relative probability for that particular value of the quantity. In Figure 8.1, the relative probability for 40 Cal/mile is very low —practically zero. On the other hand, the highest relative probability corresponds to 80 Cal/mile. The numerical value of the function output at 80 Cal/mile is 40. That particular value, 40, tells us very little on its own; it needs to be put in context. The context for a relative probability is the function output for other inputs, that is, for other possibilities of the Cal/mile value. For instance, the relative probability for 60 Cal/mile is about 5, while the relative probability for 80 Cal/mile is about 40—that is, eight times as likely as 60 Cal/mile.

When the output of a relative probability function is close to zero, the occurrence of the corresponding input value is vanishingly rare. We took advantage of this in Figure 8.1 to avoid showing the entire number line on the input axis; we centered the display on the most likely values.

`hill()` is often a good choice for modeling a probability distribution. Often, but not always. The following section explores `hill()` along with a handful of other probability distribution functions. Note that in general the core modeling functions are not used for probability distributions. `hill()` is the one exception. We will have more to say about this in Section 8.4.

8.2. Where do probability distributions come from?

Probability was invented in the 17th century by mathematicians interested in gambling. Progress was made by a mixture of experience reinforced by mathematical theorizing and algebraic derivation.

A significant development was the realization, in the late 18th and early 19th centuries, that the `hill()` function was particularly useful in the analysis of data. `hill()`-like distributions were observed frequently in data from diverse contexts, so much so that it came to be called the "☞**normal distribution**☜." Mathematical theorizing, long after the normal distribution came into use, showed that in a scenario in which many independent random numbers, from whatever source, are added together, the normal distribution, or something very much like it, results. This scenario appears frequently in statistical calculations. However, over-reliance on this scenario outside statistical methodology can lead people to wrongly discount the possibility of rare events such as droughts, floods, and financial crises.

Probability distributions like the normal distribution are often used to model uncertainty. In Chapters 11 and 12, we will add a somewhat different perspective: a probability distribution is a ☞**hypothesis**☜ which we use to make sense of our experiences and observations in the world.

Starting about 1900, statisticians looked at the question of how correctly to infer properties of the real world from the results of statistical calculations. That is, they wanted to figure out what kinds of conclusions about the world are justified by analysis of data. This matter is called ☞**statistical inference**☜. One important school of thought emphasized making mathematically provable statements about inference. To do this, they had to develop a theoretical framework which came to be called ☞**frequentism**☜. Frequentist methods play a central role among statistical theorists and remain the foundation of much statistical education. Readers who have studied statistics have likely used formulas and nomenclature—e.g., p-value, hypothesis tests—that stem from the Frequentist point of view. Another framework for inference, ☞**Bayesian**☜ inference, predates Frequentism. In the Bayesian point of view, the emphasis is not on defining things in a way suitable to support mathematical proof, but rather in a way to answer questions in a form needed for quantitative reasoning, uncertainty assessment, and decision making. Over the last half-century, the Bayesian point of view has gained many adherents among statistical theorists, partly because Bayesian calculations that were once difficult became practical by using computers. In this book, we adopt the Bayesian perspective.

8.3. Exemplary probability distributions

There is an infinity of probability distributions simply because there is an infinity of functions. Nevertheless, almost always modelers draw from a small library of about twenty named distributions. More precisely, these are ☞**named families of distributions**☜. Each of these families includes one or two parameters that, often, are equivalent to applying an *input transformation* and/or *output transformation*. In this section, we focus on three distributions that modelers use in everyday work: the *normal*, the *uniform*, and the *exponential*.

Uniform distribution The ☞**uniform distribution**☜ models a situation where the possibilities must be within a known range, but could be anywhere in that range. Figure 8.2 shows what a uniform distribution would be for a situation like this: "She texted that she will arrive in the afternoon, but I don't know exactly when."

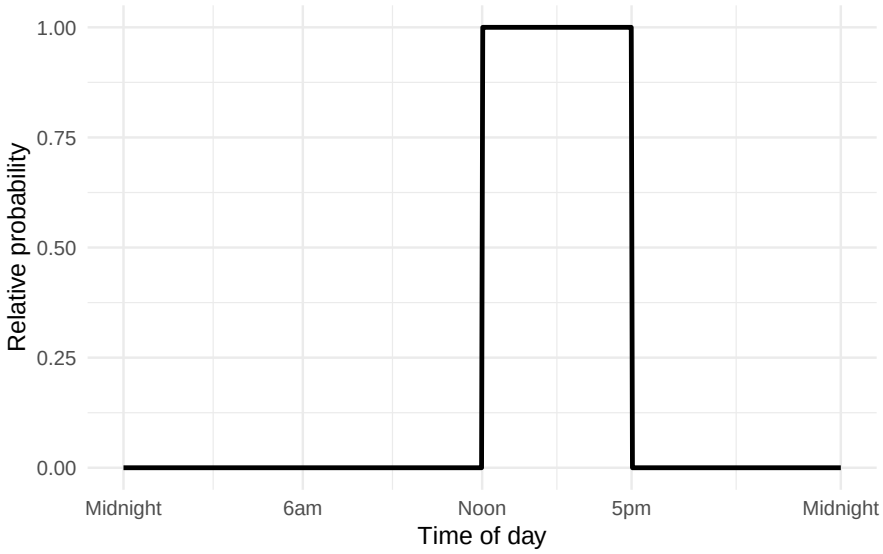


Figure 8.2.: A uniform distribution.

The parameters for the uniform distribution family are the minimum and maximum extremes of the range of possibilities.

Normal distribution We introduced the normal distribution in Section 8.1 as the standard distribution family that corresponds to the type of uncertainty often expressed using a range or the $\pm$ notation.

It surprises many people that these notations do not correspond to the *uniform distribution.* Recall that the uniform distribution labels as *impossible* any outcome outside of the range from minimum to maximum. However, in many circumstances, an outcome outside the range is merely unlikely, not impossible. For instance, is it impossible that our friend will arrive in the evening or before noon?

The normal distribution, on the other hand, softens the edges and replaces "impossible" with "not likely." The further from the interval, the less likely. The closer to the center of the interval, the more likely. The normal distribution in Figure 8.3 includes the possibility that she arrived yesterday or will arrive next week. Very, very unlikely, but possible.

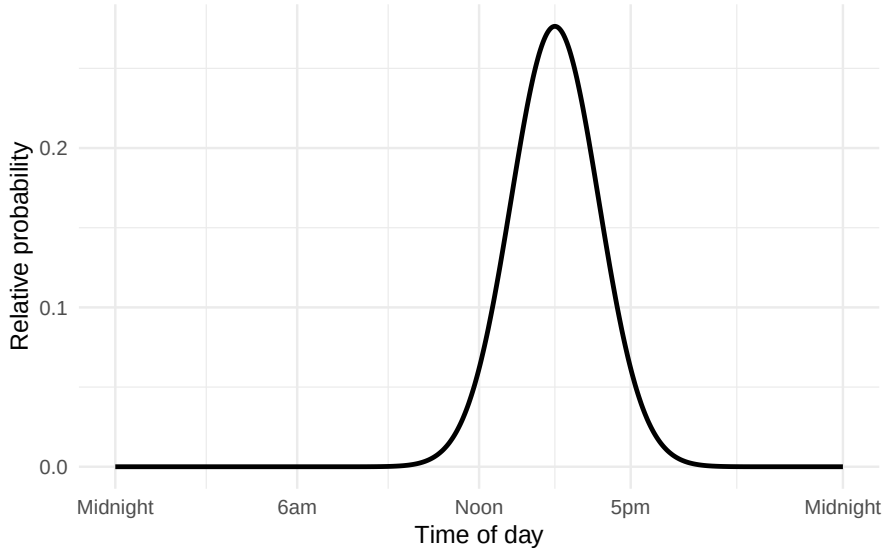


Figure 8.3.: A normal distribution smooths out the sharp corners of Figure 8.2.

The parameters for the normal distribution set the center and spread, but are usually called the "☞**mean**☜" and "☞**standard deviation**☜." (Some authors prefer to use "☞**variance**☜" instead of "standard deviation": the variance is the square of the standard deviation.) The mean can be any value; the standard deviation must be positive.

Exponential distribution The ☞**Exponential distribution**☜ corresponds to a different message from a friend. "It's noon now. I'm going to try to get there right away. I think it will take about an hour, but I do not know anything beyond that. It might be much longer." The exponential distribution expresses much greater uncertainty at the

high end. A much more typical application would be the uncertain time between 100-year storms.

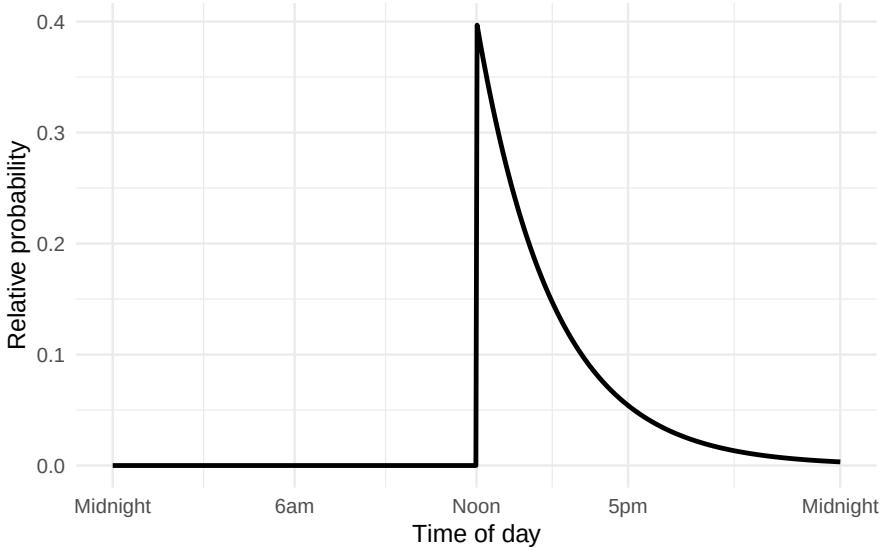


Figure 8.4.: The exponential distribution.

A concrete way to think about the exponential distribution is as a description of the intervals *between* randomly occurring events. In Figure 8.4, the first event was assumed to be at noon; the second event will be the time that the friend arrives. The exponential model is based on the idea that there is a fixed probability that the second event will happen in a short interval after the first. But if it doesn't occur in that short interval, the same probability applies to the event happening in the next short interval, and so on. Perhaps contrary to intuition, this means that the probability of the second interval happening at time t after the first event decays exponentially with t, as in Figure 8.4.

The exponential model assumes that the second event is equally likely to occur at any time. There is one parameter for the distribution typically called "lambda" (which is λ in the Greek alphabet). The mean time between events is $1/\lambda$. (In Figure 8.4, the mean corresponds to that for the distributions shown in Figure 8.3 and Figure 8.2.)

8.4. Simulation

A relative probability distribution function takes a possible value as input and returns a number. That number is an indication of how *likely* that possible value is. Relative probability distributions are useful for describing uncertainty.

Often, however, we need to reason about the consequences of input uncertainty for the uncertainty in the output of the function. One technique for such reasoning with uncertainty is 👉**simulation**👈.

At the heart of simulation is the idea of using a relative probability distribution function to generate a likely input value. The input values that are associated with a high relative probability are more likely to be generated than those associated with a low relative probability.

We call the process of generating a likely input value 👉**randomly sampling from the distribution**👈 or, more generally, 👉**random number generation**👈.

For instance, in Figure 8.1, a random sample between, say, 75 and 85 would be quite likely, while sample values less than 40 would be vanishingly unlikely.

Random number generation is an important tool for many modeling applications and even plays a role in constructing the responses from AI systems. Such generators are provided for many computer languages and are often built into the language itself.

To illustrate, let's generate a single random sample from the distribution in Figure 8.1.

```
rnorm(1, mean=80, sd=10)
```

```
[1] 82.48764
```

Generating a single number does not give much information, so modelers often generate many random numbers. We will illustrate with 20, but it's common to generate thousands or even millions of random numbers

```
rnorm(20, mean=80, sd=10) |>  # The hill() distribution.
  round(digits = 1)
```

```
 [1] 72.4 54.9 94.7 67.1 90.7 72.2 82.6 76.7 89.4 79.8 82.7
[12] 83.0 79.4 82.7 75.5 76.2 60.1 88.1 73.5 79.3
```

As a simple example of using simulation, consider the problem posed at the start of this chapter: [-3,5]×[-10,2].

```
set.seed(102)
product <-
  runif(20, min =  -3, max = 5) *
  runif(20, min = -10, max = 2)
round(product, digits = 1)
```

```
 [1]   1.7  -2.5 -11.3  -3.4  15.7  -2.1 -42.2   1.9 -
21.0
[10]  -1.6 -21.4   1.9 -32.8   0.0   2.7 -28.9 -25.2   0.0
[19]  -7.5 -20.2
```

Notice that the 20 samples of the product vary from one to the other. It is only as a group that they give us information about the uncertainty in the product. Computer graphics—with a much larger sample—makes it easy to visualize the uncertainty, as in @two-multiplied-uniforms. The distribution is not uniform, but rather has a peak in the middle and tails off to zero at the extremes.

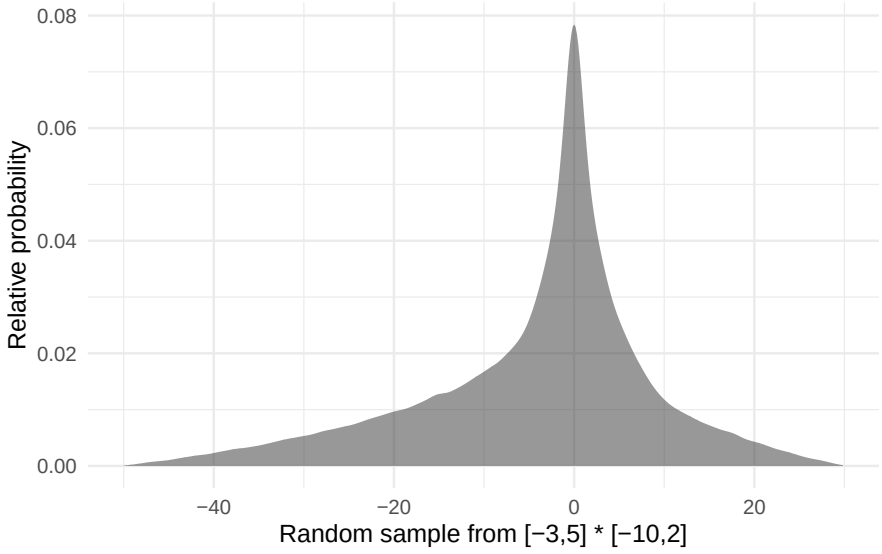


Figure 8.5.: Relative probability distribution for [-3,5] × [-10,2] computed with 200,000 random samples.

9. Composing functions

Vocabulary Review

- conversion factor (3)
- core modeling functions (7)
- data (2)
- dimension (2)
- functions (6)
- models (1)
- per (3)
- point (3)
- quantity (3)
- rate (4)
- rate of change (4)
- system (6)
- units (3)
- value (2)

New

- coefficient
- contour plot
- linear regression

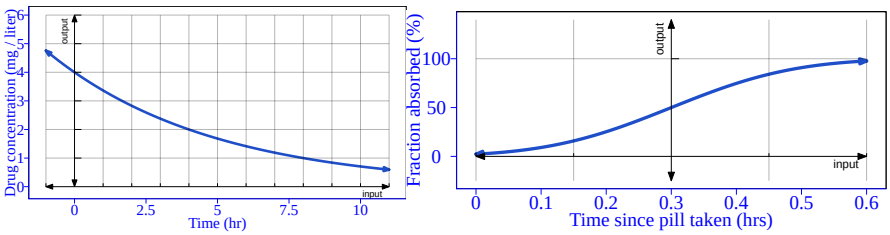
This chapter describes tools to build custom functions. The tools help the modeler compose a function that encompasses their ideas and data about a given relationship.

To illustrate, consider this situation: a person develops a sinus infection, ending up having to go to an urgent-care clinic for treatment. An antibiotic is prescribed, the patient picks up the prescription from the pharmacy and begins taking the medication as directed: one small pill every eight hours for seven days.

Wouldn't it be more convenient to take just one pill a day? Or why not just one pill that covers the entire week? A model can help us understand the implications of different dosing schedules.

To start, we need to know a bit about the physiology of drug absorption and elimination in the human body. The drug in the pill is absorbed through the gut over a short period of time, say, an hour. But even as the drug is being absorbed, the liver metabolizes it and removes it from the blood stream. The removal rate is proportional to the current concentration of the drug in the bloodstream; low concentration means a slow removal rate, while high concentration means a fast removal rate.

There's no single function among our core modeling functions that captures both absorption and removal of the drug. Still, the exponential function seems a match to drug elimination. As Figure 9.1(a) shows, the exponential function with input and output scaling shows a pattern of high rate of reduction when the drug concentration is high, and low rate of reduction when the concentration is low.



(a) Exponential function

(b) Hillside function

Figure 9.1.: These core modeling functions cover different aspects of the drug absorption and elimination processes.

The model of absorption is the hillside function with input scaling so that about half-an-hour after the pill is taken, practically all has been absorbed.

Now we are going to combine the two functions to become a reasonable model of the whole process: absorption and elimination. Drawing on past modeling experience, we're going to multiply the two functions together, giving the overall function in Figure 9.2.

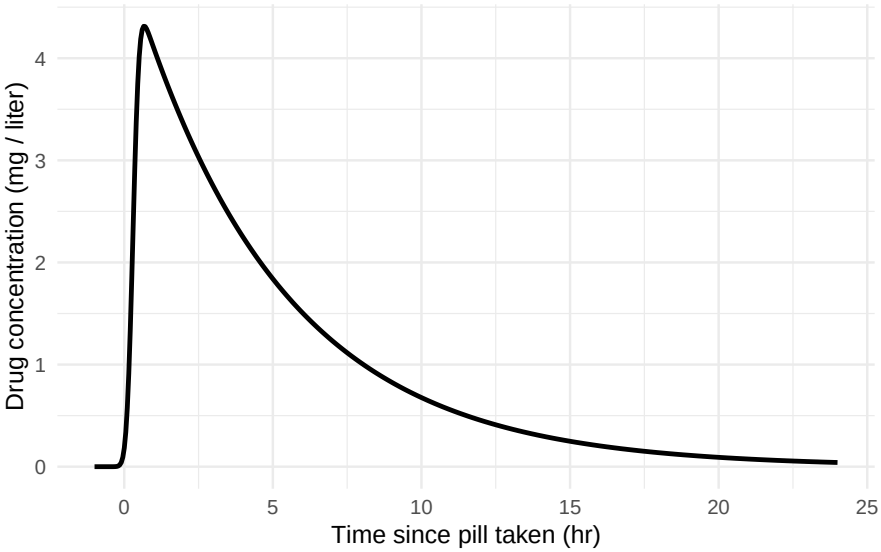


Figure 9.2.: Multiplying the two functions in `r xf("fig-one-pill-separate")` to create the overall drug concentration profile for a single pill.

IN DRAFT: Fix the figure reference in the caption.

> **Multiplication in epidemics**
>
> Recall the SIR model introduced in Section 1.5, where it was described as a book-keeping device for tracking the number of susceptible, infective, and recovered individuals in a population. For each new day, the number of new infectives is a function of the current number

of infectives (I) and of susceptibles (S).
Consider how new infectives are *produced*: a current infective comes into contact with a currently susceptible person. Each such encounter or *interaction* carries a risk that the susceptible person will become infected. Imagine that the infective encounters several other people each day. Those other people might currently have the illness, or have already recovered from the illness. But some of the contacts will be susceptible. How many? It depends on the fraction of the population who are susceptible. The fraction is S / N, where N is the total population. In terms of our core modeling functions, the fraction of people who are susceptible is `same(S)` with some output scaling. Likewise, the risk that a susceptible who has been in contact with a current infective is just another layer of output scaling on the `same(S)` function.
Naturally, each of the I current infectives interacts with other people. The bigger is I, the larger is the number of such encounters. Overall, the number of new infectives is an output scaled version of I $\times$ S. This from of function often appears in modeling, so often that it has a name: an 👉**interaction term**👈. This is an instance where the name is descriptive of the underlying mechanism that's being modeled.
Interaction terms appear in many modeling contexts, not just epidemics. For example, in chemistry the speed of a chemical reaction between two reagents, A and B, depends on the product of their concentrations: [A] [B] in the chemists' notation. But the name used in chemistry is different: the "Law of Mass Action." But an interaction, is still an interaction, whether it be between two molecules or between an infective and a susceptible, or between a fox and a rabbit.

9.1. Linear combinations of functions

Although multiplying functions is a valid and powerful way to construct models, it is much more common to encounter models built by *adding* functions together.

Figure 9.2 depicts the concentration of a drug after taking a single pill. Let's extend the model to cover situations where multiple pills are taken over time, as with the prescribed "one pill every 8 hours for seven days."

With input scaling, we can adjust the time at which the pill is taken. Let's

use `pill(t)` to refer to the function depicted in Figure 9.2. This pill was taken at hour 0. If hour 0 is mid-night, then `pill(t - 10)` is the function for a pill taken at 10 am. Similarly, `pill(t - 18)` represents a pill taken at 6 pm. And `pill(t - 26)` is the next one on the schedule. and so on for the whole seven-day prescription.

The drug from each pill adds to whatever is already in the bloodstream. So the function giving the drug concentration for pills taken at 8-hour intervals (starting at 10 am on day 1) is:

```
all_pills <- function(t) {
  one_pill(t - 10) + one_pill(t - 18) + one_pill(t - 26) +
  one_pill(t - 34) + one_pill(t - 42) + one_pill(t - 50) +
  one_pill(t - 58) # ... and so on.
}
```

`All_pills(t)` is an example of a 👉**linear combination of functions**👈. Here, the functions being combined are all input-shifted version of the same `pill(t)` function, but linear combinations can involve any functions. Figure 9.3 shows `all_pills()` over the first two-and-a-half days of the prescription.

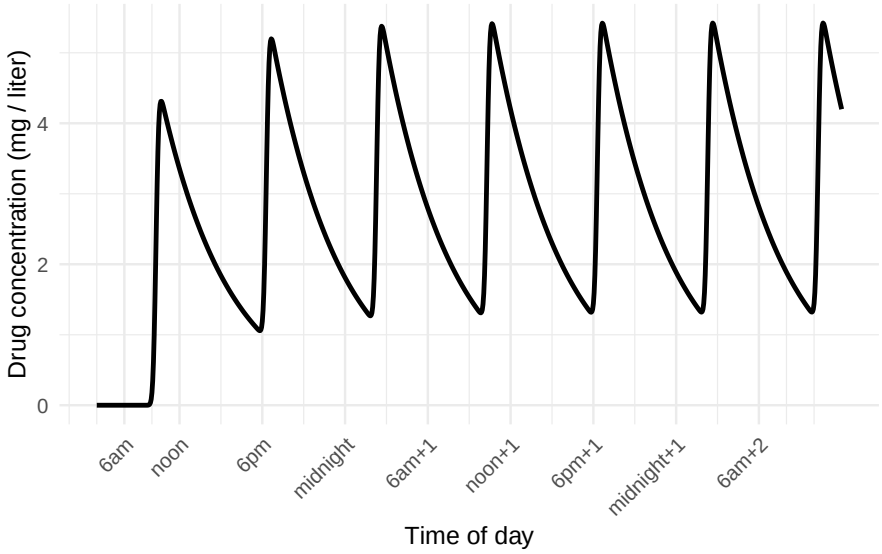


Figure 9.3.: Concentration of a medicine in the bloodstream of a person who takes a pill every 8 hours, starting at 10 am on day 1.

Figure 9.3 shows how each additional pill raises the drug concentration. Suppose the pharmacist's goal is to maintain at all times a drug concentra-

tion above the therapeutic level of 1 mg/liter, providing no respite to the bacteria causing the infection.

Compare the situation in Figure 9.4 to that of Figure 9.2. A single pill per day regimen would result in the drug concentration falling below 1 mg/liter for most of the day. That's not acceptable. The most obvious fix comes from recognizing that pill every 8 hours is three pills a day. So to be fair, the comparison the daily pill should have three times the dose of the 8-hour pill. Taking this into account, will the one-pill-per-day regimen work if we *tripled* the dose of each pill?

This situation also falls under the aegis of the *linear combination* technique. In linear combinations, the modeler is free to set the output scale the functions being combined. For the one-dose-per-day plan (at triple the dose) the model of drug concentration looks like this:

```
once_per_day <- function(t) {
  3 * one_pill(t - 10) + 3 * one_pill(t - 34) +
    3 * one_pill(t - 58) # ... and so on
}
```

Let's graph the function:

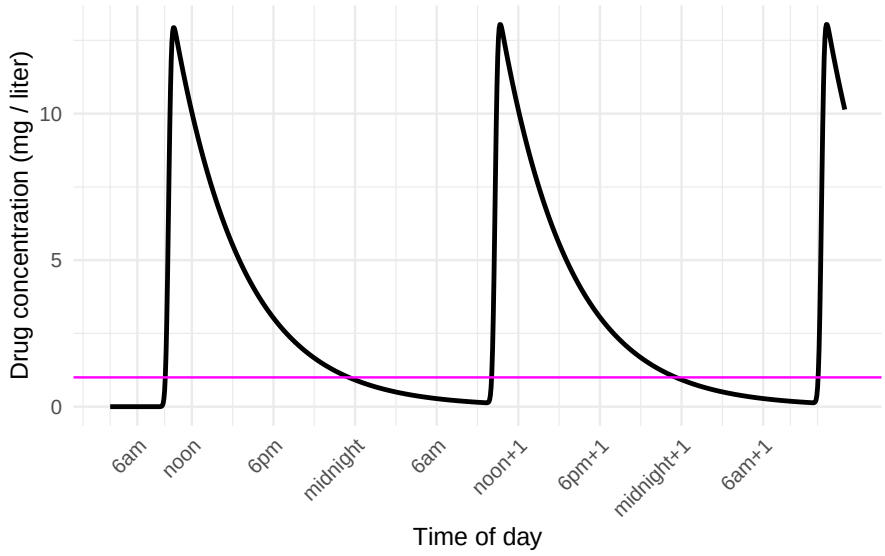


Figure 9.4.: Drug concentration over time when taking one pill per day at triple the dose. The drug concentration falls below the therapeutic dose (the magenta line) for almost half of each day.

The process of exponential decay, which we used in our model of drug concentration, is so familiar throughout science and engineering that it

was not surprising to pharmacologists to observe the pattern with drug elimination. The model used in this section uses a drug half-life—the amount of time it takes for the value of the function to fall by half—of about 4.5 hours. Naturally, the half-life can vary from drug to drug. When figuring out what dose should be used in a pill and how often it should be taken, scientists used experiment and statistics to determine the therapeutic dose and the drug half-life.

It turns out that the method of linear combinations is well suited to constructing model functions from data, not just in biomedicine but in all manner of social science and economics contexts.

9.2. Linear regression

[In everyday speech, the word "regression" means a return to a previous, less advanced, or worse state.

Definition of regression



A linear *combination* is a mathematical idea: a way to build new functions by combining existing functions. This is in not at all the meaning of the word in the phrase "linear regression." We arrived at this confusing situation by historical accident. The original usage in statistics in the 1890s reflected a misconception that treated a simple mathematical pattern as if it emerged from an underlying natural, biological, or economic process. History can be sticky; our modern understanding is better captured by the name 👉**linear modeling**👈.]{.column-margin} Very often, however, the point of building a new function is to represent *data*, constructing a function that captures the patterns seen in the data. The technique for using data to inform the construction of a linear combination is traditionally called 👉**linear regression**👈.

The `same()` and `flat()` core modeling functions are the most common building blocks for linear regression models, but in principle any function can be included in a linear regression. And, linear regression effectively automates the process of *output scaling* introduced in Section 7.2.

Let's use an example to set the stage for understanding linear regression/modeling. Suppose you are studying the factors that contribute to student performance in elementary school. Teachers like to think of themselves as the critical factor, but also point to the benefits of small class sizes and supportive family environments. It's easy to imagine a system where teacher performance in the classroom is carefully observed and scored by

experts, although genuine experts are always in short supply. But, for our example, let's operationalize teacher skill using such a performance evaluation. Class size is easy to operationalize: count the students in each class and average over each student's classes.

Supportive family environments? This is harder. We are all inspired by stories of undereducated parents pushing their children to succeed against the odds. Operationalizing this dimension (these dimensions?) requires identifying measurable indicators. Two such are the parents' level of education and the family's income. (Perhaps it's really *lack of income* that matters, since low-income families often face significant challenges.)

Operationalizing student performance can be controversial, but for this example let's simply use the `score` on a standardized test.

We've identified four input quantities: teacher `performance`, parent's `education`, family `income`, and class `size`. We want to build a function that translates these four inputs into the one output: standardized test `score`. What should that function look like?

We refer to "inputs" and "output" to be consistent with the nomenclature used in this book to describe functions. In statistics texts, the phrases "predictors" or "explanatory variables" are often used instead of "inputs," and "response variable" is often used instead of "output." In economics, it's common to use the terms "independent variables" and "dependent variable."

Linear regression involves basing the construction of the function using data. For the example, let's imagine a data frame where the specimen is an individual student in grade 5 and the columns correspond to the input quantities (`performance`, `education`, `income`, `size`) and the output (`score`).

Typically, as in this example, the input quantities have different dimension and are measured in different units. For example, in this example, `income` is measured in dollars (dimension W), `education` in years of schooling (dimension T), `performance` as a score on an evaluation (dimension [1]), and `size` as the number of students in a class (also dimension [1]). The output quantity, `score`, also has dimension [1] and its own units.

There are many scales for measuring school performance. When we talk about them, we don't typically give a unit name. For instance, a score of"600 on the SAT math test," which is roughly equivalent to a score of "25 on the ACT text," or a "gradepoint of 3.5."

A linear combination is an ideal format for combining input quantities with different dimensions. It is a generalization of the **output scaling** introduced in Chapter 7 which we used to convert the pure-number output of the core modeling functions to an output quantity with dimension and units. For our example, the function might be written this way:

```
score_model <-
  function(performance, education, income, size) {
    A * performance +
    B * education +
    C * income +
    D * size +
    E
}
```

The quantity `A` is a conversion factor that turns `performance` into a quantity that has the same units and dimension as `score`. Similarly, `B` converts years of `education` into a `score` and so on for `C`, `D`, and `E`. These conversion factors are called ☞**coefficients**☜.

Another word for "coefficient" is ☞**scalar**☜. This name is a reminder that the point is to "scale" something.

We can figure out the units and dimension of each of the coefficients using the *conversion factor* ideas from Chapter 3. For instance, the `C` coefficient has to turn `income` (dimension W) into a `score` (dimension [1]), so its dimension would be [1] / W. Even more, we can say what are the *units* of `C`: if `income` is measured in dollars and `score` is measured in SAT points, then the units of `C` would be SAT points per dollar.

Knowing the units and dimensions of the coefficients helps, but it does not directly tell what the numerical value of the quantity should be. This is where the method of linear regression comes in: to find the numerical values of the coefficients that best match the observed data. The calculations themselves are based on theory beyond the scope of this book, but that poses no problem at all since linear regression is implemented in software that anyone can use.

IN DRAFT: Some computing skill development exercises demonstrate how.

Given that software finds the values of the coefficients, the modeler can focus on two critical tasks:

1. Identifying the relevant input and output quantities for the model.
2. Interpreting the coefficients and what they tell us about the workings of the system being modeled.

These two tasks are intimately related . Let's start with (2).

Each of the coefficients on an input is not just a rate (for instance SAT-points/dollar for `C`); it is also a **rate of change**. In other words, the coefficient tells us how much the model output quantity changes when the input quantity changes by one unit. In the example, each of the four coefficients

`A`, `B`, `C`, and `D` is a rate of change that quantifies how the model output will change with a unit change in the corresponding model input. All the terms in the linear combination add together, giving the total change in the model output as a result of changes in all the inputs.

One consequence of this is that each coefficient has meaning only within the context set by the other coefficients. If the modeler were to change the input quantities (for example, leaving out teacher `performance`), the numerical value of each of the other coefficients would also likely change.

Sadly, even as of the 2020s, mainstream introductory statistics courses often fail to adequately teach students about the context-dependent nature of regression coefficients and the importance of specifying which input quantities are included in a model.

This mathematical fact accounts for many of the disputes one hears about in research results. Different researchers who use different input quantities in their models, even when using the same data for fitting, will get their own coefficients that can seem in conflict with those found by other researchers.

In reporting research results, the phrases 👉**after adjusting for**👈 or 👉**holding constant**👈, or 👉**controlling for**👈 are used to signal which input quantities are used in the model. This is critical information.

9.3. Function graphics

One function that we live with every day is the terrain around us: elevation as a function of latitude and longitude. There is a well-accepted graphical format for showing terrain: the topographical map. Figure 9.5 shows an example: the terrain of an island.

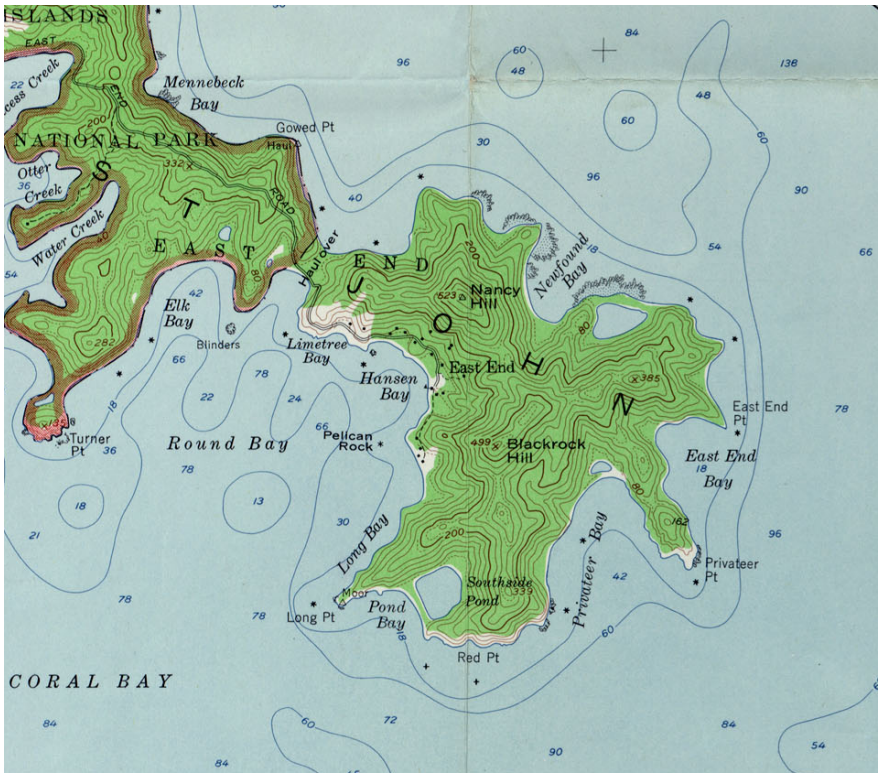


Figure 9.5.: 1958 US Geological Survey topographical map of St. John in the US Virgin Islands in the Caribbean Sea. Source

Topo map online

The topographical map has many curved lines, called *contours*. These are easiest to see for the part of the scene that is covered by water. There, the contours are thin blue lines. Each such line traces a path that maintains the same depth.

On land, the contours are brown and are very closely spaced. Here, each contour depicts where the elevation is at a constant elevation above sea level. Many of the contours are labeled. On land, the label gives the elevation; in the sea, the label gives the depth of the water. The map also labels with a number the elevation of a few outstanding points such as the peak of hills or the depth of undersea bowls.

It takes practice to learn how to extract information from topographical maps. Let's get started by focussing on the terrain around Hansen Bay, just to the left of center on the map. The shoreline is marked as a blue curve. More important for our purposes, that blue curve is a contour line that marks elevation 0, at sea level. You can trace the contour around the island: it never goes inland or out to sea. Moving inland, there are several more contours pretty much parallel to the shoreline and to each other. On this map, each successive contour is at a higher elevation than the one previous; the elevation difference between adjacent contours is 40 feet. Walking inland from Hansen Bay toward Nancy Hill, you cross four light brown contours until you arrive a darker brown one. That darker brown contour indicates an elevation of 200 feet above sea level. (If you trace the dark brown contour around the island, you will find a label on it: 200.) Continuing toward Nancy Hill, you cross another 4 light-brown contours then, having ascended another 200 feet, alight on a dark-brown one circling Nancy Hill. The dark-brown contour marks an elevation of 200 + 200 = 400 feet above sea level. Continuing to climb from the 400 foot contour toward the peak, you cross one, two, and a third contour barely visible under the marker for the peak. We have reached 200 + 200 + 120 = 520 feet when we reach the last, highest contour. The peak is just a little higher: marked as 523 feet on the map.

The contours tell you more about the terrain than the elevation at each point. For example, the steepest route up from any given point is perpendicular to the contour through that point. Steep areas of terrain have closely spaced contours. Conversely, areas with widely spaced contours are relatively flat, as you can see along the shoreline of Hansen Bay. Naturally, the steepness of a path depends on which direction the path goes. Paths perpendicular to contours are steep, paths parallel to contours are flat. With this in mind, look at the road that links Hansen Bay to Princess Creek in the far northwest corner of the map. Roads are generally designed to avoid going steeply uphill, and you can see that the Princess Creek/Hansen Bay road tends to follow the contours, crossing obliquely at shallow angles rather than cutting straight up the steepest slopes.

Or consider the narrow neck connecting the two larger landmasses. This is marked "haul over," which makes sense because the path connecting the south side of the next to the north side does not cross any contours. That path is relatively flat, a good route to haul a boat overland between the two sides.

The equivalent of a topographical map for functions is called a ☞**contour**

plot☜. The output of the function corresponds to the elevation in the map, two inputs play the roles of latitude and longitude respectively. A contour plot provides a mechanism to read off the output for given values of the inputs, but the skilled interpreter can glean more information than that. Figure 9.6 shows an example, the heat-index function introduced in Chapter 6. The inputs are temperature and humidity; the output is the heat index.

Panel (a) of Figure 9.6 shows the function in tabular form, just as it was presented in the 1979 paper that introduced the heat index. Given values for the temperature and humidity, you can easily find the corresponding output from the table.

Panel (b) shows the heat index function as a contour plot. By avoiding the table's visual clutter of a grid of numbers, some simple patterns can be seen. For instance, consider on (b) the heat index output for input temperatures lower than 30°C; the contours run in parallel with a slight upward slope from left to right.

Such parallel contours correspond to the function being a linear combination of the inputs. The slope means that the function increases with both dry-bulb temperature and relative humidity. That the slope is shallow means that—at least in the top portion of the graphical space—the function hardly depends on humidity.

Now consider dry-bulb temperatures higher than 30°C, especially for humidities greater than, say, 50%. The contours are still parallel but they are heavily inclined from left to right and closely spaced together.

The close spacing of the contours indicates, as in the topographical map of Figure 9.5, the function is very steep: a small change in either input produces a large change in the output. This relationship is what's behind the common summertime complaint: "It's not the heat, it's the humidity."

The inclination of the contours tells us something else: the heat index for high ambient temperatures is closely linked to both humidity and temperature. And the curvature of the contour lines indicates an *interaction* between temperature and humidity.

TABLE 2. Temperature-humidity scale: Apparent temperature (°C) corresponding to each combination of dry-bulb temperature and relative humidity. Values in parentheses correspond to skin humidities above 90% and are approximate.

Dry bulb temperature (°C)	Relative humidity (%)									
	0	10	20	30	40	50	60	70	80	90	100
20	16	17	17	18	19	19	20	20	21	21	21
21	18	18	19	19	20	20	21	21	22	22	23
22	19	19	20	20	21	21	22	22	23	23	24
23	20	20	21	22	22	23	23	24	24	24	25
24	21	22	22	23	23	24	24	25	25	26	26
25	22	23	24	24	24	25	25	26	27	27	28
26	24	24	25	25	26	26	27	27	28	29	30
27	25	25	26	26	27	27	28	29	30	31	33
28	26	26	27	27	28	29	29	31	32	34	(36)
29	26	27	27	28	29	30	31	33	35	37	(40)
30	27	28	28	29	30	31	33	35	37	(40)	(45)
31	28	29	29	30	31	33	35	37	40	(45)	
32	29	29	30	31	33	35	37	40	44	(51)	
33	29	30	31	33	34	36	39	43	(49)		
34	30	31	32	34	36	38	42	(47)			
35	31	32	33	35	37	40	(45)	(51)			
36	32	33	35	37	39	43	(49)				
37	32	34	36	38	41	46					
38	33	35	37	40	44	(49)					
39	34	36	38	41	46						
40	35	37	40	43	49						
41	35	38	41	45							
42	36	39	42	47							
43	37	40	44	49							
44	38	41	45	52							
45	38	42	47								
46	39	43	49								
47	40	44	51								
48	41	45	53								
49	42	47									
50	42	48									

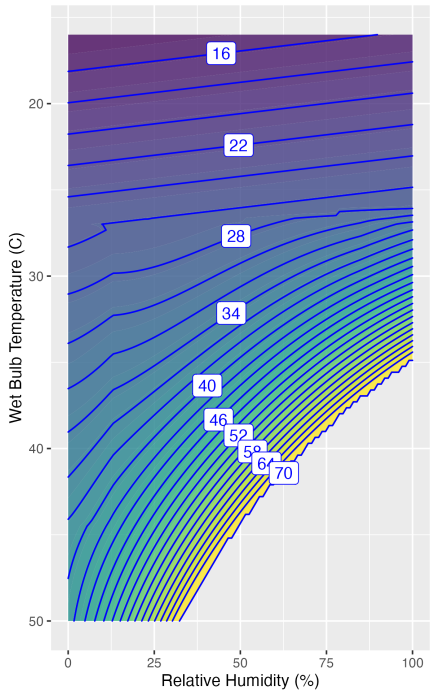


Figure 9.6.: Two graphical presentations of the heat index function. Panel (b) has been drawn with an upside-down vertical axis, the better to correspond the tabular convention of increasing values from top to bottom.

The explanation for the big change in pattern from top to bottom on the plot is physiological, not meteorological. Keep in mind that human core body temperature is 37°C. Metabolism generates heat and the core temperature would go up above 37°C if it were not for the body's cooling mechanisms. When a lot of heat is being generated, as in exercise, one of these cooling mechanisms becomes evident: sweating. Sweating is most effective at low humidities. At high humidities, the evaporation of sweat is less efficient, leading to a rapid increase in perceived heat, which is captured by the steep contours in the heat index plot.

A heat index of 37°C leaves no way for the body to dissipate metabolic heat. Consequently, the core body temperature starts to go up. You can see from the function output that a heat index of 37°C corresponds to dry-bulb temperatures around 42°C, *so long as the humidity is low.* But as humidity increases, the safety mechanism of sweating shorts out and the skin temperature rises quickly.

A heat index of 65°C or thereabouts is thought to be the limit of human tolerance. After a few hours of exposure at this level the risk of death is very high. At high humidities, this fatal threshold is reached even when the ambient dry-bulb temperature is near normal body temperature.

10. Trade-offs

The everyday verb "trade" involves exchanging one thing for another. Buying lunch is a trade: you exchange money for food. Almost always, such an exchange involves two parties, in this case a buyer and a seller. A 👉**trade-off**👈 is a kind of trade that involves only one party who has to balance competing objectives. A person who decides to buy a cheaper lunch in order to have money for a movie faces a trade-off between the quality of the food and the amount of money available for entertainment.

Trade-offs often involve money, but not always. For instance, a painter bending far from a ladder to get to that last little bit of unpainted third-story window trim is trading off efficiency for safety. Another painter, who climbs down and moves the ladder closer to the window, faces the same trade-off but evaluates it differently. The second painter is prioritizing safety over efficiency.

This chapter is about describing trade-offs and some of the ideas that can be used to deal with them quantitatively.

Vocabulary Review

- arg (6)
- data (2)
- dimension (2)
- functions (6)
- mean (8)
- per (3)
- point (3)
- relative probability (8)
- value (2)
- worth (3)

New

- argmax
- argmin
- dominates
- incommensurate
- maximum
- minimum
- objective function
- shadow price
- trade-off

10.1. Optimization

"Optimization" is a mathematical topic that deals with finding the best from a set of possible choices. There are many technical situations in which optimization techniques are useful. An example is calculating the coefficients used in linear regression. But optimization is also used, importantly, in routing and logistics, for example finding the best sequence of stops for a delivery van or the layout of transistor circuits on an integrated silicon chip.

In a trade-off, the existence or meaning of "best" is not clear-cut. Still, some of the terminology from optimization is useful for describing trade-offs quantitatively.

The structure of an optimization problem involves a function—called the 👉**objective function**👈—whose input domain is the set of possible choices. An objective function is a tool that allows systematic comparison of the available choices in terms of the outcome each one will produce.

Remember, the *domain* of a function is the space of legitimate inputs to the function.

Optimization can involve one or more objective functions, although most mathematics courses emphasize the single-objective case. Optimization with a single objective function is a good way to introduce the basic concepts and terminology, but trade-offs typically involve multiple objective functions.

An objective function is a perfectly ordinary function along the lines discussed in Chapter 6 or Chapter 9. The modeler constructs an objective function to capture a particular goal in the optimization problem. For instance, apartment hunting can be seen as an optimization problem where the input domain is the list of available apartments for rent. One typical objective is to maximize the *square footage* of the apartment, others are to minimize the rent and distance to work, and to maximize the quality of the neighborhood, etc. These are four different objectives and each is represented by its own objective function. In this case, the modeler building the objective functions would assemble a table over each of the available apartments with each objective function being a column in the table. For instance, square footage would be one column, rent another, and so on.

The term "objective function" is properly descriptive. In optimization problems one has some goal in mind—the objective, the output of the function—and seeks to select an input value that will produce the largest value of that output.

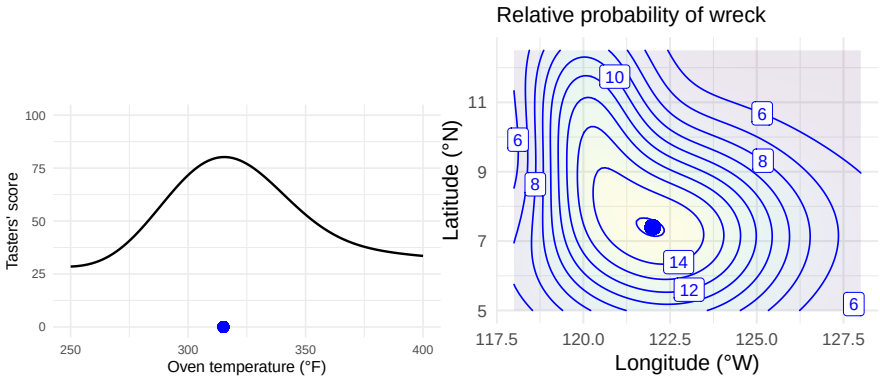
Optimization problems sometimes involve situations where the goal is to minimize rather than maximize the output. Any minimization problem can be turned into a maximization problem by putting a minus sign in front of the objective function.

The "solution" to the optimization problem is the input value that achieves this maximum (or minimum) output. The output achieved is called the objective function's 👉**maximum**👈 or 👉**minimum**👈, depending on the context, but the point is to find the input value that produces it. Such an input could be reasonably called the "optimum input," but the technical name for it is the 👉**argmax**👈. For a minimization problem, the corresponding word is 👉**argmin**👈.

"Argmax" and "argmin" are not ordinary English words; they are syllabic abbreviations, like FedEx (short for Federal Express), formed from the first

syllables of words in a phrase. The phrase relevant to optimization is "the **arg**ument that produces the **max**imum output value." Keep in mind that "argument" in this phrase is used as a synonym for "input."

Figure 10.1 shows the argmax for two example objective functions. Those functions each have an output that varies with its input. In the spirit of fun, we imagine that the output in panel (a) is the taste score assigned to a baked item and, for (b), the relative probability of locating a shipwreck. However, the output scale does not matter at all to the optimization problem. The argmax is merely the *input* that produces the maximum output.



(a) Objective function with one input.

(b) Objective function with two inputs.

Figure 10.1.: The argmax (blue dot) for each of two example objective functions.

To judge from panel (a), the tastiest baking temperature for the item is 315 °F. As for the shipwreck, its most likely position is (122 W, 7.4 N), but the relative probability there is not much higher than at areas within about ± 1 degree of longitude and latitude. In each panel, the blue dot marks the input value, not the output value.

How does one find the argmax? If you have a graph, look at it, find the point in the picture that corresponds to the peak of the hill and read off the input values for that point.

It is hard to construct useful graphics when the objective function has three or more inputs. Optimization problems with many inputs can be solved with a different technique, which we will discuss in Chapter 16. In training AI models, the objective function literally can have billions of inputs.

The technique for finding an argmax can be implemented in software. This fact lowers the priority to the student of learning mathematical theory of optimization. The real questions one faces in optimization are how to

develop a model for the objective function and whether there is an argmax at all.

10.2. Toward trade-offs

Often, the purpose of finding an argmax is to guide a decision or choice. In Figure 10.1(a), the choice involves setting the oven temperature to produce the tastiest cake. In (b), the choice is where to start searching for the sunken wreck.

Sometimes, however, the proposed objective function does not capture all relevant considerations. In such cases, the argmax of an objective function may not guide a good decision. For instance, consider the objective function in Figure 10.2 that takes an automobile's speed as input and produces as output the time until the car reaches its destination. This is a minimization problem, since we want to take the least time possible to get to the destination.

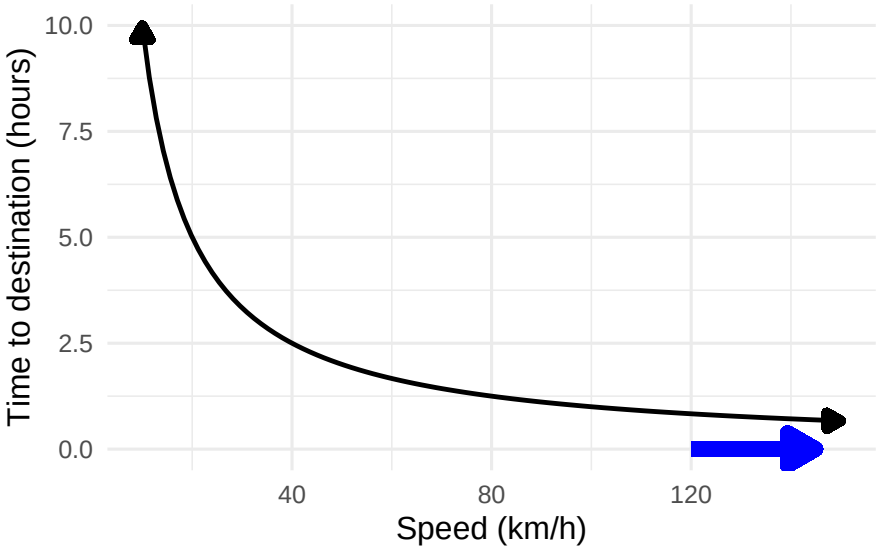


Figure 10.2.: Time in hours to a destination 100 km away as a function of car speed. To minimize time to destination, a higher speed is better.

Figure 10.2 shows the objective function. Naturally, the faster the car goes, the less time it takes to reach the destination. We've marked the argmin near 120 km/h on the graphics input scale, but mathematically any input larger than this would result in an even better (lower) output, which is the meaning of the blue arrow. Keep in mind that the blue annotation corresponds to an arg**min**, even though it is at a *high* speed.

Remember: The "min" in argmin refers to the output of the objective function, not the input. The "minimum" is the about the output of the function, the "argmin" is about the input that generates the minimum output.

The optimization result—the argmin being at a high speed, i.e., go as fast as possible—violates any driver's common sense. The problem is that the model has only one objective: minimizing time to destination. Any responsible driver will incorporate additional objectives into the decision-making process. Let's list some of these:

1. minimize time to destination
2. maximize safety
3. maximize fuel efficiency
4. minimize legal violations

Each of these objective functions, when considered in isolation, has its own argmax or argmin, depending on whether the output of the function is to be maximized or minimized.

1. Time to destinationArgmin: Go as fast as possible.
2. Safety: Argmax: Drive at a safe speed. If there is traffic, a safe speed will be similar to other cars on the road. If no traffic, heed visibility and road conditions.
3. Fuel efficiency:Argmax: Look up the most fuel-efficient speed for your car on a trustworthy web site. On a windless day, this is likely to be around 80 km/h (50 mph).
4. Lawfulness: Argmin: Keep speed at or under the legal speed limit.

Depending on conditions, the objective functions can be in conflict with one another. For instance, when traffic is exceeding the speed limit, the safety goal conflicts with the legal compliance goal. Each of the goals (2), (3), and (4) conflict with (1).

Such conflicting objectives are the origin of trade-offs.

10.3. Resolving trade-offs

A key feature of an unresolved trade-off is that the outputs of the various objective functions are 👉**incommensurate**👈 with each other. If they were commensurate, we could simply add the outputs of the various objective functions together, to produce a single objective functions, a situation giving an easy answer.

An important quantitative approach to trade-offs is to find *conversion factors* to transform the output from each objective function into a common

unit, allowing for direct comparison and aggregation. (As we will see, the conversion factors are called *shadow prices*.)

Drivers deal with the driving-speed trade-off in the previous section every day using intuition or heuristics. One important heuristic here is "obey the speed limit." By way of justification for the heuristic, a driver might argue that the speed limit is set based on safety considerations and, since the government has an interest in reducing fuel consumption, is likely close to the optimal speed for efficiency. Another heuristic is "follow the flow of traffic," which is particularly valuable when the flow of traffic exceeds the posted speed limit. And in special circumstances—snow, ice, dense fog—even the flow of traffic may not be a good guide.

It's unlikely that a careful quantitative analysis would provide better guidance than the heuristics already in use. But I want to draw on the driving situation to illustrate what mathematical approaches are available to guide decision-making in general. Using a non-controversial issue as the example—driving—frees us from the preconceptions, political, and ethical stances that often complicate discussions in other domains, such as health care, government spending policies, and environmental regulations.

Even if we do not perceive the four objectives in highway driving as active elements in our everyday choices, a simple thought experiment can reveal that each would be deemed important under changed circumstances.

For instance, suppose that fuel costs went up by a factor of 100, that is, that gasoline cost about $400 US per gallon (approximately $100 per liter). The fuel-related cost of driving a mile would increase from about 15 cents per mile to about $15 per mile. We can construct a small model to quantify the implications of this. Supposing that a highway speed reduction from 60 mph to 40 mph reduces fuel consumption by, say, 20%. Travel time on a previously 1 hour trip (60 miles at 60 mph) would increase to 1.5 hours (60 miles at 40 mph): a 30-minute increase in travel time. This time increase is offset by the savings on fuel. At $4 per gallon fuel cost, the 20% reduction in fuel use would save 3 cents per mile, or $1.80 over the 60-mile trip. Judge for yourself whether a savings of $1.80 is worth the extra 30 minutes of travel time. I think most people would say no. At the higher fuel costs, the savings would be $180. I suspect most people would gladly exchange 30 minutes of their time for $180.

You might discard this observation on the grounds that the hypothetical 100-fold increase in fuel costs is unrealistic. That people would change their behavior in this extreme scenario, however, is a signal that drivers are

sensitive to fuel costs. Think for yourself whether a $90 savings would be worth the 30 minutes. Or perhaps $45 would be worth it, or even $15. If so, we don't need the 100-fold increase to make the point; an 8-fold increase is all that's required.

Now consider a scenario in which a zero-tolerance policy to speeding is adopted. Any violation of the speed limit would lead to a suspension of your driver's license. Would this change your driving behavior? Most likely it would. (At least until the next election when the people responsible for the policy would be thrown out of office).

To simplify the discussion, Figure 10.3 imagines an optimization context with two objective functions, Score() and Savings(). The input to each is a policy measure (analogous to the speed you choose to drive). For easier comparison between the two functions, they have been overlaid on the same plot. For each function, the argmax is an input value, that is, a point on the horizontal axis. The two functions have different argmaxes. And there are different vertical scales, one applying to Score and the other to Savings. Note that the vertical scales differ not just in amount but in dimension: one is a "Score in points" and the other is a "Savings in tons."

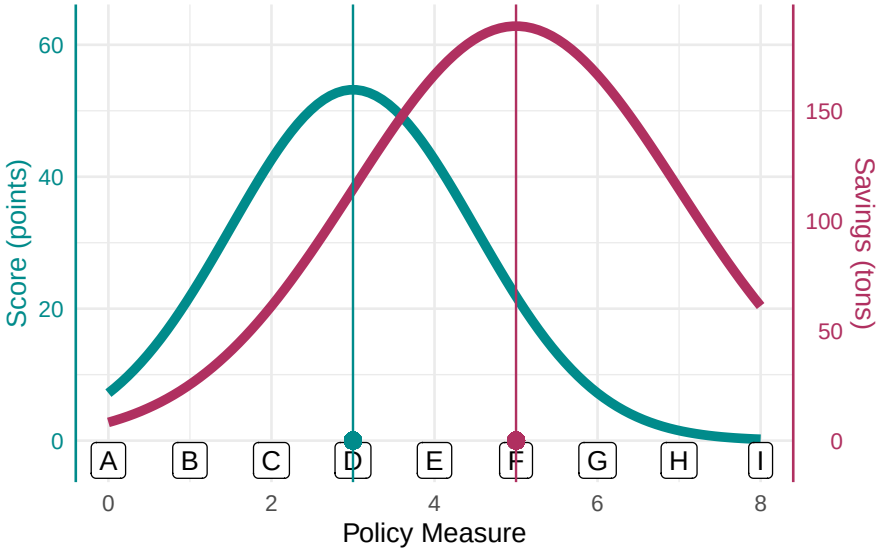


Figure 10.3.: Two objective functions whose outputs have different dimension. For later reference, a handful of values of the policy measure input are marked with letters. For instance, policy D corresponds to an input value 3.

If Score() were the only objective function, we would set the policy measure to 3, the argmax for Score(). Similarly, if Savings() were the only objective function, we would set the policy measure to 5, the argmax for Savings().

It would be handy to add the two objective functions together to produce a single function whose argmax can be conclusively found. But the outputs of the two objective functions are stated with different dimensions and units: score is in points while savings is in tons. That is, the two outputs are incommensurate. As described in Chapter 3, addition is not a valid operation when quantities have different dimensions. (A hint about what is to come: Division is valid even when the quantities have different dimensions.)

But if both objectives in Figure 10.3 are worthy of consideration, setting the policy measure at the argmax of one leads to a seriously suboptimal outcome for the other objective.

Let's switch to another graphical format to illustrate the trade-off. In Figure 10.3 the horizontal axis the the policy measure, that is, the input to the objective problems. The two vertical axes correspond to the output of the two objective functions—Score() and Savings()—one axis for each function.

In Figure 10.4 we re-arrange the axes. First, we delete entirely the horizontal input axis and represent the input value as the position along a curve. For clarity, Figure 10.3 we marked with letters nine different input values. Those same letters appear in Figure 10.4 as locations along the curve. But the axes for Figure 10.4 are re-arranged versions of the two vertical axes from Figure 10.3. Each point on the curve shows the output of the Score and Savings function for one input value. For example, for the input value A, the position on the curve marked with A is based on the Score value for input A and the Savings value for input A.

From Figure 10.4, what can we say about which choice is the best? It's important to keep in mind that each of the objective functions in Figure 10.3 is for a *maximization* problem. So bigger outputs are better. From the axes, you might translate this into a simple rule: move as far as possible to the north-east. But this isn't right.

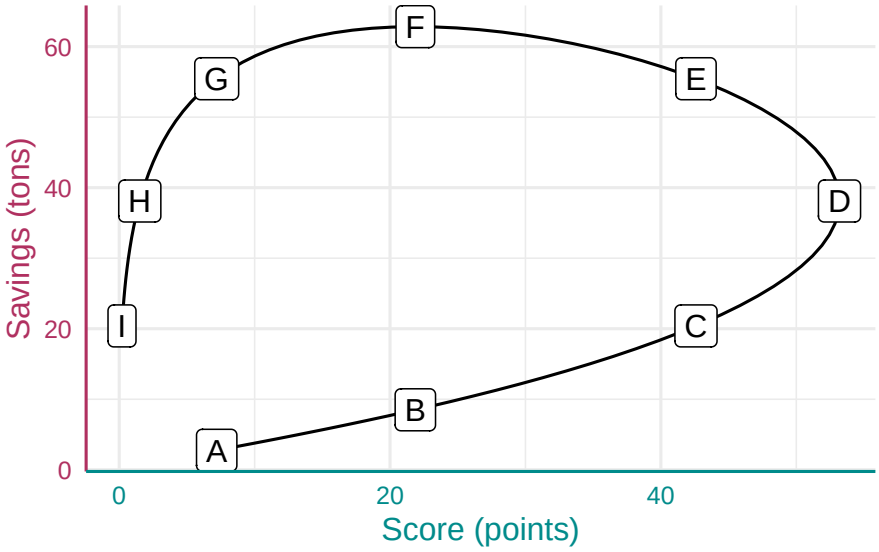


Figure 10.4.: Illustration of the trade-off between two objectives: score() and savings().

Figure 10.4 lets us compare any two input choices, say D versus E or A versus B. In order to claim that one of the labeled choices is better than the other, we need the outcome on **both** graphical axes to be better for one choice than for the other. For instance, both the Score and the Savings are larger at input B than at input A. Choice B is therefore better in all respects than choice A. In contrast, when comparing D versus E, we see that E has a larger Savings but it also has a *smaller* Score.

When comparing two inputs, as with A and B, if one is better in all respects—here this means better both in Score and in Savings—we say that one choice 👉**dominates**👈 the other. B dominates A.

We can compare any two input levels in this way. For instance, F *dominates* G because it has higher Savings and a higher Score. And D dominates C for the same reason. But we cannot demonstrate that F dominates D or vice versa; each is strong on one axis but weak on the other.

Since F dominates G, H, and I, we can eliminate those choices; they are suboptimal. Similarly, D dominates C, B, and A, eliminating those as sensible choices.

Deleting all the dominated inputs, what's left? Any choice from F through E to D.

Here is another, more human oriented way to look at the situation. Suppose we have two people, one a big fan of setting the policy measure to 8, that is, favoring option I. The other is more moderate and prefers option F.

Now they are presented with the objective functions in Figure 10.3. The F-advocate can make a compelling argument to the I-advocate. Moving from I to F will accomplish both objectives better; a win-win outcome.

If the competition were between partisans of F, E, and D, however, the win-win argument doesn't apply. To convince the F-partisans to move toward E (or further along, to D), the E-partisan would have to acknowledge that the increased outcome for Score (by about 22 points) would come at the cost of a decrease in the outcome for Savings (by about 8 tons).

Another way to say this is that there is a *price* for Score in terms of Savings. Usually, a price is a ratio of money per item, e.g. $2.89 per apple. Note that in a price, the item purchased is on the bottom of the ratio. And the top quantity does not have to be money; it could be tons of Savings. In this situation, switching between F and E entails a gain of 8 tons of Savings at the loss (that is, cost) of 22 points in Score. As a price for Score, this is 8/22 = 0.36 tons per point. This ratio is called the 👉**shadow price**👈. The word price is appropriate by analogy to the things we buy, measured as money per unit of the goods purchased, as in $4 per gallon of gasoline.

Whenever there is the possibility of an exchange, a price is the quantitative indicator to help determine whether the exchange is worthwhile. You wouldn't buy a shirt if it were, in your opinion, too expensive. You get to make this choice, but knowing the price helps to make the decision criterion more clear.

10.4. Case study: Shadow prices in tuberculosis control

Paul Farmer (1959–2022) was an anthropologist and physician, and famously a co-founder of Partners in Health, an organization dedicated to providing high-quality healthcare to impoverished communities around the world. A particular emphasis to PIH is the treatment of tuberculosis. Farmer both developed and advocated for the treatment of multi-drug resistant (MDR) tuberculosis, which is notably more difficult and expensive to treat than standard tuberculosis. He marshalled many arguments to allocate funds to MDR treatment, including ones about human rights and the risk of untreated MDR tuberculosis spreading within communities.

In 1998, at a meeting of the American Academy of Arts and Sciences, Farmer made the case to an audience of physicians, scientists, and policymakers. One member of the audience, Russian physician Alex Goldfarb, offered

another perspective on the issue, drawing on his experience managing tuberculosis programs to treat Russian prison inmates. As recounted in Tracy Kidder's 2003 biography of Farmer, *Mountains beyond Mountains* (p. 162), Goldfarb described his dilemma:

> "I want to share with you a simple reality. I have six million dollars. With three million dollars I can implement DOTS [a standard treatment, but not effective for MDR] for five thousand prison inmates. And assuming that ten percent have MDR-TB, forty-five hundred will be cured and five hundred will go down with MDR-TB and die. There's nothing much you can do. So, I have a choice. And my choice is to use another three million dollars to treat the five hundred with MDR-TB, or go to another region and treat another five thousand. I'm working with limited resources. So my choice is not involved in the human rights of five-hundred people, but five-hundred people versus five thousand people. And this is a very practical question for me, because I have six million dollars. And the second question is that if I disclose to the Russian people that I spent six thousand dollars per case in MDR-TB in the prisons with tens of thousands of people dying all around, they will tell me that I am building a golden palace for a selected few. So for those of us who *have* to make those decisions with limited resources, it's a very serious question."

Clearly, one objective in this situation is to save lives. More lives saved is better: a maximization problem. The other, competing objective in the situation is indicated by the phrase, "I'm working with limited resources." Seen one way, this is a simple statement of fact. But the fact emerges because an objective of the body that makes decisions about funding is to *minimize* the amount of money spent.

If the *only* objective was to minimize money spent, the decision would be straightforward: don't spend any money at all on MDR-TB, spend it on other needs. Or, If the *only* objective was to maximize the number of lives saved by treatment, the decision is equally straightforward: spend as much money as needed to save as many lives as possible. This would, of course, draw money away from other needs. The trade-off exists because we are trying to accomplish *both* objectives at the same time.

The numbers given by Goldfarb indicate a shadow price of $3,000,000 per forty-five hundred cured patients, or roughly $667 per life saved from

non-MDR tuberculosis. This is a tragically low price, which points to a harsh and grossly unfair allocation of resources in the broader Russian economy and prison system. Still, Goldfarb can do little about the broad situation; he has to work within the constraints of his limited budget. His choice, buy non-MDR lives at $667 per life versus buying MDR lives at $6,000 per life, still low, but much higher than $667 per life.

What would you do, instead?

11. Hypothesis and likelihood

We build models based on our understanding of the systems being modeled. Naturally, this understanding is not perfect. Indeed, there can be honest dispute about how a system operates and what factors are most important in influencing its behavior.

In many (most?) settings, it is not realistic to expect the modeler to get things right the first time, or even after several tries. Effective modelers treat their work as an application of the 👉**scientific method**👈, iteratively refining their models based on insight and evidence.

Such refinement depends on the ability to compare models to one another and pick the better one. In the language of Chapter 10, we might call this finding an "optimal" model. We will do this by developing an objective function that takes competing models as input and returns as output a numerical measure of each model's performance.

The method is standard in many areas of science, technology, commerce, and any field where decisions must be taken. The standard terminology, however, is not usually about comparing models. The word used most often is "hypothesis." The two words—"model" and "hypothesis"—are closely related. Every model is a formalized implementation of a hypothesis about how the system being modeled works.

There are many definitions for the word hypothesis. Often, the definitions emphasize the 👉**scientific method**👈, one description of which is called the 👉**hypothetico-deductive method**👈, as described here. In the context of the scientific method, a reasonable definition for "hypothesis" is along the lines of the following, offered by an AI prompted to define "hypothesis" specifically in the context of the scientific method:

> *A hypothesis is a testable explanation for a phenomenon that serves as a starting point for further investigation through experimentation or observation.*

Vocabulary Review

- bayesian (8)
- data (2)
- exponential distribution (8)
- hypothesis (8)
- magnitude (5)
- models (1)
- objective function (10)
- per (3)
- point (3)
- probability distributions (8)
- quantity (3)
- rate (4)
- system (6)
- uncertainty (8)
- value (2)

New

- likelihood
- likelihood framework
- null hypothesis
- scientific method

Hypothetico-deductive model on Wikipedia



We often think of hypotheses (and the models that implement them) as developing over time, as implied by the phrase "further investigation." But for the purposes of comparing and competing hypotheses, a better picture is that of a set of models all present at the same time, like the candidates in an election.

To choose the best among the hypotheses, we need to be able to compare them to one another in terms of their ability to account for observed data and make accurate predictions. The numerical measure, the output of our objective function, is a quantity called 👉**likelihood**👈.

11.1. The likelihood framework

Chapter 8 discussed how to use *probability distributions* to quantify uncertainty. Section 8.2 introduced a Bayesian notion of where probability distributions come from: they are hypotheses concocted by humans.

In the 👉**likelihood framework**👈, we replace ideas about individual hypotheses being correct or incorrect, right or wrong, true or false. Instead, we consider *collections* of hypotheses, putting them into *competition* with one another, just as sports playoffs pit different athletes or teams against one another. In sports, a higher score gives a team a claim to being better than the opponent.

In the likelihood framework, the competing entities are hypotheses represented by probability distributions about the relative probability of each possible observed outcome. The "score" for each competing hypothesis is the "👉**likelihood**👈." And just as sports teams compete in a particular setting—a game or match—hypotheses compete in a setting: data.

The reader who has never heard of "likelihood" in this sense may wonder if the framework is the right or best one for placing hypotheses into competition. We turn to the field of statistics to answer. As described in Section 8.2, Bayesian and Frequentist statisticians fundamentally disagree. Despite their conflicting stances, both camps put likelihood at the center of their methodologies.

To illustrate how likelihood works, consider how to model uncertainty in the time interval between major events, for example, the time between economic depressions or the time between magnitude 6.5+ earthquakes in a region.

To be specific, suppose we have data on the timing of the last five events of some kind: events occurred in 1905, 1919, 1950, 1973, and 1990. We would like to model the uncertainty in the timing of the next event based on these observations. We have information on five observed intervals between consecutive events. Four of them are definite: 14 years between 1905 and 1919, 31 years between 1919 and 1950, 23 years between 1950 and 1973, and 17 years between 1973 and 1990. The fifth interval, between 1990 and the next event, is unknown. This doesn't mean that we have no information about it. At this time of this writing, 2027, we know the fifth interval is at least 37 years long.

The standard model for the interval between random events is the exponential distribution. This has one parameter, the rate. A good choice of rate is the one most consistent with the data. Likelihood provides the measure of consistency between a hypothesized rate and the observed intervals. The exponential function Figure 11.1 shows the likelihood of any one interval observation for a hypothesized rate of 0.03. Each of the observed intervals, 14, 17, 23, and 31 years has been marked in the graph.

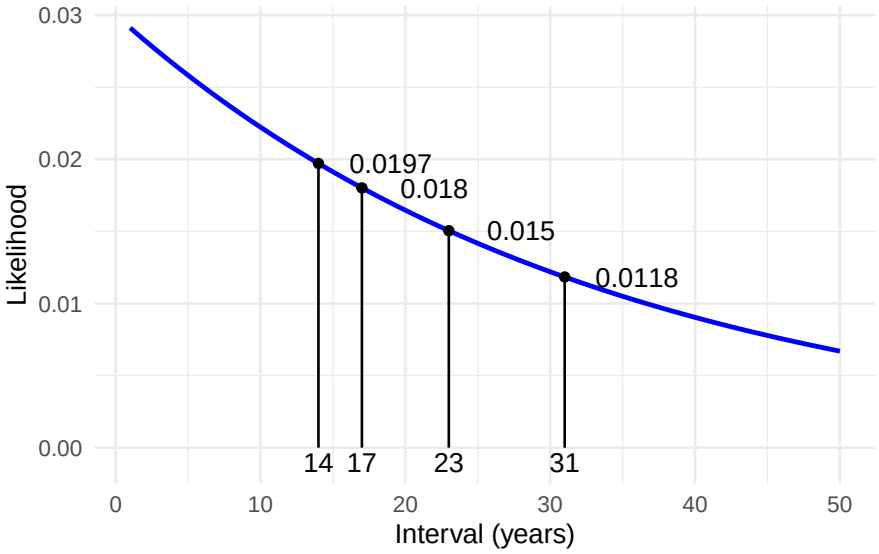


Figure 11.1.: The likelihood for any observed interval under the hypothesis of an exponential distribution with rate 0.03. The four completed intervals by vertical lines. The likelihood of each is printed beside the vertical line.

The likelihood for each individual observation is the value of the likelihood function at that observation. For instance, the likelihood of an interval of 14 years is 0.0197. When there are multiple observations, we need to combine their likelihoods. How to accomplish this?

The answer comes from the same feature of probability arithmetic that

you may have encountered in games (or in math classes about games). For instance, the probability of rolling a 5 with a die is 1/6. The probability of rolling a 5 twice in a row involves combining the individual probabilities by multiplication: 1/6 × 1/6 = 1/36.

The overall likelihood of our four observations—intervals of 14, 17, 23, and 31 years—is the product of their individual likelihoods: #$0.0197 \times 0.0180 \times 0.0150 \times 0.0118 = 6.325e-8$.$ This number is very small in magnitude. That's because there are very many other possibilities for the intervals that could have been observed, say, 12, 19, 24, 28 and such. Since there are so many similar sets of numbers, the likelihood of observing any particular one is small. As you'll see, the ultimate use of likelihoods is to compare them one to another, that is, to compare their *relative size*.

If all we had were the four completed intervals, the overall likelihood of 6.325e-8 would be the value we assign to the specific hypothesis of an exponential distribution with a rate parameter of 0.03. We are going to put this specific hypothesis into a competition for the *best* hypothesis. As described in Chapter 10, to define "best," we need an *objective function* whose input will be the rate parameter and whose output will be the overall likelihood of our observed data—14, 17, 23, and 31 years—under that hypothesis.

By carrying out the same calculations we did for a rate of 0.03, we can find the likelihood for other rates. These are displayed in Table 11.1. The workflow is this, for each row:

i. set the hypothesized rate
ii. compute the likelihood of each observation under that rate
iii. multiply these likelihoods to get the overall likelihood for the row.

This is repeated for each hypothesized rate. For the sake of conciseness, Table 11.1 shows only five hypothesized rates. See Figure 11.2 for a more complete picture.

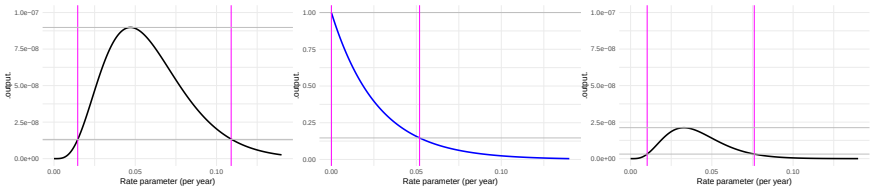
Table 11.1.: The likelihoods of the observations under five different hypothesized rates.

hypothesized rate	14	17	23	31	overall
0.03 per year	0.0197	0.0180	0.0150	0.01180	6.33e-8
0.04 per year	0.0228	0.0203	0.0159	0.01160	8.54e-8
0.05 per year	0.0248	0.0214	0.0158	0.01060	8.92e-8
0.06 per year	0.0259	0.0216	0.0151	0.00934	7.90e-8

hypothesized rate	14	17	23	31	overall
0.07 per year	0.0263	0.0213	0.0140	0.00799	6.26e-8

From the overall likelihood for each of the hypothesized rate parameters shown in Table 11.1, we can see that a rate of 0.05 /year is more likely than our original hypothesis of 0.03 / year. But we can say more. The likelihood ratio comparing the 0.05 per year hypothesis to the 0.03 per year hypothesis is 1.41. Table 11.2, to be introduced in the next section, translates likelihood ratios into verbal statements of relative support. A likelihood ratio of 1.41 is really "no evidence" in favor of 0.05 per year versus 0.03 per year. The reason for this wimpy result is that four observations is a tiny amount of data.

If we can't argue for the 0.05 per year hypothesis over the 0.03 per year hypothesis based on the four complete observations, what possible hypotheses can we rule out. To illustrate this, we're going to extend the calculations shown in Table 11.1 to a wider range of intervals and plot the results in Figure 11.2.



(a) The four completed intervals.

(b) The one incomplete interval of 37 years.

(c) All five intervals combined.

Figure 11.2.: Likelihood of the observed data as a function of the hypothesized *rate* parameter.

Panel (a) shows the likelihood function for the four complete intervals, the same calculation as in Table 11.1.

Panel (b) is the likelihood function for the incomplete interval, that is, an interval that is still ongoing after 37 years from 1990 to 2027. If the rate parameter were very low, say one event per 100 years, an incomplete interval of at least 37 years is very likely. If the rate parameter were higher, say one even per 10 years, the likelihood of seeing the 37-year incomplete interval is very low.

Panel (c) puts together the data from the complete and incomplete intervals. The likelihood function in (c) is the *product* of the two likelihood functions in (a) and (b).

The amount of data available dictates how precise a statement is justified by the evidence. Statisticians often quantify precision in this context by using the $\pm$ notation. For instance, an estimate of 0.03 ± 0.01 would indicate "95-percent confidence" that the true rate parameter lies within the interval 0.02 to 0.04. In each of panels (a), (b), and (c) we have indicated this 95-percent confidence interval with the vertical magenta lines.

A good rule of thumb is that quadrupling the amount of data roughly halves the width of the confidence interval. But there is little reason to expect the new interval to be centered on the old estimate. The most one can say is that the new interval will probably overlap the original interval. That may be counter-intuitive, but uncertainty is sometimes that way!

11.2. Putting numerical evidence into words

A *likelihood* ratio is a measure of the strength of evidence. In interpreting likelihood, verbal ideas like "weak" evidence, "moderate," and "strong" turn out to correspond to the *magnitude* of the likelihood ratio, where "magnitude" is used in the sense of Chapter 5.

The translation of a continuous quantity into a small set of verbal descriptions necessarily involves imposing some breakpoints. Table 11.2 provides a widely accepted set of breakpoints for translating the magnitude of the likelihood ratio into a verbal interpretation.

Table 11.2.: A scale for translating the likelihood ratio into verbal categories.

Magnitude	Likelihoodratio	Verbal interpretation
0	1	No evidence.
1	10	Limited evidence to support the hypothesis.
2	100	Moderate evidence to support the hypothesis.
3	1000	Moderately strong evidence to support the hypothesis.
4	10000	Strong evidence to support the hypothesis.

In accepting the verbal interpretations in Table 11.2 we are relying on the experience of past researchers. There's nothing wrong with that. But it would be nice to have a definite interpretation of what "limited" or "moderate" evidence constitutes. The mathematics of likelihood provide a means to do this.

First, there's general consensus among statisticians and scientists that a likelihood ratio of 10,000 is sufficient to satisfy the criterion for "strong evidence" in favor of a hypothesis. Now, imagine that your observations have yielded a likelihood ratio of 100. How much more evidence would you need to collect to reach a ratio of 10,000. One answer is, you can't know for sure because you don't know in what direction the new evidence will point.

But assume you are excited enough about your existing evidence to believe it will be worthwhile to continue collecting data. Your excitement leads you to suspect that the additional data will point in approximately the same direction as your current evidence. (But there is never a guarantee!) If your suspicion were correct, you would have to plan to **double** the amount of data you use to calculate the likelihood ratio. A nice way to do this would be to have some other researcher confirm your earlier results while doubling the amount of data overall.

Suppose you flip a coin 10 times and get 6 heads. Are you justified in believing the coin is biased toward heads? To answer this question, you need to compare two hypotheses. One hypothesis is that the coin is not biased, that is, the likelihood of getting a head is 0.5. Statisticians refer to such a "nothing-out-of-the-ordinary" hypothesis as a 👉**Null hypothesis**👈. You need another hypothesis to compare it to, a hypothesis that specifies what is the *amount* of bias. Let's call this the "research hypothesis." " For instance, you might consider a research hypothesis that the likelihood of getting a head is 0.6.

Under the Null hypothesis the likelihood of getting 6 heads and 4 tails is:

$$\underbrace{0.5 \cdot 0.5 \cdot 0.5 \cdot 0.5 \cdot 0.5 \cdot 0.5}_{\text{For the heads}} \cdot \underbrace{0.5 \cdot 0.5 \cdot 0.5 \cdot 0.5}_{\text{For the tails}} = 0.0010$$

Under the research hypothesis, the likelihood of getting 6 heads and 4 tails is:

$$\underbrace{0.6 \cdot 0.6 \cdot 0.6 \cdot 0.6 \cdot 0.6 \cdot 0.6}_{\text{For the heads}} \cdot \underbrace{0.4 \cdot 0.4 \cdot 0.4 \cdot 0.4}_{\text{For the tails}} = 0.0012$$

The likelihood ratio comparing the research hypothesis to the Null hypothesis is 1.2. This provides no reason to favor the research hypothesis. But, like a detective hunting for the criminal, you can try to collect more evidence, that is, more data.

For fun, imagine a dialog with the chief investigator (your boss) and with a likelihood expert present:

Detective: Chief, I only have very preliminary evidence that the coin is biased (at 0.6), but I have a strong hunch and I'm sure it will become more convincing if I collect more data.

Chief: How much time have you spent on this?

Detective: Just two days. But enough to feel it in my gut that the coin is biased.

Chief: It would be nice to put this case to bed, one way or the other. Take another two days and update me then.

Detective: Will do. Thanks, Chief.

Two days later

Detective: I have been following up on the case. What I'm seeing is a match to my previous hunch, I think it's more than a hunch.

Chief: Do you think you can wrap it up in another two days?

Detective: I'm confident.

Department Likelihood expert: Hold on. Your first two days investigation produced a likelihood ratio of 1.2. If your new data is consistent with your previous data, the combined likelihood ratio will be the square of 1.2, which is 1.44. If you were to spend another 4 days, the ratio would be $1.44 \times 1.44 = 2.07$. That would be laughed out of court!

Chief: What would it take for the evidence to be convincing?

Department Likelihood expert: For evidence "beyond a reasonable doubt", we would need a likelihood ratio of 10,000 or thereabouts. That would take about 100 days worth of data. And even after that, there's no reason to think, at this point, that things would come out the way you want.

Chief: How much data would we need to convince the perp to accept a plea deal?

Department Likelihood expert: For a plea deal, the required likelihood ratio would be much lower, perhaps around 100 to 1000. We're talking 50 to 75 days of evidence collecting.

Chief (to Detective): I want you off the case. We've got better things to do.

12. Bayesian reasoning

We are in the midst of a real-world experiment involving millions of people in a life-or-death situation. Perhaps that is an overly dramatic description. The experiment we have in mind concerns the testing of self-driving taxis in urban areas. Companies like Waymo have deployed fleets of self-driving cars that offer a taxi service.

That such testing is allowed is itself notable. How much experience will we need with the cars to be sure that we are making a proper judgment?

Vocabulary Review

- analysis (3)
- bayesian (8)
- data (2)
- functions (6)
- hypothesis (8)
- likelihood (11)
- mean (8)
- operationalization (2)
- per (3)
- point (3)
- range (8)
- rate (4)
- uncertainty (8)
- value (2)

New

- posterior
- prior

12.1. An uncertain situation

Companies currently run pilot programs deploying self-driving taxis in a handful of cities. They have already collected substantial amounts of data and reported summaries such as this Waymo bulletin. Instead of commenting on this data, let us imagine ourselves at the time when the safety experiment is just beginning. Then, we will gradually accumulate data to form and update our beliefs about fleet safety. We will keep it realistic, but because the data will be made-up, do not use this example to draw conclusions about the real world.

Waymo bulletin



"Safety" is a dimension, an aspect of the transportation system. Modeling safety starts by defining it clearly, to *operationalize* it. Our operationalization in this discussion will be the *number of miles driven* by each car from the time it enters service until the first accident. We can define an "accident" as an incident that generates a claim on insurance or requires a police report. For other purposes, a different operationalization might be appropriate, for instance the number of close calls or near-misses

A car without an accident to date is not necessarily safe. Even accident-free cars provide information about fleet safety; the duration of the no-accident-yet interval carries that information. (See Figure 11.2.) A car that goes a large number of miles without an accident provides pro-safety evidence; A car that has an accident after a small number of miles does the opposite.

Chance influences whether a car is in an accident. This play of chance can be modeled with a probability distribution. An appropriate family of distributions for this was used earlier in Chapter 11: the exponential family. In Chapter 11 we used the *time* between incidents. For driving, the mileage of each car is a better representation of that car's exposure to the risk of accidents.

Exponential distributions model the time interval, or mileage, in our case, between incidents. The interval here is the mileage from the start of the experiment until the first accident. For an individual car, this interval is modeled as random, a draw from the exponential distribution used to model the individual cars.

Even though each accident (or lack of accident) is random, the fleet safety plays a role. This fleet-level safety is operationalized by the particular *rate* parameter used for the individual-car randomness. The appropriate value for the *rate* parameter is uncertain. This uncertainty is large at the start of the experiment. As the experiment proceeds and more data arrives, the uncertainty will decrease.

12.2. The fleet-wide likelihood function

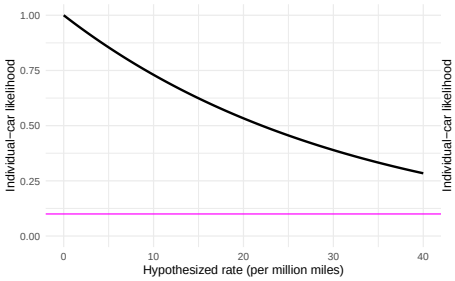
Fleet-safety uncertainty is a probability distribution on the *rate* parameter itself. Unlike individual-car uncertainty, this need not be exponential in form. At first, we don't even know the proper form for the distribution should be. We will have to discover that form as we go along.

Suppose that at the end of each month the mileage and accident status of every car in the fleet are recorded. For any assumed value of the rate parameter, the likelihood of the observed data can be computed: multiply together the likelihoods for each individual car's observed outcome.

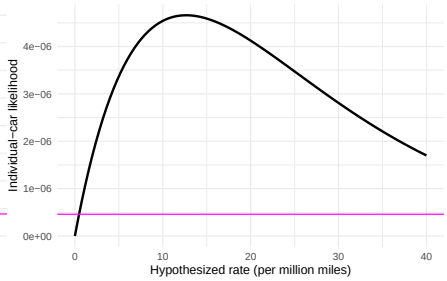
The rate parameter is uncertain. Repeating the likelihood calculation for each possible value of the *rate* allows us to see how plausible each value is given the observed data. In other words, we create a *likelihood function* for the rate parameter.

Imagine that you are the person responsible for constructing the likelihood function that incorporates the latest data. You have a stack of cards, each recording the current mileage of an individual car or, if the car had an accident, the mileage at which the accident occurred. Each card's information corresponds to an individual likelihood function for that car.

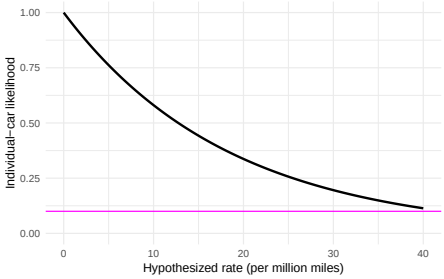
Plotting the individual-car likelihood functions requires choosing appropriate range of values for the rate parameter. Historical data on human-driven cars suggests a rate of about five serious accidents per million vehicle miles. A sensible plotting range is 0 to 40 accidents per million miles. The upper end lets you consider hypothetically very dangerous cars. The low end hypothesizes that self-driving cars are safer than human-driven. The middle hypothesizes that self-driving cars are about as safe as human-driven cars. Figure 12.1 shows three individual-car likelihood functions based on the most recent data for that car. The cars plotted in panels (a) and (c) have not yet had an accident. The car in (b) had an accident at mileage 78,954.



(a) 31,448 miles with no accident



(b) 78,954 miles with an accident



(c) 54,365 miles with no accident

Consider the likelihood functions graphed in Figure 12.1. The two accident-free cars' likelihood functions differ only slightly in shape.

The accident-free cars favor a low hypothesized rate parameter. Their likelihood functions peak at a rate of 0 accidents per million miles and fall steadily as the hypothesized rate increases. This evidence hardly proves that the cars are safe. Recall that a likelihood ratio of 10 is only "limited evidence" in favor of the more likely hypothesis. (See Table 11.2.) Only hypothesized rates where the likelihood falls *below* the magenta can be considered unlikely. So the accident-free cars don't rule out a very high hypothesized rate of 40 accidents per million miles. This isn't surprising,

since the rate 40 per million miles amounts to an average of one accident every 25,000 miles! Reaching 54,000 miles isn't shocking even under high risk.

The car in panel (b) experienced an accident. Its likelihood function has an argmax at a hypothesized rate of 12.7 accidents per million miles. This means an average of one accident every 80,000 miles, entirely consistent with the car having had an accident at 78,954 miles. This car only weakly argues against rates below 0.5 accidents per million miles.

Any individual car's evidence does not precisely identify a specific hypothesized rate parameter. Each car here is entirely consistent with the historical rate for human-driven cars.

But your job is not yet complete. Combine the evidence from all the individual cars to assess what they have to say about fleet safety. Fortunately, this is easy. Just multiply the functions together. The result, from just these three cars is shown in Figure 12.2.

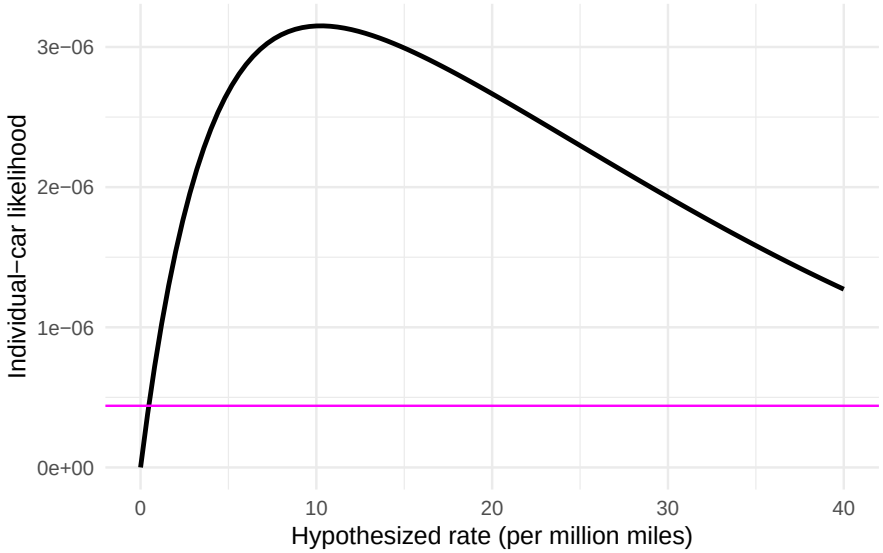


Figure 12.2.: Combined likelihood for all three cars

Data from just three cars cannot resolve whether the self-driving cars are safer or more dangerous than human-driven cars. You need data from hundreds or perhaps thousands of cars.

Since individual-car likelihoods are weak on their own, you decide to adopt a work flow where, as you pick up each car's data card, you immediately

update the combined likelihood, by multiplying it by the new card's likelihood.

12.3. The Prior

There's a small technical glitch to take care of. The first card has no combined likelihood to multiply it with. Two simple work-arounds are available:

a. Use the likelihood function from the first card as the initial combined likelihood.

b. Start with a combined likelihood function of that is uniform, which effectively says "I have no prior evidence for one hypothesized rate over another."

Note that (a) and (b) amount to the same thing. In practice you are starting out with the likelihood function from the first card. But designers already did substantial work to make the cars safe. For instance, the government approved the test cars for use on the public roads in the first place.

The Bayesian framework has a slot for inserting such before-the-first-data-card knowledge into the analysis. This is called the 👉**prior**👈 and reflects what you know *before* looking at the data. The "you" here is important. The experiment aims to convince others, no just yourself, that enough evidence exists for a meaningful conclusion. So the prior should reflect the beliefs of both the skeptic and the enthusiast.

US Supreme Court Justice Ruth Bader Ginsburg had a helpful view about debate. She said, "Fight for the things that you care about, but do it in a way that will lead others to join you." One way to lead others to join you is to acknowledge the possibility that they might be right. In the Bayesian framework, this means broadening your prior to support the views of all (reasonable) people engaged in the debate. You do not have to assign much probability to those views, but you must assign some.

Sensible priors should be broad, not narrow minded. This point is not unique to quantitative analysis, but applies to reasoning and debate more generally.

> *To have doubted one's own first principles is the mark of a civilized man.* — Supreme Court Justice, Oliver Wendell Holmes, Jr (1841-1935)
>
> *Doubt is an uncomfortable condition, but certainty is a ridiculous one.* — Voltaire (1694-1778)
>
> *The whole problem with the world is that fools and fanatics are always so certain of themselves, and wiser people so full of doubts.* — Bertrand Russell (1872-1970)

12.4. The Posterior and Bayesian updating

Let's re-consider the work-flow presented in the previous section. You started with an appropriately vague prior, then multiplied it successively by the likelihood from each data card as you picked it up. In the end, you have a distribution that reflects all the data. This distribution is called the ☞**posterior**☜, the state of your knowledge after taking all the data into account.

Calling the likelihood function from the first card L_1_() and so on for the subsequent cards, the posterior distribution after n cards is:

posterior() ≡ prior() × L_1_() × L_2_() × ... × L_n_()

Since the prior() in the previous section was vague, perhaps even `flat()`, it's reasonable to wonder why even to give it a name; just multiply the cards' likelihood functions together. Indeed, for a hundred years statistical theory was dominated by exactly this point of view, named "Frequentist statistics."

Consider, however, a slightly different scenario for processing the data on vehicle mileage and accidents. Rather than a batch of cards coming in all at once, suppose that they arrive sequentially, one by one. For the sake of definiteness, let's assume that they arrive once per day and that the likelihood function for each day's data is L_day_().

On day zero, before any cards arrive, all you have is your vague prior, which we'll call prior_0_(). In comes the first card, day 1, and you perform your calculation, multiplying that day's likelihood by the prior:

posterior_1() $≡L_1_() × prior_0_()

On day two, the second card arrives. The Bayesian multiplication for day two will be

posterior_2() $≡L_2_() × posterior_1_()

And for day three:

posterior_3() $≡L_3_() × posterior_2_()

The general pattern is that yesterday's posterior becomes today's prior. The prior is your state of knowledge in the morning. The posterior is your state of knowledge after the day's work is done.

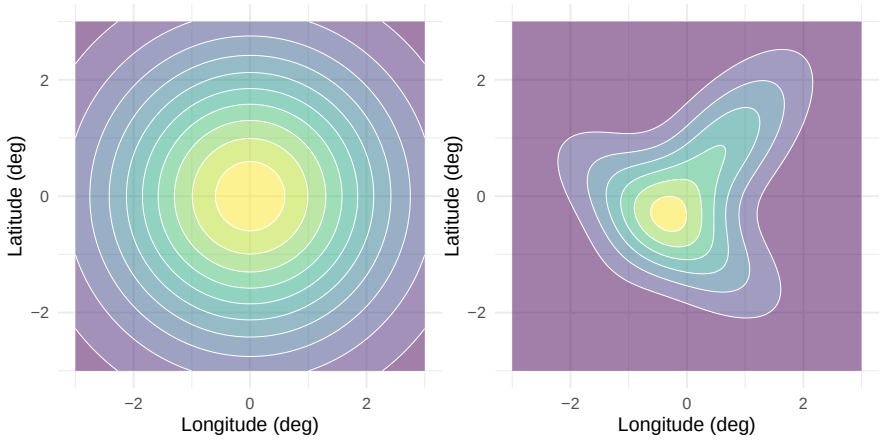
Even if prior_0_() is vague, the prior for successive days will become less and less vague as more data is incorporated.

To illustrate, consider this dramatic example. A small ship has caught on fire. The crew has mere moments to send an SOS which includes a rough latitude and longitude for their location. "Rough" means that there is uncertainty.

The SOS is received by far-away radio stations. The content of the SOS contains the rough coordinates, but the *direction* from which the signal is received is also noted. These directions, one from each station help to define the search target.

Before the SOS, the Coast Guard's prior for the sinking ship is more or less uniform, covering a broad expanse of ocean. (Indeed, the Coast Guard doesn't yet know there is a sinking ship.)

Receipt of the SOS allows the Coast Guard to calculate a likelihood for the ship's location based on the Coast Guard's previous experience with the accuracy of such signals. Similarly, the direction-detecting equipment gives a likelihood function for the direction of the sinking ship relative to the Coast Guard's stations' locations. Multiplying these three likelihoods—from the SOS contents, and the two directions—allows the Coast Guard to calculate a *posterior probability* for the location where the ship sank. Figure 12.3 shows the posterior probability as a function of latitude and longitude.

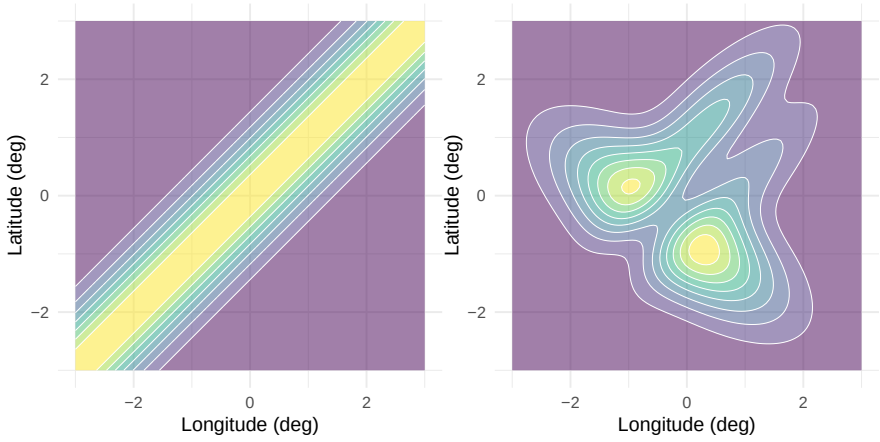


(a) Based only on the SOS content.

(b) Adding in directional data.

Figure 12.3.: Posterior probability for the ship's location based on the SOS and direction-detecting equipment.

Anxious to find the ship's crew, one of the Coast Guard stations sends out a search and rescue plane. It flies in a straight line toward the most likely location in Figure 12.3(b). It can be hard to spot a life raft from a quick-moving plane. Again, based on the plane's course and the Coast Guard's experience they can construct a likelihood for the plane's finding the raft. This is shown in Figure 12.4(a).



(a) The likelihood for the plane to spot the life raft.

(b) The updated posterior probability after the plane's flight.

The plane complete's its mission, but unfortunately it does not spot the life raft. Taking this information into account, the new posterior probability for the raft's position is shown in Figure 12.4(b).

The sensible thing at this point is to send out another plane to fly another course. Your job is to decide what the course should be to maximize the chances of spotting the raft. The current posterior provides a lot of guidance here. What would you do?

13. Dynamics: Modeling change

Section 1.5 introduced a model of an epidemic: the SIR model. SIR tracks the numbers of infective and susceptible people from day to day, week to week, and month to month. Such models, and the systems the models represent, are called 👉**dynamical systems**👈.

In simple language, a dynamical system is one that *changes over time*. A pendulum, swinging back and forth, is a dynamical system. A skyscraper shifting in the wind is a dynamical system. A flying airplane or rocket is a dynamical system. So are a building's heating and cooling systems. Engineers and others often model these real-world dynamical systems to determine the strength, stability, and motion of the objects and, importantly, to design the control systems that govern their behavior. In a modern airplane, the pilot flies a model, and the control system brings the airplane into compliance with the model.

We have to be careful in understanding "changes over time." The pendulum, airplane, and skyscraper are not changing; they are fixed *mechanisms*. It is their state that changes. The angle and speed of the pendulum bob oscillate up and down. An airplane's control surfaces change with every whisper of air turbulence, and with them, the pitch, roll, and yaw of the plane change second by second. Builders may have installed a home heating system a decade ago; the "changes in time" consist of the furnace turning on and off in response to fluctuations in indoor temperature.

Historically, we can trace the mathematical foundations for modeling dynamical systems to the decades around 1700. The "Laws" of science are generally written in the language of dynamical systems. The earliest of these are Newton's Laws, but much more modern frameworks—quantum physics, general relativity, electromagnetism, fluid mechanics—are all formulated in terms of dynamical systems.

We will consider only simple physical systems, such as the swinging pendulum, which suffices to lay out the mechanics of modeling dynamical systems. Mostly, we will emphasize theoretical dynamical systems used to model social, ecological, national security, and similar phenomena, such

Vocabulary Review

- dimension (2)
- operationalization (2)
- value (2)

New

- analysis
- base dimensions
- base units
- capita
- controlling for
- conversion factor
- coordinate plane
- derived dimensions
- derived units
- dimensionless
- flavors of one
- number line
- partial change
- per
- point
- pure number
- quantity
- si
- space
- units
- with respect to
- worth

as epidemics. These theoretical systems are story-like descriptions, not intended to be taken literally as "laws." Despite their approximations and simplifications, the theories serve as frameworks for developing understanding and anticipating the outcomes of interventions in real-world systems. Conveniently, for the person starting to learn what dynamical systems are, the theoretical systems allow a reader to work with familiar mechanisms, such as competition and disease transmission, rather than first having to learn the scientific esoterica of torque, angular momentum, viscosity, space-time curvature, quantum wave functions, and such. However, if the reader develops an interest in such things, he or she will already have a grasp on the mathematics shared by the scientific theories.

13.1. The Euler framework

We will use the ☞**Euler framework**☜ for introducing the quantitative description of dynamical systems. The Euler framework involves three mathematical components:

State The ☞**State**☜ is a collection of one or more quantities that give a *complete description* of the instantaneous configuration of the system. It can be helpful to consider the instantaneous state to be a *point* in a space, just as we did in Section 3.7. We use the term ☞**state space**☜ as the set of possible states for a dynamical system.

Dynamical rule The ☞**Dynamical rule**☜ specifies how the state changes over time. The dynamical rule is a *function* that takes a state as input and produces as output a description of the near future. That is, the dynamical rule tells, for each possible state, how the state changes over the next *instant*. Since the dynamical rule is a *function*, for each input there can be only *one output value*.

As an example, consider the position of a ball dropped from a tower. At any instant, the ball's altitude quantifies how far it has moved since it was released. The altitude is, of course, changing. The ☞**velocity**☜ is the rate of change of the position. Similarly, the output of the dynamical rule is the *rate of change* of the state.

Thinking of the picture in terms of a *state space*, the state at any instant is a position in the state space. The output of the dynamical rule is the direction and speed with which the state moves.

Accounting system The dynamical rule tells how the state changes, but only over the next instant. Often, we want to trace the state's movement over a substantial interval of time. The accounting system keeps track of the state's instantaneous rate of change, accumulating it to compute the state as a function of time. This function is called the 👉**trajectory**👈 of the state.

There are different ways to do the accounting. The methods traditionally taught in college-level math involve algebra that many students find daunting. The method we will use is arithmetical, invented by Leonhard Euler (1707-1783), one of the greatest mathematicians of all time.

The accounting method is entirely mechanical. To start, the user gives an initial state for the system. This state is fed into the dynamical rule which produces the instantaneous rate of change for the state, much like a "velocity" for the state. The method multiplies the velocity by a small time increment, producing a change of state, then adds the change to the initial state. This process of

$$\text{state} \;\rightarrow\; \text{change of state} \;\rightarrow\; \text{new state}$$

is repeated over and over, leading to a sequence of states: the trajectory of the system. The calculations involved are arithmetic, but tedious. Typically, software is used to do the work.

We will return to the accounting method and give it a dignified mathematical name in Chapters 14, 15, and 16. Since it is automatic, the modeler need not be particularly concerned with how it works. Think of it as an appliance, say, a toaster.

Dividing the mathematical work between the dynamical rule and the accounting method greatly simplifies the dynamical rule. The rule needs only to tell the change in the state over the briefest imaginable interval of time. This feature of dynamical rules enables the concise description of scientific theories of nature's operation.

13.2. Defining the state and rule

It is convenient to start with a system whose dynamical rule is straightforward and where we can easily keep the overall trajectory in mind. With this in mind, imagine a pendulum, such as a tire hanging by a rope from a tall tree branch, or the swinging mechanism of a grandfather clock.

Animated pendulum online

The animation in Figure 13.1 consists of a series of frames. Figure 13.1 shows the first five frames individually, making it straightforward to measure the bob's position in each.

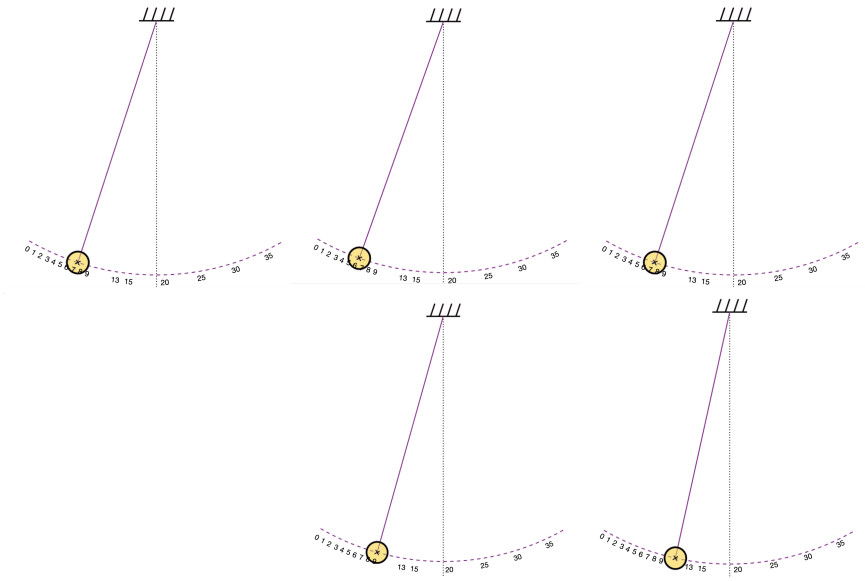


Figure 13.1.: Five successive frames from the movie. The spacing between tick marks is 1 degree. Between frames 1 and 2, the pendulum is swinging slightly to the left. Thereafter, the swing is to the right.

What is the instantaneous *state* of the pendulum system? Let us start the discussion with a proposition (that will turn out to be wrong): the state is merely the position of the bob at each instant, as marked in the snapshots shown in Figure 13.1.

The position changes from frame to frame:

$$7.2^\circ \to 6.3^\circ \to 7.2^\circ \to 8.9^\circ \to 11.0^\circ \to \ldots \ .$$

In both frames 1 and 3, the position is 7.2 degrees. Following frame 1, the position moves to 6.3 degrees, then back to 7.2 degrees for frame 3. Continuing in that direction, the position arrives at 8.9 degrees for frame 4.

Recall from Section 13.1 that a dynamical rule is a *function*. A function can produce only one output for each particular input. Consequently, the dynamical rule for a pendulum cannot produce two different outputs for the input state of 7.2 degrees.

Tempting though it might be to deal with this problem by allowing the dynamical rule to be something other than a function, there is a good reason for the (arbitrarily sounding) restriction, introduced in Chapter 6, that there be a unique output for any given input.

Instead of changing the *framework*, we need to redefine the state. For a pendulum, this change is straightforward: add another state component. An excellent way to do this is to add the bob's instantaneous *velocity* to the state, so that the state space becomes the coordinate pair (position, velocity).

The markings in Figure 13.1 suffice to mark the position, but the individual frames say nothing about velocity. For velocity, we need to do a calculation. The position at frame 1 is 7.2 degrees, then 6.3 degrees at frame 2. So the change in position from frame 1 to 2 is $6.3 - 7.2 = -0.9$ degrees. Velocity, however, is not just a change; velocity is a *rate of change* with respect to time.

The *frame rate* of the video in Figure 13.1 is 10 per second. That is, the time between successive snapshots in Figure 13.1 is 0.1 seconds. Consequently, the rate of change of position with respect to time—in other words, the velocity—from frame 1 to 2 is -0.9 / 0.1 = -9.0 deg per sec.

In contrast, from frame 3 to 4, the change of position is $8.9^\circ - 7.2^\circ = 1.7^\circ$. The *velocity* over the 0.1 seconds between frames is therefore 17 degrees per second.

Defining the state as the pair (position, velocity) allows us to construct a dynamical rule that has a unique output for each state input.

Since the state has two components, the output of the dynamical rule must also have two components: the rate of change for each of the components in the state.

The rule for the first component is straightforward because the velocity is the rate of change of position. So the rule (in part) looks like this:

$$\text{pendulum_rule(pos, vel)} \equiv (\text{vel}, \textbf{???}) \tag{13.1}$$

We placed ??? in the slot for the rate of change of the "vel" component of the state in Math expression 13.1. The rate of change of the velocity—that is, the acceleration—is given by the physics of gravity, the extent to which gravity tugs with or against the current velocity. Still, we can model the acceleration as a function of position very simply. Recognize that at the very bottom of the pendulum's swing—position 19—gravity is exactly countered by the vertical rod, so at position 19, the acceleration is zero. That lets us specify more of the dynamical rule for the pendulum:

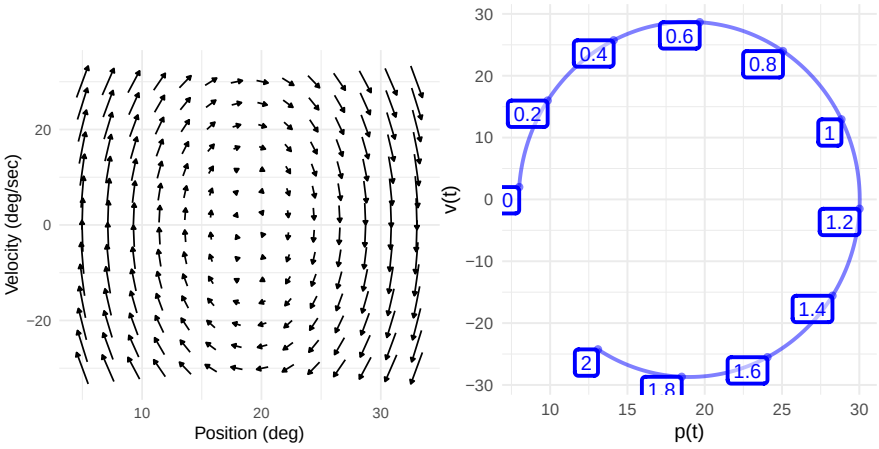
$$\text{pendulum_rule}(\text{pos} = 19, \text{vel}) \equiv (\text{vel}, 0) \tag{13.2}$$

To construct a complete dynamical rule, we still need to generalize this to apply to any position, not just 19 degrees. A straight-line function of position will be a good start. We know the output of the straight-line function will be zero when the position is 19. By the argument in Note 1, the acceleration at position 7.2 is 80°per sec^2. The linear function that connects velocity 0 at position 19 and velocity 80°per sec^2 at position 7.2 is $-6.78(\text{pos} - 19)$. Slotting this formula in place of the ??? in Math expression 13.2 gives the complete dynamical rule:

$$\text{pendulum_rule}(\text{pos}, \text{vel}) \equiv (\text{vel},\ -6.78(\text{pos} - 19)) \tag{13.3}$$

Rules in this form are suitable for calculation, but to get a better understanding of how the system works, Figure 13.2(a) shows the state space with the two-component output of the dynamical rule drawn as a vector. The graphic displays the dynamical rule at only about 200 points in a row-and-column grid in state space. In reality, the dynamical rule applies to *all* points in the space. We cannot show them all because the graphic would become a solid mass of black arrows.

Applying the accounting method turns the dynamical rule into a trajectory. Figure 13.2(b) shows the trajectory. The initial state (time = 0) is marked 0. The other boxed numbers indicate the time (in seconds) at which the state reached that point.



(a) Dynamical rule

(b) Trajectory

Figure 13.2.: The dynamical rule for the system of `r('fig-pendulum-gif')`. The rate of change at selected points in the state space is a vector since both components of the state are changing.

We draw the dynamical rule at the various points in the state space with a straight arrow. The arrow has two components, horizontal and vertical. Similarly, there is an *instantaneous rate of change* for each component of the state. The arrows show both instantaneous rates of change simultaneously. In contrast, the trajectory traces out the evolving state over a period of time: two seconds in Figure 13.2(b). The trajectory is a smooth curve. The accounting method does the mathematical work to turn the straight arrows of the dynamical rule into a curved trajectory.

Note 1: Estimating acceleration

We can estimate the acceleration at position 7.2 by examining the short three-frame movie $6.3^\circ \rightarrow 7.2^\circ \rightarrow 8.9^\circ$. The velocity represented by the first arrow is, as before,

$$\frac{7.2^\circ - 6.3^\circ}{0.1 \text{ sec}} = 9^\circ\text{per sec}$$

Likewise, the velocity at the second arrow is

$$\frac{8.9^\circ - 7.2^\circ}{0.1 \text{ sec}} = 17^\circ\text{per sec}$$

The acceleration is the rate of change of the velocity:

$$\frac{17^\circ\text{per sec} - 9^\circ\text{per sec}}{0.1 \text{ sec}} = 80^\circ\text{per sec}^2$$

13.3. The SIR model

A key feature of the Euler framework is that the dynamical rule describes the rates of change in the state only *instantaneously*. Although the *trajectory* may curve over extended durations of time, at each instant the dynamical rule needs to account for only a tiny change in the state corresponding to a small fraction of the vector. The accounting method moves this small amount, then evaluates the dynamical rule at the new state, then moves again, and so on. Any curve in the trajectory arises from repeated evaluations of the dynamical rule at state points that are very close together. Since the accounting method handles any curvature in the trajectory, the dynamical rule is much simpler than the extended trajectory.

We can now explore the dynamical rule for the SIR model of epidemics first presented in Chapter 1. To start, we need to specify the *state* of the SIR system.

A simple but effective representation of the state is two numbers: the number of susceptible people (S) at any instant and the number of infective people (I) at that instant.

To keep the model simple, we will not account for the infection's origin. To start an infection, we set I to a nonzero value and let the dynamical rule take over.

To start, we will draw the state space and the dynamical rule at many individual points in the state space, as in Figure 13.3. Then we will interpret the dynamical rule in terms of familiar aspects of infectiousness and recovery from infection, and we will show the modeling techniques used to implement it.

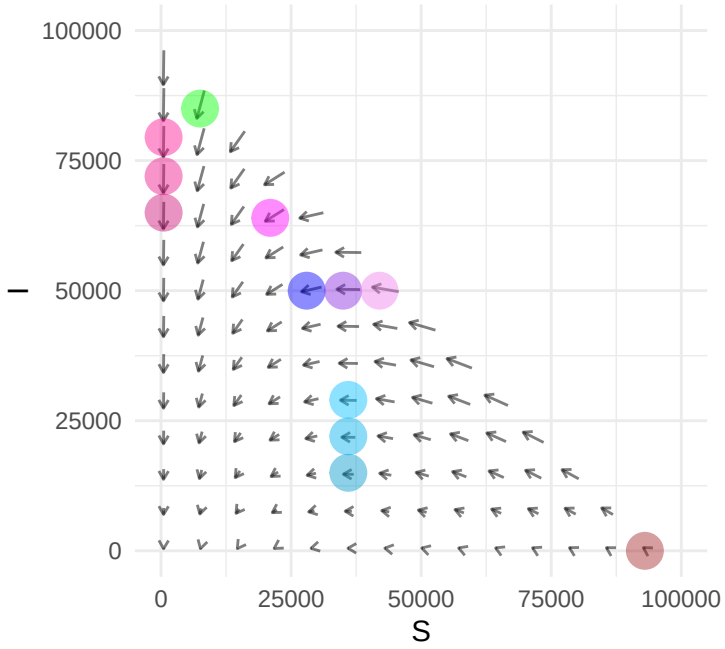


Figure 13.3.: The state space of the SIR model with the dynamical rule shown with arrows.

Figure 13.3 shows a state space suitable for tracking an epidemic in a population of 100,000 or fewer. At any instant of time, the sum of S and I cannot be greater than 100,000. So, we need only draw the dynamical rule in the area of state space where $S + I < 100,000$, the lower triangular region of the state space.

The first thing to notice is that the dynamical rule arrows tend to have a leftward-heading component. The leftward trend indicates that S is decreasing; the model does not include the production of new susceptibles by birth or immigration.

In contrast, the rate of change in I can be either upward or downward. For low values of S, the rate of change in I is downward because fewer people catch the disease than the number who are *recovering from infection*. Such recovery occurs at high S as well, but there the susceptible population is so large that many new individuals become infective.

We have marked a few state-space positions with colored dots to help us consider how the dynamics are different depending on S and I.

The dynamical rule comes from the modeler's knowledge of the system. Even those who may not have previously thought about the quantitative description of epidemics likely understand some basic principles.

i. New infectives are created when an existing infective meets a susceptible, converting the susceptible to an infective.

ii. Existing infectives recover at some rate, lowering the number of infectives without changing the number of susceptibles.

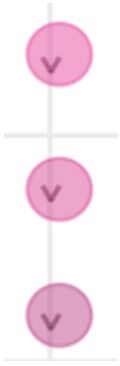
iii. If we ignore births, immigration, and deaths, as we are doing here, the only route for a change in the number of susceptibles is conversion via an infective.

iv. All other things being constant, a larger number of infectives leads to a larger number of conversions of susceptibles. Similarly, all other things equal, a larger number of susceptibles results in more conversions.

These principles play out differently across regions of the state space, since the balance between susceptibles and infectives varies with position. Consider these (color-coded) regions in state space:

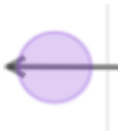


Few susceptibles, many infectives The flow is mainly southward, a little to the west. This corresponds to a slow decline in susceptibles as they run into any of the many infectives. Net decline in infectives: more are recovering than are being created by conversion of susceptibles.



No susceptibles The flow is slowly to the south. No new infectives, but old infectives are gradually recovering.

A few infectives and many susceptibles The flow is slowly to the northwest. This corresponds to slow growth in infectives and a corresponding decrease in susceptibles.



Many infectives and many susceptibles The flow is rapidly to the west. This is a rapid decline in susceptibles.

13.4. Trajectories

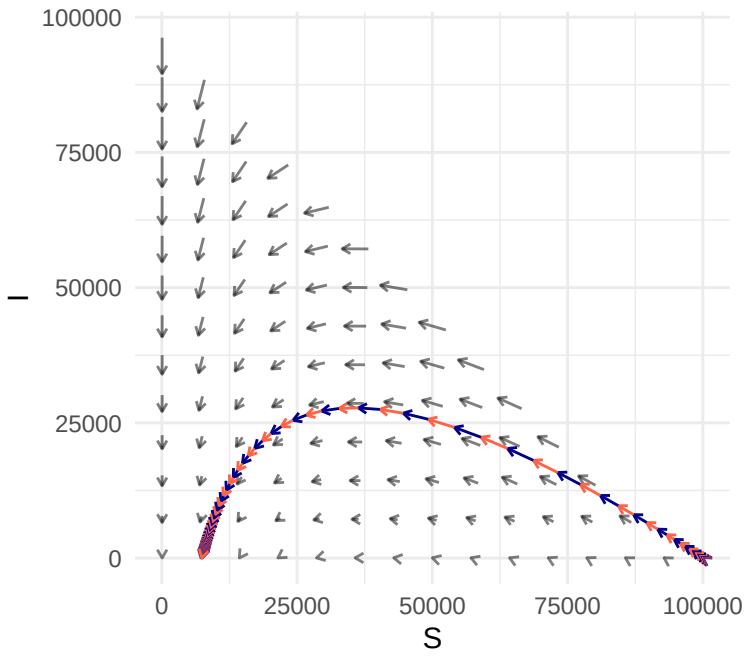


Figure 13.4.: A chain of steps for the SIR model starting with $S \approx 100,000$ and a handful of infectives. Each step covers 12 hours. The steps are plotted in alternating colors to make it clear where one step ends and the next one begins. The path indicated by the chain of steps is the *trajectory* of the dynamical system from the initial state.

In Figure 13.4, each step is a short line segment. The overall path accumulated through many steps, however, is curved. The curve is the result of re-evaluating the dynamical rule at the end of each short, simple step. Since the state moves only a small distance during one step, the output of the dynamical rule will be only slightly different when evaluated at the end of the step than it was at the start of the step. The overall trajectory, the accumulation of many small steps, approximates a gradual curve, just as we perceive smooth motion in a video, which is just one still frame after another. The time between the frames is very small, so successive frames are *almost* identical. The slight frame-to-frame difference produces the sensation of continuous motion.

Compare the trajectory in Figure 13.5 with the corresponding graph in Figure 1.1. They look completely different because the graphs are made in different spaces. In Figure 13.5 the axes are I vertically and S horizontally. But in Figure 1.1 the horizontal axis is time. The levels of I and S are both plotted along the vertical axis. The time of each position in the trajectory in Figure 13.5 is implicit and can be calculated by counting the number of steps needed to bring the state to that position. In contrast, Figure 1.1 shows

the time is explicitly.

13.5. SIR dynamical rule

As mentioned previously, a dynamical rule is a *function* that takes the state as input and produces the instantaneous change of state as output. In Chapter 9, we introduced ways to build custom functions to represent a given situation. We are going to put those techniques to work, now.

The two state quantities in SIR are the number of susceptibles (S) and the number of infectives (I). Even without thinking much about the mechanism of epidemics, we can write down a skeleton form for the instantaneous rate of change. We will use the abbreviation ROC to stand for "instantaneous rate of change," and the ROC of S will be named $ROC_S()$ and likewise for I. Both $ROC_S()$ and $ROC_I()$ will take the state (S, I) as input. Each will be a linear combination of four simple functions.

$$ROC_S(S, I) \equiv a_s + b_s\, S + c_s\, I + d_s\, S\, I \tag{13.4}$$

and

$$ROC_I(S, I) \equiv a_I + b_I\, S + c_I\, I + d_I\, S\, I \tag{13.5}$$

The first thing to notice about the proposed functions in Expressions 13.4 and 13.5 is that they are *almost* the same. That might be surprising, since they describe the ROC of two different quantities, S and I. Keep in mind that we are building a ☞**skeleton**☜ for the functions. We will flesh out the skeleton in different ways.

The "flesh" on the skeleton is the coefficients used in the linear combination. We combine four simple functions, so each expression has four coefficients. We have (uncreatively) named these coefficients a, b, c, and d.

We need a dynamical-rule function for each of the two state variables. For simplicity, we will use the same skeleton for each function, but endow each with its own flesh. That is, we will use different coefficient values for each dynamical rule. A simple notation helps make the skeletons' similarities clear while at the same time signalling that the values of the corresponding coefficients in each skeleton can differ. We do this by adding a subscript, I or s, to each coefficient, depending on whether it is in the dynamical rule for the infectives or for the susceptibles. For instance, a_I is the coefficient

on flat() for the ROC_I-skeleton. Similarly, d_S is the coefficient on the *interaction* between S and I for the ROC_S-skeleton.

Ultimately, we will want to give quantitative values for each of the eight coefficients $a_s, b_s, c_s, d_s, a_I, b_I, c_I, d_I$.

Experienced modelers have a trick to facilitate the initial construction of a model. Rather than specifying a value for each coefficient, they state, based on their understanding of the system they are modeling, the *sign* for each coefficient: negative, positive, or zero. Going slightly further, they explore a limited set of scenarios: the state variables are either "large" or "small." (The state variables in SIR cannot be negative, since there is no such thing as a negative number of infectives or susceptibles.)

Let us start with the coefficients in ROC_I.

- a_I: Think about what this term would indicate if both S and I were zero, the hypothetical world with no infectives or susceptibles. It seems reasonable to suppose that new infectives (that is, the rate of change of infectives) would not come out of the air. So $a_I = 0$.

If the model is to allow influx of infectives from other places, the model would call for a positive a_I.

- b_I: This is the coefficient on S. Imagine that S is large, but I is zero. What will happen to the ROC of I? Without any infectives, there is no way for a susceptible person to become sick. So b_I must be zero.

If the model is to allow *spontaneous* generation of infection, b_I would be positive.

- c_I: This is the coefficient on I. Imagine, contrary to the discussion of b_I, that I is large and S is zero. In such a hypothetical world, there are no susceptibles who can become sick so that no new infectives will arise from the susceptibles. However, there is a mechanism for a change in I even when S is zero: Infectives can recover and stop being infectives. The more infectives there are, the larger the rate (say, infectives/day) at which they recover. In consequence, c_I is a negative number. Without susceptibles, the number of infectives decreases over time.

- d_I: The coefficient on the interaction term $S\,I$ for $ROC_I()$. This is a case where standard nomenclaturee gets things right: "Interaction" is an excellent name for this term. A nice way to think about interaction is that it refers to what happens when an S meets an I: they interact. With infectious illness, the possible outcomes of the interaction are either no change in the situation or, with some probability, the S catches I's infection, and the S becomes infectious. So when an S meets an I, the number of infectives increases. Thus, d_I must be positive; it is creating new infectives.

The analysis of a_s, b_s, c_s, and d_s follows the same kind of logic. a_s would correspond to immigration of susceptibles (if positive) or emigration (if negative). The SIR model does not consider this. SIR also does not consider births or deaths. The number of susceptibles only changes when an S meets an I. When that happens, the number of susceptibles decreases as susceptibles are converted into infectives. So d_s must be negative. What is more, since every susceptible individual who is infected becomes a new infective, it must be that $d_s = -d_I$.

A nice convention when writing rate-of-change coefficients is to stipulate that all are positive. When a coefficient enters negatively into the rate-of-change function, we will use a *minus sign* instead of the $+$ used in Equations 13.5 and 13.4

Taking into account all the coefficients that we argued are zero, we end up with two formulas for the rate of change functions:

$$ROC_S(S, I) \equiv -d\,S\,I$$

and

$$ROC_I(S, I) \equiv -bI + d\,S\,I$$

There are only two coefficients, not three, since the decrease in susceptibles balances any increase in infectives!

13.6. Exponential growth and decay

Demographers estimate that a fertility rate of 2.1 children per woman keeps the population steady; the number of births matches the number of deaths. In many wealthy countries, however, fertility rates are much lower. To dramatize the consequences, some pundits have extrapolated what would happen if the worldwide fertility rate were roughly one child per woman.

Assuming equal proportions of males and females, this low fertility rate results in half as many births in each generation as in the preceding generation. Over ten generations, successive halvings would result in the tenth generation having only one-thousandth as many births as the starting generation.

These successive halvings are an example of 👉**exponential decay**👈. Likewise, if the fertility rate were roughly four children per woman, the number

of births would *double* every generation. These successive doublings, each taking approximately one generation illustrate ☞**exponential growth**☜.

Demographers would put the rate for doubling every generation at 4.2 children per woman. Why? One reason is that not all of those children will themselves have children. Certainly, this is true of the boys, who constitute roughly half the population of child-bearing age. So, 4.2 per woman is 2.1 per person.

The phrase "exponential growth" appears often in the press, almost always in reference to a quantity growing rapidly, e.g., a population of insects or a national debt. The news media use synonyms for "exponential growth" to convey drama: skyrocketing, explosive, rampant, mushrooming, and so on.

The characteristic of all exponential growth is successive doublings, each taking the same time as the previous doubling. For example, at an annual compounding rate of 6%, it takes about 12 years for an investment to double. An investment starting at 1000 USD would grow to 2000 USD in 12 years, then to 4000 USD at the 24-year point, and to 8000 USD at the 36-year point. All forms of exponential growth work like this, whether they be about money, bacteria, growing vegetation, or anything else. They differ only in *how long* it takes for the doubling to occur. The ☞**doubling time**☜ is the parameter used to describe exponential growth.

Exponential decay is the same. Imagine an abandoned bank account with 1000 USD, where the bank takes a "service fee" of only 4% per year. At this rate, it takes the bank 18 years to extract half the account balance. So, an account with 1000 USD in 2000 will have only 500 USD by 2018, then 250 USD by 2036, then 125 USD by 2054, and so on. By analogy to the *doubling time* of exponential growth, exponential decay is characterized by a ☞**halving time**☜.

Some people point to the world's human population (Figure 13.5) as a compelling example of "skyrocketing," that is, exponential growth. However, a careful examination reveals a more nuanced story. After all, even skyrockets run out of fuel at some point.

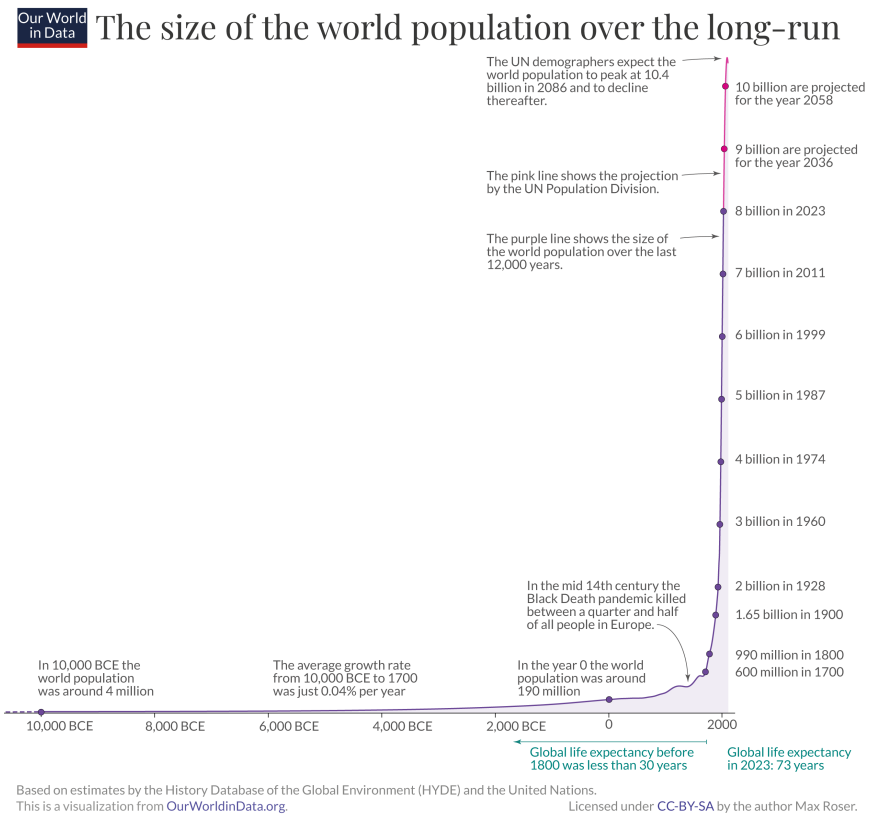


Figure 13.5.: Source: Our World in Data

World population data

To evaluate patterns like that shown in Figure 13.5, it pays to drop the dramatic descriptions and carefully examine what exponential growth is and the mathematical assumptions that underlie it. Knowing these assumptions lets us judge to what extent those assumptions are justifiable and when they are not.

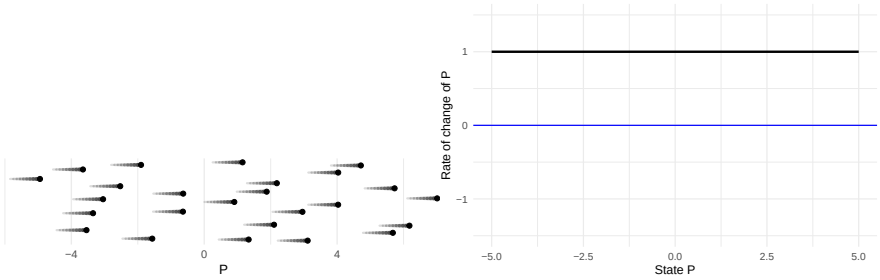
We can start with dynamical systems that are, mathematically, extremely simple. What do we mean by "simple?" Two things: 1) the state space consists of just a single quantity, and 2) the dynamical rule is free from complexity. In honor of Figure 13.5, we will call the state quantity P. (Of course, any other name would do: x, pop, S for state, and so on.)

The simplest function in our core modeling collection is flat(), where the output is always 1 regardless of the input. We will start from there and gradually add complexity.

Figure 13.6 shows the dynamics of the flat() rule in three different ways. On top (panel (a)) is the flow field. In principle, since the state space is

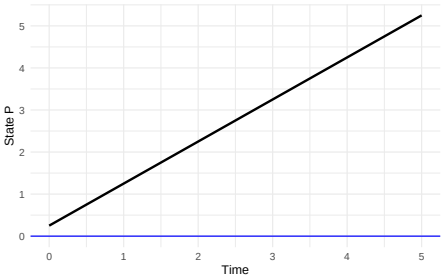
one-dimensional, all the flow arrows should be on the number line. To avoid the arrows overlapping, however, we have spread them out at random vertically. Panel (b) shows the dynamical rule, namely, the function flat(P). Panel (c) shows how the state P changes with time. (The name given to how something changes with time is 👉**time series**👈.)

Keep in mind the substantial difference between the dynamical rule and the corresponding pattern of how P changes with time, that is, the time series. The dynamical rule is a function that takes P as input and produces the rate of change of P as output. The time series is also a function, but instead of taking P as input, it takes *time* (t) as the input. Moreover, in the time series, the output of the function is P itself, not the rate of change in P.



(a) Flow field

(b) Dynamical rule



(c) State vs time

Figure 13.6.: Flow field when dynamical rule is flat().

The flow arrows tell a simple story. The flow is the same for every value of P: steady, rightward movement. Tracking along the flow would give us P as a function of *time*, that is, the trajectory (panel (c)). The dynamical rule (panel (b)) corresponds with the flow: the rate of change of P with respect to time is constant. Of course, this is also what the trajectory shows: the rate of change of P with respect to time is constant.

A good name for the patterns in Figure 13.6 is 👉**steady growth**👈. A domestic example of steady growth is the accumulation of water in a bathtub as it flows in at a steady rate from the faucet.

Following the strategy of gradually adding complexity to the dynamical rule, consider the second simplest of the core modeling functions: same(). Figure 13.7 shows the dynamics. Notice that the flow arrows (Figure 13.7(a)) differ markedly in length. For positive P, the larger P is, the longer the right-pointing flow arrow. (For negative P, the more negative the larger the left-pointing flow arrow.) The dynamical rule, same(), is straightforward (panel (b). However, the consequences of the dynamical rule are not at all steady. The state grows at an ever-increasing rate; the larger P is, the faster P changes with respect to time.

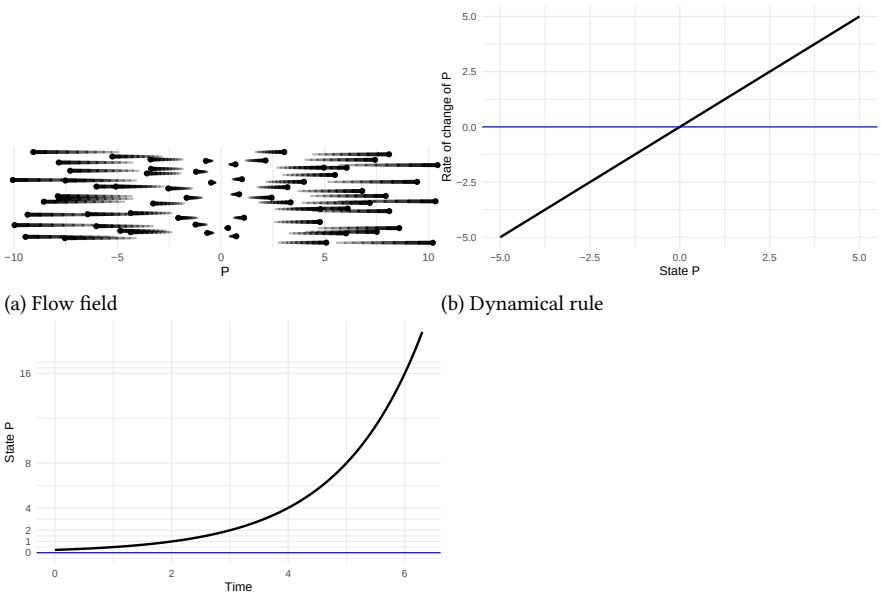


Figure 13.7.: Flow field when dynamical rule is same().

(a) Flow field

(b) Dynamical rule

(c) State vs time

Figure 13.7 shows *exponential growth*. It stems from using same() as the dynamical rule. With same() as the *dynamics*, the state itself follows *exponential* growth.

It is important to point out a distinctive feature of exponential growth, the same feature that we see in the exp2(t) function. Referring to the time series (panel (c) in Figure 13.7), we see the doubling phenomenon. For instance, track P as it increases from $P = 0.5$ to $P = 1$, a change that occurs in one time unit. Similarly, to grow from $P = 1$ to $P = 2$ takes one time unit. Same for the growth from $P = 2$ to $P = 4$: one time unit. And so on.

It is always the same: exponential growth corresponds to the state doubling every unit of time. In Figure 13.7, we arranged things so that the doubling

time is 1. However, it is still exponential growth even when the doubling time is different. For instance, under ideal conditions, a bacterial population doubles every 20 minutes.

Doubling every 20 minutes means that there are three doublings in one hour. Starting with a population of p, the three doublings produce, successively, at intervals of 20 minutes, population 2p, 4p, and, at the end of the hour, 8p; the population increases 8-fold in one hour. In two hours, the 8-fold increase will repeat, giving a 64-fold increase over the two hours! When the dynamics are same(), the 64-fold increase will occur every two hours.

The pattern of a 64-fold increase every two hours will produce unbelievable growth, as indicated in Table 13.1.

Table 13.1.: Bacterial growth under the hypothesis that the population doubles every 20 minutes.

time	population	bacterial mass
Midnight	1	roughly 1e-12 gm or 1 picogram
2 am	64	
4 am	64^2, or 4096, that is, 4.096e3	
6 am	64^3, or 262,144, that is, 2.621e5	
8 am	64^4, or 16,777,216, that is, 1.678e7	
10 am	64^5, or 1073741824, that is, 1.074e9	
Noon	64^6, or 68,719,476,736, that is 6.872e10	roughly 0.007 gm
...	...	... changing units ...
6 pm	64^9, or 1.801e16	roughly 18 kg
8 pm	64^{10}, or 1.153e18	roughly 1100 kg
...	...	...
Next Midnight	64^{12}, or 4.722e11	roughly 47,000 tons
2 am	> 64^{13}, or 3.022e23	roughly 300,000 tons

According to Table 13.1, after a little more than a day, the bacterial mass will be larger than its human host. Unbelievable!

A good rule of thumb is not to believe the unbelievable without challenging it. In reality, of course, the mass of bacteria reaches a steady state estimated at 300 gm.

The reason for the striking divergence between theory and reality is that the theory is wrong. Steady() dynamics will indeed produce exponential

growth; that part of the mathematics is correct. However, the same() dynamical rule is not an adequate model.

The reality is that the bacterial mass reaches a steady state, so we know that the dynamical rule must give zero growth at the steady-state mass. The simple dynamical rule same() does not do this. The model is wrong—like all models!—but same() is wrong in a way that fundamentally misstates the pattern of bacterial growth.

In Figure 13.8, we compare the exponential growth dynamics (black) to a more realistic pattern that produces exponential growth when the bacterial mass is small, but brings the growth to zero at the steady state.

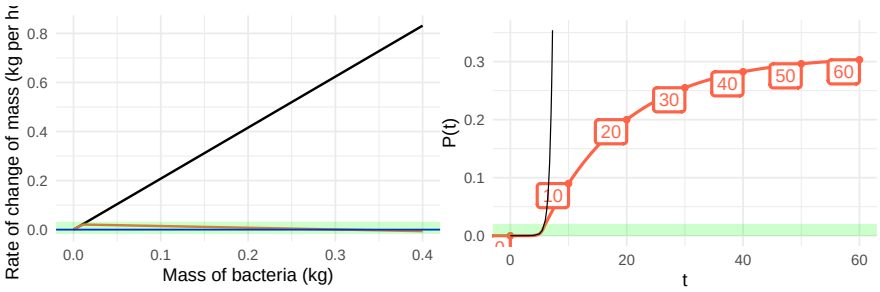


Figure 13.8.: Dynamics of bacterial growth. Black is the same(P) model, red is a more realistic model.

(a) Dynamics (vertical axis is mass growth rate) (b) Time series (vertical axis is mass)

In steady growth, as t increases by one unit, the value of P increases by a fixed quantity. So, in Figure 13.6(c), the value of P increases by the same amount when t goes from 0 to 1 as it does when t goes from 1 to 2 or from 2 to 3.

14. Rates of change

Chapter 4 introduced the distinction between a *rate* and a *rate of change*. A rate is one quantity divided by another. A rate of change is also a rate. But it is a special kind of rate.

The word "special" when talking about rates of change is relevant in two different senses. One definition is "different from usual." Indeed if a rate is the usual thing, then a rate of change is different from usual because the quantities that form the *ratio* are always *changes* in a quantity. And not just changes, but **simultaneous changes**. By photographic analogy, you can measure a rate by looking at a snapshot: read off the instantaneous values of the two quantities of interest, then divide one by the other. In contrast, measuring a *rate of change* requires a movie: a *series* of instantaneous snapshots. The minimum length of the movie is small, just two frames. You measure the two quantities in one of the frames, then in the other. Take the change in each quantity between the two frames, then form a ratio of those *changes*.

Another meaning of "special" is captured by synonyms: exceptional, extraordinary, unique, important, significant, rare. Although the distinction between a *rate* and a *rate of change* may seem trivial at first glance, the invention of the concept (1600's) and the gradual realization of the many important applications (starting about 1700 and speeding up over the next century and more) shaped many aspects of the modern world. This is most evident in science, technology, and engineering and the reason that a large fraction of STEM students are *required* to study rates of change and learn to do calculations with them. But it is just as important in economic matters, whether it be theories of how markets and firms work, or business decisions about prices and production.

Even if you have no interest in science, business, or economics, rates of change almost certainly apply to matters of importance to you. Knowing about rates of change, and how they are special, provides a major tool for understanding and making decisions about important matters. Later in this

Vocabulary Review

- analysis (3)
- data (2)
- functions (6)
- mean (8)
- models (1)
- operationalization (2)
- per (3)
- quantity (3)
- rate (4)
- rate of change (4)
- state (13)
- total change (3)
- tradeoffs (10)
- value (2)
- velocity (13)
- with respect to (3)
- worth (3)

New

- calculus
- continuity
- derivative
- differentiation
- diminishing marginal returns
- first derivative
- margin
- second derivative

chapter, we'll illustrate using death and taxes. First, though, let's motivate the how and why of rates of change.

14.1. Taxes on the margins

1913 US Federal tax brackets



Income taxes in the US are calculated according to "brackets." Each bracket specifies the tax *rate*, that is the fraction of income that is to be paid as taxes. Table 14.1 gives the brackets for taxes in 1913, the first year after the 16th Amendment to the US Constitution that empowered Congress to impose income tax.

Table 14.1.: 1913 US Federal tax brackets. Source: The 1913 1040 form

Income	Tax rate
$0	0%
$3000	1%
$20,000	2%
$50,000	3%
$75,000	4%
$100,000	5%
$250,000	6%
$500,000	7%

Care must be taken in interpreting Table 14.1. Despite appearances, it does not say, for instance, that someone earning $3000 must pay 1% (that is, $30) in taxes. The correct interpretation is that the $3K earner pays zero tax. But on any incremental income amount *above* $3K, the tax rate will be 1%. For instance, imagine a taxpayer who earned $4000. They will pay 0% on the first $3000 and 1% on the next $1000, so a $10 payment in total.

Using a standard cost-of-living conversion, $4000 in 1913 was equivalent to about $140,000 in the mid-2020s. The $10 payment in 1913 corresponds to about $44 in 2020s money.

In 1913, only about one-percent of households had an income of $3000 or greater, so taxpayers were a wealthy group of people.

Figure 14.1 shows lines 7 and 8 from the tax form for 1913. The form is blank, but we have filled in the numbers for a person earning the very large (in 1913) income of $90,000 per year.

7. TAXABLE INCOME on which the normal tax of 1 per cent is to be calculated. (See Instruction 3) .					$		87	000	00
8. When the net income shown above on line 3 exceeds $20,000, the additional tax thereon must be calculated as per schedule below:											
					INCOME.				TAX.		
1	per cent on amount over $20,000 and not exceeding $50,000 . .				$	30	000	00	$		300	00
2	"	"	50,000	"	" 75,000 . .		25	000	00			500	00
3	"	"	75,000	"	" 100,000 . .		15	000	00			450	00
4	"	"	100,000	"	" 250,000 . .								
5	"	"	250,000	"	" 500,000 . .								
6	"	"	500,000										
					Total additional or super tax				$		1	250	00
					Total normal tax (1 per cent of amount entered on line 7) . .				$			870	00
					Total tax liability				$		2	120	00

Figure 14.1.: Part of the 1913 US income-tax form. It has been filled in as appropriate for a total income of $90,000.

Figure 14.1 is pretty confusing. There are a lot of numbers filled in. Ironically, even though the income is $90,000, that number does not appear on the form!

The 2024 tax table simplifies the calculation by giving a table. You need only look up your income. Figure 14.2 shows the appropriate entry from page 9 (of 11) of the closely-spaced pages. Less arithmetic, but still the income of $90,000 doesn't appear.

A tax rate, as you know, is the fraction of income owed as tax. The tax table (Figure 14.2) does not indicate the rate, just the tax owed.

The quantities presented in the 1913 form in Figure 14.1 are not actually tax rates, although they look like it. In the form, they don't even give a name to those quantities. The correct name is ☞**marginal tax rates**☜. A marginal tax rate is a *rate of change*; it compares a small change in income to the corresponding change in taxes owed. The tax policy expressed on line 8 of the 1913 form is a function: for any level of income, it tells the marginal tax rate for that income.

The advantage of specifying taxes in terms of marginal rates is two-fold. First, just a handful of numbers can convey the entirety of the matter. This is much easier to make sense of than the 11-pages of numbers given in the modern tax tables.

Figure 14.2.: The format used today for a

75,000

75,000	75,050	11,559
75,050	75,100	11,570
75,100	75,150	11,581
75,150	75,200	11,592
75,200	75,250	11,603
75,250	75,300	11,614
75,300	75,350	11,625
75,350	75,400	11,636
75,400	75,450	11,647
75,450	75,500	11,658
75,500	75,550	11,669
75,550	75,600	11,680
75,600	75,650	11,691
75,650	75,700	11,702
75,700	75,750	11,713
75,750	75,800	11,724
75,800	75,850	11,735
75,850	75,900	11,746
75,900	75,950	11,757
75,950	76,000	11,768

The second advantage is one of fairness, avoiding sharp breaks in the amount of taxes owed. Income $3000, taxes (in 1913) zero. Income $3001, taxes (in 1913) one cent, at least if marginal rates are being used. If the numbers gave overall rate, if it said that people with incomes of $3001 have to pay a tax rate of 1%, their taxes owed would be $30.01. A one-dollar increase in income leads to a $30.01 increase in tax owed. Stated another way, the take home earnings of the person with a gross income of $3000 is $3000. The take-home income for a gross income of $3001 will be $2970.99. Not very fair.

Author Charles Dickens expressed his concern about a lack of such fairness in the Victorian-era laws about debt. In *David Copperfield*, Mr. Micawber famously stated:

> "*Annual income twenty pounds, annual expenditure nineteen pounds nineteen and six, result happiness. Annual income twenty pounds, annual expenditure twenty pounds ought and six, result misery.*"

A modern reader, used to decimal currency, may find this hard to follow. The difference between "nineteen pounds nineteen (shillings) and six (pence)" and twenty pounds ought and six amounts to 1 shilling out of a total of 400 shllings. That is, one-quarter of a percent in income separates happiness from misery.

The overall principle here is that small differences in input should lead to small differences in output. In mathematical language, this is a principle called ☞**continuity**☜. In terms of the graph of a function, continuity means you can trace the function with a pencil without having to lift it off the table.

Having dealt with fairness in taxes, it is time to turn to fairness in deaths. First, however, we look at the meaning of "marginal."

14.2. On the margin

The word ☞**margin**☜ comes from the Latin *margo*, meaning "edge." On a printed book, the margins are at the edge of the text. "Marginal" is an adjective meaning situated on the edge of something else. When we talk about *marginal income*, the thing with an edge is the income itself. For instance, for the hypothetical person in 1913 with a $3000 income, the edge is at $3000. The *marginal income* sits at the edge; it is a small increment of income, say $1. The marginal tax rate is the rate that applies to that small increment, but not necessarily to the part of the income on the other side of the edge, the part below $3000.

In exploring systems whose state changes in time in Chapter 13, we considered rates of change of the state quantities. In particular, we worked with the concept of an *instantaneous* rate of change of state, the rate of change that applies for just an instant as the trajectory passes through each state in turn.

Marginal rate conveys the same sense as instantaneous, both are concerned with examining a tiny change on the edge of where you are. Of course, "instant" applies to time, while "marginal" applies to other kinds of quantities, for example income or expenditure or tax rate.

Economists often speak of ☞**diminishing marginal returns**☜. Consider, for instance, the situation of buying a car. They buyer is spending money and expects to get benefits in return: convenience and transportation services. For this, it suffices to buy the base model of the car. Many people go further, adding a marginal expense to buy an "options package," with, say, heated seats and cruise control. In response to this marginal expense, the buyer anticipates getting a somewhat better car, that is, marginal benefits. But these marginal benefits are typically small in comparison to the benefits provided by the base model of the car. After all, by far the most important benefit of the car is providing transportaion services. When the ratio of the marginal benefits to the marginal expense is smaller than the benefit/cost ratio of the base model, one says that buying car options has *diminishing marginal returns*.

The next section looks at diminishing marginal returns in the context of setting health policy.

14.3. Competitive allocation

Many important matters in public and private decision making involve setting priorities. Compelling examples are provided by life-and-death issues. Consider this made-up setting which can illustrate important decision-making principles and is still somewhat realistic:

> There are two innovations in public health care being considered. We will imagine them to be pre-natal clinic and improved cancer treatment, but give them very short names—"natal" and "cancer"—as a reminder that this is a made-up example.

> The potential benefits from each innovation play out over the years. The Natal program will cost $5-billion, the cancer program $10-billion. Which one should the investment go to?

Different people can have different, reasonable views. Families of cancer victims will understandably place emphasis on the cancer initiative. People whose interests and lives center on young children might lean toward the natal initiative. Still others will advocate a compromise: split the $10-billion between both initiatives. And, inevitably, there will be another group who ask, "Why limit the budget to $10-billion? Fund both!"

The funding dilemma is an example of a *trade-off*. (Chapter 10) The new thing in this situation is that both objective functions—lives saved by spending on "natal" versus lives saved by "cancer" spending—are both quantified with the same dimension.

But see the discussion of QALYs below.

The bald debate between advocates of spending on "natal" versus "cancer" does not point to a resolution. But since the objectives are measured in the same dimension, we do not need to think carefully about the "shadow costs" introduced in Chapter 10. Instead, it's likely that the dualing advocates have a common goal: maximize the efficiency of the spending.

Consider briefly what should be the operationalization of the goals of the optimization problem. Perhaps it is obvious that a meaningful operationalization of the positive impact of each initiative would be the *number of lives saved*. Laudable though this be, this operationalization does not capture some important considerations. For instance, does deferring death for one day or one month count equally to deferring death for one year? "Number of lives saved" has dimension [1]. But [T] is also important.

A more nuanced operationalization is *quality adjusted life years* (QALYs). The "years" gives QALYs dimension [T].We will measure benefits in QALYs, but the same quantitative principles apply to other measures of benefit. And, since the reader is probably unfamiliar with QALYs, please accept for the purposes of discussion that QALYs are a better match to many people's intuitive values than "lives saved." Here are the (made-up) facts:

> The Natal program will save 500,000 QALYs each year. The Cancer program will save 750,000 QALYs yearly.

Here are two quantitative arguments that lead to different outcomes:

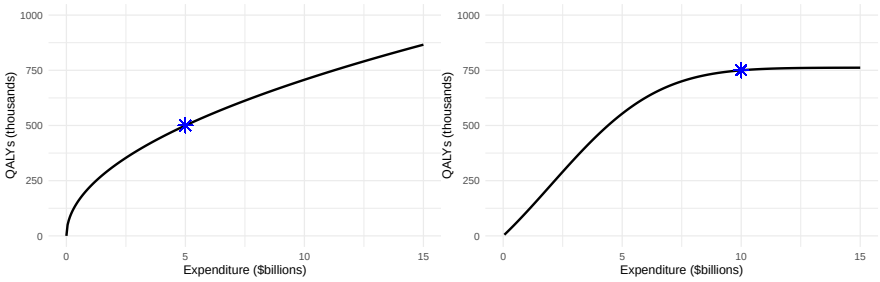
a. The Cancer program saves more QALYs, so fund Cancer.

b. The Natal program costs $10,000 per QALY, while the Cancer program costs more: about $13,000 per QALY. Natal is a more economical choice.

Previous examples in this book have pointed out situations where *rates*, as in dollars-per-QALY, are more informative that single quantities. But it does not seem obviously wrong to favor the save-more-QALYs (a) argument over the save more dollars-per-QALY (b).

Arguments (a) and (b)—amount vs rate—are not the only way to compare the programs. Let's consider one more useful quantitative perspective: *rate of change*.

In order to use rates of change, we need more information than provided above. We have to imagine that each program could be launched with a lower level of funding or perhaps even a higher level. Each potential level of funding corresponds to a QALY savings. Figure 14.3 shows two hypothetical functions: QALYs saved as a function of expenditure for the natal and cancer programs separately.



(a) Natal initiative

(b) Cancer initiative

Figure 14.3.: QALYs saved as a function of expenditure on each health initiative. The "facts" presented above are represented by the *. Plausible functions are sketched in for each initiative.

At a glance, the Natal and Cancer functions sketched in Figure 14.3 are very similar. Nevertheless, the two functions have different implications for how to allocate the funds available. An appropriate analysis using *rates of change* will let us see this.

To start, we zoom in more closely to the domain near to the proposed funding level for each initiative. Figure 14.4 shows the competitive situation. In both graphs, the vertical scale covers 4000 QALYs; the horizontal scale covers $100 million.

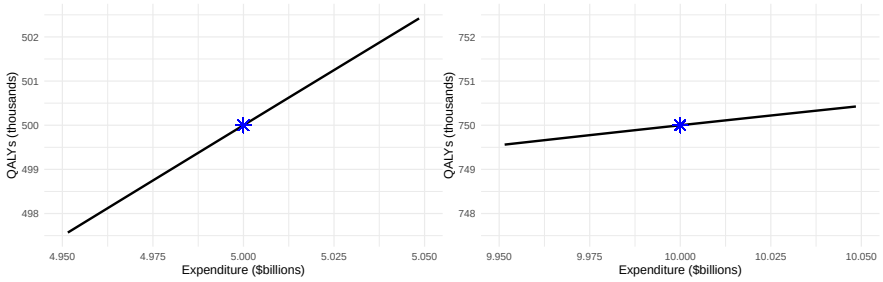


Figure 14.4.: Zooming in on the *'s in Figure 14.3.

(a) Natal (b) Cancer

The rate of change of QALYs with respect to expenditure corresponds to the slope of the curve. Figure 14.3 shows that the slopes for the two programs are very different at the proposed expenditure levels. Figure 14.4 makes it easier to calculate the slope. For the Natal program, an increase in expenditure of $50 million corresponds to an increase in QALYs of 2,400. In contrast, for the Cancer program, a $50 million increase corresponds to only about 400 QALYs. The rate of change for Natal is $\frac{2400}{50} = 48$ QALYs per $million. For Cancer, the rate of change is $\frac{400}{50} = 8$ QALYs per $million.

The important implication here is that a million dollars is worth more QALYs if spent on Natal than on Cancer. This suggests that we should transfer money from Cancer to Natal. But how much?

Suppose we moved $5 billion from Cancer to Natal. The picture will look like Figure 14.5. The red stars mark the new spending proposal, re-allocating $5 billion from Cancer to Natal. The new proposal buys 1260 QALYs, the original proposal buys 1250. In reality we would never have such detailed information to distinguish a model output of 1260 from 1250; they are effectively the same. The implication: we are indifferent between the two proposals.

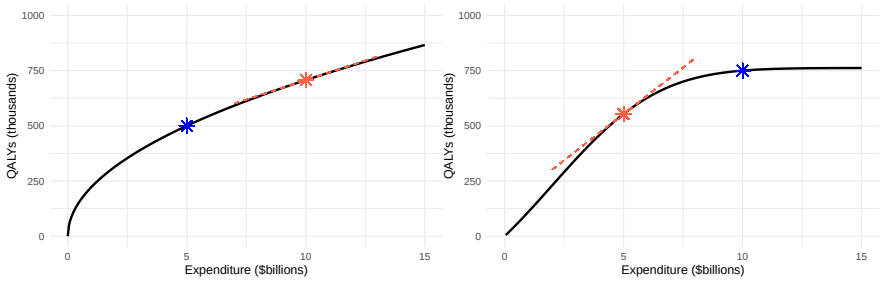


Figure 14.5.: QALYs gained versus expenditure

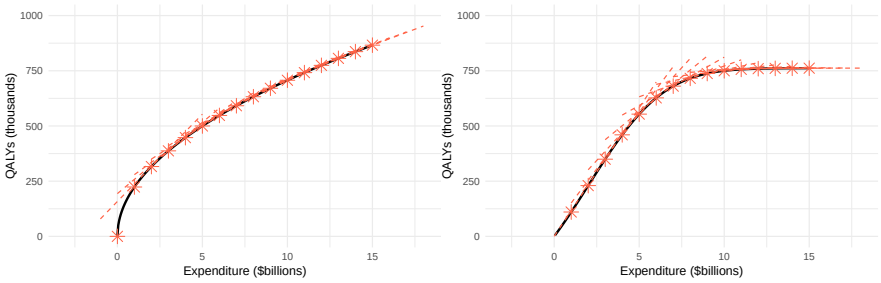
(a) Natal (b) Cancer

Does indifference between the two spending proposals—$5 billion for natal

and $10 billion for cancer, versus $10 billion for natal and $5 billion for cancer—mean that it doesn't really matter how we split the money between the two programs? Before rashly jumping to this conclusion, we should do a bit more analysis.

In Figure 14.5 we have annotated the graph with the rate of change (QALYs per million dollars) at the red proposal. This rate of change is, of course, the slope of the function at the selected expenditure level. We would need to do a little calculation to find the numerical value for the rate of change, but we can visualize it by drawing a line segment that has the same slope as the function at the selected expenditure level. The slope is steeper for Cancer than Natal, suggesting that we ought to shift money back to Cancer from Natal. This may seem futile, switching money back and forth, but that's because we are shifting a lot of money: $5 billion. Suppose we shift less.

Whatever spending level we choose, there will be a QALY level marked by the star *and* a rate of change marked by the slope of the line segment. Figure 14.6 plots some of these line segments and stars at many different levels of expenditure.



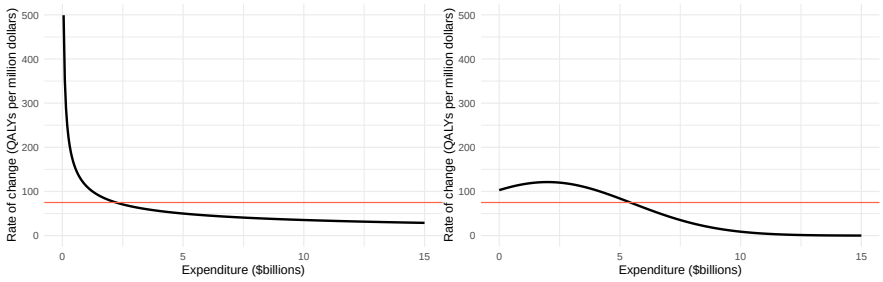
(a) Natal

(b) Cancer

Figure 14.6.: QALYs saved and rate of change at many expenditure values. The graphs are too crowded to read clearly.

Figure 14.6 is too crowded to serve its main purpose: showing the rates of change at many different values of expenditure. We need to simplify things.

We can accomplish this by changing the format for representing rates of change from slope liness into a quantity that can be plotted. The units for the values will be QALYs per million dollars, a different dimension than shown by the vertical axis in Figure 14.6. So we need to make a new graphics frame to show QALYs per million dollars. We do that in Figure 14.7.



(a) Natal (b) Cancer

Figure 14.7.: The rate of change, QALYs per million dollars, as a function of expenditure level.

Keep in mind the purpose of the analysis: to find out how to allocate the total of $15 billion to Natal and Cancer to make the QALYs saved as large as possible. The information in Figure 14.7 is not yet in the form that lets us see the answer we seek. We will have to do some work in order to get an answer that serves our purpose.

Whenever our expenditure policy corresponds to different rates of change (QALYs per million dollars) for the two policies, we would want to shift money from the program with the lower rate of change to the program with the higher rate of change. When such shifting brings the expenditures to levels where the two rate of changes are equal, no further shifting is called for: the earlier imbalance has been corrected.

To illustrate, we have annotated the graphs with two red lines, both drawn at the level of 75 QALYs per million dollars. For Natal, this rate of change occurs at an expenditure level of about $2 billion. For Cancer, 75 QALYs per million dollars occurs at an expenditure level of about $5.5 billion. Those two values of expenditure are in balance in a very important sense: at those values, there is no reason to swap funds between Natal and Cancer.

Even so, at the $2 billion (Natal) and $6 billion (Cancer) levels we are spending only (!) eight billion dollars. But there is $15 billion available. To find out where the spending is in balance at a total level of $15 billion, we need to do a little work. This is purely mathematical work, so ideally suited to delegate to a computer. But let's follow the process.

At 75 QALYs per million dollars, we spent only $8 billion. If we spend more, we will be at an operating point with fewer QALYs per million dollars. That sounds bad, but remember that the QALYs per million dollars is a *rate of change*. So long as the rate of change is *positive*, spending more produces more QALYs overall.

In Chapter 13 explored systems whose state changes in time. There we spoke of the *dynamical function*, that is, the function that gives the *instantaneous* rate of change (with respect to time) of the state quantities.

Suppose we were to increase spending to bring the system into balance at a level of 50 QALYs per million dollars. This happens at an expenditure level of $6 billion (Natal) and $7 billion (Cancer), so $13 billion in total. Still too little spending, but closer to the target of $15 billion. Evidently, there is more mathematical work to do: keep trying different levels of QALYs per million dollars until we hit upon one that brings us to a total expenditure of $15 billion. Doing so produces the information we need: At a marginal benefit of about 40 QALYs per million dollars, total expenditures are $15 billion, split $7.8 billion for Natal and $7.2 billion for Cancer. Outcome from the spending: 1,313,000 QALYs.

Undeniably, reaching the result involves an intricate argument in which a neophyte is likely to get frustrated and make errors. First, we imagined there to be a production function that relates expenditures to QALYs saved, not just the two pairs given originally as facts.

- Natal: $5 billion produces 500,000 QALYs
- Cancer: $10 billion produces 750,000 QALYs

Next, we introduced a new perspective: A spending level is in balance only when the *rate of change* of QALYs with respect to dollars is the same for both Natal and Cancer.

We then looked at the two production functions, calculating their rates of change at many expenditure input values. Finally, we found expenditure levels totaling $15 billion split between Natal and Cancer in a way that produced an equal marginal benefit for each.

Without spending a single dollar above the $15 billion total expenditure, we have re-allocated spending in a way that increased output from 1,250,000 QALYs to 1,313,000 QALYs. That is an increase of 63,000 QALYs with no extra costs. 63,000 QALYs corresponds to saving the lives of 1000 18-year olds, giving each of them 63 extra years of full-quality life. That seems worthwhile d.

Of course, the QALYs don't have to go to 18-year olds. 63,000 QALYs might also correspond to saving the lives of 10,000 75-year olds, or 100,000 80-year olds

Everyone understands why prospective medical professionals—doctors and nurses—go through difficult years of schooling. It helps them to save lives and increase the quality of life. It is the same thing for quantitative reasoning: a few years of difficult schooling to master an approach to analysis that can save lives and increase the quality of those lives.

It is understandable if the reader is not completely convinced that the analysis described above is correct. Partly, that's because many of the concepts are new. And there are some difficult to believe aspects of the argument. For instance, In determining the best allocation of expenditures, we looked only at marginal benefits, not the total change in QALYs. In Chapter 15 we will see why the marginal-benefit function has latent in it the critical information about the total change. And in Chapter 16 we will introduce a few mathematical operations, each of which is pretty simple, that can be used to build processes such as the one by which we figured out the optimal allocation of spending.

14.4. Differentiation and the derivative

The *marginal*—or, equivalently, *instantaneous*—rate-of-change functions plotted in Figure 14.7 for Natal and Cancer are intimately related to the functions presented in Figure 14.3. Indeed, all the information needed to calculate the Figure 14.7 is already present in Figure 14.3. One can fairly say that Figure 14.7 is ☞**derived from**☜ Figure 14.3.

Let's explore how the derivation is done. For the demonstration, we will use a generic name for a function, say $f()$. The function $f()$ could be just about anything: a core modeling function, a linear combination, product, or pipelining of multiple functions. Data could have been involved in composing the function, for example using linear *regression* or *machine learning*.

It will be handy to have a name for the marginal-change function that we derive from $f()$. There are all sorts of possibilities. Isaac Newton (1643-1727) would have called $f()$ the *fluent* and the derived function the *fluxion*, although nobody uses that nomenclature now. Newton's contemporary, Wilhelm Leibniz (1646-1716) described the derived function as "the *ratio of the differentials*." For him, a ☞**differential**☜ was just a small *change* in a quantity. The *ratio* involves two quantity: take the small change in one and divide it by the corresponding change in the other. We no longer use the name "ratio of differentials," but it is easy to trace the evolutionary link to "rate of change."

Why "would have called?" The mathematical concept of "function" was introduced only some decades after Newton's death. Newton's terminology for what we now call a function was variously "a changing quantity" or "a curve."

The word that used universally today for the kind of function we have in mind, *derived* from a function $f()$, is rather pedestrian: the ☞**derivative**☜ of $f()$. The word used to name the *process of calculating* the derivative

of $f()$ or any other function is 👉**differentiation**👈. The field that has "differentiation" as a major object is called 👉**calculus**👈.

Despite the off-putting name "calculus," differentiation can be accomplished very simply. We start with a function $f()$ that has one argument. It doesn't matter what that argument is named, but having *some* name is helpful. We will name the argument t. By convention, t is often used to denote "time," but differentiation does not need to be about time. For example, in Figure 14.3 the argument—the input to the function—was "expenditure" of money.

The nature of a function is that it turns the input into another quantity, the output. We have been graphing functions for a while, but merely as a reminder, in a slice plot the input quantity is represented by position on the horizontal axis, while the output corresponds to position along the vertical axis. Just for illustration, Figure 14.8 shows a picture of a made-up $f(t)$.

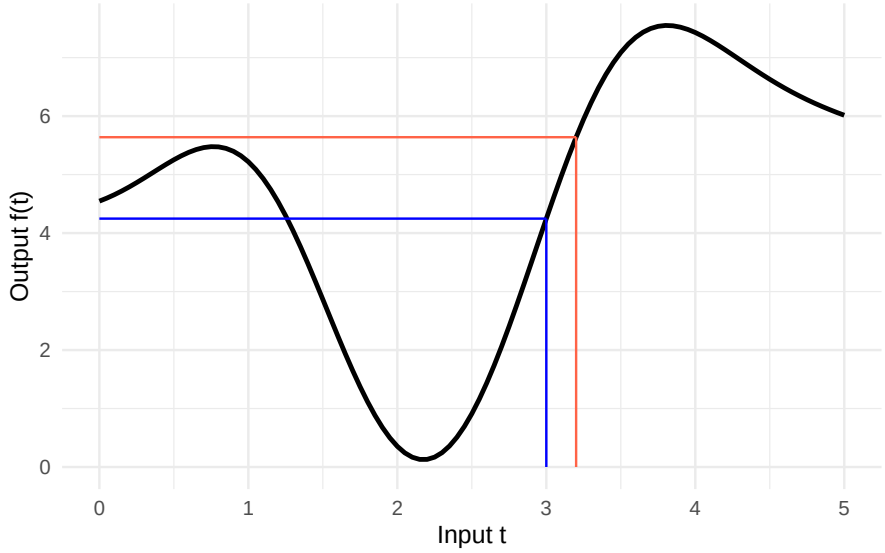


Figure 14.8.: Illustrating the rate of change of the output of the function as the inut changes from 3.0 (blue) to 3.2 (red).

Section 4.4 introduced the "rate of change" as the ratio of two changes. Figure 14.8 shows a function evaluated at two nearby input values. From this, we can calculate the rate of change of the output value with respect to the input value, that is, the change in output divided by the change of input. In Figure 14.8, read the change of input off the horizontal axis: 3.2 - 3 = 0.2. Read the change of output off the vertical axis: $f(3.2) - f(3) = 5.64 - 4.25 = 1.39$. The rate of change is the ratio of the two changes 1.39 / 0.2 = 6.96.

14.5. Instant? Instantaneous? Tiny input change?

In the previous example, we used input values of 3 and 3.2 as two inputs and calculated the rate of change as the corresponding difference in output—$f(3.2) - f(3.0)$—divided by the difference in input. In the language of *marginal change*, these inputs corresponding to looking at an increment of 0.2 from the "edge" 3.0.

The word "instant" and the phrase *instantaneous change* implicitly refer to *time*. This nomenclature reflects the history of development of these concepts, where the applications were to motion of objects over time. But application areas are much broader now, for instance, in the QALY/expenditure example the input is *spending level*. An "instant of spending" doesn't make literal sense; the real meaning is the "current spending level." A so-called "instantaneous" change of spending isn't meant to refer to time but to a change in spending that is tiny. How tiny? For spending at the level of a billion dollars, a change of 1 cent is certainly tiny. So is one dollar or even a thousand or a million dollars.

Is it fair to take a change of spending of one-million dollars as tiny, that is, "instantaneous"? As long as we are looking for *tiny*, why not set the change of spending to zero, as small as it can get.

A input change of zero throws a wrench into the mechanism for calculating rates of change. Looking to the example of Figure 14.8, in response to zero change in input the change in output would be $f(3) - f(3) = 0$. The rate of change over the instant would be

$$\frac{f(3) - f(3)}{3 - 3} = \frac{0}{0} = \text{???}$$

The straightforward resolution to this problem is not to throw the wrench in the first place. Instead of exactly zero duration, use a small but non-zero duration.

One perspective on small reflects our interest in looking at the rate of change as it varies for different inputs. Consider Figure 14.6: the slope of the function varies as the input changes. In the Natal vs Cancer allocation problem, Figure 14.7, the changing slope enabled us to find inputs that balanced the marginal benefits of spending between Natal and Cancer. If we are going to compare the slope of QALYs per million dollars at two

different spending points, say 3.1 million USD and 3.2 million USD, "small" means "smaller than 0.1 million USD."

A second perspective is that our choice of "small," so long as it is small enough, should not affect the result of the calculation. Or, more precisely, a change of input is "small" when making it even smaller would give the same result. If there were many people calculating a rate of change, using their own choices for "small," we would like them nevertheless to get the same answer!

14.6. Vanishingly small

Newton and Leibniz and the people who used their techniques for the next 200 years did not have modern computers available. Verifying, by trial and error, that a proposed h is small enough would have been tedious work. And the phrase "small enough" doesn't carry enough gravitas to honor a technique that was becoming more and more central to mathematics. Understandably, they sought a more dignified word, a word that would draw attention away from the need to search for a "small enough" change in input. The first ever calculus textbook, from 1694, had the title, "Analyse des Infiniment Petits," that is, "Analysis into the infinitely small." In a somewhat later book, Newton wrote and explanation which is only clear if you already understand it: "Those ultimate ratios … are not actually ratios of ultimate quantities, but limits … which they can approach so closely that their difference is less than any given quantity". The phrase "vanishingly small" for the change in input was used by proponents of calculus, and "evanescent" in one famous early critique (philosopher George Berkeley, 1734). Instead of "infinitely small," some preferred the word "infinitesimal." A famous image for the hard-to-grasp notion of something that stays even as it shrinks toward zero was provided by Charles Lutwidge Dodgson (1832-1898) who, under the pen-name Lewis Carroll, introduced the Cheshire Cat. The cat's smile corresponds to the rate of change. You cannot have a smile without a body surrounding it, just as you cannot have a rate of change of a function with a change in the input. In *Alice in Wonderland*, the Cat's smile remains even as the body disappears.

In calculus textbooks from the last 150 years, derivatives are defined as "limits" with a notation like

$$\lim_{h \to 0} \frac{f(x+h) - f(x)}{h}.$$

None of this need concern the person using calculus to create and interpret models. The practical test that `h` is small enough is readily implemented on the computer without any need for philosophical reflection.

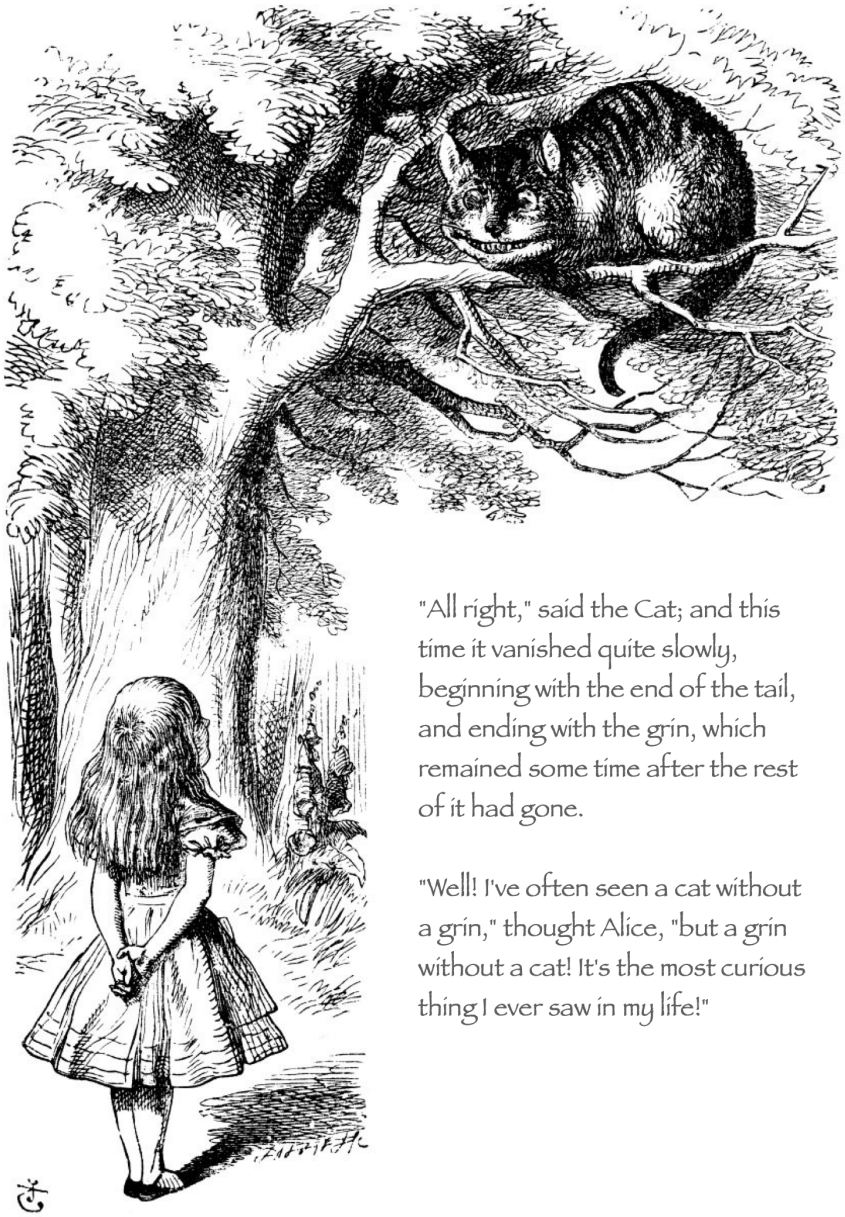


Figure 14.9.: The Cheshire Cat from *Alice in Wonderland.*

14.7. The derivative is a function

In Section 14.4 we calculated the rate of change of the function graphed in Figure 14.8 at the input value `t <- 3`. But we could have equally well performed the calculation for any other input value. The output of the calculation is a quantity.

When we can describe the algorithm for a calculation at any in a set of input values, we have a function. In other words, the derivative of a function is also a function. The new function takes the same inputs as the original. An analogy is the relationship between mother and child. Both are people. And children can become mothers with their own children, that is, grandchildren of the original mother.

To keep track of this generational ladder of functions, it helps to have a notation that shows the family relationship explicitly, such as a surname. We will use a function name as the surname, e.g., $f()$ or $osc()$ or rainfall(). To indicate a derivative of the named function, we will use the symbol ∂, which is usually pronounced "d." We need something more: an indication of which is the with-respect-to input. When a function has a single input, naming the with-respect-to input is not strictly necessary. Still, it can be a good habit for clarity.

Consider a function with two inputs, however, such as $f(x, t)$. $f()$ is the surname. The derivative *with respect to* x is written, $\partial_x f(x, t)$. Similarly, the derivative with respect to t is written $\partial_t f(x, t)$. Both $\partial_x f(x, t)$ and $\partial_t f(x, t)$ are descendents of the same parent function $f(x, t)$. But the two different derivatives are related in a manner analogous to half-siblings, sharing one mother but different fathers. Such children can have very different personalities, but are still closely related.

Remember that the derivative of a function is itself a function, so think about $\partial_x f()$ and $\partial_t f()$ as being names. This notation for names reminds us where the function came from, or, to use an older English grammar, *whence they were derived.*

$\partial_x f(x, t)$ and $\partial_t f(x, t)$, being functions themselves, they also can be differentiated with respect to either of their inputs. It would be long-winded to write $\partial_x \partial_x f()$ as the name of the function which is derived with respect to x from $\partial_x f()$. For concision, we write instead $\partial_{xx} f()$. Generically, such derivatives are called ☞**second derivatives**☜. $\partial_x f()$ is the "first derivative of $f()$ with respect to x," analogous to the child of $f()$, that is, the first generation of descendents. $\partial_{xx} f()$ is the grandchild of $f()$: the second

generation of descendents. $\partial_{xx} f()$ has siblings and cousins such as $\partial_{tt} f()$ and $\partial_{xt} f()$.

There is one everyday situation where second-derivatives can be felt viscerally. Suppose we know where an object (such as a car, plane, or bike) is at any time t. That is, position is a function of t, which we could call *pos(t)*. The velocity of the object is the derivative of position. The acceleration is the second derivative of position, or, if you prefer, the first derivative of velocity. You can feel acceleration, although we typically label the feeling as "force." Newton's *Second Law of Motion* famously states that force equals mass $\times$ acceleration: $F = ma$.

The second derivative can play an important role in many fields outside of physics or mathematics. There is rarely a reason, except in technical work, to consider the third or higher derivatives.

Notations for derivatives

The reader who has studied calculus will be familiar with other notations for derivatives. Naturally, the ones that appeared early in the history of calculus became standards, such as Leibniz's notation $\frac{dy}{dx}$ and Newton's dot notation $\dot{y}$, invented in the late 1600s. Unfortunately, these commonly used notations pre-date the modern concept of a function and nothing in the notation hints that the objects are *functions* rather than *quantities*. (The use of y, unadorned with the parentheses that are every function's birthright, stems from an algebraic convention learned by beginning students, such as $y = mx + b$, where we would write $y(x) \equiv mx + b$.)
Another standard calculus notation does use a function's name, as in f' for the first derivative of $f()$ with respect to its single input. Second derivatives are denoted f''
For functions with a single input, all of the various notations—$\frac{df}{dx}$, f', and $\dot{f}$—have the same meaning as $\partial_x f()$.

The "Two Cultures"

In 1959, C.P. Snow, a British writer and scientist, proposed a model of what he saw as an increasing split between the sciences and the humanities. He called this the "two cultures." Any observer of the contemporary academic world will be sympathetic to Snow's thesis that science and the humanities operate in mutual incomprehensibility.

> Increasingly, decisions in government policy and industry are driven by quantitative analysis of models and data. If such decisions, or support or opposition to such decisions, become incomprehensible even to highly educated people, we are on the verge of descending into conflict and irrationality.
> One of Snow's examples is particularly germane to the issue of *instantaneous change*. He wrote:
>
> > "*A good many times I have been present at gatherings of people who, by the standards of the traditional culture, are thought highly educated .... If I had asked—What do you mean by mass, or acceleration, which is the scientific equivalent of saying, Can you read?—not more than one in ten of the highly educated would have felt that I was speaking the same language.*"

Students who take a calculus course memorize the derivatives of a handful of families of functions: e.g. sin(), sqrt(), exp(), ln(), x^n, and so on. They learn *rules* for algebraically computing from this handful the derivatives of even more obscure functions. And they spend a lot of time calculating second-, third-, and high-order derivatives. These rules along with the algebraic framework for them, and the insistence on a mathematically precise definition of "small enough," make calculus, as it is commonly taught, a hard subject. And aside from future teachers of college-level math and physics, few graduates of a calculus course will use what they learned in their career. The important, useful ideas in calculus are that one function can be related to another function by differentiation, that the derived function has the same inputs as the original but an output that differs both in value and *dimension*, and that important problems of 👉**equilibrium**👈, as in the useful balance of funding between Natal and Cancer, are more easily thought of when the idea of differentiation can be applied.

In Chapter 16 we will see another important use for derivatives: performing mathematical tasks.

15. Accumulation

Here are a couple of definitions of "accumulation" from the Oxford Languages dictionary.

> *The acquisition or gradual gathering of something. Example: "the accumulation of wealth"*
>
> *A mass or quantity of something that has gradually gathered or been acquired. Example: "the accumulation of paperwork on her desk"*

The previous chapter focused on turning a function into another function that gives the instantaneous rate of change. That process is called *differentiation*. This chapter is about going the other way: starting out with a function that is the derivative of something, and figuring out what that something is. Consider these scenarios:

1. A beachcomber starts with an empty basket and walks north along a beach. Roughly once every five minutes, she sees a collectible shell. She picks it up, places the shell in her basket, and walks on. At the end of a long, two-hour excursion, how many shells has she accumulated?
2. An intern works at a large accounting firm during rush season. He gets paid $25/hour, increasing to $37.50 if he works "overtime," that is, more than 40 hours in a week. At the end of the week, he has worked 68 hours. What are his accumulated earnings for the week?
3. An epidemic spreads through a community. On the first day, only a couple of people catch the virus. But as more people become infected, the infection rate increases, reaching 20 new cases on the 7th day. The health department keeps a daily list of new cases. Overall, how many names accumulate during the first ten days?

Each of these three scenarios involves a quantity accumulating over time: shells, money, and people falling ill. And each involves one or more *rates of change* and a *duration*

Vocabulary Review

- calculus (14)
- data (2)
- derivative (14)
- differentiation (14)
- dynamical rule (13)
- euler framework (13)
- exponential growth (13)
- functions (6)
- margin (14)
- per (3)
- point (3)
- quantity (3)
- rate (4)
- rate of change (4)
- state (13)
- time series (13)
- value (2)
- with respect to (3)

New

- accumulation
- anti-differentiation
- incidence
- integration

1. One shell per 5 minutes over a two-hour walk.
2. $25 per hour, going up to the "time-and-a-half" rate of $37.50 per hour after 40 hours worked over 68 total hours.
3. Two new cases per day, increasing to a rate of 20 new cases per day at the end of the first seven days.

Solving the *shell-collection* problem is straightforward: *Multiply the rate times the duration*:

$$\frac{1}{5}\frac{\text{shell}}{\text{minute}} \times 2\,\text{hours} = \frac{2}{5}\,\text{shell} \times \frac{\text{hour}}{\text{minute}} = \frac{2}{5}\text{shell} \times \frac{60\ \text{minute}}{\text{minute}} = \frac{120}{5}\text{shell} = 24\,\text{shell}$$

The wages problem involves more bookkeeping. The first 40 working hours are paid at a rate of $25/hour. Multiply the rate times the duration,

$$40, \text{hours} \times \$25/\text{hour} = \$1000\ ,$$

giving $1000 for that part of the week. The remaining 68 - 40 = 28 hours are paid at $37.50/hour:

$$28, \text{hours} \times \$37.50/\text{hour} = \$1050.$$

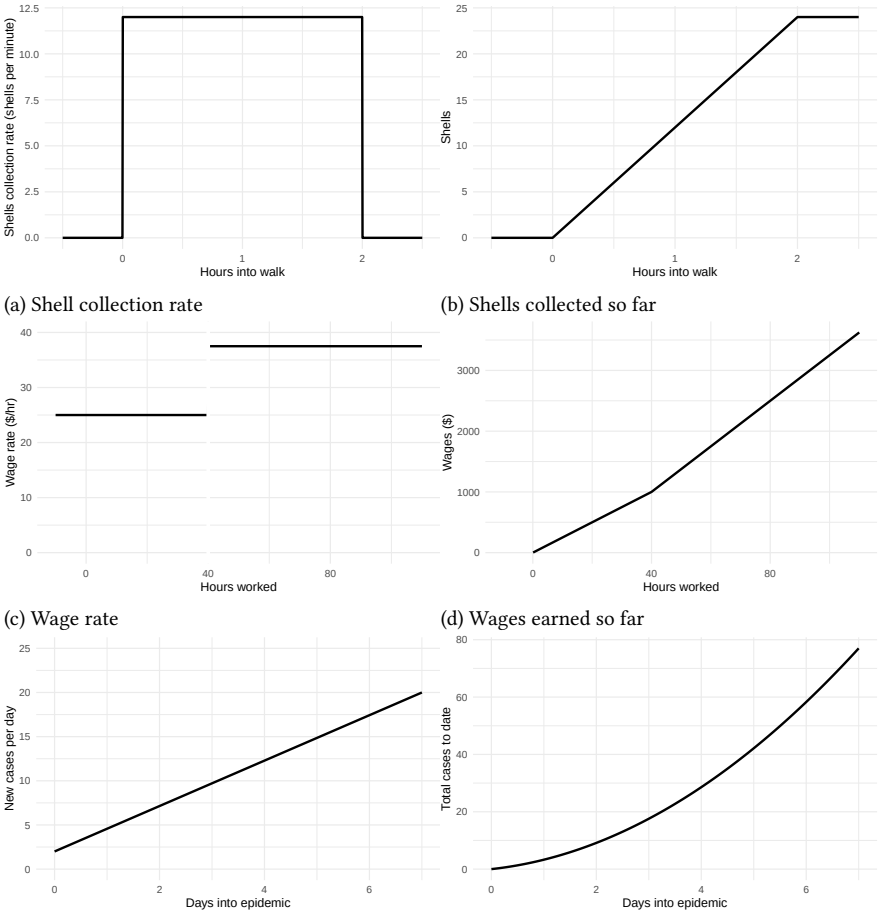
Adding these two components gives $1000 + $1050 = $2050 in overall earnings for the week.

The epidemic problem is more difficult because the infection rate (new cases per day) is changing during the week under consideration. This change in rate is also seen in the wages problem, where the wage rate changed from $25 to $37.50 during the period of interest.

Even though the three problems appear to differ in complexity, they can all be solved by the same mathematical method. We call this method "accumulation," but it is more widely known by the standard mathematical term "integration".

15.1. The rate-of-change framework

To understand why the three problems posed above can be solved by a single method, let's restate them in a common framework. For each, we'll depict the instantaneous rate of change (with respect to time) as a function of time. The output of the three functions is variously shells/hour, dollars/hour, and cases/day. The left column in Figure 15.1 shows the three rate-of-change functions.



(a) Shell collection rate

(b) Shells collected so far

(c) Wage rate

(d) Wages earned so far

(e) New cases rate

(f) Total cases to date

Figure 15.1.: The rate of change function for each of the three scenarios.

The right-hand column of Figure 15.1 shows accumulations of the rate-of-change functions in the left column. Calculating the accumulation from the rate-of-change function is a fully automated process called "anti-differentiation" that requires little or no creative input.

15.2. Example: Figuring taxes owed

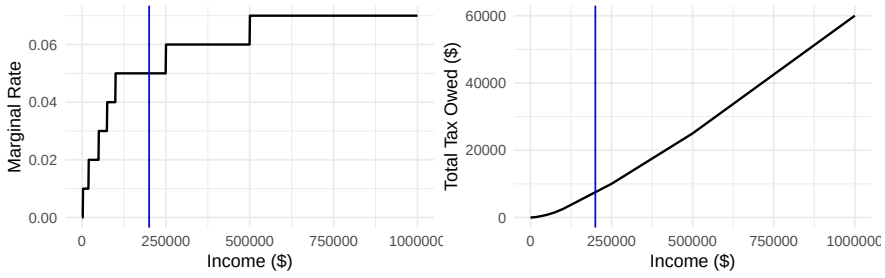
Section 14.1 discussed the calculation of income taxes *on the margin*. The tax law specified the *marginal* tax rate at each level of income. (See Table 14.1.)

As discussed in Chapter 14, *differentiation* is the name given to finding a marginal function from a "mother" function. The function so derived is called a *derivative*. These names point to a critically important habit of mind for quantitative reasoners: we think about relationships such as Table 14.1 as *functions*. Why is this important? Knowing that the relationship is a function reminds us of a set of operations that can be performed on it to put it in a more useful form for whatever purpose is at hand.

Consider the purpose originally served by the tax-bracket table: to inform members of Congress the exact policy they were imposing on taxpayers. The tax-bracket table does not serve another purpose very well: enabling taxpayers to easily figure out what they owe. The result is tax forms like that in Figure 14.1 which are potentially off-putting.

Quantitative reasoners who know about differentiation and derivatives have a powerful framework for simplifying the presentation of marginal tax-rate information to taxpayers. Since the *marginal* tax rate is a function, we can graph it. (Figure 15.2(a))

Figure 15.2.: Tax-rate functions



(a) Marginal rate

(b) Total rate

The two functions in Figure 15.2 convey the same information, but in different forms. The *marginal rate* function (Figure 15.2(a)) literally the Congressional policy. Using it, you can look up the *marginal* tax rate at any income. For example, a blue vertical line has been drawn at an income of $200,000. The marginal tax rate at that income is exactly 5%. Even so, the tax-payer does *not* owe $200,000 × 5% = $10,000. The 5% refers to the additional tax owed on a small incremental increase in income.

The *total tax owed* function (Figure 15.2(b)) is set up to suit people calculating how much to pay. The output is the amount of tax owed. At the blue line for a $200,000 income, the tax owed is $860.

This is not a book about income tax calculations. The important point is about the connection between the two functions shown in Figure 15.2. This sort of connection is fundamental to quantitative reasoning. Figure 15.2(a) is the *derivative* of Figure 15.2(b). No new information is needed to construct Figure 15.2(a) from Figure 15.2(b); they are merely different ways of representing the same information.

As we said, function (a) is the *derivative* of (b). The derivation of (a) from (b) is done by the process of *differentiation*, that is, calculating marginal rates of change

It is equally possible to compute function (b) from function (a). Math textbooks call the computational process 👉**integration**👈 or, almost the same thing, 👉**anti-differentiation**👈. A good, everyday word to describe the process is 👉**accumulation**👈. Function (b) accumulates the marginal tax payments over the entirety of the taxpayer's income.

The connection between *differentiation* and *anti-differentiation* (which is so clear from the words used) is important in practice. Many of our core modeling functions are derivative/anti-derivative pairs.

Calculus students memorize such patterns. We defer to the exercises a description of the processes by which anti-derivatives can be computed. This is for two reasons: 1) the process is automatic, 2) the main point is to know, whenever you are working with a function, whether its derivative or anti-derivative might be a better form than the function itself for the purpose a model is being built for.

15.3. Anti-differentiation

There are various algorithms for calculating the anti-derivative of a rate-of-change function. Traditional calculus students learn—or, at least, study!—algebraic techniques that can become puzzle-solving problems of considerable difficulty. Sometimes, there is no algebraic solution. But finding the anti-derivative need not be a puzzle. Here, we will demonstrate the idea graphically. To illustrate, we will use an important pair of functions related by differentiation and anti-differentiation.

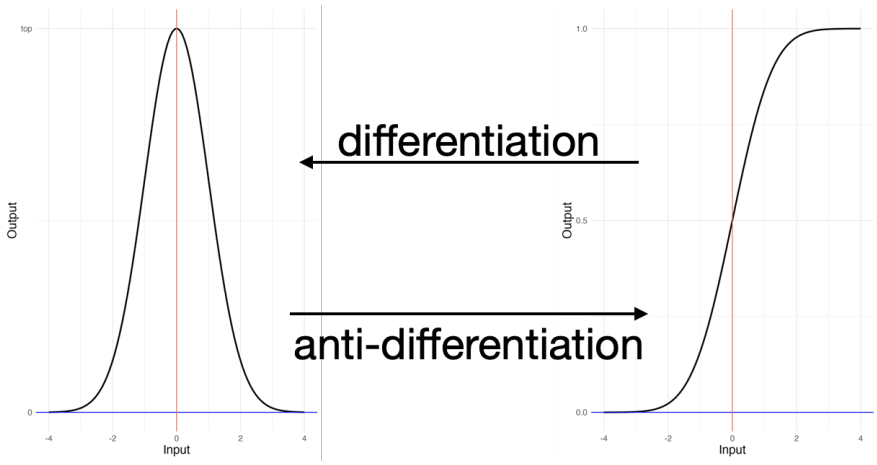


Figure 15.3.: `hill()` and `hillside()` are a derivative/anti-derivative pair.

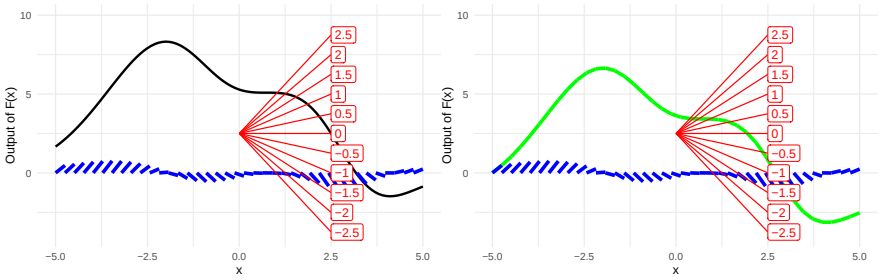
The hill() function gives the *derivative*—that is, the rate-of-change function—of the hillside() function. Conversely, the hillside() function is the *anti-derivative* of the hill() function. Differentiation turns a function into its rate-of-change function. Anti-differentiation undoes differentiation.

A homey metaphor may help here. Imagine a family of functions mutually related by differentiation or anti-differentiation. We'll label the functions Francis, Georgia, Heather, Ulrika, and Victoria: F(), G(), H(), U(), V(). The relationships among these functions are that G() is the derivative of F(), H() is the derivative of G(), H() is the derivative of H(), and so on. In human terms, these functions correspond to succesive generations in a family. Francis is the great-grandmother, Georger the grandmother, Heather the mother, Ulrika the daughter, and the grand-daughter is Victoria. Differentiation is the function that takes a person and produces their direct offspring. Anti-differentiation takes a function and produces their mother.

For demonstration purposes, to see how an anti-derivative can be calculated, let's break convention and use a special graphical format to display a rate-of-change function.

Keep in mind that differentiation contructs an new "daughter" function from a "mother" function. Finding the daughter is straightforward; it is the function whose **value** at any input corresponds to the **slope** of the mother at that same input. Figure 15.4 shows a function F(x) in black, and a kind of compass for measuring slopes. The slope of F() varies across the input values. We can calculate that slope for many different input values. Ordinarily we would plot the derivative—that is, the daughter function—using the vertical axis for the measured slope.

But Figure 15.4(a) shows the daugher function (in blue) not in the usual way as a value, but instead as a small, sloped line segment. The reader can confirm this by matching each blue line segment to the slope of the black function directly above or below it.



(a) Differentiation gives the slope for any input.

(b) Anti-differentiation accumulates the slopes to reconstruct the function (green).

Figure 15.4.: A function (black) with its rate-of-change function depicted as short inclined segments (blue).

Anti-differentiation is the process of accumulating the sloped segments to reconstruct a function with those slopes. The accumulation itself is child's play. Start with the leftmost segment. Raise the segment next to it so that the two segments match. Continue the process by moving to the next segment, and so on.

Playing this child's game of connecting the segments by moving each segment vertically traces out the green curve. Evidently, the green curve has the same **shape** as the original mother function F().

The green curve runs parallel to the original F() but can have a different vertical position. Why? The game-playing child is free to place the first segment at any vertical position she likes. The reconstructed function will be moved up or down accordingly. When we accumulate, whether seashells or inclined segments, we are free to specify how much was "in the basket" when we started.

15.4. Example: QALYs versus cost

Figure 14.3 showed functions for the number of quality-adjusted life-years saved and the expenditure on each of two hypothetical programs, which we called Natal and Cancer. Where did these functions come from? The quick answer is that the functions are the anti-derivatives of another function.

That other function, the derivative of QALYs(expenditure), has a simple logic that can easily be illustrated. Such a situation is common across many areas of science or quantitative reasoning. Indeed, we saw it in Chapter 13. The SIR model is merely a description of the rates of change of susceptibles and infectives as functions of the numbers in each group. Tracking the course of the epidemic over time was a mechanical operation ("Euler's method") that integrated the rate-of-change functions.

Consider now the QALY(expenditure) function for the Cancer program. From ongoing public health data collection, we know the 👉**incidence**👈 of cancer cases, that is, how many new cases arise in a year. We also have a good idea, from medical experience, of *survival times* for people with cancer and how survival is different with and without treatment. To illustrate, Figure 15.5 shows "survival curves" for colon cancer victims, comparing those who underwent radical surgery to those where the surgical intervention was merely palliative.

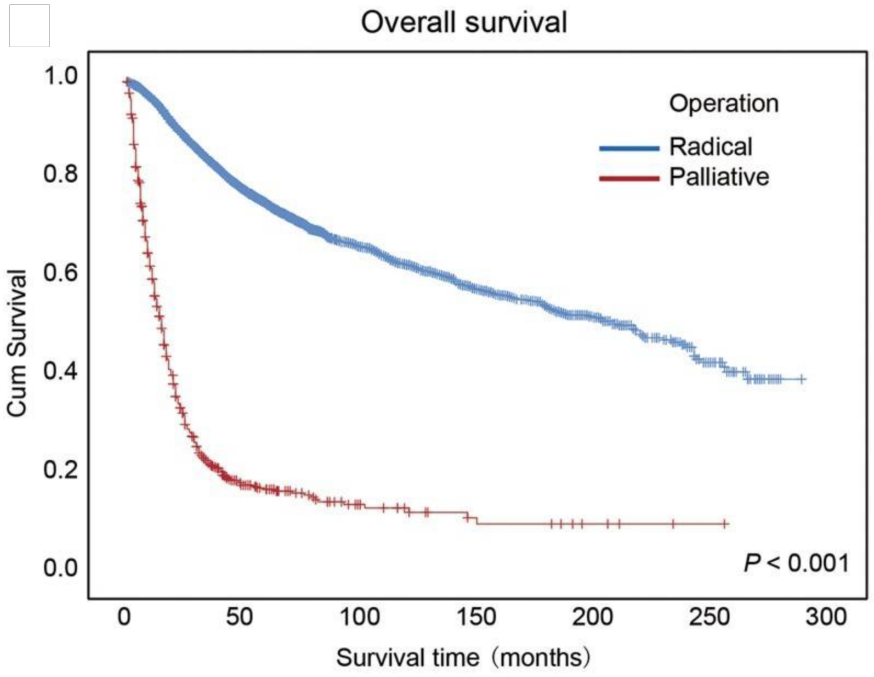


Figure 15.5.: Comparative survival curves for people undergoing surgery for colon cancer. Source

Source for Figure 15.5

Figure 15.5 shows two survival "curves" constructed from patients' records in a clinical study. Each curve is a function that takes months of survival as input and returns as output the fraction of all patients who survived that long. For instance, about 60% of patients who underwent radical surgery

survived for 100 months, whereas only about 15% of palliative-treatment patients survived that long.

Colon-cancer calculator

The advantage of thinking of survival time as a function is that we can imagine what other inputs might go into the function. For instance, we might have patient age, sex, and stage of cancer at the time of diagnosis, or other information. Such functions are often constructed by researchers. An example, for colon cancer, is linked to in Figure 15.6.

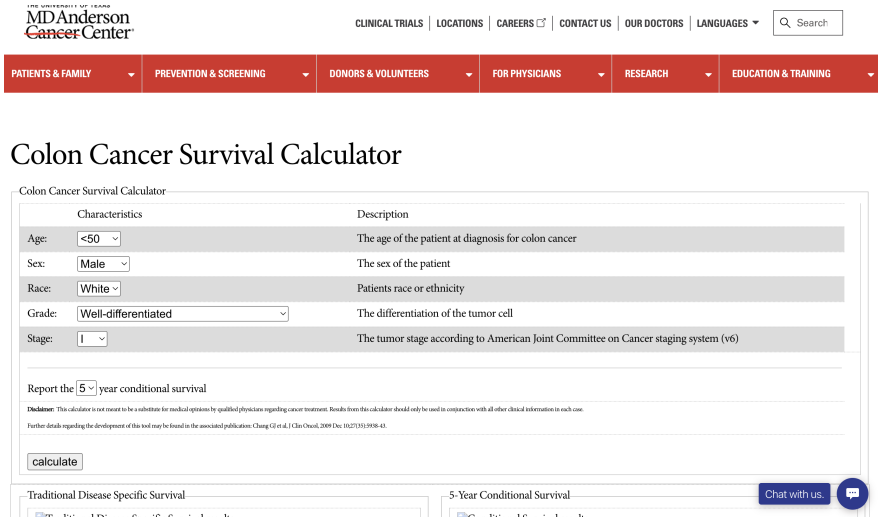


Figure 15.6.: A calculator for (average) survival time with colon cancer. Link.

Using such functions, we can compute the expected survival time for each age or cancer-stage class of patients diagnosed with colon cancer. More specifically, we can compute the average *extra* survival time for those who undergo radical surgery.

This illustrates an important distinction between public health and medicine. For an individual patient, a medical doctor might prefer to present the survival curve to show the range of possible outcomes. But for social policy making, we are not dealing with individuals but *groups* of people. This is the public health perspective. Average outcome gives a good indication for the group as a whole.

For the sake of argument, we will imagine that the average *extra* survival time is a decreasing function of age. That is, younger patients are more likely to survive longer. Perhaps this is more easily seen in reverse. An elderly patient is, for various reasons, unlikely to survive the 20-year period covered by the survival curve. Figure 15.7 sketches out what the *extra* survival time verus age function might look like.

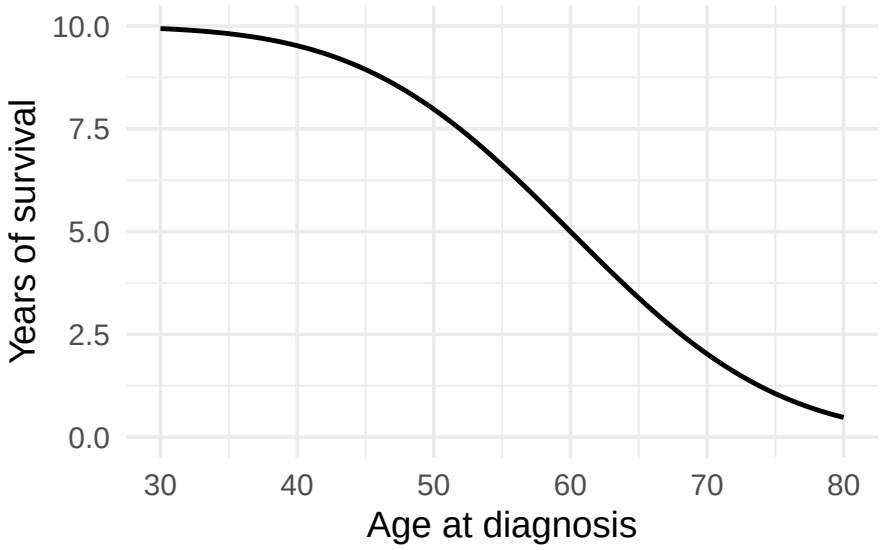


Figure 15.7.: A sketch of an imagined function giving *extra* survival time versus age.

The detailed calculation needed to translate Figure 15.7 into a function of QALYs saved per person treated is hard to understand without extensive modeling experience. But the general outline is accessible. Using only the treatment cost and the age distribution of incidence, we can translate the horizontal axis into the total cost of treating all people up to any given age. The result will be a graph showing the marginal years saved at each cost. Accumulating (anti-differentiating) that marginal years saved function over the cost produces the curve shown in Figure 14.3(b).

The point for us is that figuring out relationships like that in Figure 14.3(b) is based on several habits of experienced quantitative reasoners:

1. Thinking of years saved *on the margin*, that is, the function that describes how many years are saved for each additional unit of cost.

2. Knowing that the on-the-margin function corresponds to another function, total years saved versus cost, that is an ordinary anti-derivative of the function in (1).

3. Realizing that the function in (1) is much easier to justify, a good format for representing the relationship between years saved and cost, so putting our effort into modeling (1) and then using the automatic process of anti-differentiation to produce the function in (2).

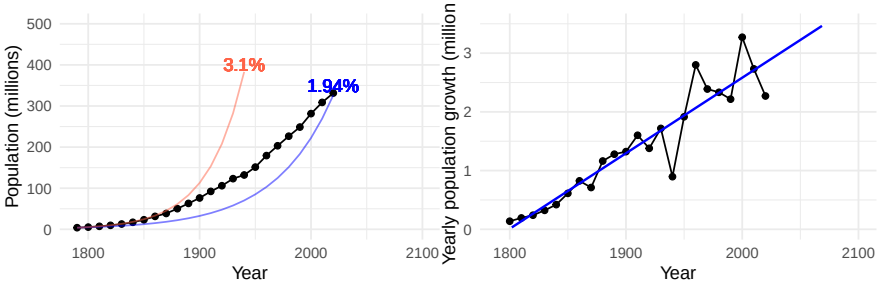
15.5. Example: Predicting peak population

Sometimes, clear patterns in the rate-of-change function can be easily modeled, but then need to be translated via anti-differentiation to display the information of interest. An instance of this is seen in the population of the US, as it has grown over the centuries.

Every 10 years, starting in 1790, the US Census Bureau carries out a constitutionally mandated census: a count of the current population. The overall count as a function of year is shown in Figure 15.8.

In the 230 years spanned by the census data, the US population has grown 100-fold, from about 4 million in 1790 to about 330,000,000 in 2020. Figure 15.8 shows the growth. Note that there is a lot of fluctuation in the data. These are not just random. The deviations correspond to historical events. There is a relatively low growth rate from 1860 to 1870, due to the US Civil War. The Great Depression is evident in the very low growth rates from 1930 to 1940. Baby Boom: look at the growth from 1950 to 1960. The bump from 1990 to 2000? Not coincidentally, the 1990 Immigration Act substantially increased the yearly rate of immigration.

Data source for Figure 15.8



(a) Population

(b) Rate of change in population with respect to time

Figure 15.8.: Two views of population growth in the US.

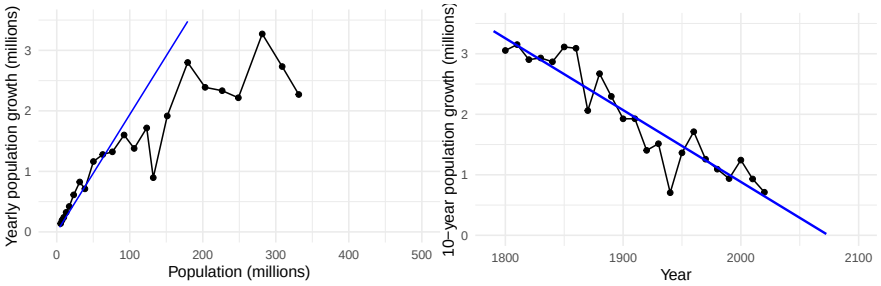
Both panels in Figure 15.8 are based on the Census data. Panel (a) plots the actual population. The upward curving pattern of the data (in black) in panel (a) is reminiscent of exponential growth. We annotated the data with two curves, both with strict exponential growth at the year percentage rates indicated. In the context of actual exponential growth, the data do not convincingly show exponential growth.

The rate of change in population (with respect to time) is the slope of the segments in panel (a). Panel (b) translates that slope into position along the vertical axis. The blue line in panel (b) guides the eye to see an ascending,

straight-line pattern in the data, admittedly with some sharp, decade-long deviations from the line.

One is tempted to conclude from Figure 15.8 that, if the US population follows the historical pattern, it will continue to grow at an ever-faster rate. However, that conclusion is not well-founded. The problem is the lack of a theoretical framework for interpreting the data.

Chapter 13 has some lessons to teach about growth, particularly exponential growth. Remember that the Euler framework for dynamics does not start with a time series, such as those in Figure 15.8. Instead, the Euler framework starts with a *dynamical rule*, the function that takes the *state* of the system (population, here) and turns it into a rate of change with respect to time. The plots in Figure 15.8 have *time* on the horizontal axis, not population. Figure 15.9(a) presents the data in dynamical rule format: change in state (with respect to time) versus state.



(a) Dynamical rule

(b) Proportional change (%) of state

Figure 15.9.: Plotting dynamics of growth

Figure 15.9(a) plots the yearly rate of change (with respect to time) versus the population. The format is exactly that of a dynamical rule in the Euler framework. True exponential growth corresponds to a same() dynamical rule. (See Figure 13.8.) The particular dynamical rule shown corresponds to 1.94%. That is, the blue line shows the dynamical rule that is

$$\text{ROC population}(P) \equiv 1.0194\, P\,. \tag{15.1}$$

The 1.94% per year dramatically overstates population growth when the population exceeds 100 million, a level reached by 1920. At 100 million and above, the data *suggest* a lower rate of growth. Owing to the peaks and dips in the data, it is hard to see a straight line above 100 million population. Nonetheless, the dynamics are clearly different above 100 million than they were below that.

One way to incorporate *changing dynamics* into Math expression 15.1 is to replace the 1.0194 with a function of year. We will call it r(year), and we would like the data to show us what it is. For this, we need to plot r(year) versus *year*, not population. Math expression 15.2 re-arranges Math expression 15.1 to emphasize that both ROC population and population themselves are functions of *year*.

$$\text{ROC population}(year) \equiv r(year)\, P(year)$$

or

$$r(year) \equiv \frac{\text{ROC population}(year)}{\text{P}(year)}. \tag{15.2}$$

To see the form of r(year) from the data, plot $\frac{\text{ROC population}(year)}{\text{P}(year)}$ versus year. Figure 15.9(b) does this. It is much easier to infer from the data in Figure 15.9(b) a pattern of a linearly decreasing (with respect to time) yearly growth rate (r(year)).

If the trend (blue line in Figure 15.9(b)) in the growth rate continues, the US will reach zero net growth around 2070, then enter negative growth. Of course, negative growth is just decline. A simple prediction from Figure 15.8(b) is that the argmax of the US population—that is, the year that the growth rate reaches zero—will occur around 2070.

Using the tools of mathematical accumulation, we can accumulate the rate of change (that is, r(year) P(year)) over future years to estimate the future population. Figure 15.10 shows the result.

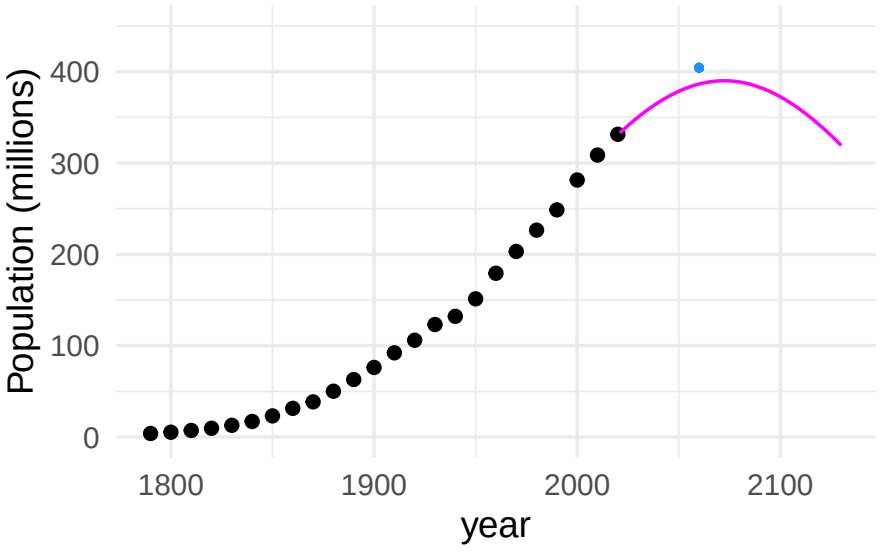


Figure 15.10.: The Census population data along with the function forecasting future population based on the rate-of-change model in Figure 15.9(b)

According to the accumulation model, the population peaks in 2075 at 390 million. We will be back down to the present population level in about 100 years.

Professional demographers construct much more sophisticated models using detailed data from many sources and, as needed, take into account the age structure of the population to predict future births. The demographers at the US Census Bureau predict that the population will reach a maximum of 404 million in 2060, shown by the little blue dot in Figure 15.10. That is not too different from what we got by analyzing the raw census numbers alone. Importantly, we analyzed those numbers in the context of the Euler framework (Chapter 13), formulating a model dynamical rule and using the *accounting method* (that is, *accumulation*) to translate the dynamical rule into a time series.

16. Moving Past Two Inputs

Previous chapters introduced a framework for quantitative thinking, highlighting dimensionality, which relates aspects of the world through mathematical relationships, such as between location and velocity or between energy and power.

- Functions and their input spaces describe how different aspects of the world are related. They also provide a way to express uncertainty and likelihood, and to extract information from data in a Bayesian style.
- magnitudes as a device for relating the very large to the very small;
- Rates of change and accumulation are versatile tools. They represent dynamics, guide decisions, and capture common relationships among aspects of the world.

This framework developed over centuries, spurred by ingenuity, insight, and technological advances like computing.

In this final chapter, we bring together these components to show how the framework enables new forms of quantitative reasoning. These new approaches take on many forms; for example, the emergence of Large Language Models that attempt to capture meaning in text.

Vocabulary Review

- algorithm (6)
- data (2)
- dimension (2)
- functions (6)
- margin (14)
- maximum (10)
- mean (8)
- models (1)
- per (3)
- point (3)
- rate (4)
- rate of change (4)
- space (3)
- system (6)
- value (2)
- with respect to (3)

New

- alignment between two vectors
- dot product
- gradient
- large language models
- learning
- length of a vector
- optimization
- plane
- projecting
- projection
- subspace
- unsupervised learning
- vector

16.1. Revising high-school algebra

To start, we examine two familiar high school algebra problems. Techniques for these were developed when paper and pencil were students' only tools. Unsurprisingly, such methods are ineffective with several variables—and even less so with large quantities.

1. Find "solutions", "x-intercepts," and "roots." These words signal the same kind of problem: finding function inputs that result in a specified output. Here, a "solution" is a value of x that makes the equation true, an "x-intercept" is a point where the graph crosses the x-axis,

and a "root" is another term for a solution. For example, you might be asked to find the roots of a polynomial such as $x^2 - 2x - 3$. This is the same as "solving" $x^2 - 2x - 3 = 0$.

2. Solve ☞**simultaneous equations**☜. There is more than one function, each sharing the same inputs. You want to find the input values that make each function reach a specified value. In textbooks, a simultaneous equation problem usually appears like this: there are always the same number of equations as input variables.

$$\begin{array}{rcrcr} 5\,x & - & 2\,y & = & 4 \\ 2\,x & + & 3\,y & = & -2 \end{array}$$

A third problem setting, ☞**optimization**☜, is important across applications of quantitative reasoning. It is just beyond the reach of high school algebra methods. An example in the style of high-school algebra is: Given that $x, y = 100$, what values of x and y give the smallest value for $x + y$?[1]

These three problem settings are important in practice. Imagine yourself controlling a complex system: adjusting valves, pumps, and burners in a chemical plant, or managing logistics to source components for a product like a cell phone. Figure 16.1 shows a generic image of this situation.



Figure 16.1.: A control panel for managing a complex system.

Perhaps your task for the day is to adjust the controls so the tank's temperature and pressure stay at the right levels. You need to adjust dials, flip switches, and so on. Or, in a bigger task, you might have to maximize plant productivity while staying within limits on quality and safety.

[1] Or, given that the area of a rectangle is 100 in^2, what should the side lengths be so the perimeter is as short as possible.

These tasks fit into a framework for finding solutions or optimizing. You don't have the system's "equations," but you have some basic knowledge, outputs from various gauges, and switches and dials to set inputs. Quantitative tools here differ from the symbol manipulation found in high school problems. Let's examine these tools and how they relate to topics from previous chapters.

16.2. Walking uphill

Every child knows how to get to the top of a hill: walk uphill! In this section, we look at the basic mathematics of walking uphill. We then extend it to the realistic case where the entire scene is hidden by impenetrable fog.

We want to solve quantitative problems by walking uphill (or downhill) to find solutions and optimize. For illustration, we create mathematical hills and explore ways to move in the right direction.

Our mathematical hills will be functions. The function's output is the hill's elevation. The inputs are the location where you are standing, and the latitude and longitude. For simplicity, we start with functions with a single input, as if you were obliged to walk along a road to get to the top of the hill. The function's input is the position along the road.

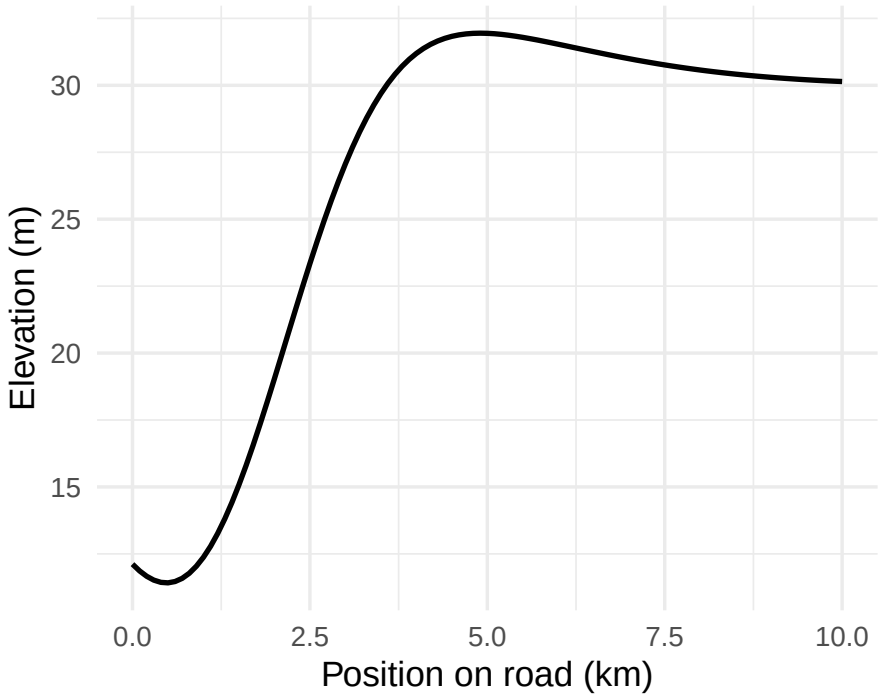


Figure 16.2.: Elevation vs position along a road

Figure 16.2 provides an example of such a function. Where should you walk on the road to reach the highest elevation? A glance is enough to find the answer: about 5 miles.

Figure 16.2(b) shows a more realistic view. You can't see the whole function. You only know the elevation at your current spot and whatever you note in your notebook. We start the walker at position 2 on the road, where the elevation is about 12 m.

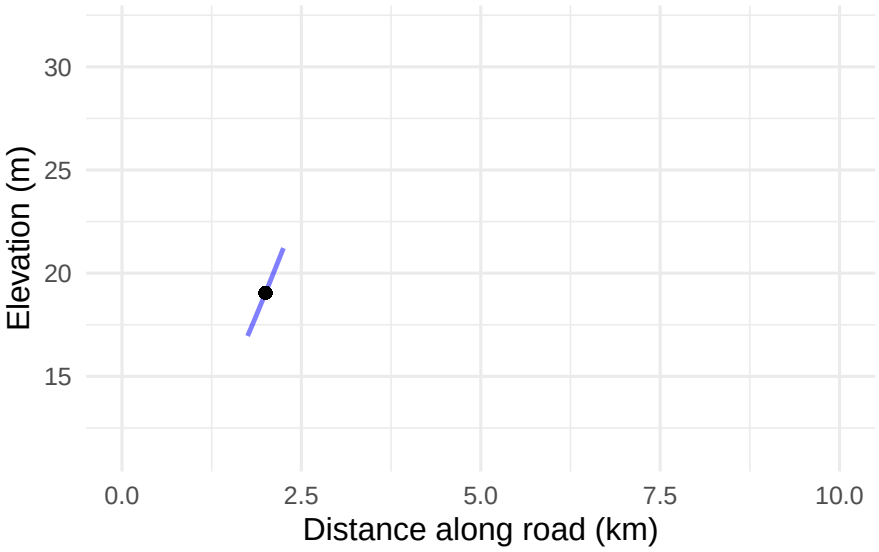


Figure 16.3.: Elevation vs position along a road, displaying the value of the output only at the places where you have already been. Rate of change is the slope of the blue segment.

The key tool for navigating the foggy road is the rate-of-change function: the derivative. You can find its value at your position by taking a small step right, then measuring the elevation again. If the new elevation is higher, the rate of change is positive.

The algorithm for reaching the top can be simple: if the function's rate of change is positive, step to the right. Otherwise, go left. Keep on stepping in this fashion.

The interesting problems are those with many inputs and expensive evaluations. For example, setting inputs might require shutting down an entire chemical plant. Another example comes from AI: optimizing a large-language model (LLM) like GPT-5 involves about 500 billion inputs. In both cases, finding a good maximum output yields the desired intelligence or outcome.

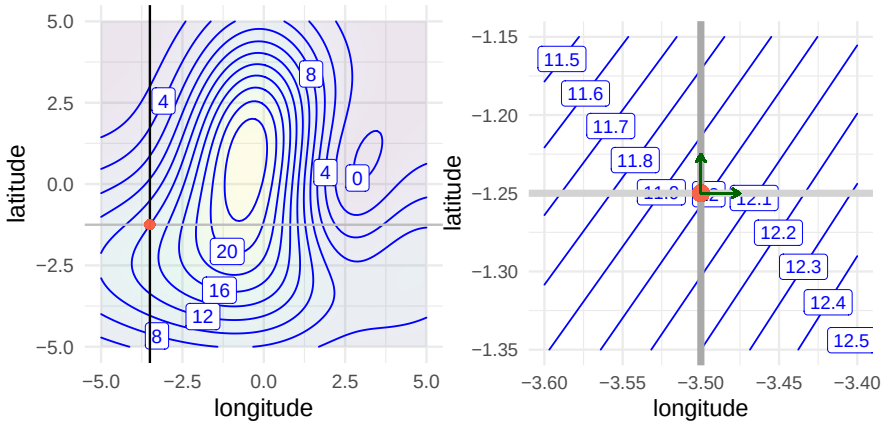
While building an LLM is far beyond our scope—we would need significant resources for that—we can still develop intuition for strategies that help us find an optimum without graphing the function.

Interactive web app

> **An interactive web app.**
>
> There is a collection of interactive web apps for QR-A at https://dtkaplan.shinyapps.io/QR-A-shiny/ including one that lets you evaluate a function without graphing it. Go to the Optimization/g(x,y) menu and experiment. It's hard to find the maximum of a function with two inputs without a graph. But if you turn on the "Follow Gradient" feature, you'll always know which direction is up. Take small, successive steps along the arrow; you'll trace an uphill path. Until you develop intuition, you may get stuck. Use the "training wheels" option for help when needed.

This strategy of walking uphill also applies to functions with 2 or more inputs, including functions with hundreds, thousands, or millions of inputs. With two inputs, we can create a visualization that guides our intuition for dealing with functions with many inputs.

Recall that for a function with two inputs, say $f(x, y)$, there are two different rate-of-change functions. Imagine standing at your position on the hill. That position is the intersection of two roads. One road runs east-west, the other runs north-south, as in Figure 16.4.



(a) Zoomed out to show a large region (b) Zoomed in close to the intersection of the roads

Figure 16.4.: A landscape with two roads. You are standing at the red dot.

To start, you are at elevation 11.98 m. Take a step to the east, that is, right on the gray road. For the sake of definiteness, each step will length of the green arrows. The eastward step will take you to an elevation of about 12.06 m, that is, an elevation gain of 0.08 m. In contrast, the step to the

north will bring you downhill to an elevation of about 11.93, a loss of 0.05 m.

From this information, you can calculate the direction of the step that will take you in the steepest uphill direction. To judge from Figure 16.4, the steepest step will be in the east-south-east direction.

The step to the east gives you what's needed to calculate the rate of change of the output with respect to longitude evaluated at the red dot. The step to the north gives you the rate of change of the output with respect to latitude. Each of these is a quantity with units of meters (of elevation) per degree (of latitude) for the north step or meters (of elevation) per degree (of longitude) for the east step. Combining the two rates of change gives the direction of steepest ascent on the hill and the amount of elevation gain achieved by taking a unit-length step. This information—direction and amount—is called the 👉**gradient**👈.

The word "gradient" comes from a Latin root gradi, meaning "to step" or "to advance." The gradient is a 👉**vector**👈. The word "vector" stems from the Latin vehere, which means "to carry." The gradient vector carries you uphill. The algorithm for optimizing by following the gradient vector dates back to 1847.

16.3. Another description of the gradient

The rate-of-change functions give important information about the way uphill. We use the plural "functions" because each direction has its own rate of change function. The goal for the mathematical hill climber is to find the direction with the greatest rate of change.

The problem setup is this:

1. There is a function $f(x, y, \ldots)$ for which you seek to find the value for each input that will give the greatest possible output. We will denote these input values as $x^\star, y^\star, \ldots$, but we do not yet know their values.
2. We are standing at a position $(x_0, y_0, \ldots)$. This is our starting position. It is often chosen at random, as if we drifted downward into the terrain like a leaf or raindrop.
3. We want to find a direction that points uphill and, ideally, uphill in the steepest possible direction.

4. Upon finding this direction, take a small step forward to a new position $(x_1, y_1, \ldots)$. The new position will be higher than the previous one, since the gradient always points uphill. Repeat (3) and (4) many times. In Figure 16.5, these successive steps are marked by circles.

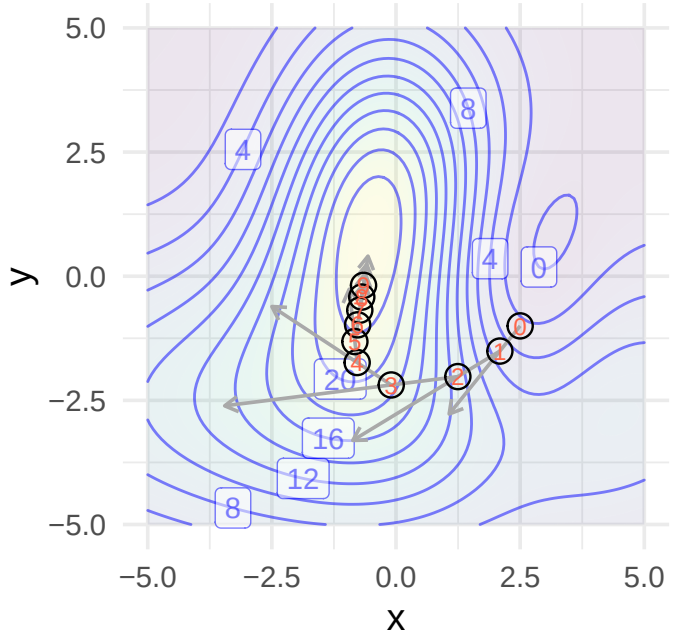


Figure 16.5.: Success steps climbing up the gradient from initial position 0. Each gray arrow is the gradient vector from the current position.

The walk started at a random position, marked by a circle with the number 0. From position 0, we find the uphill direction. This is the gray arrow from 0. We go a small distance that way and reach position 1. Find the uphill direction again and walk a small distance to point . Continue the process. As we approach the maximum, the steps become shorter. We can stop whenever the steps are too small to matter.

To implement this simple algorithm, we need to find the (steepest) uphill direction. We can do this with purely local information about the rate of change around our current position. For a function with two inputs, there are two rates of change: one with an east-west orientation, the other oriented north to south. These rates of change are easy to calculate:

For the east-west direction, the rate of change is

$$\partial_x f(x_0, y_0) = \frac{f(x_0 + h, y_0) - f(x_0, y_0)}{h}.$$

Similarly, the north-south rate of change is

$$\partial_y f(x_0, y_0) = \frac{f(x_0, y_0 + h) - f(x_0, y_0)}{h}.$$

As discussed in Chapter 14, we can set h to be any small value. The most uphill direction at an input location (x_0, y_0) turns out to be simply

$$\left(\partial_x f(x_0, y_0),\ \partial_y (x_0, y_0)\right)\ . \tag{16.1}$$

This coordinate pair can be interpreted as a direction and length in the (x, y) space. In mathematics, a coordinate pair such as (3, 4) can be interpreted either as a position in Cartesian coordinates or, equally well, as the direction and distance to that position from the point (0, 0). When we interpret a coordinate pair as a direction/distance, we call the pair a *vector*.

Vectors serve many important purposes throughout science and technology. The "gradient vector" at the input location (x_0, y_0) is an ordinary vector, which we create by calculating Math Expr 16.1 for the inputs x_0 and y_0.

The gradient ascent algorithm works with almost any function with any number of inputs. It doesn't matter if the function is expressed as a formula or as a real-world setting, such as a chemical plant or a logistics operation. The only information needed is local. Following the chemical plant analogy, imagine a set of dials that control the plant, as in Figure 16.6.

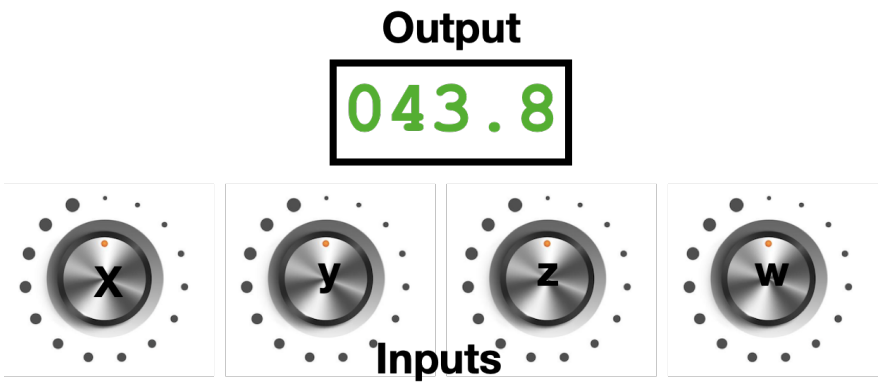


Figure 16.6.: Control inputs and a readout of an output.

Each dial's setting specifies one input. All input dials together determine a position in what we might call dial space. The space in Figure 16.6 is four-dimensional. For any setting of the dials—that is, any position in dial space—the function produces an output. The readout in Figure 16.6 displays this output value.

In Figure 16.6, all the dials are set to 1.

To calculate the rate of change for a dial, record the output displayed: 43.8. Then turn the X dial a small amount, say to 3. This is a change of 2 in the dial position. Suppose, in consequence of the change, that the output now reads 42.2. Then the rate of change is $\frac{43.8-42.2}{+2} = 0.8$. This tells us that $\partial_X f(1,1,1,1) = 0.8$.

Turn the X dial back to its original position. Then repeat the procedure with the Y dial. Suppose the rate of change with respect to Y is $\partial_Y f(1,1,1,1) = 1.3$. Turn Y back to its original position, then again repeat the procedure with Z, and once again with W. Suppose that the manipulation of Z gives $\partial_Z f(1,1,1,1) = -0.9$ and W gives $\partial_Z f(1,1,1,1) = 2.2$. You now assemble the gradient vector:

$$(\partial_X f(1,1,1,1),\ \partial_Y f(1,1,1,1),\ \partial_Z f(1,1,1,1),\ \partial_W f(1,1,1,1)) = (0.8, 1.3, -0.9, 2.2)\,.$$

This four-component vector is a direction in the four-dimensional dial space. Take a small step in dial space by turning the dials by 0.08 for X, 0.13 for Y, -0.08 for Z, and 0.22 for W. At the new setting, the output will be higher.[2]

16.4. Example: Training a neural network

One very widely used function type used in AI and other areas of technology is the "neural network." A neural network is a function that maps input data to output predictions. It consists of layers of interconnected nodes, or "neurons," each performing a seemingly simple computation to produce an output that is a single numerical value.

Figure 16.7 shows a relatively simple neural network provided for demonstration purposes by the organization <tensorflow.org>. Since the network is small, <tensorflow.org> can provide an excellent graphical depiction. (The neural networks used in AI are vastly more complex and not well understood even by the people who build them.)

[2] We can actually estimate how much higher the output will be. Taking the step (0.08, 0.13, -0.08, 0.22), the output will increase by about 0.8

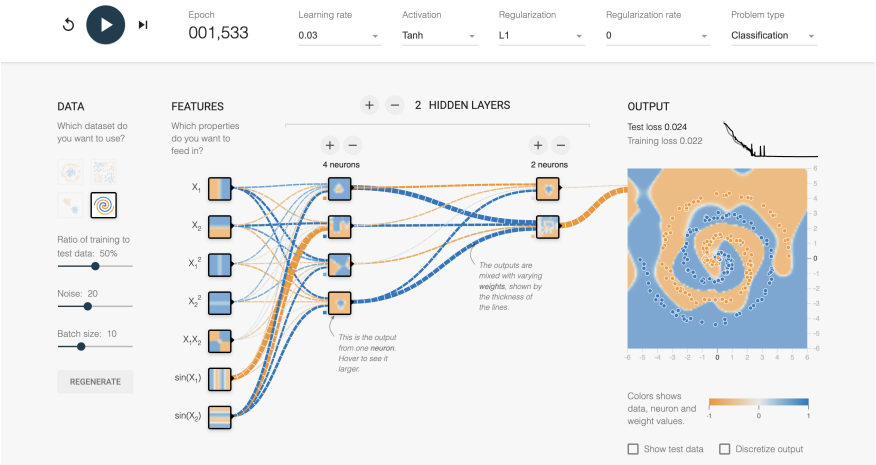


Figure 16.7.: A snapshot from a demonstrator web app showing training for a simple neural network model. You can interact with the demonstrator at https://playground.tensorflow.org/

The *interactive* form of the demonstration displays the process of building a network to serve a particular purpose. To start, let's work through the structure from left to right in Figure 16.7.

The area labeled "Data" on the left represents a data frame. Each row of the data frame records the x-y position of a point, together with a column holding the group to which the point belongs, like this:

X	Y	Group
0.43	-0.21	orange
0.12	0.34	blue
-0.56	0.78	blue

The rows are displayed graphically on the right-side of the panel; each row is one dot.

Next to the "Data" area is a stack of seven small squares labeled "Features." Each of those squares stands for a function that takes x-y coordinates as input and produces as output a value between -1 and 1. For instance, the topmost "feature" outputs a value close to 1 for points in the right half of the plane and a value close to -1 for points in the left half. The other features perform similar transformations, capturing different aspects of the input space.

The result of applying those seven functions to the input data is seven new columns for the data frame, each of which holds the output from

the corresponding function. For simplicity, we will show just the top two functions, `f1()` and `f2()`:

X	Y	Group	f1	f2	⋯ to f7
0.43	-0.21	orange	0.92	-0.64	⋯
0.12	0.34	blue	0.31	0.75	⋯
-0.56	0.78	blue	-0.87	0.84	⋯

To the right of the "features" is a stack of four functions. Each function takes as input a linear combination of the f1(), f2(), ..., f7() output values. The output of each of those four functions transforms the input so that the output ranges from -1 to 1.

As always, a linear combination is a sum of the form

$$a_1 f_1 + a_2 f_2 + \cdots + a_7 f_7$$

, where $a_1, a_2, \ldots, a_7$ are coefficients that determine the contribution of each feature. In Figure 16.7 the coefficients are displayed as dashed curves. The color of each curve indicates whether the coefficient is positive (blue) or negative (orange). The width of each curve is proportional to the size of the corresponding coefficient.

Ultimately, the input to the four functions is the x-y coordinate of the corresponding row of the data frame. The output—as a function of x-y position—is shown by the color in the small square representing each of the four functions. You can see that the four functions are more complicated than the original feature functions. Next in the display is a stack of two neurons. These take as input a linear combination of the outputs from the four neurons to the left. Again, the coefficients for this linear combination are displayed by the color and thickness of the dashed curves leading into the two neurons.

Finally, the outputs of the two neurons are combined linearly to produce the overall function shown in the larger square to the right of the two neurons. Since there are only two neurons being combined, its easy to read the coefficients in the final linear combination from the graph. The top neuron has a small, slightly negative coefficient. The bottom neuron has a stronger negative coefficient.

Focus now on the large square display on the right side of the graphic. As mentioned before, the individual rows of the data frame are represented

by dots. The end-function implemented by the seven features and the six neurons is shown by the background color in the square. That end-function is intended to be a classifier for new data, that is, to take any x-y coordinate as input and provide as output the predicted class (i.e., orange or blue) for that coordinate.

As you can see, the neural network has learned enough about the pattern in the data to provide a reasonable classification for any new x-y coordinate. The orange dots lie in the orange area of the end-function: likewise, for the blue dots.

None of the original features resemble the pattern of the data points. Yet the coefficients of the various linear combinations turn these features into an end-function that captures the data pattern. How did that happen?

In abstract, we can say that the neural network was *trained* on the data from the data frame. Here, the goal of training is to establish the value for each of the coefficients in the network.

We can count the coefficients that have to be determined during training. Each of the seven features provides an input to each of the 4 neurons in the first hidden layer, resulting in $7 \times 4 = 28$ coefficients. Each of the 4 neurons in the first hidden layer provides an input to each of the 2 neurons in the second hidden layer, resulting in $4 \times 2 = 8$ coefficients. Finally, each of the 2 neurons in the second hidden layer provides an input to the single output neuron, resulting in $2 \times 1 = 2$ coefficients. In total, there are $28 + 8 + 2 = 38$ coefficients that need to be determined during training.

Here's how the training works. To start, the coefficients are set to 38 random values. Naturally, the resulting function is unlikely to capture the pattern in the data. The training process looks at the function output for each of the dots in turn. Suppose the output from the initial end-function for row 1 of the data frame is 0.37 which can be translated to words: slightly blue. Since row one is actually in the orange class, the function provides an incorrect prediction. The goal of training is to adjust the coefficients so that the end-function gives a more appropriate output for row 1 (and all the other rows in the training data).

The end-function, as we've said, has 38 coefficients. The output of the end-function depends on each of these coefficients. We can differentiate the end-function with respect to each coefficient, producing a gradient vector in the 38-dimensional space of the coefficients. If we changed the coefficients slightly in the direction of this gradient, the output of the end-function would increase slightly. That would make the end-function output

worse with respect to the data in row 1 of the data frame. To make the end-function output better, we simply need to change the coefficients slightly in the opposite direction of the gradient.

This change in the 38 coefficients might make the output of the end-function better or worse with respect to the other rows of data. The training process repeats this adjustment for each row in the data frame, each repetition making the end-function output slightly better with respect to that row.

The observed behavior of this iterative adjustment process is, very gradually, to improve the overall performance of the end-function with respect to all the rows in the data frame.



Consider now the upper right corner of the display, shown here. This shows the progress of the training over time, as each successive move is taken along or opposite to the gradient. At each step, the "training loss," that is, the total error of the end-function is calculated. Over thousands of adjustment steps, the training loss fluctuates up and down but overall, decreases. When further adjustment steps don't reduce the training loss, the training algorithm stops, giving us the best set of 38 coefficients that was found.

Link to the neural network playground

The interactive, online version of the demonstrator shows the training process over time. As training proceeds, you can observe at each step the current end-function and how it changes in response to the adjustments of the coefficients. The behavior is very complex, yet it amounts to many, many steps along the gradient vector.

16.5. Space exploration

Chapters 11 and 12 introduced vectors in two-dimensional space, where simple tasks can be performed intuitively. For example, to measure the *length* of a vector, use a ruler! To find out whether two vectors are parallel, just look!

Now we want to extend our capabilities into higher-dimensional space. The motivation for this is, for example, making sense of complex data or constructing a model of language.

We start with three-dimensional space, where the geometry can be conveyed by moving the viewing perspective. This will let us explore how pairs of vectors can be used to specify a unique ☞**plane**☜. This leads to a new

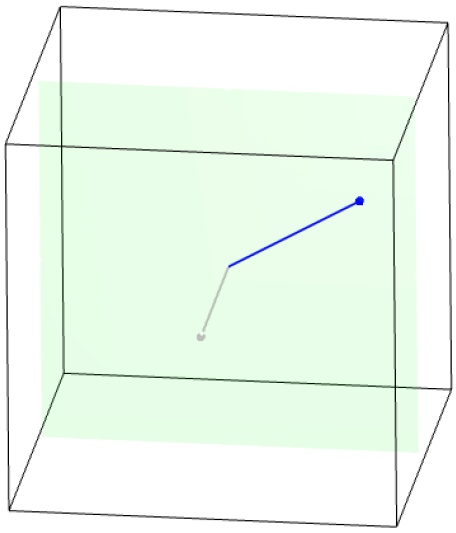
operation, ☞**projection**☜, which casts the shadow of a vector onto another vector or onto a plane.

The projection operation is key to working with data, for instance, building statistical models or identifying patterns in the data. But projection is especially powerful when working in high-dimensional spaces.

In four- and higher-dimensional spaces, it is practically impossible to draw a meaningful graphic. But the operations we can see in three dimensions can be calculated in any dimension using arithmetic. This includes vector length, angles between vectors, and projections onto planes or higher-dimensional extensions of planes.

To build intuition, let's return to 3-dimensional space. Consider Figure 16.8, which shows two vectors, one in blue and the other gray. They are drawn on a green background. At a glance, one can see that the angle between the vectors is obtuse, perhaps about 130 degrees. Also, the gray vector is shorter than the blue vector. Easy!

In fact, each vector is 3-dimensional. And the green "background" is actually a plane that contains both the vectors. Try rotating the scene to see this more clearly.



As you change the viewing perspective in Figure 16.8, the apparent lengths of the vectors as well as the angle between them can change. And, you'll see

Several of the following diagrams represent 3-dimensional space, starting with Figure 16.8 and through Figure 16.11. In this printed edition, the diagrams are static and, flattened onto the page, do a poor job of representing three-dimensional space. You will have a better experience of the space by following the link to the online interactive diagrams that allow you to rotate the space. Doing this is extremely helpful being able to see the objects from different directions in space.

Link to interactive diagrams

Figure 16.8: Interactive 3D view of two vectors: blue and gray. You can position the cursor over the scene and press/drag to rotate the vectors in three dimensions.

that the green "background" is actually a plane, with both vectors contained in the plane.

Indeed, if you change the perspective to view the plane edge-on, the blue and gray vectors will appear to lie on the same line. And depending on which angle you take (keeping the plane edge on), the two vectors will be arranged variously in opposite directions or in the same direction.

To measure the proper *length* of a vector, you need to lay down a ruler on top of the vector, parallel to it. Similarly, the measure the proper *angle* between the two vectors, you need to lay a protractor in the green plane that contains both vectors.

We can represent the situation *arithmetically*. Since the vectors live in three-dimensional space, the tip of each vector can be written in (x,y,z) coordinates. (For simplicity, the base of each vector will always be at the origin, that is, (0,0,0).) We could have put tick marks along the x-, y-, and z-axes to let you measure the coordinates of each vector, but for convenience, here they are.

$$\vec{v} \equiv \begin{pmatrix} 7 \\ 4 \\ 0 \end{pmatrix} \quad \text{and} \quad \vec{w} \equiv \begin{pmatrix} 1.5 \\ -4 \\ 0 \end{pmatrix}$$

Note that, for future reference, we have given names, $\vec{v}$ and $\vec{w}$ to the two vectors.

With the arithmetic representation, we don't need a ruler to measure the 👉**length of a vector**👈; the Pythagorean formula does the job.

$$\sqrt{7^2 + 4^2 + 0^2} = 8.06 \quad \text{and} \quad \sqrt{1.5^2 + 4^2 + 0^2} = 4.27\,. \tag{16.2}$$

We won't do very much arithmetic here. But for what we will do, it will help to have a much more concise notation. Here are two parts of that notation:

- A function, called the 👉**dot product**👈 which involves simple multiplication and addition. Here's an example, the dot product between $\vec{v}$ and $\vec{w}$:

$$\vec{v} \odot \vec{w} = 1.5 \times 7 \; + \; -4 \times 4 \; + \; 0 \times 0 = -5.5.$$

The dot product takes two vectors as input and produces an ordinary number as output.

Sometimes it is helpful to dot a vector with itself. For instance:

$$\vec{v} \odot \vec{v} = 7^2 + 4^2 + 0^2 = 65$$

and

$$\vec{w} \odot \vec{w} = 1.5^2 + (-4)^2 + 0^2 = 18.25 \ .$$

This use of the dot product captures a large part of the Pythagorean formula for vector length (Math Expr 16.2).

- The **length** of $\vec{v}$ is written $\|\vec{v}\|$. There is a simple formula for the length of a vector that uses the dot product:

$$\|\vec{v}\| = \sqrt{\vec{v} \odot \vec{v}} \quad \text{and} \quad \|\vec{w}\| = \sqrt{\vec{w} \odot \vec{w}} \ .$$

This dot-produce notation pays off with slightly more elaborate calculations. For instance,

- The ☞**alignment between two vectors**☜ is calculated like this:

$$\text{alignment}(\vec{v}, \vec{w}) \equiv \frac{\vec{v} \odot \vec{w}}{\sqrt{\|\vec{v}\| \, \|\vec{w}\|}} \tag{16.3}$$

16.6. Spaces with dimension greater than 3

Understandably, people have difficulty visualizing scenes in 4- or higher dimensions, or even imagining that such scenes might exist. Our cognitive abilities are tuned to perception in the familiar 3-D space of everyday life.

With the arithmetic representation of vectors, however, it's easy to see how to construct vectors in 4-space, 5-space, or any-dimensional space. A vector in such a space is merely a column of numbers. Here are a few examples.

$$\vec{a} \equiv \begin{pmatrix} 1.5 \\ 3.0 \end{pmatrix} \quad \vec{b} \equiv \begin{pmatrix} 4.2 \\ -7.1 \\ 2.3 \end{pmatrix} \quad \vec{c} \equiv \begin{pmatrix} 0.3 \\ 4.1 \\ 1.9 \\ -3.2 \end{pmatrix}$$

$$\vec{d} \equiv \begin{pmatrix} 1.0 \\ -7.0 \\ 2.9 \\ 1.2 \\ -4.1 \end{pmatrix} \quad \vec{g} \equiv \begin{pmatrix} -0.1 \\ -8.8 \\ -4.6 \\ 6.5 \\ 5.2 \\ -4.2 \end{pmatrix}$$

The count of components in each vector tells the dimension of the space where the vector exists or "lives in."

The reader may object. "Of course, you can make a column of numbers with as many components as you like. What does that have to do with 'space?' "

What makes these simple columns of numbers into an "object in space" is the calculations we can do with them, for instance, the "length" of a vector or the alignment between two vectors. Both of these calculations are rooted in the dot product, whose formula is a simple extension of that already shown. For instance, here's a dot product in 5-dimensional space:

$$\begin{pmatrix} 1.0 \\ -7.0 \\ 2.9 \\ 1.2 \\ -4.1 \end{pmatrix} \odot \begin{pmatrix} 2.6 \\ 8.3 \\ 6.4 \\ 1.6 \\ 2.7 \end{pmatrix} = 1.0 \times 2.6 - 7.0 \times 8.3 + 2.9 \times 6.4 + 1.2 \times 1.6 - 4.1 \times 2.7 = -46.09$$

16.7. Alignment, projection, and decomposition

Three important spatial operations in modern technology are

i. Finding the **alignment** between two vectors.

ii. Defining a ☞**subspace**☜ of a larger space. As you will see, subspaces are a way of encoding information about patterns.

iii. ☞**Projecting**☜ a vector in the larger space down into a subspace. Since subspaces encode patterns, projection determines to what extent a vector displays those patterns.

The alignment quantifies how *similar* the two vectors are, that is, whether they have information in common. An alignment of 1 indicates that the vectors share all information. An alignment of zero means that they have no information in common. (An alignment of -1 means that the vectors share information completely, but that one vector has been reversed in direction.) The reader who has studied statistics may recall the "correlation coefficient." This is essentially the same as the alignment.

In a two-dimension space, a subspace is a line: an infinite set of points that all lie within the bigger space. The line is a *one-dimensional subspace*. In three-dimensional space, there are two different kinds of subspaces: a line (1-dimensional) and a plane (2-dimensional). In higher-dimensional spaces, say one with n dimensions, there can be 3-, 4-, and up to $(n-1)$-dimensional subspaces.

A one-dimensional subspace is defined by a single vector. A two-dimensional subspace is defined by two vectors.[3] Similarly, an $(n-1)$-dimensional subspace is defined by $n-1$ vectors.

Figure 16.9 shows a two-dimensional subspace (that is, a plane) of the three-dimensional space. The two vectors drawn in the same color as the subspace have been used to define the subspace. We have colored the subspace-defining vectors to be inconspicuous, so the reader can focus attention on the subspace itself.

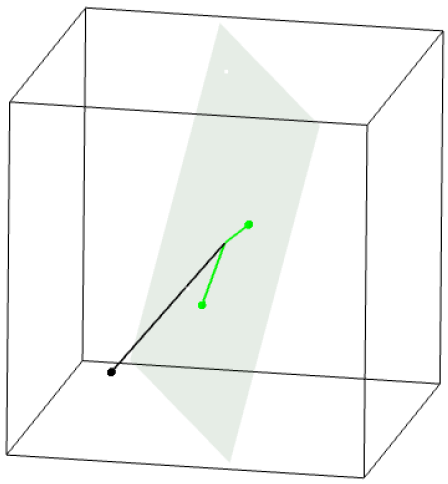


Figure 16.9 A two-dimensional subspace of a three-dimensional space. The black vector is **not** in the subspace.

[3]The vectors must not be exactly aligned. Otherwise, the subspace is one-dimensional.

You can verify that the two green vectors are in the subspace by rotating the scene. Whatever way you rotate it, you will never see the tips of the green vectors outside of the subspace. The black vector, however, does not lie in the subspace. (Of course, the green vectors must be in the subspace. We defined the subspace *using* the two green vectors.)

Projection of a vector onto a subspace means to find a new vector *in the subspace* that is a close as possible to the vector being projected. Figure 16.10 shows the projection of the black vector onto the green subspace. We have drawn in gray the "new" vector, that is, the one that lies in the subspace and is as close as possible to the black vector.

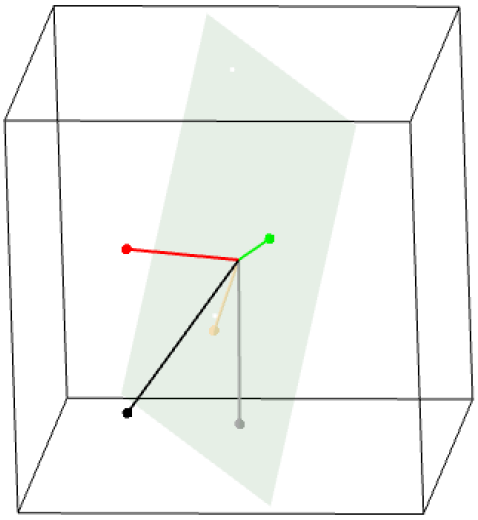


Figure 16.10: The black vector consists of a part entirely *within* the subspace and a part entirely outside the subspace. (See the online, interactive version to explore this space.)

Link to interactive diagrams

To confirm that the gray vector really is the one in the subspace that is as close as possible to the black vector, observe these facts by rotating the scene:

a. The gray vector is in the green subspace. No matter how you rotate the scene, the gray vector never emerges from the subspace.

b. The tip of the black vector is directly above the tip of the gray vector. Admittedly, it can be hard to be sure of this just by observing the co-movement of the black and gray vectors.

To allow greater precision of observation, we have added a red that is perpendicular to the green subspace. You will want to confirm that this statement is accurate. Do so by rotating the space until you are looking at

the subspace edge on. Whichever such perspective you take, the red vector will be perpendicular to the vestigial line of the subspace.

Now rotate the space so you are looking straight down the red vector. When you have this right, the red vector will look like a single dot. You will also see that the tip of the gray vector is exactly covered by the tip of the black vector.

Go back to looking at the subspace edge on. You should be able to see the black, red, and gray vectors. Notice that if the red vector were moved so that its base is at the tip of the gray vector, it would reach directly to the tip of the black vector. That is,

$$\text{gray} + \text{red} = \text{black} .$$

Writing a vector as a sum of two other vectors is a form of decomposition. This decomposition of the black vector involves a part *directly in* the subspace (gray) and a part entirely outside the subspace (red).

In three-dimensional space, this is just fun and games. But in higher-dimensional space, where we need to perform the projection using arithmetic, it is the core technique of statistical modeling, a method that underlies many published scientific findings and that gives a more nuanced view of relationships than the simple rates described in Chapter 4.

16.8. Example: Learning to identify different species

We are going to take on a 👉**learning**👈 task: figuring out how to classify the species of a penguin. For people, such learning comes from experience. Spend enough time with penguins, and you'll figure out how to discern one type of penguin from another.

Consider now the problem of getting a computer to discern the different species from one another using physical measurements. Our motivation here is to demonstrate a process of *learning*. I don't know that there is a practical, ecological motivation behind this problem.

In this learning task, we will start by sending human experts out to collect data from penguins: the species, but also quantitative data such as the penguin's mass, beak length, and width, and so on. The data collected

constitute a 👉**training data set**👈 that we will use to establish quantitative rules for identifying a penguin's species.

Figure 16.11 shows training data from 344 penguins across three species: Adélie, Chinstrap, and Gentoo. Each penguin is one dot in the picture. The judgement of the human experts about the species is indicated by the *color* of the dot: Adelie is blue, Chinstrap green, and Gentoo red. Five different quantitative measurements were taken for each penguin, so the dots rightfully belong in a five-dimensional space. But we can only show three dimensions graphically, so we work with those.

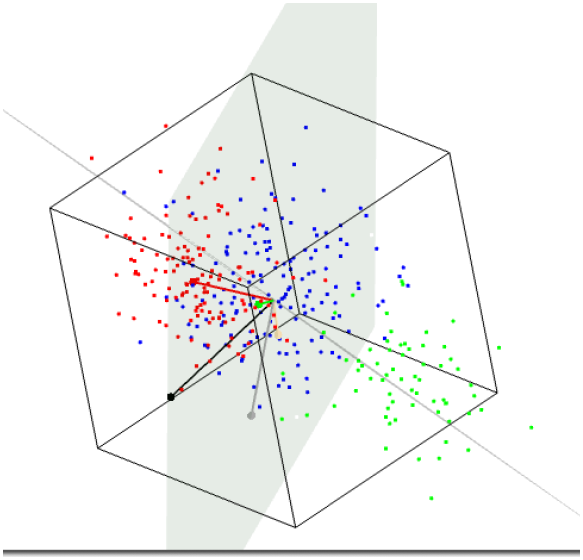


Figure 16.11:Representing the penguin data as points in space. (See the online, interactive version to explore this space.)

From the perspective shown initially, it's easy to distinguish the Chinstrap penguins from the other two species. But the Adelie and Gentoo species are all mixed up; we can't divide the space into distinct regions that separate one species from another.

Now rotate the scene until each of the three species occupies a distinct region. By rotating, you are determining how to combine the various measurements to obtain the cleanest results from the training data. Once that learning has been accomplished, you'll know how to classify any new penguin that comes along without a species label.

Think about how you learned the pattern. You rotated the space more or less at random, keeping track of which perspective best separates the groups of different-colored dots. Such a perspective optimization can also be automated, without human involvement. In fact, that is how the subspace

shown in Figure 16.11 was determined. The information about species corresponds to the pattern encoded by the subspace.

This was a pretty easy learning task since it can be done in three dimensions. The next example demonstrates a more impressive feat performed in a space of more than 100 dimensions. What's more, there is no expert providing the training data; the method will figure out how to perform classification without knowing the goal!

16.9. Example: Voting patterns

In high-dimensional space, random vectors tend to have an alignment close to zero. This is an important clue to help us spot patterns in data. As described previously, patterns are encoded by a choice of subspace. To search for patterns, we look for subspaces where projecting onto the space tends to produce vectors that are more closely aligned.

To avoid unnecessary abstraction and the mystique that can arise from leaving the familiar realm of three dimensions, we have selected an easy-to-understand setting: votes by legislators in a council. In particular, we will examine a series of 773 votes by 135 lawmakers in the Scottish Parliament in 1999-2000.[4] This might seem an obscure topic, but the advantage for us is that few readers will have direct knowledge of the parliament's political structure, for instance, the number and make-up of political parties. Similarly, the reader will not know how the 773 recorded ballots are related to one another.

This is an example of ☞**unsupervised learning**☜. In *supervised* learning, as with the penguin species assignment, we have *training data* about the patterns we seek to uncover: the different colored points in Figure 16.11 that enable us to find an advantageous projection onto a low-dimensional subspace. In *unsupervised* we have no such knowledge. The problem is then to find a low-dimensional subspace without knowing in advance what we are looking for.

To orient the reader, consider some of the data for one of the 135 members of parliament, Bruce Crawford. Mr. Crawford voted "For" 235 ballots, "Against" 264, and didn't register a vote in 274. For instance, Bill S1M-786 was entitled "Ethical Standards in Public Life." Mr. Crawford was in favor,

[4]This example was provided soon after the session of Parliament by then-undergraduate Caroline Ettinger and Professor Andrew Beveridge at Macalester College.

as were 98 of his colleagues. (It is likely that most members of parliament are in favor of ethical standards in public life, but evidently, 17 members objected to something or another in the bill. There were 2 abstentions and 16 absences.)

A diligent researcher can read through the minutes of Parliament, which record which party each member belongs to and the content of each bill. But, in the spirit of unsupervised learning, we will not use this information.

Figure 16.12 shows the votes and absences for each of the 134 members of parliament on each of the 773 bills, giving somewhat over 100,000 data points. Each point is one member voting for or against one bill (or abstaining or being absent). The point's color shows whether the vote was for or against.

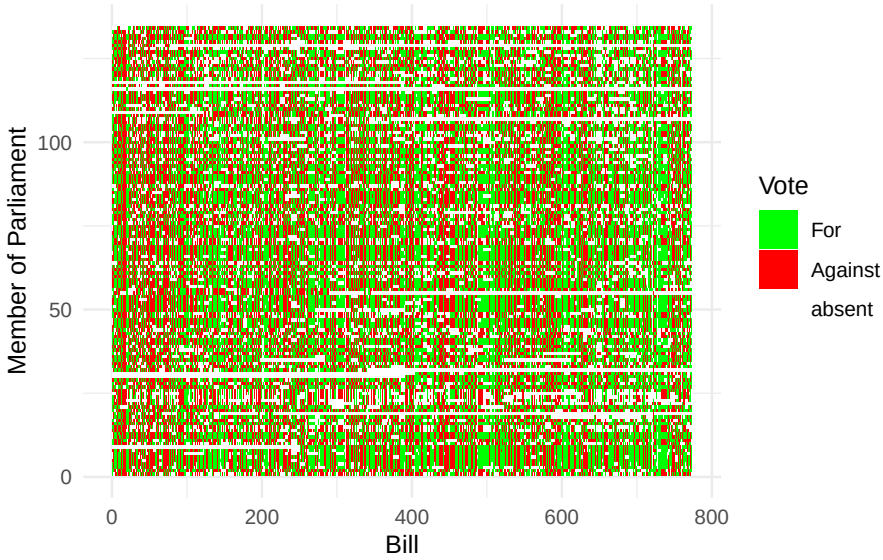


Figure 16.8.: Votes by 134 members of the Scottish Parliament on 773 bills considered in the 1999-2000 session. The "Ethical Standards" bill is column 183. Mr. Crawford is row 100.

Our goal is to find and interpret patterns in the data. There are many ways to define "pattern." For instance, the reader will, possibly, see the overall pattern in Figure 16.12 as a *plaid*, a common fabric pattern. (Satisfyingly, plaid is culturally associated with Scotland, although that has nothing to do with the matter at hand.) Here, however, we will stick with the sorts of patterns that can be seen through the lens of quantitative reasoning, particularly the alignment of vectors and the detection of low-dimensional subspaces that provide a good approximation to the raw data.

We start with alignment, a useful measure that can be calculated using the formula in Math Expr 16.3, which involves simple arithmetic. We start by

measuring the alignment among the various members of parliament. For each of the 133 members of parliament, we have a vector of votes on the 773 bills. For example, Bruce Crawford's voting record is represented by the vector we have named $\vec{M}_{100}$ in Math Expr 16.4. We also show vectors for a randomly selected few of his colleagues: Colin Campbell ($\vec{M}_{99}$), Paul Martin $\vec{M}_{57}$, and Scott Barrie ($\vec{M}_{25}$). Each of the vectors has 773 components; we're leaving out most of them for human readability.

$$\vec{M}_{100} = \begin{pmatrix} 1 \\ -1 \\ -1 \\ 1 \\ 1 \\ 1 \\ 1 \\ -1 \\ \vdots \\ -1 \\ -1 \\ 1 \end{pmatrix} \quad \vec{M}_{99} = \begin{pmatrix} -1 \\ 1 \\ 1 \\ -1 \\ -1 \\ -1 \\ -1 \\ 1 \\ \vdots \\ 1 \\ 1 \\ 0 \end{pmatrix} \quad \vec{M}_{57} = \begin{pmatrix} 1 \\ -1 \\ -1 \\ -1 \\ 1 \\ 1 \\ 1 \\ -1 \\ \vdots \\ 0 \\ 0 \\ 0 \end{pmatrix} \quad \vec{M}_{25} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ -1 \\ 1 \\ 1 \\ \vdots \\ 0 \\ 1 \\ 0 \end{pmatrix} \tag{16.4}$$

Note that the first component of all the vectors in Math Expr 16.4 corresponds to the vote on the same bill, S1M-1, and similarly for all the other bills up to S1M-4064.

Observe that $\vec{M}_{100}$ is almost entirely the opposite voting pattern of $\vec{M}_{99}$. To find the quantitative alignment, first take the dot product of the two vectors,

$$(-1) \times 1 + (-1) \times 1 + 1 \times (-1) + 1 \times (-1) + \cdots = -8$$

then divide by the square root of the lengths of the two vectors, $\sqrt{11}$ and $\sqrt{10}$ respectively. This arithmetic operation gives an alignment of -76%.

The point here is not to emphasize this particular calculation, but to show that the alignment is a straightforward statistic to compute. And, in any event, we should take into account all 773 bills when computing the alignment between two members of parliament. Table 16.1 shows the result.

Table 16.1.: Alignments among the voting patterns of four members of parliament.

	Scott	Paul	Colin	Bruce
Scott	1	-0.33	0.68	-0.31
Paul	-0.33	1	-0.39	0.85
Colin	0.68	-0.39	1	-0.37
Bruce	-0.31	0.85	-0.37	1

Notice that (of course!) each MP is *perfectly aligned* with themself. And, the alignments are symmetrical: the alignment between Paul and Scott is the same as the alignment between Scott and Paul. Such patterns tell us nothing about voting patterns. They are merely an inevitable consequence of the symmetry in the dot-product calculation.

The alignment array, however, shows some patterns. Notice that Scott and Colin are strongly aligned. Paul and Bruce are even more so. But Scott and Paul are somewhat negatively aligned, as are Colin and Bruce.

Re-arranging the names to place strongly aligned ones near each other and negatively aligned ones far away gives the following order:

Scott, **Colin**, **Bruce**, **Paul**

Table 16.2 puts Table 16.1 into this order. The re-arrangement suggests that among these four MPs there are two voting blocks, blue and green.

Table 16.2.: Re-ordering the members of parliament to put those closely aligned near each other shows two voting blocks.

	Scott	Colin	Bruce	Paul
Scott	1	0.68	-0.31	-0.33
Colin	0.68	1	-0.37	-0.39
Bruce	-0.31	-0.37	1	0.85
Paul	-0.33	-0.39	0.85	1

Another way to look at the voting data is as 773 vectors, each having 134 components. That is, we make 134 measurements on each of the 773 bills. Looking at the alignments among the 773 vectors lets us organized the bills into blocks (Figure 16.13).

Figure 16.13 shows the entire data set from Figure 16.12 with both the rows and columns re-arranged to enhance the "blockiness" of the data.

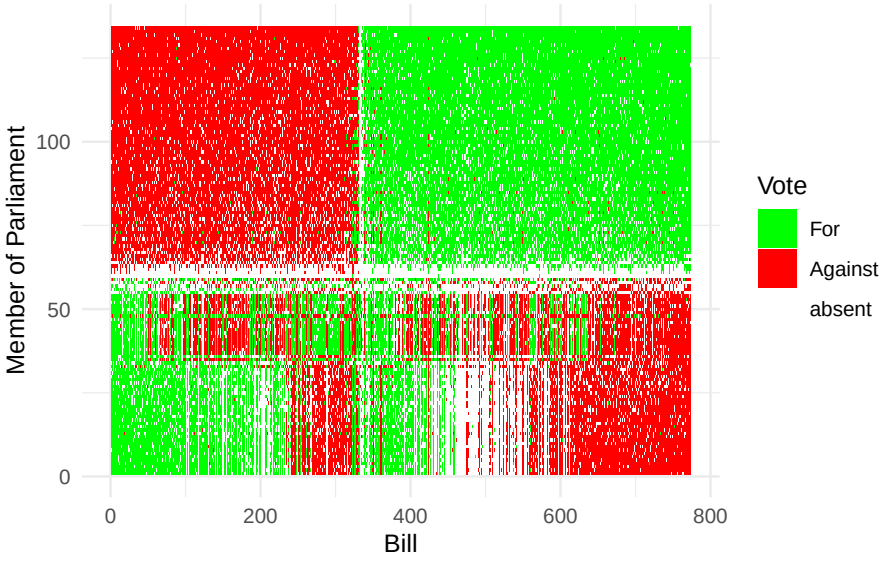


Figure 16.9.: The same data as in Figure 16.11, but the rows and the columns have been reordered so that closely aligned vectors are placed nearby along the axes.

The strong block structure in Figure 16.13 is evident. Looking across the *columns*, there are four factions. The largest, by far, includes the members numbered 60 and higher. The second largest faction consists of members numbered 1 through 36. This second faction is distinguished from the first by voting "for" bills 1-230 and against bills 580+. They also have many absences for bills 450-550.

Having identified two major factions, we can compare the voting patterns of members of parliament who are not in either faction. A third faction, contrasting with the first two, occupies 37-55. Meanwhile, a small fourth "faction," rows 56 to 59, is a mix of opinions and absences.

To facilitate further discussion, let's call the three substantial factions the "Top," "Middle," and "Bottom." It turns out, by referring to the known party registration of each MP, the Top faction consists of a mix of Scottish Labour and Scottish Liberal Democrats, the Bottom faction is the Scottish National Party, and the Middle faction is the Scottish Liberal Democrats. The party registration information, however, played no role in the construction of Figure 16.13.

With these names for the horizontal bands in Figure 16.13, we can discuss the vertical bands: the types of bills considered by Parliament. Bills 1 to 230 have the Top faction voting "for" and the Bottom faction voting "against."

Bills 231 to 325 are opposed by both factions. Bills numbered 600+ are opposed by the Bottom faction but supported by the Top faction. Bills 326 to 450 receive support from both factions.

This is a lot to find out about a session of Parliament from a set of 100,000 "for" and "against" votes. It is not, of course, everything to know about the session. For instance, although we have identified four types of bills, we would need to read the actual bills or the debate record to determine which social, economic, or administrative features characterize each type. Here, modern quantitative reasoning provides a tool that can inform other types of description and argument.

16.10. Toward AI

Starting about 2015, the quantitative methods described in this chapter were combined to create the 👉**Large Language Models**👈, which helped launch the AI revolution. These approaches include:

i. Gradient ascent for setting optimal parameters.

ii. Representing words and word contexts as vectors in high-dimensional spaces.

iii. Finding projections that collapse words and their contexts so that sets nearby geometrically are also similar semantically.

This chapter does not equip the reader to use such methods in practice. Realizing current AI capabilities required many billions in research on quantitative methods. Researchers also discovered new quantitative methods, unknown to previous generations. For example, the 2017 paper Attention is all you need made a major step toward effective AI.

Attention is All You Need

Readers who are daunted by the richness of the quantitative concepts in this book should keep in mind that mastery comes only after years of study and effort. This is true across many fields, including the humanities, medicine, the social sciences, and engineering. Even experts of one decade can be challenged by innovations in the next. For instance, this author has not yet understood the "Attention" method enough to describe it or outline its key ideas.

A. Chapter Summaries

A.1. Chapter 1: Models

Quantitative argumentation uses quantitative **models** to strengthen everyday reasoning, support decisions, and connect real-world situations with formal reasoning.

Reasoning transforms **givens**—–facts, observations, beliefs, assumptions, or knowledge—–into new beliefs or conclusions. **Heuristic reasoning** is broadly useful but not guaranteed; **formal reasoning** is reliable when its rules are applied correctly.

A model is a **representation** of an object, process, or setting. Models serve as bridges: heuristic reasoning helps *construct* a model, which then becomes the **givens** used for formal reasoning. .

Models are not intended to match the endless complexities of the real world. Instead, a good model captures essential features of the world in a way that is useful for reasoning and decision-making. **Modeling** is the iterative *process* of creating, testing, and refining models to improve their usefulness.

The example of the **SIR epidemic model** illustrates how a verbal account of a system can be translated into quantitative bookkeeping that supports simulation and the exploration of possible interventions.

A.2. Chapter 2: Systems

A **system** is a structure of interconnected elements. In this book's account of modeling, each element is a feature or **dimension** of the system. In a quantitative model, dimensions are represented by **quantities**.

Chapter 3 will introduce another, closely related meaning for "dimension" in the context of quantities

Operationalization is the procedure used to translate a dimension into a measurable quantity. Operationalizations do not necessarily capture all

aspects of a system, they can be a compromise between the modeler's mental image of a dimension (example: "comfort") and the practicalities of measurement. Modelers should check that a dimension's operationalization adequately represents the intended aspect of the system to suit the purpose intended for the model.

A **value** is the specific numerical level of a quantity, together with its appropriate unit. A changing value does not necessarily mean that the dimension or its operationalization has changed. For instance, the temperature of a room may change from 20°C to 22°C, but the dimension (temperature) and its operationalization (measured in °C) remain the same.

Data are recorded facts, findings, or measurements. A **data frame** organizes such data into rows and columns: each row represents a specimen of a common object type, each column represents a system dimension, and each cell contains a value.

A **sample** is a collection of specimens drawn from a larger population; **relationships observed in data** are collective properties inferred by comparing multiple specimens to one another.

A.3. Chapter 3: Quantities

A **quantity** combines a pure number with **units** and, optionally, a description of what is being measured.

Dimensions describe the kind of quantity being measured; **units** are standardized ways of measuring it. Multiplying or dividing base units produces derived units, while **dimensional analysis** tracks the corresponding relationships among dimensions.

Dimensional analysis was invented originally for work in science, especially physics and engineering. But it is useful for quantitative work generally. For such general work, this book adds the dimensions "worth" [W] and "people" [C] (for "capita") to the standard scientific base dimensions of length [L], mass [M], time [T], and thermodynamic temperature [Θ]. The square brackets, as in [T], can be read "the dimension of …." (Other scientific dimensions—electric current [I], amount of substance [N], and luminous intensity [J]—are not used in this book.) We will also consider the dimension of pure numbers [1] as discussed below.

A **pure number** is a number without any associated units. It is useful to consider pure numbers as having their own dimension, denoted [1] in the bracket-format used in the chapter. This allows pure numbers to be treated consistently within the general dimensional analysis framework for quantities. Using the word "dimensionless" to refer to the dimension [1] is standard.

Quantities can be added, subtracted, or compared only when they are commensurate, meaning they have the same dimension. Unit conversion puts commensurate quantities into a common unit; a **conversion factor**, or "flavor of one," is a dimensionless ratio of equal quantities whose units show the direction of conversion.

A *quantitative change* is the to value minus the from value, so it can be positive or negative. It compares a primary quantity across a context-setting quantity (such as time), expressed as change *with respect to* that context. The dimension of a quantitative change is that of the primary quantity. . A **partial change** examines one context-setting quantity while holding others constant.

"Rates of change" (Chapter 4) are different. Their dimension reflects both the primary quantity and the context-setting quantity.

A **space** represents a collection of quantity types; a set of possible values. The space for a single quantity is a number line. Two quantities together form a coordinate plane. Larger collections form higher-dimensional spaces. Each point in the space a value for each of the quantities that make up the space.

A.4. Chapter 4: Rates

A **rate** consists of one quantity divided by another and is an important way of describing the relationship between two **dimensions**. The division is signified by the word **per**.

"Dimensions" here refers to attributes of systems.

Being a quantity, a rate has units and is of the corresponding dimension.

"Dimension" here refers to the type of quantity the units measure.

A **conversion factor**, or flavor of one, is a dimensionless ratio of equal quantities. But even though a conversion factor is dimensionless, it has units. The units determine the direction of conversion: to convert from one unit to another, use units of to/from.

A **rate of change** compares a *change* in a primary quantity with the corresponding *change* in a context-setting quantity. It is a rate, so its units identify both what is changing and what that change **with respect to**.

A.5. Chapter 5: Magnitudes

Order of magnitude provides an alternative way of representing a number that is well suited when comparing numbers that differ by large factors. Magnitude describes the size of a number by focusing on the position of its first nonzero digit relative to the decimal point. Numbers that differ by factors of ten differ by one order of magnitude; numbers that differ by factors of 100 differ by two orders of magnitude.

Unit prefixes and scientific notation make magnitude easier to see without counting zeros. In scientific notation, the mantissa gives the digits and the exponent gives the approximate magnitude.

A logarithm provides a more precise, single-number representation of magnitude by incorporating both the mantissa and exponent. This book uses base-10 logarithms because they correspond directly to decimal scientific notation.

Logarithms apply only to pure numbers. Before comparing quantities with logarithms, convert them to the same unit; the units do not belong inside the logarithm.

Magnitude-based comparisons and graphics are useful when quantities span large ranges and the relevant comparison is a ratio. For quantities that differ by small amounts, such as ordinary changes in temperature, height, or blood pressure, direct differences are often more informative.

A.6. Chapter 6: Functions

Recall that systems are collections of interconnected elements. A **function** is an important way to represent such a connection. A function takes as input *one or more* quantities and produces as output a *single* quantity. (By using multiple functions, we can represent more complex relationships and interactions within a system.)

The process by which a function turns the inputs into its output is called an **algorithm**. The several functions on calculators correspond to algorithms; you invoke the algorithm with a button press. The computer programming notation to invoke a function is different: the **function name** followed by parentheses containing the values for inputs, for example `sultriness(temperature=28, humidity=0.75)`.

When a modeler creates a new function, she does so by putting together existing functions. (See, e.g., Chapter 9 for more on building complex functions from simpler ones, such as the core modeling functions introduced in Chapter 7.)

A.7. Chapter 7: Core modeling functions

As a set of basic building blocks for constructing more complex functions, this book selects eight **core modeling functions**. Each core modeling function gives a name to a simple functional pattern—a **shape** that relates a single input to the output.

Each core modeling function is like a simple plot in a collection of stories. Putting those plots together creates a richer narrative that captures more complex relationships within a system.

Every core modeling function takes a single pure-number input and produces a single pure-number output. We use **input scaling** to translate a dimensioned quantity into a pure number suitable for the core function. Similarly, **output scaling** translates the pure-number output back into a quantity with meaningful units.

that provide simple, reusable patterns that capture common types of relationships in quantitative models. A core modeling function is a named, reusable relationship that transforms one pure-number input into one pure-number output. The chapter provides a small library of eight simple functions whose shapes commonly occur in quantitative models.

The graph of a core function shows how its output changes as its input changes. The input and output are read from the grid, and the function's shape provides a qualitative story about the relationship: constant, steadily increasing, oscillating, rising and falling, or changing at an increasing or decreasing rate.

The eight functions are same(), flat(), hill(), hillside(), exp2(), log2(), recip(), and osc(). Their names help communicate the relationship they represent and make the functions easier to use in computer implementations.

Input and output scaling translates the pure-number input and output of a core function into quantities with meaningful units. Shifting and stretching the axes preserves the function's shape while adapting it to a particular modeling situation.

A modeler can use a core function as a starting pattern, then scale and interpret it for a real system. The chapter illustrates this process by modeling global oil production with exponential growth and by using familiar temperature stories to interpret each function's shape.

A.8. Chapter 8: Uncertainty

Often, a quantitative value is uncertain. Such uncertainty is an important feature of the quantity and needs to be represented formally. A simple way to communicate the amount of uncertainty is with a **range**: a lower and upper value.

A **probability distribution** is a richer and more reliable way to represent uncertainty. There are differently shaped probability distributions to reflect different patterns of uncertainty. A probability distribution is a function whose input is a possible value and whose output indicates how likely that value is relative to other possibilities.

The chapter introduces three useful **distribution families**: the uniform distribution for equally possible values within a known range, the normal distribution for values concentrated around a mean with decreasing likelihood farther away, and the exponential distribution for waiting times or other quantities with greater uncertainty toward the high end. (There are many more families used by modelers. This chapter focuses on these three.) One selects a specific probability distribution from a family by choosing appropriate **parameters** that specify its shape and scale.

Simulation is a useful computational technique that propagates uncertainty through a a system by randomly sampling inputs from their probability distributions, applying the system functions, and examining the resulting collection of outputs.

A.9. Chapter 9: Composing functions

Just as a music composer can put together individual notes to form a melody, a modeler can put together simple functions (such as the core modeling functions) to **compose** a more complex representation of a system.

The chapter emphasizes two main ways of composing functions: **multiplication**, which captures interactions between quantities, and **linear**

combinations, which capture additive effects of multiple influences. There are many other ways to compose new functions, which we don't consider in any detail.

A **linear combination** of functions adds scaled functions. The scaling is accomplished by simple multiplication of the function with a number, called a **coefficient**. In general, coefficients are conversion factors that translate the units of the original function's output into the desired units for the combined function.

Linear regression uses **data** to select good coefficients in a linear combination. The coefficients selected in this way are called **regression coefficients**.

Regretably, conventional introductory statistics courses cover only one simple aspect of regression (calling it "simple regression") which amounts to what we are calling *output scaling* of a single function. But combining multiple functions, especially functions with different inputs, is a core quantitative technique throughout the social and natural sciences called "multiple regression." Multiple regression is a foundation for understanding complex systems through data. One use of multiple regression is to model *partial change* (Chapter 3) and is associated with phrases like "after adjusting for ..." or "controlling for ..." or "holding other variables constant."

A.10. Chapter 10: Tradeoffs

A **tradeoff** occurs when a decision-maker must balance competing **objectives**. **Optimization** provides a way to compare possible choices using an **objective function**, whose input is a possible choice and whose output measures the consequence of that choice in terms of the objective.

The **argmax** is the *input* that produces the maximum output of an objective function; the **argmin** produces the minimum output. A single objective function can identify a mathematically best choice, but that choice may be poor if the function omits other goals that matter.

When several objectives conflict, their outputs may have *different dimensions* and cannot simply be added. With multiple objectives, each policy choice produces an output from each objective function. Those several output values can be plotted as a point in the *objective space*; a space with an axis for each output objective. A graph of the objective outputs can identify choices that are dominated, meaning another choice is at least as good

on every objective and better on at least one. Dominated choices can be eliminated; the remaining choices represent genuine tradeoffs.

A **shadow price** quantifies the exchange between competing objectives. It expresses how much of one objective must be given up to gain more of another, helping decision-makers clarify whether a proposed trade is worthwhile within real-world resource constraints.

A.11. Chapter 11: Likelihood

A **hypothesis** is a claim about something, that might be off target to a greater or a lessor degree. The modeler's understanding of how a system works, together with relevant data, inspire the modeler to formulate a hypothesis about how a system works, and a model is a formal implementation of such a hypothesis.

Likelihood is a quantitative measure of how well a hypothesis accounts for observed data. It is constructed by composing a probability distribution function corresponding to uncertainties in a model. The output of that probability distribution is a probability. The input to the function is the entire set of observations/data.

Technically, this may be a probability **density**. This book sidesteps that technicality When a hypothesis asswith the concept of *relative probability*.

Likelihoods are meaningful only in comparison to each other. The simplest comparison concerns competing hypothetical hypotheses for a *parameter* or *set of parameters*. Each parameter value corresponds to a specific probability distribution. The same input—the entire set of observations/data—is used as the input to the probability distribution function for each of the parameter values, one at a time. A **likelihood function** maps each parameter value to the likelihood of the observed data under that parameter value.

A **likelihood ratio** compares the support for one hypothesis to the support for another. If the likelihood ratio is greater than one, the hypothesis in the numerator of the ratio is preferred to the hypothesis in the denominator. However, likelihood ratios should be much greater than 1 to constitute any evidence at all for the preference. Likelihood ratios greater than 1000 provide strong evidence. A **likelihood interval** (much the same as a "confidence interval" in classical statistics) describes a range of parameter values that are well-supported by the observed data.

The likelihood depends on the amount of observations/data available. Collecting more data generally narrows the interval—quadrupling the data

roughly halves the width of the interval—though it need not center the new estimate at the old one.

A.12. Chapter 12: Bayes

Bayesian reasoning provides a way to translate a *likelihood function* into a probability distribution that takes into account other information, such as **prior** knowledge or beliefs, that is outside the observations/data used when evaluating the likelihood function. The resulting probability distribution is called a **posterior** probability distribution for the set of hypotheses under consideration (such as the possible values for a parameter).

The **prior** is a probability distribution function whose input domain is the set of hypotheses under consideration. The prior assigns a relative probability to each hypothesis, reflecting the degree of belief in that hypothesis before observing the data. The **likelihood** function reflects only the observations/data, not the prior belief.

Multiplying the **prior** function by the *likelihood* function produces the *posterior* function, which is the updated state of belief *after* taking into account the observed data.

There can be reasonable disagreement about the best choice of a prior, just as there can be reasonable disagreement about beliefs in general. A sensible analysis considers the extent to which the posterior is influenced by the specific prior selected. The more the observations/data, the less the influence of the prior; evidence overcomes prior belief.

At any point, a prior or a posterior can be multiplied by a number without changing their meaning. A favored normalization is to select the number such that the resulting distribution sums (or integrates) to one, ensuring it is an absolute (not relative) probability distribution. (See Chapter 15).

A.13. Chapter 13: Dynamics

"Change" will be the main theme of Chapters 13, 14, and 15. Up to now, we have emphasized "with respect to" change, for instance a system like A -> B where a change in A produces a change in B. Here, A is an *exogenous quantity* whose change is determined outside the system, while B is an *endogenous quantity* whose change is determined by the system.

Another important perspective on change comes from a system where the connections between elements form a loop. A simplistic example is A -> B -> A. This is simplistic because it doesn't specify which comes first, A or B: A chicken and egg problem.

We can avoid this ambiguity by inserting an intermediary in the form of a system element describing the "change in A with respect to time." For example:

A -> B -> 'change in A with respect to time'

This is called a **dynamical system** and turns the exogenous quantity A in A -> B into a quantity whose change is determined within the system.

The **Euler framework** represents such a system with three components: the state, a dynamical rule that gives the state's instantaneous rates of change, and an accounting system that accumulates those changes into a time-based trajectory.

A new graphics representation helps visualize the situation. The state quantities themselves (A and B in the above) are represented by a **state space** which has a dimension for each state quantity. The instantaneous state is represented as a point in this space. The change of state (the "change in A with respect to time" in the A -> B example) is represented as a vector originating from that point. The length of the vector indicates the speed of change, what we call the **instantaneous rate of change**.

A model of a dynamical system assigns a vector to each point in the state space; that vector is a function of the position of the point, that is, the values of A and B. This collection of vectors is called the **dynamical rule** or, informally, the "flow field."

Following the flow produces a **trajectory**, the sequence of values taken up by the state quantities as they change with respect to time. A trajectory may be curved even though each individual vector is straight.

Exponential growth or decay arises from system where the change in the state is proportional to the current state, for instance, A -> 'change in A with respect to time'. Perhaps surprisingly, such simple systems with multiple state variables can exhibit not just exponential behavior but also oscillations.

In a finite world, exponential growth cannot continue indefinitely. This truth emphasizes that models are always approximations which may not include all factors that eventually become relevant.

A.14. Chapter 14: Rates of change

A **rate of change** compares simultaneous changes in two quantities, for instance, how B changes with respect to A.

It turns out to be useful to consider **instantaneous** rates of change, which looks at how B changes with respect to a very small change in A. "Very small" can be treated as common sense; a change that is barely noticeable. There is also a mathematical abstraction which formalizes the idea of an infinitesimally smallchange change in A and its corresponding infinitesimally small change in B. Mathematics courses such as calculus tend to emphasize the abstraction, but for practical purposes the "barely noticeable" change is often sufficient.

The system A -> B can also be described as a *function* B(A), where A is the input and B is the output. The **derivative** of a function is a *new function* that gives the original function's instantaneous rate of change. We might name this function change_in_B(A), but there are a variety of mathematical notations for it such as $\partial_A B(A)$ where ∂_A means "change with respect to A of ...." **Differentiation** is the process of calculating a derivative, and **calculus** is the broader field that studies this relationship. The derivative has the same input as the original function but a different output and dimension.

Calculus students will recognize $\frac{dB}{dA}$ or $\frac{\partial B}{\partial A}$ which emphasizes that the derivative is a *rate* of change, but obscures that the derivative is a function of the with-respect-to input.

In economics, business, and most other disciplines outside of math and physics, the term **marginal rate** is used instead of "instantaneous rate of change" but has the same meaning of purtaining to a small, incremental change. Marginal tax rates, for example, describe the additional tax owed on a small *increase* in income rather than the average tax owed on all income.

Tradeoff situations, with two objective functions, have two rates of change with respect to the input policy quantity. The ratio of these two rates of change is effectively the rate of change of one objective output with respect to the other objective output. This can be interpreted as a *shadow price*. An optimization example about health care in the chapter seeks a balance between two health programs at a point where the marginal benefit per dollar spent is equal across the programs.

A.15. Chapter 15: Accumulation

Accumulation is the process of finding a total from a rate of change over an interval. When the rate is constant, accumulation is calculated by multiplying rate by duration; when the rate changes, the same idea is carried out by **integration**, or **anti-differentiation**.

An accumulation function and its rate-of-change function contain the same information in different forms. The rate-of-change function is the **derivative** of the accumulation function, while the accumulation function is the anti-derivative of the rate. An initial value is needed to determine the vertical position of the accumulated function.

The relationship between marginal and total quantities can make models easier to construct and interpret. For example, a marginal tax-rate function can be accumulated to obtain total tax owed. Similarly, modeling QALYs saved per additional dollar can be easier to justify than directly modeling total QALYs saved; accumulation then produces the total-benefit function.

Accumulation is also at the core of the accounting system used in the *Euler framework* for dynamical systems.

The chapter illustrates this framework with population data. Rather than extrapolating the population time series directly, modelers can estimate how the population's rate of change varies with population or time, then accumulate that rate to forecast future population. This approach makes the model's assumptions visible and provides a basis for checking whether predicted growth or decline is plausible.

A.16. Chapter 16: Toward higher dimensions

Many quantitative problems involve finding inputs that produce a desired output or maximize an objective. When the function cannot be graphed or described algebraically, a modeler can estimate local rates of change and **walk uphill** by taking small steps in a direction that increases the output.

For a function with several inputs, the **gradient** is a vector made from the rates of change with respect to each input. It points in the direction of steepest ascent. Repeatedly evaluating the gradient and taking small steps

produces a practical **gradient ascent** algorithm for optimization, even when the function has thousands or millions of inputs.

Vectors provide a way to represent points, directions, and data in spaces with any number of dimensions. The **dot product** can be used to calculate vector length as well as the angle of **alignment** between two vectors. Alignment measures similarity: positive alignment indicates similar direction, zero indicates no shared direction, and negative alignment indicates opposing direction. AI systems use such alignment to quantify similarlity of meaning, allowing them to exploit patterns of meaning expressed in different terms.

A **subspace** is a lower-dimensional structure within a larger space. **Projection** represents a vector by the closest vector in the subspace, separating the part of the data captured by a pattern from the remaining residual. These operations provide the mathematical foundation for statistical modeling and for finding patterns in high-dimensional data.

The chapter illustrates **supervised learning** by using labeled penguin measurements to identify species and **unsupervised learning** by finding voting patterns without using party labels or bill descriptions. Reordering and projecting high-dimensional voting data reveals factions and types of bills, showing how quantitative methods can discover patterns that require other forms of reasoning to interpret.